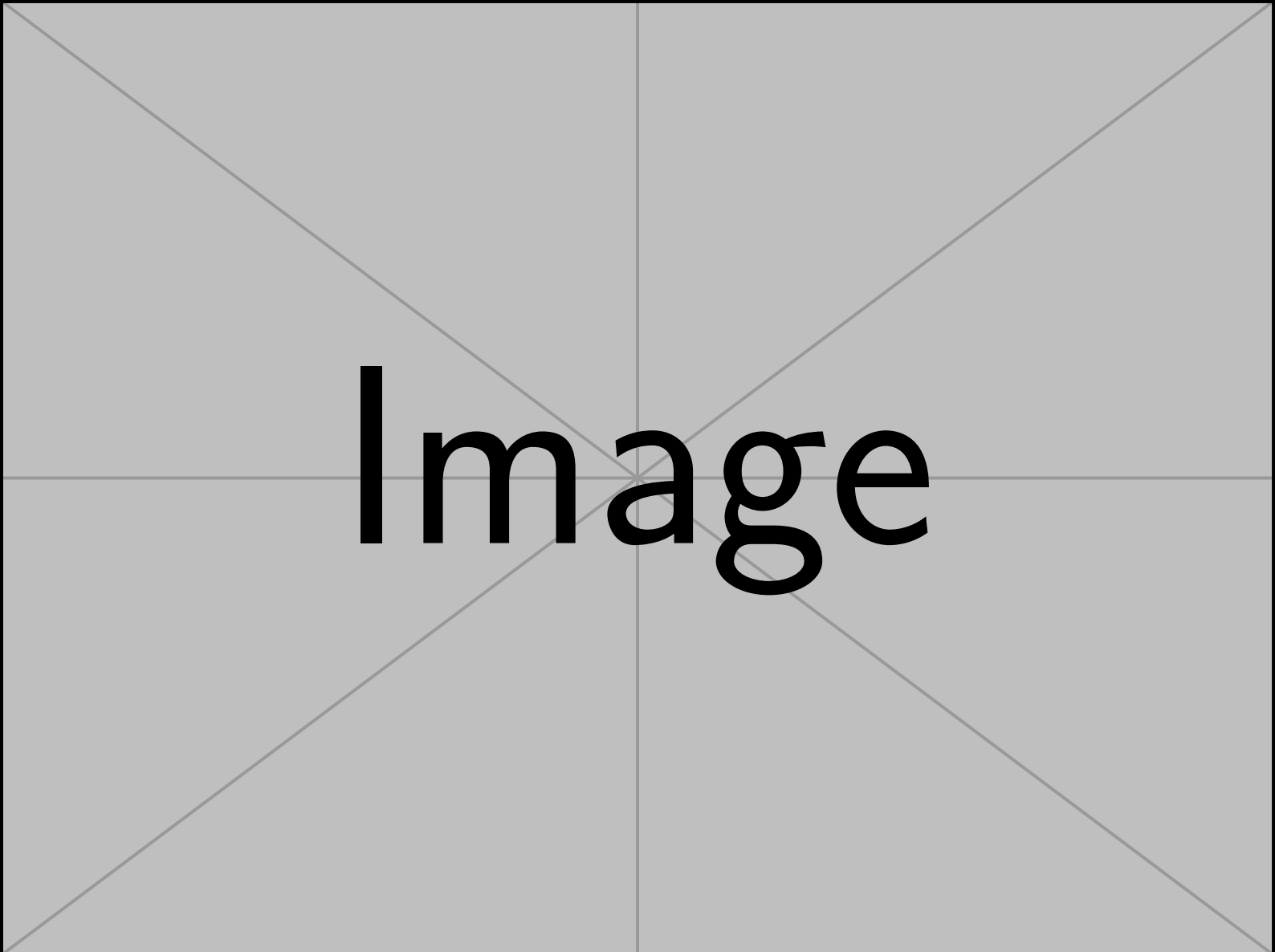




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Abel's Theorem in Problems and Solutions

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Preface to the English Edition by V. I. Arnold

Abel's Theorem, claiming that there exists no finite combination of radicals and rational functions solving the generic algebraic equation of degree 5 (or higher than 5), is one of the first and the most important impossibility results in mathematics.

I had given to Moscow High School children in 1963–1964 a (half-year-long) course of lectures containing the topological proof of Abel's theorem.

Starting from the definition of complex numbers and from geometry, the students were led to Riemannian surfaces in a sequence of elementary problems. Next came the basic topological notions, such as the fundamental group, coverings, ramified coverings, their monodromies, braids, etc.

These geometrical and topological studies implied such elementary general notions as transformation groups and group homomorphisms, kernels, exact sequences, and relativistic ideas. The normal subgroups appeared as those subgroups which are relativistically invariant, that is, do not depend on the choice of the coordinate frame, represented in this case as a numbering or labelling of the group elements.

The regular polyhedra symmetry groups, seen from this point of view, had led the pupils to the five Kepler cubes inscribed into the dodecahedron. The 12 edges of each of these cubes are the diagonals of the 12 faces of the dodecahedron.

Kepler had invented these cubes in his *Harmonia Mundi* to describe the distances of the planets from the Sun. I had used them to obtain the natural isomorphism between the dodecahedron rotation group and the group of the 60 even permutations of 5 elements (the Kepler cubes). This elementary theory of regular polyhedra provides the non-solubility proof of the permutation group on 5 elements: it cannot be constructed from commutative groups by a finite sequence of extensions with commutative kernels.

The situation is quite different for the permutation groups of fewer than five elements, which are soluble (and responsible for the solvability of equations of degree smaller than 5). This solubility depends on the inscription of two tetrahedra inside the cube (similar to the inscription of the 5 Kepler cubes inside the dodecahedron and mentioned also by Kepler).

The absence of non-trivial relativistically invariant symmetry subgroups of the group of rotations of the dodecahedron is an easy result of elementary geometry. Combining these high-school geometry arguments with the preceding topological study of the monodromies of ramified coverings, one immediately obtains the topological proof of Abel's Theorem: the monodromy group of any finite combination of radicals is soluble, since radical monodromy is a cyclical commutative group, whilst the monodromy of the algebraic function $x(a)$ defined by the quintic equation

$$x^5 + ax + 1 = 0$$

is the non-soluble group of the 120 permutations of the 5 roots.

This theory provides more than Abel's Theorem. It shows that the insolvability argument is topological. Namely, no function having the same topological branching type as $x(a)$ is representable as a finite combination of rational functions and radicals.

I hope that my topological proof of this generalized Abel's Theorem opens the way to many topological

insolvability results. For instance, one should prove the impossibility of representing generic abelian integrals of genus higher than zero as functions topologically equivalent to elementary functions.

I attributed to Abel the statements that neither generic elliptic integrals nor generic elliptic functions (which are inverse functions of these integrals) are topologically equivalent to any elementary function.

I thought that Abel was already aware of these topological results and that their absence in the published papers was, rather, owed to the underestimation of his great works by the Paris Academy of Sciences (where his manuscript had been either lost or hidden by Cauchy).

My 1964 lectures had been published in 1976 by one of the pupils in the high-school audience, V. B. Alekseev. He has somewhere algebraicized my geometrical lectures.

Some of the topological ideas of my course had been developed by A. G. Khovanskii, who had thus proved some new results on the insolvability of differential equations. Unfortunately, topological insolvability proofs are still missing in his theory (as well as in Poincaré's theory of the absence of a holomorphic first integral and in many other insolvability problems in the theory of differential equations).

I hope that the description of these ideas in the present translation of Alekseev's book will help the English-reading audience to participate in the development of this new topological insolvability theory, started with the topological proof of Abel's Theorem and involving, say, the topologically non-elementary nature of the abelian integrals as well as the topological non-equivalence to the integral combinations of complicated differential-equation solutions.

The combinatorial study of the Kepler cubes, used in the proof of Abel's theorem, is also the starting point of the development of the theory of finite groups. For instance, the five Kepler cubes depend on the 5 Hamilton subgroups of the projective version $\mathrm{PSL}_2(\mathbb{Z}_5)$ of the group of matrices of order 2 whose elements are residues modulo 5.

A Hamilton subgroup consists of 8 elements and is isomorphic to the group $\{\pm 1, \pm i, \pm j, \pm k\}$ of quaternionic units.

The peculiar geometry of finite groups includes their squaring monads, which are oriented graphs whose vertices are the group elements and whose edges connect every element directly to its square.

The $\mathrm{PSL}_2(\mathbb{Z}_5)$ monad theory leads to unexpected Riemann surfaces (including the monads as subgraphs), relating Kepler's cubes to the peculiarities of the geometry of elliptic curves.

The $\mathrm{PSL}_2(\mathbb{Z}_7)$ extension of the Hamilton subgroups and of Kepler's cubes leads to the extended four-colour problem (for the genus-one toroidal surface of an elliptic curve), the 14 Hamilton subgroups providing the proof of the necessity of 7 colours for the regular colouring of maps of a toroidal surface.

I hope that these recent theories will be developed further by the readers of this book.

V. Arnold

Preface

In high school, algebraic equations in one unknown of first and second degree are studied in detail. One learns that for solving these equations there exist general formulae expressing their roots in terms of the coefficients by means of arithmetic operations and radicals. But very few students know whether similar formulae exist for solving algebraic equations of higher order. In fact, such formulae also exist for equations of the third and fourth degree. We shall illustrate the methods for solving these equations in the introduction. Nevertheless, if one considers the generic equation in one unknown of degree higher than four, one finds that it is not solvable by radicals: there exist no formulae expressing the roots of these equations in terms of their coefficients by means of arithmetic operations and radicals. This is exactly the statement of Abel's theorem.

One of the aims of this book is to make this theorem known. Here we will not consider in detail the results obtained a bit later by the French mathematician Évariste Galois. He considered some special algebraic equations, i.e., equations having particular numbers as coefficients, and for these equations found the conditions under which the roots are representable in terms of the coefficients by means of algebraic operations and radicals¹.

From the general Galois results one can, in particular, also deduce Abel's theorem. But in this book we proceed in the opposite direction: this will allow the reader to learn two very important branches of modern mathematics, group theory and the theory of functions of one complex variable. The reader will be told what a group (in mathematics) is, what a field is, and which properties they possess. He will also learn what the complex numbers are and why they are defined in such a manner and not otherwise. He will learn what a Riemann surface is and what the "basic theorem of the algebra of complex numbers" consists of.

The author will accompany the reader along this path, but he will also give him the possibility of testing his own powers. For this purpose he will propose to the reader a large number of problems. The problems are posed directly within the text, and so represent an essential part of it. The problems are labelled by increasing numbers in bold figures. Whenever a problem might be too difficult for the reader, the chapter "Hints, Solutions, and Answers" will help him.

The book contains many notions which may be new to the reader. To help him orient himself among these new notions, we put at the end of the book an alphabetical index of notions, indicating the pages where their definitions are to be found.

The proof of Abel's theorem presented in this book was given by Professor Vladimir Igorevich Arnold during his lectures to the students of the 11th course of the physics-mathematics school of Moscow State University in the years 1963–64. The author of this book, who at that time was one of the pupils of that class, organized during the years 1970–71 a special seminar for the pupils of that school dedicated to the proof of Abel's theorem. This book consists of the material collected during these activities. The author is very grateful to V. I. Arnold for having made a series of important remarks during the editing of the manuscript.

V. B. Alekseev

Introduction

We begin this book by examining the problem of solving algebraic equations in one variable from the first to the fourth degree. Methods for solving equations of first and second degree were already known to the ancient mathematicians, whereas methods for solving algebraic equations of third and fourth degree were invented only in the sixteenth century.

An equation of the type

$$a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n = 0,$$

in which $a_0 \neq 0$, is called the generic algebraic equation of degree n in one variable.

For $n = 1$ we obtain the linear equation

$$a_0x + a_1 = 0.$$

This equation has the unique solution

$$x = -\frac{a_1}{a_0}$$

for any value of the coefficients.

For $n = 2$ we obtain the quadratic equation

$$ax^2 + bx + c = 0, \quad a \neq 0$$

(in place of a_0, a_1, a_2 we write a, b, c , as learnt in school). Dividing both members of this equation by a and putting $p = b/a$ and $q = c/a$, we obtain the reduced equation

$$x^2 + px + q = 0. \tag{1}$$

After some transformations we obtain

$$x^2 + px + \frac{p^2}{4} = \frac{p^2}{4} - q, \quad \text{and} \quad \left(x + \frac{p}{2}\right)^2 = \frac{p^2}{4} - q. \tag{2}$$

In high school one considers only the case $\frac{1}{4}p^2 - q \geq 0$. Indeed, if $\frac{1}{4}p^2 - q < 0$, then one says that Eq. (1) cannot be satisfied and that Eq. (2) has no real roots. In order to avoid these exclusions, in what follows we shall not restrict ourselves to algebraic equations over the field of real numbers, but will consider them over the wider field of complex numbers.

We shall examine complex numbers in greater detail (together with their definition) in Chapter 2. In the meantime it is sufficient for the reader to know, or to accept as true, the following propositions about complex numbers:

1. The set of complex numbers is an extension of the set of real numbers, i.e., the real numbers are contained in the complex numbers, just as, for example, the integer numbers are contained in the real numbers.
2. Complex numbers may be added, subtracted, multiplied, divided, and raised to a natural power; moreover, all these operations possess all the basic properties of the corresponding operations on real numbers.
3. If z is a complex number different from zero, and n is a natural number, then there exist exactly n roots of n th degree of z , i.e., n complex numbers w such that $w^n = z$. For $z = 0$ we have $\sqrt[n]{z} = 0$. If w_1 and w_2 are square roots of the number z , then $w_1 = -w_2$.

In the following we shall be interested not only in complex roots of equations as well as real ones, but will

also consider arbitrary complex numbers as coefficients of these equations. Hence, the arguments previously expounded about linear and quadratic equations remain true by virtue of property 2 of complex numbers.

Let us continue to study the quadratic equation. In the field of complex numbers, for any values of p and q , Eq. (2) is equivalent to

$$x + \frac{p}{2} = \pm \sqrt{\frac{p^2}{4} - q},$$

where by $\sqrt{\frac{1}{4}p^2 - q}$ is indicated whichever of the defined values of the square root. In so doing,

$$x_{1,2} = -\frac{p}{2} \pm \sqrt{\frac{p^2}{4} - q}. \quad (3)$$

Going back to the coefficients a, b, c , we obtain

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \quad (4)$$

For what follows we need to recall two properties related to equations of second degree.

1. Viète's Theorem²: The complex numbers x_1 and x_2 are the roots of the equation $x^2 + px + q = 0$ if and only if $x_1 + x_2 = -p$ and $x_1x_2 = q$. Indeed, if x_1 and x_2 are roots of the equation $x^2 + px + q = 0$, then Eq. (3) is satisfied, from which $x_1 + x_2 = -p$ and $x_1x_2 = q$. Conversely, if $x_1 + x_2 = -p$ and $x_1x_2 = q$, then, substituting p and q in the equation $x^2 + px + q = 0$ by their expressions in terms of x_1 and x_2 , we obtain

$$x^2 - (x_1 + x_2)x + x_1x_2 = (x - x_1)(x - x_2) = 0,$$

and therefore x_1 and x_2 are roots of the equation $x^2 + px + q = 0$.

2. The quadratic trinomial $ax^2 + bx + c$ is a perfect square, i.e.,

$$ax^2 + bx + c = [\sqrt{a}(x - x_0)]^2$$

for some complex number x_0 , if and only if the roots of the equation $ax^2 + bx + c = 0$ coincide (they must both be equal to x_0). This happens if and only if $b^2 - 4ac = 0$ (see formula (4)). The expression $b^2 - 4ac$ is called the discriminant of the quadratic trinomial.

We now consider the reduced cubic equation

$$x^3 + ax^2 + bx + c = 0. \quad (5)$$

The generic equation of third degree is reduced to Eq. (5) by dividing by a_0 . After the substitution $x = y + d$ (where d will be chosen later) we obtain

$$(y + d)^3 + a(y + d)^2 + b(y + d) + c = 0.$$

Removing the brackets and collecting terms of the same degree in y , we obtain the equation

$$y^3 + (3d + a)y^2 + (3d^2 + 2ad + b)y + (d^3 + ad^2 + bd + c) = 0.$$

The coefficient of y^2 in this equation is equal to $3d + a$. Therefore, if we put $d = -a/3$, after substituting $x = y - a/3$ we transform the equation into

$$y^3 + py + q = 0, \quad (6)$$

where p and q are some polynomials in a, b, c .

Let y_0 be a root of Eq. (6). Representing it in the form $y_0 = \alpha + \beta$ (where α and β are temporarily

unknown), we obtain

$$\begin{aligned}\alpha^3 + 3\alpha\beta(\alpha + \beta) + \beta^3 + p(\alpha + \beta) + q &= 0, \\ \alpha^3 + \beta^3 + (\alpha + \beta)(3\alpha\beta + p) + q &= 0.\end{aligned}\tag{7}$$

We check whether it is possible to impose that α and β satisfy

$$\alpha\beta = -\frac{p}{3}.$$

In this case we obtain two equations for α and β :

$$\begin{cases} \alpha + \beta = y_0, \\ \alpha\beta = -p/3. \end{cases}$$

By Viète's theorem, for any y_0 such α and β (which may be complex) indeed exist, and they are the roots of the equation

$$w^2 - y_0w - \frac{p}{3} = 0.$$

If we take such (still unknown) α and β , then Eq. (7) is transformed into

$$\alpha^3 + \beta^3 + q = 0.\tag{8}$$

Raising both sides of the equation $\alpha\beta = -p/3$ to the third power, and comparing the obtained equation with Eq. (8), we have

$$\begin{cases} \alpha^3 + \beta^3 = -q, \\ \alpha^3\beta^3 = -p^3/27. \end{cases}$$

By Viète's theorem, α^3 and β^3 are the roots of the equation

$$w^2 + qw - \frac{p^3}{27} = 0.$$

In this way,

$$\alpha^3 = -\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}} \quad \text{and} \quad \beta^3 = -\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}},$$

where again $\sqrt{(q^2/4) + (p^3/27)}$ indicates one defined value of the square root. Hence the roots of Eq. (6) are expressed by the formula

$$y_{1,2,3} = \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}},$$

in which, for each of the three values of the first cube root⁴, one must take the corresponding value of the second in such a way that the condition $\alpha\beta = -p/3$ is satisfied.

The obtained formula is named Cardano's formula⁵. Substituting in this formula p and q by their expressions in terms of a, b, c and subtracting $a/3$, we obtain the formula for Eq. (5). After the transformations $a = a_1/a_0$, $b = a_2/a_0$, $c = a_3/a_0$, we obtain the formula for the roots of the generic equation of third degree.

Now we examine the reduced equation of fourth degree

$$x^4 + ax^3 + bx^2 + cx + d = 0.\tag{9}$$

(the generic equation is reduced to the previous one by dividing by a_0). By making the change of variable $x = y - a/4$, similarly to the change made in the case of the equation of third degree, we transform Eq. (9) into

$$y^4 + py^2 + qy + r = 0,\tag{10}$$

where p, q, r are some polynomials in a, b, c, d .

We shall solve Eq. (10) by a method called Ferrari's method⁶. We transform the left member of Eq. (10) in the following way:

$$\left(y^2 + \frac{p}{2}\right)^2 + qy + \left(r - \frac{p^2}{4}\right) = 0,$$

and

$$\left(y^2 + \frac{p}{2} + \alpha\right)^2 - \left[2\alpha\left(y^2 + \frac{p}{2}\right) + \alpha^2 - qy + \frac{p^2}{4} - r\right] = 0, \quad (11)$$

where α is an arbitrary number. We now try to determine α such that the polynomial of second degree in y

$$2\alpha y^2 - qy + \left(\alpha p + \alpha^2 + \frac{p^2}{4} - r\right)$$

within square brackets becomes a perfect square. As was noticed above, in order for it to be a perfect square it is necessary and sufficient that the discriminant of this polynomial vanish, i.e.,

$$q^2 - 8\alpha\left(\alpha p + \alpha^2 + \frac{p^2}{4} - r\right) = 0. \quad (12)$$

Eliminating the brackets, to find α we obtain an equation of third degree which we are able to solve. If in place of α we put one of the roots of Eq. (12), then the expression in the square brackets of (11) will be a perfect square. In this case the left member of Eq. (11) is a difference of squares and therefore it can be written as the product of two polynomials of second degree in y . After that it remains to solve the two equations of second degree obtained.

Hence the equation of fourth degree can always be solved. Moreover, as in the case of the third order, it is possible to obtain a formula expressing the roots of the generic equation of fourth order in terms of the coefficients of the equation by means of the operations of addition, subtraction, multiplication, division, raising to a natural power, and extracting a root of natural degree.

For a long time mathematicians tried to find a method of solution by radicals of the generic equation of fifth order. But in 1824 the Norwegian mathematician Niels Henrik Abel (1802–1829) proved the following theorem.

定理 0.1 (Abel)

The generic algebraic equation of degree higher than four is not solvable by radicals; that is, formulae do not exist for expressing roots of a generic equation of degree higher than four in terms of its coefficients by means of operations of addition, subtraction, multiplication, division, raising to a natural power, and extraction of a root of natural degree.



We will be able to prove this theorem at the end of this book. But for this we need some mathematical notions, such as those of group, soluble group, function of a complex variable, Riemann surface, etc. The reader will become familiar with all these and other mathematical instruments after reading what follows in the forthcoming pages of this book. We start by examining the notion of group, a very important notion in mathematics.

第一章 Groups

1.1 Examples

In arithmetic we have already met operations which put two given numbers in correspondence with a third. Thus, addition puts the pair $(3, 5)$ in correspondence with the number 8 and the pair $(2, 2)$ with the number 4. Also, the operation of subtraction, if considered inside the set of all integer numbers, puts every pair of integers in correspondence with an integer; in this case, however, the order of the numbers in the pair is important. Indeed, subtraction puts the pair $(5, 3)$ in correspondence with the number 2, whereas it puts the pair $(3, 5)$ in correspondence with the number -2 . The pairs $(5, 3)$ and $(3, 5)$ thus have to be considered different.

When the order of elements is specified, a pair is said to be ordered.

Definition. Let M be a set of elements of arbitrary nature. If every ordered pair of elements of M is put into correspondence with an element of M , we say that a binary operation is defined in M .

For example, addition in the set of natural numbers and subtraction in the set of integer numbers are binary operations. Subtraction is not a binary operation in the set of natural numbers because, for example, it cannot put the pair $(3, 5)$ in correspondence with any natural number.

Problem 1. Consider the following operations: (a) addition; (b) subtraction; (c) multiplication; in the following sets: (1) all even natural numbers; (2) all odd natural numbers; (3) all negative integer numbers. In which cases does one obtain a binary operation¹?

Let us consider some further examples of binary operations. We shall often return to these examples in the future.

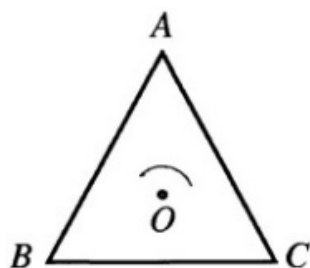


图 1.1

Example 1. Let A , B , and C be the vertices of an equilateral triangle (Figure 1). We rotate the triangle by an angle of 120° around its centre O in the direction shown by the arrow. Then vertex A goes to vertex B , B to C , and C to A . In this way the final triangle coincides with the initial triangle (if we neglect the labels of the vertices). We say that the rotation by 120° around the point O is a transformation which sends the triangle into itself. We denote this transformation by a . We can write it in the form

$$a = \begin{pmatrix} A & B & C \\ B & C & A \end{pmatrix},$$

where the first row contains all vertices of the triangle, and the second row indicates where each vertex is sent. A rotation by 240° in the same direction around the point O is also a transformation sending the triangle into itself. Denote this transformation by b :

$$b = \begin{pmatrix} A & B & C \\ C & A & B \end{pmatrix}.$$

There still exists one transformation sending the triangle into itself which is different from a and b : it is the rotation by 0° . We denote it by e ; thus

$$e = \begin{pmatrix} A & B & C \\ A & B & C \end{pmatrix}.$$

It is easy to see that there are only three different rotations of the plane² transforming an equilateral triangle ABC into itself, namely e, a, b .

Let g_1 and g_2 be two arbitrary transformations of the triangle. We denote by $g_1 \cdot g_2$ (or simply $g_1 g_2$) the transformation g_3 obtained by carrying out first the transformation g_2 and later the transformation g_1 ; g_3 is called the product or composition of the transformations g_2 and g_1 .

It is possible to make a multiplication table (Table 1) where every row, as well as every column, corresponds to some rotation transforming the triangle ABC into itself. We put the transformation corresponding to $g_1 \cdot g_2$ at the intersection of the row corresponding to the transformation g_1 and the column corresponding to the transformation g_2 . So, for example, in the selected cell of Table 1 we have to put the transformation $a \cdot b$, which is obtained by first rotating the triangle by 240° and later by 120° more. Hence $a \cdot b$ is a rotation by 360° , i.e., it coincides with e . We obtain the same result by the following reasoning: transformation b sends vertex A onto vertex C , and a later sends C onto A . In this way the transformation $a \cdot b$ sends A onto A . In exactly the same way we obtain that B is sent onto B , and C onto C . Hence

$$ab = \begin{pmatrix} A & B & C \\ A & B & C \end{pmatrix},$$

i.e., $ab = e$.

表 1.1

	e	a	b
e			
a			e
b			

Problem 2. Complete Table 1.

Any transformation of some geometrical figure into itself which maintains the distances between all its points is called a symmetry of the given figure. Thus the rotations of the equilateral triangle considered in Example 1 are symmetries of it.

Example 2. Besides rotations, the equilateral triangle possesses 3 further symmetries, namely the reflections with respect to the axes l_1, l_2, l_3 (Figure 2). We denote these transformations by c, d, f , so that

$$c = \begin{pmatrix} A & B & C \\ A & C & B \end{pmatrix}, \quad d = \begin{pmatrix} A & B & C \\ C & B & A \end{pmatrix}, \quad f = \begin{pmatrix} A & B & C \\ B & A & C \end{pmatrix}.$$

Here it is possible to imagine the composition of two transformations in two different ways.

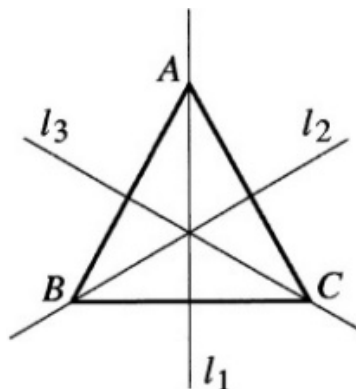


图 1.2

Consider, for example, the composition $c \cdot d$. We can imagine that the axis l_1 is sent by the transformation d into a new position (i.e., into the original position of the axis l_3), and after this consider the transformation c as the reflection with respect to the new position of the axis l_1 (i.e., with respect to the original axis l_3). On the other hand, it is also possible to consider that the axes are not rigidly fixed to the figure and that they do not move with it; therefore, in the example which we examine, after the transformation d the transformation c is done as the reflection with respect to the original axis l_1 . We will consider the compositions of two transformations in exactly this way. With this choice the reasoning about the vertices of the figure, analogous to the arguments presented immediately before Problem 2, is correct. It is convenient to use such arguments to calculate the multiplication table.

Problem 3. Write the multiplication table for all symmetries of the equilateral triangle.

Example 3. Let e, a, b, c denote the rotations of a square by $0^\circ, 180^\circ, 90^\circ$, and 270° , respectively, in the direction shown by the arrow (Figure 3).

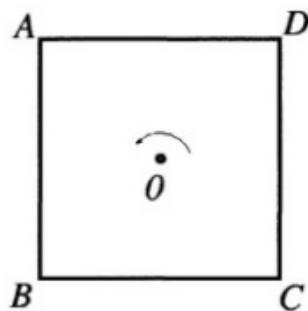


图 1.3

Problem 4. Write the multiplication table for the rotations of the square.

Example 4. Let d, f, g, h denote the reflections of the square with respect to the axes shown in Figure 4.

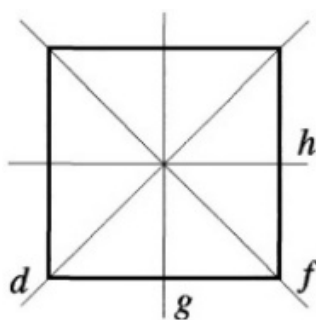


图 1.4

Problem 5. Write the multiplication table for all symmetries of the square.

Example 5. Let $ABCD$ be a rhombus which is not a square.

Problem 6. Find all symmetries of the rhombus and write their multiplication table.

Example 6. Let $ABCD$ be a rectangle which is not a square.

Problem 7. Find all symmetries of the rectangle and write their multiplication table.

1.2 Groups of Transformations

Let X and Y be two sets of elements of arbitrary nature, and suppose that every element x of X is put into correspondence with a defined element y of Y . Thus one says that there exists a mapping ϕ of the set X into the set Y , $\phi : X \rightarrow Y$. The element y is called the image of the element x , and x the pre-image of the element y . One writes $\phi(x) = y$.

Definition. The mapping $\phi : X \rightarrow Y$ is called surjective (or, equivalently, a mapping of the set X onto the set Y) if for every element y of Y there exists an element x of X such that $\phi(x) = y$, i.e., every y of Y has a pre-image in X .

Problem 8. Let the mapping ϕ put every capital city in the world in correspondence with the first letter of its name in English (for example, $\phi(\text{London}) = \text{L}$). Is ϕ a mapping of the set of capitals onto the entire English alphabet?

Definition. The mapping $\phi : X \rightarrow Y$ is called a one-to-one (or bijective) mapping of the set X into the set Y if for every y in Y there exists a pre-image in X and this pre-image is unique.

Problem 9. Consider the following mappings of the set of all integer numbers into the set of non-negative integer numbers:

- (a) $\phi(n) = n^2$;
- (b) $\phi(n) = |n|$;
- (c) $\phi(n) = \begin{cases} 2n, & \text{if } n \geq 0, \\ 2|n| - 1, & \text{if } n < 0. \end{cases}$

Which among these mappings are surjective, and which are bijective?

Let M be an arbitrary set. For brevity we shall call any bijective mapping of M into itself a transformation of the set M .

Two transformations g_1 and g_2 will be considered equal if $g_1(A) = g_2(A)$ for every element A of M . Instead of the term “transformation,” the term “permutation” is often used. We shall use this term only when the transformation is defined on a finite set. A permutation can thus be written in the form

$$\begin{pmatrix} A_1 & A_2 & \dots & A_n \\ A_{i_1} & A_{i_2} & \dots & A_{i_n} \end{pmatrix},$$

where the first row contains all the elements of the given set, and the second row indicates all the corresponding images under the permutation.

Since the transformation is one-to-one, for every transformation there exists an inverse transformation g^{-1} , which is defined in the following way: if $g(A) = B$, then $g^{-1}(B) = A$. Thus in Example 1

$$a = \begin{pmatrix} A & B & C \\ B & C & A \end{pmatrix}.$$

Therefore

$$a^{-1} = \begin{pmatrix} A & B & C \\ C & A & B \end{pmatrix},$$

i.e., $a^{-1} = b$.

Problem 10. Find the inverse transformations of all symmetries of the equilateral triangle (see Examples 1 and 2).

Problem 11. Consider the transformation of all real numbers given by $g(x) = 2x$. Find the inverse transformation.

The multiplication of the transformations g_1 and g_2 is defined by

$$g_1 g_2(A) = g_1(g_2(A)),$$

where the transformation g_2 is done first and g_1 afterwards. If g_1 and g_2 are transformations of the set M , then $g_1 g_2$ is also a transformation of the set M .

Definition. Suppose that a set G of transformations possesses the following properties: (1) if two transformations g_1 and g_2 belong to G , then their product $g_1 g_2$ also belongs to G ; (2) if a transformation g belongs to G , then its inverse transformation g^{-1} belongs to G . In this case we call such a set of transformations a group of

transformations.

It is not difficult to verify that the sets of transformations considered in Examples 1–6 are, in fact, groups of transformations.

Problem 12. Prove that any group of transformations contains the identity transformation e such that $e(A) = A$ for every element A of the set M .

Problem 13. Prove that $eg = ge = g$ for any transformation g .

Problem 14. Prove that for any three transformations g_1, g_2, g_3 the following equality holds³:

$$(g_1g_2)g_3 = g_1(g_2g_3).$$

1.3 Groups

To solve Problems 6 and 7 we wrote the multiplication tables for the symmetries of the rhombus and of the rectangle. It has turned out that in our notation (see the solutions) these tables coincide. For many purposes it is natural to consider such groups of transformations as coinciding. Therefore we shall consider abstract objects rather than sets of real elements (in our case, transformations). Furthermore, we shall consider those binary operations on arbitrary sets which possess the basic properties of the binary operation in a group of transformations. Thus any binary operation will be called a multiplication; if to the pair (a, b) there corresponds the element c , we call c the product of a and b , and write $ab = c$. In some special cases the binary operation will be called differently, for example, composition, addition, etc.

Definition. A set G of elements of an arbitrary nature, on which one can define a binary operation such that the following conditions are satisfied, is called a group:

1. Associativity: $(ab)c = a(bc)$ for any elements a, b, c of G .
2. In G there is an element e such that $ae = ea = a$ for every element a of G ; such an element is called the unit (or neutral element) of the group G .
3. For every element a of G there is in G an element a^{-1} such that $aa^{-1} = a^{-1}a = e$; such an element is called the inverse of the element a .

From the results of Problems 12–14 we see that any group of transformations is a group (in some sense the converse statement is also true; see Problem 55). In this way we have already seen many examples of groups. All these groups contain a finite number of elements; such groups are called finite groups. The number of elements of a finite group is called the order of the group. Groups containing an infinite number of elements are called infinite groups.

Let us give some examples of infinite groups.

Example 7. Consider the set of all integer numbers. In this set we shall take as binary operation the usual addition. We thus obtain a group. Indeed, the role of the unit element is played by 0, because $0 + n = n + 0 = n$ for every integer n . Moreover, for every n there exists the inverse element $-n$ (which is called in this case the

opposite element), because $n + (-n) = (-n) + n = 0$. Associativity follows from the rules of arithmetic. The obtained group is called the group of integers under addition.

Problem 15. Consider the following sets: (1) all real numbers; (2) all real numbers without zero. Say whether sets 1 and 2 form a group under multiplication.

Problem 16. Say whether all positive real numbers form a group under multiplication.

Problem 17. Say whether all natural numbers form a group (a) under addition; (b) under multiplication.

Problem 18. Prove that in every group there exists a unique unit element.

Problem 19. Prove that for every element a of a group there exists a unique inverse element a^{-1} .

Problem 20. Prove that (1) $e^{-1} = e$; (2) $(a^{-1})^{-1} = a$.

If a and b are elements of a group, then by the definition of binary operation the expression $a \cdot b$ gives some defined element of the group. Hence expressions like $(a \cdot b) \cdot c$, $a \cdot (b \cdot c)$, and $(a \cdot b) \cdot (c \cdot d)$ also give some defined elements of the group. Any two of the obtained elements can be multiplied again, obtaining again an element of the group, and so on. Therefore, in order to specify uniquely at every step which operation will be performed next, we shall put into brackets the two expressions which have to be multiplied (we may omit brackets around expressions containing only one letter). We call all expressions that we can write in this way well-arranged expressions. For example, $(a \cdot b) \cdot (c \cdot (a \cdot c))$ is a well-arranged expression, whereas $(a \cdot b) \cdot c \cdot (c \cdot d)$ is not well arranged, because it is not clear in which order one has to carry out the operations. When we consider the product $a_1 \cdot a_2 \cdot \dots \cdot a_n$ of real numbers $a_1, a_2, \dots, a_n$, we do not put in any brackets, because the result does not depend on the order in which the operations are carried out—i.e., for every arrangement of the brackets giving a well-arranged expression, the result corresponding to this product is the same. It turns out that this property is satisfied by any group, as follows from the result of the next question.

Problem 21. Suppose that a binary operation $a \cdot b$ possesses the associativity property, i.e., $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for any elements a, b, c . Prove that every well-arranged expression in which the elements from left to right are $a_1, a_2, \dots, a_n$ gives the same element as the multiplication

$$(\dots(((a_1 \cdot a_2) \cdot a_3) \dots a_{n-1}) \cdot a_n).$$

In this way, if the elements $a_1, a_2, \dots, a_n$ are elements of a group, then all the well-arranged expressions containing elements $a_1, a_2, \dots, a_n$ in this order and distinguished only by the disposition of brackets give the same element, which we will denote by $a_1 \cdot a_2 \cdot \dots \cdot a_{n-1} \cdot a_n$ (eliminating all brackets).

The multiplication of real numbers possesses yet another important property: the product $a_1 \cdot a_2 \cdot \dots \cdot a_{n-1} \cdot a_n$ does not change if the factors are permuted arbitrarily. However, not all groups possess this property.

Definition. Two elements a and b of a group are called commuting if $ab = ba$. (One also says that a and b commute.) If in a group any two elements commute, the group is said to be commutative or abelian.

There exist non-commutative groups. Such a group is, for example, the group of symmetries of the triangle (see Example 2, where $ac = f$ and $ca = d$, i.e., $ac \neq ca$).

Problem 22. Say whether the following groups are commutative (see Problems 2 and 4–7): (1) the group of rotations of the triangle; (2) the group of rotations of the square; (3) the group of symmetries of the square; (4) the group of symmetries of a rhombus; (5) the group of symmetries of a rectangle.

Problem 23. Prove that in any group:

$$(1) \quad (ab)^{-1} = b^{-1}a^{-1}; \quad (2) \quad (a_1 \cdots a_n)^{-1} = a_n^{-1} \cdots a_1^{-1}.$$

Remark. The jacket is put on after the shirt, but is taken off before it.

If a certain identity $a = b$ holds in a group G (a and b being two expressions giving the same element of G), then one obtains a new identity by multiplying the two members of the initial identity by an arbitrary element c of the group G . However, since in a group the product may depend on the order of its factors, one can multiply the two members of the identity by c either on the left (obtaining $ca = cb$) or on the right (obtaining $ac = bc$).

Problem 24. Let a, b be two arbitrary elements of a group G . Prove that each of the equations $ax = b$ and $ya = b$ has one and only one solution in G .

The uniqueness of the solution in Problem 24 can also be enunciated in this way: if $ab_1 = ab_2$ or $b_1a = b_2a$, then $b_1 = b_2$.

Problem 25. Suppose that $a \cdot a = e$ for every element a of a group G . Prove that G is commutative.

Let a be an arbitrary element of a group G . We will denote by a^m the product $a \cdot a \cdots a$, where m is the number of factors, all equal to a .

Problem 26. Prove that $(a^m)^{-1} = (a^{-1})^m$, where m is an integer.

In this way $(a^m)^{-1}$ and $(a^{-1})^m$ for every integer m indicate the same element, which we will denote by a^{-m} . Moreover, $a^0 = e$ for every element a .

Problem 27. Prove that $a^m \cdot a^n = a^{m+n}$ for any integers m, n .

Problem 28. Prove that $(a^m)^n = a^{mn}$ for any integers m, n .

1.4 Cyclic Groups

The simplest groups are the cyclic groups. They are, however, very important.

Definition. Let a be an element of a group G . The smallest positive integer n such that $a^n = e$ is called the order of the element a . If such an integer does not exist, one says that a is an element of infinite order.

Problem 29. Find the order of all elements of the groups of symmetries of the equilateral triangle, the square, and the rhombus (see Problems 3, 5, and 6).

Problem 30. Let the order of an element a be equal to n . Prove that (1) the elements $e, a, a^2, \dots, a^{n-1}$ are all distinct; (2) for every integer m , the element a^m coincides with one of the elements listed above.

Definition. If an element a has order n and in a group G there are no other elements but $e, a, a^2, \dots, a^{n-1}$, the group G is called the cyclic group of order n generated by the element a , and the element a is called a generator of the group.

Example 8. Consider a regular n -gon (a polygon with n sides) and all rotations of the plane that transform the n -gon into itself.

Problem 31. Prove that these rotations form a cyclic group of order n .

Problem 32. Find all generators in the group of rotations of the equilateral triangle and in the group of rotations of the square (see Examples 1 and 3 in §1.1).

Problem 33. Let the order of an element a be equal to n . Prove that $a^m = e$ if and only if $m = nd$, where d is any integer.

Problem 34. Suppose that the order of an element a is equal to a prime number p and that m is an arbitrary integer. Prove that either $a^m = e$ or a^m has order p .

Problem 35. Suppose that d is the greatest common divisor of the integers m, n and that a has order n . Prove that the element a^m has order n/d .

Problem 36. Find all generators of the group of rotations of the regular dodecagon.

Problem 37. Let a be an element of infinite order. Prove that the elements $\dots, a^{-2}, a^{-1}, e, a, a^2, \dots$ are all distinct.

Definition. If a is an element of infinite order and the group G has no other elements but $\dots, a^{-2}, a^{-1}, e, a, a^2, \dots$, then G is called an infinite cyclic group and a its generator.

Problem 38. Prove that the group of integers is a cyclic group under addition (see Example 7, §1.3). Find all generators.

Example 9. Let n be an integer different from zero. Consider all the possible remainders of the division of integers by n , i.e., the numbers $0, 1, 2, \dots, n-1$. Let us introduce in this set of remainders the following binary operation. After adding two remainders as usual, we keep the remainder of the division by n of the obtained sum. This operation is called addition modulo n . Thus, summing modulo 4, we have $1 + 2 = 3$, but $3 + 3 = 2$.

Problem 39. Write the multiplication table for addition modulo (a) 2; (b) 3; (c) 4.

Problem 40. Prove that the set of remainders with addition modulo n forms a group, and that this group is a cyclic group of order n .

Consider again an arbitrary cyclic group of order n : $e, a, a^2, \dots, a^{n-1}$.

Problem 41. Prove that $a^m \cdot a^r = a^k$, where $0 \leq m < n$, $0 \leq r < n$, and $0 \leq k < n$, if and only if modulo n one has $m + r = k$.

From the result of the preceding problem it follows that multiplication of the elements in an arbitrary cyclic group corresponds to addition of the remainders modulo n . Similarly, multiplication of two elements in an infinite cyclic group corresponds to addition of integers (see Example 7). We come in this way to an important notion in the theory of groups: the notion of isomorphism.

1.5 Isomorphism

Definition. Let two groups G_1 and G_2 be given with a bijective mapping ϕ from G_1 into G_2 (see §1.2) with the following property: if $\phi(a) = a'$, $\phi(b) = b'$, $\phi(c) = c'$, and $ab = c$ in group G_1 , then $a'b' = c'$ in group G_2 . In other words, to the multiplication in G_1 there corresponds under ϕ the multiplication in G_2 . The mapping ϕ is thus called an isomorphism between groups G_1 and G_2 , and the groups G_1 and G_2 are said to be isomorphic. The condition for a bijective mapping ϕ to be an isomorphism can also be expressed by the following condition: $\phi(ab) = \phi(a) \cdot \phi(b)$ for all elements a and b of group G_1 ; here the product ab is taken in the group G_1 and the product $\phi(a) \cdot \phi(b)$ in the group G_2 .

42. Which amongst the following groups are isomorphic: 1) the group of rotations of the square; 2) the group of symmetries of the rhombus; 3) the group of symmetries of the rectangle; 4) the group of remainders under addition modulo 4?

43. Let $\phi : G_1 \rightarrow G_2$ be an isomorphism. Prove that the inverse mapping $\phi^{-1} : G_2 \rightarrow G_1$ is an isomorphism.

44. Let $\phi_1 : G_1 \rightarrow G_2$ and $\phi_2 : G_2 \rightarrow G_3$ be two isomorphisms. Prove that the compound mapping $\phi_2\phi_1 : G_1 \rightarrow G_3$ is an isomorphism.

From the two last problems it follows that two groups which are isomorphic to a third group are isomorphic to each other.

45. Prove that every cyclic group of order n is isomorphic to the group of the remainders of the division by n under addition modulo n .

46. Prove that every infinite cyclic group is isomorphic to the group of integers under addition.

47. Let $\phi : G \rightarrow F$ be an isomorphism. Prove that $\phi(e_G) = e_F$, where e_G and e_F are the unit elements in groups G and F .
48. Let $\phi : G \rightarrow F$ be an isomorphism. Prove that $\phi(g^{-1}) = [\phi(g)]^{-1}$ for every element g of group G .
49. Let $\phi : G \rightarrow F$ be an isomorphism and let $\phi(g) = h$. Prove that g and h have the same order.
- If we are interested in the group operation and not in the nature of the elements of the groups (which, in fact, does not play any role), then we can identify all groups which are isomorphic. So, for example, we shall say that there exists, up to isomorphism, only one cyclic group of order n (see 45), which we denote by $\mathbb{Z}_n$, and only one infinite cyclic group (see 46), which we indicate by $\mathbb{Z}$. If a group G_1 is isomorphic to a group G_2 , then we write $G_1 \cong G_2$.
50. Find (up to isomorphism) all groups containing: a) 2 elements; b) 3 elements.
51. Give an example of two non-isomorphic groups with the same number of elements.
52. Prove that the group of all real numbers under addition is isomorphic to the group of the real positive numbers under multiplication.
53. Let a be an arbitrary element of a group G . Consider the mapping ϕ_a of a group G into itself defined in the following way: $\phi_a(x) = ax$ for every x of G . Prove that ϕ_a is a permutation of the set of the elements of group G (i.e., a bijective mapping of the set of the elements of G into itself).
54. For every element a of a group G , let ϕ_a be the permutation defined in Problem 53. Prove that the set of all permutations ϕ_a forms a group under the usual law of composition of mappings.
55. Prove that group G is isomorphic to the group of permutations defined in the preceding problem.

1.6 Subgroups

In the set of the elements of a group G consider a subset H . It may occur that H is itself a group under the same binary operation defined in G .

In this case H is called a subgroup of the group G . For example, the group of rotations of the regular n -gon is a subgroup of the group of all symmetries of the n -gon.

If a is an element of a group G , then the set of all elements of type a^m is a subgroup of G (this subgroup is cyclic, as we have seen in §1.4).

56. Let H be a subgroup of a group G . Prove that: a) the unit elements in G and in H coincide; b) if a is an element of subgroup H , then the inverse elements of a in G and in H coincide.
57. Prove that in order for H to be a subgroup of a group G (under the same binary operation) the following conditions are necessary and sufficient: 1) if a and b belong to H , then the element ab (product in group G) belongs to H ; 2) e (the unit element of group G) belongs to H ; 3) if a belongs to H , then also a^{-1} (taken in group G) belongs to H .

Remark. Condition 2 follows from conditions 1 and 3.

58. Find all subgroups of the following groups: 1) of symmetries of the equilateral triangle; 2) of symmetries of the square.

59. Find all subgroups of the following cyclic groups: a) $\mathbb{Z}_5$; b) $\mathbb{Z}_8$; c) $\mathbb{Z}_{15}$.
60. Prove that all subgroups of $\mathbb{Z}_n$ have the form $\{e, a^d, a^{2d}, \dots, a^{n-d}\}$, where d divides n and a is a generator of the group $\mathbb{Z}_n$.
61. Prove that all subgroups of an infinite cyclic group are of the type $\{\dots, a^{-2r}, a^{-r}, e, a^r, a^{2r}, \dots\}$, where a is a generator and r is an arbitrary non-zero integer number.
62. Prove that an infinite cyclic group has an infinite number of subgroups.
63. Prove that the intersection of an arbitrary number of subgroups⁴ of a group G is itself a subgroup of group G .

Example 10. Consider a regular tetrahedron, with vertices marked with the letters A, B, C , and D . If we look at the triangle ABC from the point D , then the rotation defined by the cyclic order of points A, B, C may be a clockwise or counterclockwise rotation (see Figure 1.5). We shall distinguish these two different orientations of the tetrahedron.



图 1.5

64. Is the orientation of the tetrahedron preserved by the following permutations: $a = \begin{pmatrix} A & B & C & D \\ B & C & A & D \end{pmatrix}$ (rotation by 120° around the altitude); $b = \begin{pmatrix} A & B & C & D \\ D & C & B & A \end{pmatrix}$ (rotation by 180° around the axis through the middle points of the edges AD and BC); $c = \begin{pmatrix} A & B & C & D \\ A & C & B & D \end{pmatrix}$ (reflection with respect to the plane containing edge AD and the middle point of edge BC); and $d = \begin{pmatrix} A & B & C & D \\ B & C & D & A \end{pmatrix}$ (cyclic permutation of the vertices)?

All symmetries of the regular tetrahedron obviously form a group, which is called the group of symmetries of the tetrahedron.

65. How many elements does the group of symmetries of the tetrahedron contain?
66. In the group of symmetries of the tetrahedron find the subgroups isomorphic to: a) the group of symmetries of the equilateral triangle; b) the cyclic group $\mathbb{Z}_4$.
67. Prove that all symmetries of the tetrahedron preserving its orientation form a subgroup. How many elements does it contain?

The group of symmetries of the tetrahedron preserving its orientation is called the group of rotations of the tetrahedron.

68. Find in the group of rotations of the tetrahedron the subgroups isomorphic to: a) $\mathbb{Z}_2$; b) $\mathbb{Z}_3$.

1.7 Direct product

Starting from two groups one may define a third group.

Definition. The direct product $G \times H$ of groups G and H is the set of all the ordered pairs (g, h) , where g is any element of G and h any element of H , with the following binary operation:

$$(g_1, h_1) \cdot (g_2, h_2) = (g_1 g_2, h_1 h_2),$$

where the product $g_1 g_2$ is taken in the group G , and $h_1 h_2$ in the group H .

69. Prove that $G \times H$ is a group.

70. Suppose that a group G has n elements, and that a group H has k elements. How many elements does the group $G \times H$ contain?

71. Prove that the groups $G \times H$ and $H \times G$ are isomorphic.

72. Find the subgroups of $G \times H$ isomorphic to the groups G and H .

73. Let G and H be two commutative groups. Prove that the group $G \times H$ is also commutative.

74. Let G_1 be a subgroup of a group G and H_1 a subgroup of a group H . Prove that $G_1 \times H_1$ is a subgroup of the group $G \times H$.

75. Let G and H be two arbitrary groups. Is it true that every subgroup of the group $G \times H$ can be represented in the form $G_1 \times H_1$, where G_1 is a subgroup of the group G and H_1 a subgroup of the group H ?

76. Prove that the group of symmetries of the rhombus is isomorphic to the group $\mathbb{Z}_2 \times \mathbb{Z}_2$.

77. Is it true that: 1) $\mathbb{Z}_2 \times \mathbb{Z}_3 \cong \mathbb{Z}_6$; 2) $\mathbb{Z}_2 \times \mathbb{Z}_4 \cong \mathbb{Z}_8$?

78. Prove that $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$ if and only if the numbers m and n are relatively prime.

1.8 Cosets. Lagrange's theorem

For every subgroup H of a group G there exists a partition of the set of the elements of G into subsets. For each element x of G consider the set of all elements of the form xh , where h runs over all elements of a subgroup H . The set so obtained, denoted by xH , is called the left coset of H (or left lateral class of H) in G , generated by the element x .

79. Find all left cosets of the following subgroups of the group of symmetries of the equilateral triangle: a) the subgroup of rotations of the triangle; b) the group generated by the reflection with respect to a single axis $\{e, c\}$ (see Examples 1 and 2, §1.1).

80. Prove that given a subgroup H of a group G , each element of G belongs to one left coset of H in G .

81. Suppose that an element y belongs to the left coset of H generated by an element x . Prove that the left cosets of H generated by elements x and y coincide.

82. Suppose that the left cosets of H , generated by elements x and y , have a common element. Prove that these left cosets coincide.

Hence left cosets generated by two arbitrary elements either are disjoint or coincide. In this way we have obtained a partition of all elements of a group G into disjoint classes. Such a partition is called the left partition of the group G by the subgroup H .

The number of elements of a subgroup is called the order of the subgroup. Let m be the order of a subgroup H . If h_1 and h_2 are two different elements of H , then $xh_1 \neq xh_2$. Every left coset thus contains m elements. Hence if n is the order of group G and r is the number of the left cosets of the partition of G by H , then $m \cdot r = n$ and we have proved the following theorem.

Theorem 1 (Lagrange's theorem). The order of a subgroup H of a group G divides the order of the group G .⁵

83. Prove that the order of an arbitrary element (see definition in §1.4) divides the order of the group.

84. Prove that a group whose order is a prime number is cyclic and that every element of it different from the unit is its generator.

85. Suppose that a group G contains exactly 31 elements. How many subgroups does it contain?

86. Let p be a prime number. Prove that all groups having the same order p are isomorphic to each other.

87. Suppose that m divides n . Obtain a group of order n containing a subgroup isomorphic to a given group G of order m .

88. Suppose that m divides n . Is it possible that a group of order n does not contain any subgroup of order m ?

One can obtain as well the right cosets Hx and the right partition of a group G by a subgroup H . If the order of a subgroup H is equal to m , then each right coset contains m elements and the number of cosets is equal to the integer n/m , where n is the order of the group. Hence the number of right cosets coincides with the number of the left cosets.

89. Find the left and the right partitions of the group of symmetries of the equilateral triangle by the following subgroups: a) the subgroup of rotations $\{e, a, b\}$; b) the subgroup $\{e, c\}$, generated by the reflection with respect to one axis.

90. Find the left and right partitions of the group of symmetries of the square by the following subgroups: a) the subgroup $\{e, a\}$, generated by the central symmetry; b) the subgroup $\{e, d\}$, generated by the reflection with respect to one diagonal.

91. Find the partition of the group of all integers (under addition)⁶ by the subgroup of the numbers divisible by 3.

92. Find all groups (up to isomorphism) of order: a) 4; b) 6; c) 8.

1.9 Internal automorphisms

Let us start with an example. Consider the group of symmetries of the equilateral triangle. If the letters A , B , and C denote the vertices of the triangle, then each element of the group defines a permutation of the three letters. For example, the reflection of the triangle with respect to the altitude drawn from the vertex A to the base BC will be written in the form $\begin{pmatrix} A & B & C \\ A & C & B \end{pmatrix}$. To multiply two elements of the group of symmetries of the triangle it suffices to carry out the corresponding permutations one after the other. In this way we obtain an isomorphism between the group of symmetries of the triangle and the group of permutations of the letters A , B , and C .

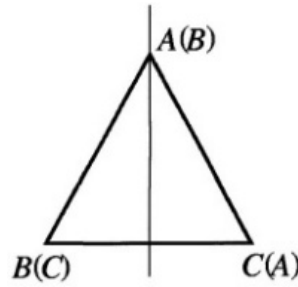


图 1.6

Now we observe that this isomorphism is not uniquely defined: it depends on which vertex is named A , which B , and which C . The change of notations of the three vertices of the triangle may be also considered as a permutation of the three letters A , B , and C . For example, $g = \begin{pmatrix} A & B & C \\ B & C & A \end{pmatrix}$ corresponds to the following change of notations:

	A	B	C
old notation	A	B	C
new notation	B	C	A

By the new notations of vertices each element of the group of symmetries of the triangle obtains a new notation in terms of permutations of the letters A , B , and C . For example, the reflection of the triangle with respect to its vertical altitude (Figure 1.6) is written in the following way:

old notation	$\begin{pmatrix} A & B & C \\ A & C & B \end{pmatrix}$
new notation	$\begin{pmatrix} A & B & C \\ C & B & A \end{pmatrix}$

93. Consider an element of the group of symmetries of the triangle, to which, for some notation of vertices, there corresponds a permutation h . What permutation will correspond to the same element of the group of symmetries of the triangle after the change of notation g of the vertices?

We observe now that by the change of notation g , an element h of a group of transformations is sent to ghg^{-1} not only in the example considered of the group of symmetries of the triangle, but also in the more general case. So the study of the changes of notations leads to the following definition.

Definition. Let G be a group and g one of its elements. Define the mapping ϕ_g of the group G into itself by the formula $\phi_g(h) = ghg^{-1}$ (where h is any element of the group). This mapping is called the internal automorphism of group G generated by the element g .

94. Prove that an internal automorphism of a group is an isomorphism of the group into itself.

95. Which is the image of the reflection of the triangle with respect to its altitude under all possible internal automorphisms of the group of symmetries of the triangle?

96. Which is the image of the rotation by 120° of the triangle under all possible internal automorphisms of the group of symmetries of the triangle?

97. Which are the pairs of elements of the group of symmetries of the tetrahedron that can be sent one onto the other by an internal automorphism? Which elements cannot? The same question for the group of rotations of the tetrahedron.

98. Prove that in any group the orders of elements ab and ba are equal.

Note that the image of a subgroup under any internal automorphism of the whole group (as well as under any isomorphism) is, in general, different (for example, the reflection with respect to one altitude of the triangle is sent to the reflection with respect to another altitude). However, some subgroups, “super-symmetric”, are invariant under all internal automorphisms (for example, the subgroup of rotations of the triangle in the group of symmetries of the triangle). We will study these subgroups in the next section.

1.10 Normal subgroups

Definition. A subgroup of a group is called a normal subgroup if it is mapped onto itself by all internal automorphisms of the group. In other words, a subgroup N of a group G is called a normal subgroup of G if for every element a of N and for every element g of G the element gag^{-1} belongs to N .

Hence in the group of symmetries of a triangle ABC the subgroup of rotations is normal, whereas the group generated by the reflection with respect to the altitude drawn from A to the base BC (containing two elements) is not normal.

99. Prove that in a commutative group every subgroup is normal.

100. Is it true that the subgroup of the group of symmetries of the square, consisting of elements $\{e, a\}$, where a is the central symmetry (see Examples 3 and 4, §1.1), is normal?

Theorem 2. A subgroup N of a group G is normal if and only if the left and the right partitions of group G by the subgroup N coincide.⁷

101. Prove Theorem 2.

102. Let n be the order of a group G , m the order of a subgroup H , and $m = n/2$. Prove that H is a normal subgroup of the group G .

103. Prove that the intersection (see footnote to Problem 63) of an arbitrary number of normal subgroups of a group G is a normal subgroup of the group G .

Definition. The set of elements of a group G which commute with all elements of the group is called the centre of group G .

104. Prove that the centre of a group G is a subgroup and, moreover, a normal subgroup of the group G .

105. Let N_1 and N_2 be two normal subgroups of two groups G_1 and G_2 , respectively. Prove that $N_1 \times N_2$ is a normal subgroup of the group $G_1 \times G_2$.

The following example shows that a normal subgroup of a subgroup of a group G can be a non-normal subgroup of the group G .

Example 11. Consider the subgroup of the group of symmetries of the square, generated by the reflections with respect to the diagonals and to the centre (see Examples 3 and 4, §1.1, the subgroup $\{e, a, d, f\}$). This subgroup contains one half of the elements of the group of symmetries of the square, and it is therefore a normal subgroup (see 102). The subgroup $\{e, d\}$, generated by the reflection with respect to one of the diagonals, contains one half of the elements of the subgroup $\{e, a, d, f\}$, and it is therefore a normal subgroup of this subgroup. But the subgroup $\{e, d\}$ is not a normal subgroup of the group of symmetries of the square, because d is sent by an internal automorphism to the reflection with respect to the other diagonal: $bdb^{-1} = f$.

1.11 Quotient groups

Let us start with an example. Consider the partition of the group of symmetries of the square by the normal subgroup $\{e, a\}$, generated by the central symmetry a (see Examples 3 and 4, §1.1). It is easy to see that the partition of our group into four cosets has the form shown in Table 1.2.

表 1.2

e	b	d	g
a	c	f	h
E	A	B	C

Denote each coset by a letter, for example E , A , B , and C . If one multiplies an arbitrary element of coset A by an arbitrary element of coset B , then the result belongs to one and only one coset C , independently of the particular elements chosen in cosets A and B . From the solution of the next problem it follows that this result is not casual.

106. Let there be given the partition of group G by the normal subgroup N . Suppose that the elements x_1 and x_2 belong to one coset and that the elements y_1 and y_2 belong to another coset. Prove that the elements x_1y_1 and x_2y_2 belong to the same coset.

In this way, multiplying in a given order two representative elements of two different cosets, one obtains an element of a coset which does not depend on the chosen representative elements. As a consequence, given a partition of a group by a normal subgroup, we can define a binary operation in the following way: whenever $A = xN$, $B = yN$, we write $A \cdot B = (xy)N$ ($A \cdot B$ is also denoted by AB). The result of Problem 106 shows that this operation is uniquely defined and does not depend on the elements x and y which generate the cosets A and B . So in the above example we have $A \cdot B = C$.

In Problems 107–109 the subgroups have to be considered as normal.

107. Let T_1, T_2, T_3 be three cosets. Prove that $(T_1T_2)T_3 = T_1(T_2T_3)$.

108. Let E be a normal subgroup. Prove that $ET = TE = T$ for every coset T .

109. Prove that for every coset T there exists a coset T^{-1} such that $TT^{-1} = T^{-1}T = E$.

From Problems 107–109 it follows that the set of all cosets with the binary operation just defined forms a group. This group is called the quotient group of the group G by the normal subgroup N and is denoted by G/N .

It is evident that $G/\{e\} \cong G$ and $G/G \cong \{e\}$. It is evident as well that the order of the quotient group is equal to the integer n/m , where n is the order of the group and m the order of the normal subgroup. For example, the quotient group of the group of symmetries of the square by the subgroup generated by the central symmetry contains 4 elements.

110. Calculate whether the quotient group of the group of symmetries of the square by the subgroup generated by the central symmetry is isomorphic to the group of rotations of the square or to the group of symmetries of the rhombus.

111. Find all normal subgroups and the corresponding quotient groups of the following groups⁸: a) the group of symmetries of the triangle; b) $\mathbb{Z}_2 \times \mathbb{Z}_2$; c) the group of symmetries of the square; d) the group of quaternions (see solution of Problem 92).

112. Describe all normal subgroups and the corresponding quotient groups of the following groups: a) $\mathbb{Z}_n$; b) $\mathbb{Z}$.

113. Find all normal subgroups and the corresponding quotient groups of the group of rotations of the tetrahedron.

114. Consider the subgroup $G_1 \times \{e\}$ in the direct product $G_1 \times G_2$. Prove that it is a normal subgroup and that the corresponding quotient group is G_2 .

1.12 Commutant

Recall that two elements a and b of a group G are said to be commuting if $ab = ba$. The degree of noncommutativity of two elements of a group can be measured by the product $aba^{-1}b^{-1}$, which is equal to the unit element if and only if a and b commute.

Definition. The element $aba^{-1}b^{-1}$ is called the commutator of the elements a and b . The set of all possible products of a finite number of commutators of a group G is called the commutant of the group G and it is denoted by $K(G)$.

115. Prove that the commutant is a subgroup.

116. Prove that the commutant is a normal subgroup.

117. Prove that the commutant coincides with the unit group $\{e\}$ if and only if the group is commutative.

118. Find the commutant in the following groups: a) the group of symmetries of the triangle; b) the group of symmetries of the square; c) the group of quaternions (see solution of Problem 92).

119. Prove that the commutant in the group of symmetries of the regular n -gon is isomorphic to the group $\mathbb{Z}_n$ if n is odd and to the group $\mathbb{Z}_{n/2}$ if n is even.

120. Find the commutant in the group of symmetries of the tetrahedron.

121. Prove that if a normal subgroup of the group of rotations or of the group of symmetries of the tetrahedron contains just a sole rotation around one axis through a vertex, then it contains all rotations of the tetrahedron.

122. Find the commutant in the group of symmetries of the tetrahedron.

Consider two groups: the group of rotations of the cube and the group of rotations of the regular octahedron.

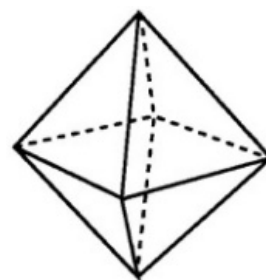
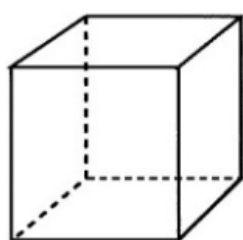


图 1.7

123. How many elements are contained in these groups? Calculate the elements of the group of rotations of the cube.

124. Prove that the groups of rotations of the cube and of the octahedron are isomorphic.

125. In how many different ways is it possible to colour the surface of a cube with 6 colours (a different colour for each face) if one considers two coloured cubes as different if they do not coincide even after some rotation? The same question for a box of matches.

126. Which group amongst those you know is isomorphic to the group of rotations of a box of matches?

To calculate the commutant of the group of rotations of the cube we inscribe in the cube a tetrahedron (see Figure 1.8). Joining the remaining vertices B , D , A_1 , and C_1 one obtains a second tetrahedron. Any rotation of the cube either sends each tetrahedron onto itself or exchanges the tetrahedra with each other.

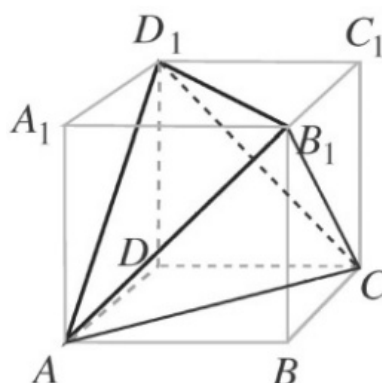


图 1.8

127. Prove that all rotations of the cube sending each tetrahedron onto itself form: a) a subgroup; b) a normal subgroup of the group of rotations of the cube.

128. Prove that the commutant of the group of rotations of the cube is isomorphic to the group of rotations of the tetrahedron.

We now prove three properties of the commutant which will be of use later on.

129. Prove that the quotient group of an arbitrary group G by its commutant is commutative.

130. Let N be a normal subgroup of a group G and let the quotient group G/N be commutative. Prove that N contains the commutant of the group G .

131. Let N be a normal subgroup of a group G and $K(N)$ the commutant of the group N . Prove that $K(N)$ is a normal subgroup of G (compare with Example 11, §1.10).

1.13 Homomorphisms

Let G and F be two groups. A mapping $\phi : G \rightarrow F$ such that $\phi(ab) = \phi(a)\phi(b)$ for all elements a and b of the group G (here the product ab is taken in G and $\phi(a) \cdot \phi(b)$ in F) is called a homomorphism of G into F . Homomorphisms are distinguishable from isomorphisms because homomorphisms are not necessarily bijective.

Example 12. Let G be the group of rotations of the cube, and $\mathbb{Z}_2$ the group of permutations of the two tetrahedra inscribed inside the cube (see §1.12). To each rotation of the cube there corresponds a well-defined permutation of tetrahedra. When we carry out two rotations of the cube one after the other, the permutation of the tetrahedra so obtained is the product of the permutations of the tetrahedra corresponding to these rotations. Therefore the mapping of the group of rotations of the cube into the group of permutations of two tetrahedra is a homomorphism.

132. Let $\phi : G \rightarrow F$ be a surjective homomorphism of a group G onto a group F . Prove that if the group G is commutative, then F is commutative. Is the converse proposition true?

133. Prove that a homomorphism of a group G into a group F sends the unit of the group G onto the unit of the group F .

134. Prove that $\phi(a^{-1}) = [\phi(a)]^{-1}$, where $\phi : G \rightarrow F$ is a homomorphism. Note that the inverse element appearing in the left member of the equation is taken in the group G , whereas in the right member it is taken in the group F .

135. Let $\phi_1 : G \rightarrow F$ and $\phi_2 : F \rightarrow H$ be two homomorphisms. Prove that $\phi_2 \circ \phi_1 : G \rightarrow H$ is a homomorphism.

Important examples of homomorphisms are obtained by means of the construction of the “natural homomorphism”. Let N be a normal subgroup of a group G . Consider the mapping ϕ of the group G into the quotient group G/N which sends each element g of the group G to a coset T of N containing the element g .

136. Prove that $\phi : G \rightarrow G/N$ is a surjective homomorphism of G onto G/N .

Definition. The surjective mapping ϕ is called the natural homomorphism of a group G into the quotient group G/N .

We have proved that to every normal subgroup there corresponds a homomorphism. We shall now prove that, inversely, every surjective homomorphism of a group G onto a group F can be seen as a natural homomorphism of G onto the quotient group G/N by a suitable normal subgroup N .

Definition. Let $\phi : G \rightarrow F$ be a homomorphism. The set of elements g such that $\phi(g) = e_F$ is called the kernel of the homomorphism ϕ and is denoted by $\ker \phi$.

137. Prove that $\ker \phi$ is a subgroup of group G .

138. Prove that $\ker \phi$ is a normal subgroup of group G .

Consider the partition of G by the kernel $\ker \phi$.

139. Prove that g_1 and g_2 belong to the same coset if and only if $\phi(g_1) = \phi(g_2)$.

Theorem 3. Let $\phi : G \rightarrow F$ be a surjective homomorphism of a group G onto a group F . The mapping $\psi : G/\ker \phi \rightarrow F$ sending each coset to the image by ϕ of a certain element of the coset (and thus of an arbitrary element; see 139) is an isomorphism.

The proof of this theorem is contained in the solutions of the following problems.

140. Prove that ψ is surjective.

141. Prove that ψ is bijective.

142. Prove that ψ is an isomorphism.

We will consider some applications of this theorem.

Example 13. Problem 110 asked whether the quotient group of the group of symmetries of the square by the normal subgroup generated by the central symmetry is isomorphic either to the group of rotations of the square or to the group of symmetries of the rhombus. To each element of the group of symmetries of the square there corresponds some permutation of the axes of symmetry l_1, l_2, l_3, l_4 (Figure 1.9). This permutation can just exchange between each other the diagonals l_1 and l_3 , as well as the axes l_2 and l_4 .

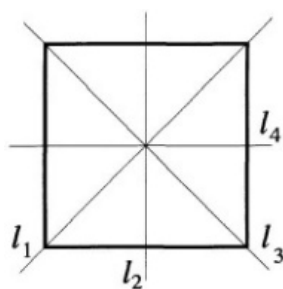


图 1.9

We thus obtain a mapping of the group of symmetries of the square into a group of permutations of four elements l_1, l_2, l_3 , and l_4 . This mapping is a homomorphism surjective onto the group of those permutations which send $\{l_1, l_3\}$ to $\{l_1, l_3\}$, and $\{l_2, l_4\}$ to $\{l_2, l_4\}$ (verify). This group consists of four permutations and is isomorphic to the group of symmetries of the rhombus $L_1L_2L_3L_4$ (Figure 1.10).

The kernel of the homomorphism so obtained contains all symmetries of the square sending each axis of symmetry onto itself. It is not difficult to verify that these transformations are just e and the central symmetry

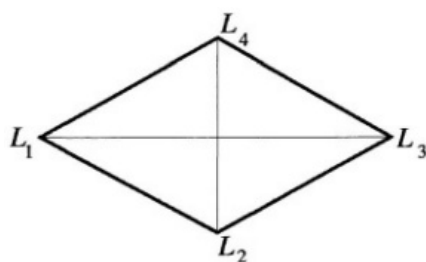


图 1.10

a. Therefore by Theorem 3 the subgroup $\{e, a\}$ is a normal subgroup of the group of symmetries of the square, and the corresponding quotient group is isomorphic to the group of symmetries of the rhombus.

The following problems may be solved in a similar way.

143. Prove that the rotations of the tetrahedron by 180° around the axes through the middle points of opposite edges form, together with the identity, a normal subgroup of the group of symmetries of the tetrahedron. Find the corresponding quotient group.

144. Prove that the rotations of the cube by 180° around the axes through the centres of opposite faces form, together with the identity, a normal subgroup of the group of rotations of the cube. Find the corresponding quotient group.

145. Let there be given on the plane a regular n -gon with centre O . Let R be the group of rotations of the plane around the point O . Consider the subgroup $\mathbb{Z}_n$ of the rotations of the plane sending the regular n -gon to itself. Prove that this subgroup is a normal subgroup of R and that $R/\mathbb{Z}_n$ is isomorphic to R .

146. Let N_1 and N_2 be two normal subgroups of groups G_1 and G_2 , respectively. Prove that $N_1 \times N_2$ is a normal subgroup of $G_1 \times G_2$ and that

$$(G_1 \times G_2)/(N_1 \times N_2) \cong (G_1/N_1) \times (G_2/N_2).$$

147. Is it possible that two normal subgroups of two non-isomorphic groups are isomorphic to each other, and that the corresponding quotient groups are isomorphic?

148. Is it possible that two normal subgroups of the same group are isomorphic and that the corresponding quotient groups are not isomorphic?

149. Is it possible that two normal subgroups of the same group are not isomorphic and that the corresponding quotient groups are isomorphic?

We now observe what happens to subgroups, to normal subgroups, and to commutants under the action of a homomorphism. Let $\phi : G \rightarrow F$ be a homomorphism. Choose in G a subset M . The set of the elements of F having at least a pre-image in M is called the image of the set M by the homomorphism ϕ (denoted by $\phi(M)$). Conversely, let P be a subset of F ; the set of all elements of G having an image in P is called the pre-image of P (denoted by $\phi^{-1}(P)$). Note that the symbol ϕ^{-1} here does not denote an inverse mapping: a homomorphism, in general, has no inverse mapping. Note also that if $\phi(M) = P$, then M is contained in $\phi^{-1}(P)$, but it does not necessarily coincide with $\phi^{-1}(P)$ (see Figure 1.11).

150. Prove that the image of a subgroup H of a group G under a homomorphism $\phi : G \rightarrow F$ is a subgroup of the group F .

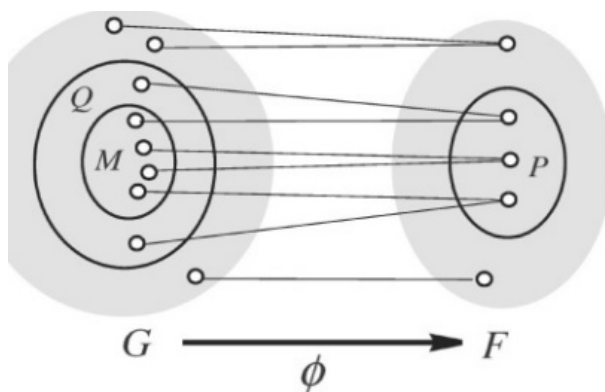


图 1.11

- 151.** Let H be a subgroup of F and $\phi : G \rightarrow F$ a homomorphism. Prove that $\phi^{-1}(H)$ is a subgroup of G .
- 152.** Let N be a normal subgroup of a group F and $\phi : G \rightarrow F$ a homomorphism. Prove that $\phi^{-1}(N)$ is a normal subgroup of the group G .
- 153.** Let $\phi : G \rightarrow F$ be a homomorphism, and let K_1 and K_2 be the commutants of G and F . Prove that $\phi(K_1)$ is contained in K_2 and that K_1 is contained in $\phi^{-1}(K_2)$.
- 154.** Let N be a normal subgroup of a group G and $\phi : G \rightarrow F$ a surjective homomorphism of group G onto a group F . Prove that $\phi(N)$ is a normal subgroup of F .
- 155.** Let K_1 and K_2 be the commutants of groups G and F , and let $\phi : G \rightarrow F$ be a surjective homomorphism of G onto F . Prove that $\phi(K_1) = K_2$. Is it true that $K_1 = \phi^{-1}(K_2)$?

1.14 Soluble groups

There exists an important class of groups which is similar to the commutative groups: that of soluble groups. This appellation comes from the fact that the possibility of solving algebraic equations by radicals depends on the solubility of some groups, as we will see in the next chapter.

Let G be a group and $K(G)$ its commutant. The commutant $K(G)$ is itself a group, and one can consider its commutant $K(K(G))$ as well. In the group obtained one can again consider the commutant, and so on. One obtains the group

$$\underbrace{K(K(\dots(K(G))\dots))}_r,$$

in short $K_r(G)$. Thus $K_{r+1}(G) = K(K_r(G))$.

Definition. A group G is said to be soluble if the sequence of groups $G, K(G), K_2(G), K_3(G), \dots$ ends, for a finite n , with the unit group, i.e., for some n one has $K_n(G) = \{e\}$.

For example, all commutative groups are soluble, because if G is commutative, then at the first step one already has $K(G) = \{e\}$. A group G is also soluble whenever its commutant is commutative, because in this case $K_2(G) = \{e\}$.

- 156.** Say whether the following groups are soluble or not: a) the cyclic group $\mathbb{Z}_n$; b) the group of symmetries of the equilateral triangle; c) the group of symmetries of the square; d) the group of quaternions (see 92); e) the

group of rotations of the tetrahedron; f) the group of symmetries of the tetrahedron; g) the group of rotations of the cube.

All groups considered in Problem 156 are soluble. It is thus natural to ask whether there exist in general non-soluble groups. We will prove that the group of rotations of the regular dodecahedron (Figure 1.12) is not soluble.

157. How many elements are contained in the group of rotations of the dodecahedron?

All rotations of the dodecahedron can be divided into four classes: 1) the identity transformation; 2) rotations around the axes through the centres of opposite faces; 3) rotations around the axes through opposite vertices; 4) rotations around the axes through the middle points of opposite edges.

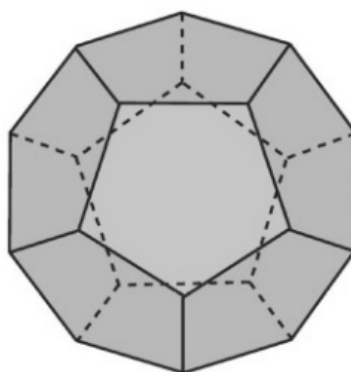


图 1.12

158. How many elements are contained in each class (without counting the identity transformation in classes 2–4)?

159. Let N be an arbitrary normal subgroup of the group of rotations of the dodecahedron and suppose that N contains at least one element of one among classes 1–4. Prove that N contains the entire class of this element.

As a consequence each one of classes 1–4 either belongs entirely to N or has no elements in common with N .

160. Prove that in the group of rotations of the dodecahedron there are no other normal subgroups except $\{e\}$ and the whole group.

161. Suppose that a group G is not commutative and that it has no normal subgroups other than $\{e\}$ and G . Prove that G is not soluble.

From Problems 160 and 161 it follows that the group of rotations of the dodecahedron is not soluble.

We shall consider some more problems whose results will be of use later on.

162. Prove that every subgroup of a soluble group is soluble.

163. Let $\phi : G \rightarrow F$ be a surjective homomorphism of a group G onto a group F and suppose that group G is soluble. Prove that the group F is also soluble.

164. Give an example in which the group F is soluble whereas the group G is not (see the preceding problem).

165. Let G be a soluble group and N a normal subgroup of G . Prove that the quotient group G/N is soluble.

166. Prove that if the groups N and G/N are soluble, then the group G is soluble.

167. Let G and F be two soluble groups. Prove that the group $G \times F$ is soluble.

168. Let G be a soluble group. Prove that there exists a sequence of groups $G_0, G_1, \dots, G_n$ such that: 1) $G_0 = G$; 2) each group G_i ($1 \leq i \leq n$) is a normal subgroup of the group G_{i-1} , and all quotient groups G_{i-1}/G_i are commutative; 3) the group G_n is commutative.

169. Suppose that for a group G there exists a sequence of groups with the properties described in the preceding problem. Prove that the group G is soluble.

The results of Problems 168 and 169 show that for a group G the existence of a sequence of groups with the properties described in Problem 168 is equivalent to the condition of solubility and can as well be considered as a definition of solubility. One may obtain yet another definition of solubility using the results of the next two problems.

170. Let G be a soluble group. Prove that there exists a sequence of groups $G_0, G_1, \dots, G_n$ such that: 1) $G_0 = G$; 2) every group G_i ($1 \leq i \leq n-1$) contains a commutative normal subgroup N_i such that the quotient group $G_i/N_i \cong G_{i+1}$; 3) the group G_n is commutative.

171. Suppose that for a group G there exists a sequence of groups with the properties described in Problem 170. Prove that the group G is soluble.

1.15 Permutations

We consider now, more attentively, the permutations (i.e., the transformations) of the set of integers $1, 2, \dots, n$; these permutations are called permutations of degree n . We observe that any permutation in an arbitrary set of n elements can be considered as a permutation of degree n : it suffices to enumerate the elements of the set by the integers $1, 2, \dots, n$. Every permutation of degree n can be written in the form

$$\begin{pmatrix} 1 & 2 & \cdots & n \\ i_1 & i_2 & \cdots & i_n \end{pmatrix},$$

where i_m is the image of the element m under the permutation. Recall that a permutation is a bijective mapping; as a consequence the elements of the second row are all distinct.

172. Which is the number of all permutations of degree n ?

Definition. The group of all permutations of degree n with the usual operation of multiplication (i.e., composition) of permutations⁹ is called the symmetric group of degree n and is denoted by S_n .

173. Prove that for $n \geq 3$ group S_n is not commutative.

A permutation can interchange some elements and fix the others. It may also happen that the permuted elements change their position cyclically. For example, the permutation

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 4 & 2 & 6 & 3 & 5 & 1 & 7 \end{pmatrix}$$

fixes the elements 2, 5, and 7, and permutes the other elements cyclically: $1 \rightarrow 4, 4 \rightarrow 3, 3 \rightarrow 6, 6 \rightarrow 1$. Permutations of this kind are called cyclic permutations, or simply cycles. For cyclic permutations we will even use another notation. For example, the expression (1436) will denote the permutation sending $1 \rightarrow 4, 4 \rightarrow 3,$

$3 \rightarrow 6, 6 \rightarrow 1$ and fixing the other elements of the set we deal with. So if our permutation has degree 7 then it coincides with the permutation we had above considered.

Permutations are not all cyclic. For example, the permutation

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 5 & 4 & 1 & 2 & 6 \end{pmatrix}$$

is not cyclic, but can be represented as a product of two cycles:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 5 & 4 & 1 & 2 & 6 \end{pmatrix} = (134) \cdot (25).$$

The cycles obtained permute different elements. Cycles of such a kind are said to be independent. It is easy to see that the product of two independent cycles does not depend on the order of the factors. If we identify those products of independent cycles that are distinguished only by the order of their factors, then the following proposition holds.

Problem 174. Every permutation can be uniquely represented (up to different orderings of factors) by a product of independent cycles. Prove this proposition.

A cycle of type (i, j) , containing only two elements, is called a transposition.

Problem 175. Prove that every cycle can be represented as a product of transpositions (not necessarily independent).

The transpositions $(1, 2), (2, 3), \dots, (n-1, n)$ are called elementary transpositions.

Problem 176. Prove that every transposition can be represented as a product of elementary transpositions.

From the results of Problems 174–176 it follows that every permutation of degree n can be represented as a product of elementary transpositions. In other words, the following theorem holds.

Theorem 4. If a subgroup of the group S_n contains all elementary transpositions, then it coincides with the whole group S_n .

Suppose that the numbers $1, 2, \dots, n$ are written in a row in an arbitrary order. We say that the pair i, j is an inversion in this row if $i < j$ but j appears before i in this row. The number of inversions in a row characterizes the disorder with respect to the usual order $1, 2, \dots, n$.

Problem 177. Find the number of inversions in the row $3, 2, 5, 4, 1$.

In the sequel we shall no longer be interested in the number of inversions, but in its parity.

Problem 178. Prove that the parity of the number of inversions in a row changes if one exchanges any two numbers.

Definition. The permutation

$$\begin{pmatrix} 1 & 2 & \cdots & n \\ i_1 & i_2 & \cdots & i_n \end{pmatrix}$$

is called even or odd according to the parity of the number of inversions in the lower row. For example, the identity permutation

$$\begin{pmatrix} 1 & 2 & \cdots & n \\ 1 & 2 & \cdots & n \end{pmatrix}$$

is even because the number of inversions in the lower row is zero.

Problem 179. Determine the parity of the permutation

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 5 & 4 & 1 & 3 \end{pmatrix}.$$

Problem 180. Prove that by multiplying an even permutation by an arbitrary transposition one obtains an odd permutation, and, conversely, by multiplying an odd permutation by an arbitrary transposition one obtains an even permutation.

Problem 181. Prove that an even permutation can be decomposed only into a product of an even number of transpositions, and an odd permutation only into an odd number of transpositions.

Problem 182. Determine the parity of an arbitrary cycle of length: (a) 3; (b) 4; (c) m .

Problem 183. Prove that the result of the multiplication of two permutations of the same parity is an even permutation, whereas the result of the multiplication of two permutations of opposite parities is an odd permutation.

Problem 184. Let a be an arbitrary permutation. Prove that a and a^{-1} have the same parity.

From the results of Problems 183 and 184 it follows that the set of all even permutations forms a subgroup of the group S_n .

Definition. The group of all even permutations of degree n is called the alternating group of degree n and is denoted by A_n .

Problem 185. Prove that for $n \geq 4$, A_n is not commutative.

Problem 186. Prove that the alternating group A_n is a normal subgroup of the symmetric group S_n and find the partition of S_n by A_n .

Problem 187. Calculate the number of elements of the group A_n .

Problem 188. Prove that the groups S_2 , S_3 , and S_4 are soluble.

We now prove that the alternating group A_5 is not soluble. One of the possible proofs uses the following construction. We inscribe in the dodecahedron five regular tetrahedra, numbered by the numbers 1, 2, 3, 4, and 5 in such a way that to every rotation of the dodecahedron there corresponds an even permutation of the tetrahedra, and that to different rotations there correspond different permutations. So we have defined an isomorphism between the group of rotations of the dodecahedron and the group A_5 of the even permutations of degree 5. The non-solubility of the group A_5 will thus follow from the non-solubility of the group of rotations of the dodecahedron.

Problem 189. Inscribe in the dodecahedron five tetrahedra as explained above¹⁰.

Another proof of the non-solubility of the group A_5 consists in repeating the argument of the proof of the non-solubility of the group of rotations of the dodecahedron. To do this one must solve the next problem.

Problem 190. Prove that every even permutation of degree 5, different from the identity, can be decomposed into independent cycles in just one of the following ways: (a) $(i_1 i_2 i_3 i_4 i_5)$; (b) $(i_1 i_2 i_3)$; (c) $(i_1 i_2)(i_3 i_4)$.

Problem 191. Let N be a normal subgroup of the group A_5 . Prove that if N contains at least one permutation which splits into independent cycles in one of the ways indicated in Problem 190, then N contains all the permutations splitting into independent cycles in this way.

Problem 192. Prove that the group A_5 does not contain normal subgroups except the identity and the whole group.

From the results of Problems 192 and 161, and from the group A_5 being noncommutative, it follows that the group A_5 is not soluble.

Problem 193. Prove that the symmetric group S_n , for $n \geq 5$, contains a subgroup isomorphic to A_5 .

From the results of Problems 193 and 162 we obtain the following theorem.

Theorem 5. For $n \geq 5$ the symmetric group S_n is not soluble.

The proof of this theorem, as well as the other results of this chapter, will be needed in the next chapter to demonstrate the non-solvability by radicals of algebraic equations of degree higher than four¹¹.

第二章 The Complex Numbers

When studying mathematics in school, the set of numbers considered was progressively extended. The reason for this was that these extensions allowed us to operate on numbers with more freedom. Thus, on passing from the natural numbers to the integers it became possible to subtract any two numbers; on passing to the rational numbers it became possible to divide any two numbers, etc. But the most important result of such extensions consists in the properties of the extended system often allowing us to discover some new properties of the initial system. For example, many difficult problems of number theory, concerning only integers, were solved using the real numbers as well as the complex numbers.

Historically, the complex numbers appeared just as a way of solving certain problems in the real numbers. So, for example, the Italian mathematician Cardano (1501–1576) devised a correct procedure for determining the roots of an equation of third degree using, in intermediate steps of calculations, the “non-existing” roots of negative numbers.

Afterwards the complex numbers played an increasingly important role in mathematics and applications. They were introduced for the first time in the theory of algebraic equations, because the domain of complex numbers turned out to be a more convenient setting for the study of such equations.

For example, every algebraic equation of degree n ($n \geq 1$) with real or complex coefficients has at least one complex root (see below the “fundamental theorem of algebra,” §2.8), whereas not all algebraic equations with real coefficients have at least one real root.

Since an interpretation of complex numbers was found in terms of vectors in the plane, geometrical notions such as that of continuity and geometrical transformation became applicable to the study of complex numbers. The relation between complex numbers and vectors also allows us to rewrite several problems of mechanics in terms of complex numbers and their equations—in particular, in hydrodynamics and aerodynamics, the theory of electricity, thermodynamics, etc.

2.1 Fields and Polynomials

Real numbers can be added and multiplied, and the inverse operations are also allowed: subtraction and division (the latter, however, not by zero). In any addition of several numbers the terms can be permuted in any way, and they can be collected arbitrarily within brackets without changing the result. The same holds for the factors of any product. All these properties, as well as the relation between addition and multiplication, can be summarized as follows.

The real numbers possess the following three properties:

1. They form a commutative group (see §1.3) under addition (the unit element of this group is denoted by 0 and is called zero).
2. If one excludes 0, then the real numbers form a commutative group under multiplication.
3. Addition and multiplication are related by distributivity: for any numbers a, b, c ,

$$a(b + c) = ab + ac.$$

The existence of these three properties is very important because they allow us to simplify the arithmetic of algebraic expressions, to solve equations, etc. The set of real numbers is not the only set to possess these three properties. In order to single out all these sets the following notion is introduced.

Definition. A set in which two binary operations (addition and multiplication) possessing the above properties are defined is called a field.

Problem 194. Verify whether the following subsets of the set of real numbers, with the usual operations of addition and multiplication, are fields: (a) all natural numbers; (b) all integer numbers; (c) all rational numbers; (d) all numbers of the type $r_1 + r_2\sqrt{2}$, where r_1 and r_2 are two arbitrary rational numbers.

Problem 195. Prove that in every field the identity $a \cdot 0 = 0 \cdot a = 0$ holds for any element a .

Problem 196. Prove that in every field: (1) $(-a) \cdot b = a \cdot (-b) = -(a \cdot b)$; (2) $(-a) \cdot (-b) = a \cdot b$ for any elements a and b .

Problem 197. Let a, b be two elements of an arbitrary field and let $a \cdot b = 0$. Prove that either $a = 0$ or $b = 0$.

Example 14. Suppose that in the set $\{0, 1, 2, \dots, n-1\}$, besides the operation of addition modulo n (see Example 9, §1.4), there is also given multiplication modulo n , which associates to two numbers the remainder of the division by n of their usual product.

Problem 198. Construct the tables of multiplication modulo 2, 3, and 4.

Problem 199. Prove that the remainders modulo n with the operations of addition and multiplication modulo n form a field if and only if n is a prime number.

Definition. By the difference $(b - a)$ of the elements a and b in an arbitrary field one denotes the element which solves the equation $x + a = b$ (or $a + x = b$). One calls the quotient of the division of the element b by a , for $a \neq 0$, denoted by b/a , the element which solves the equation $ya = b$ (or $ay = b$).

From the result of Problem 24 and from addition and multiplication in a field being commutative, it follows that the elements $b - a$ and b/a (for $a \neq 0$) are uniquely defined in all fields.

Since a field is a group under addition as well as, if one excludes zero, under multiplication, the equation $x + a = b$ is equivalent to the equation $x = b + (-a)$ and the equation $ya = b$, for $a \neq 0$, is equivalent to the equation $y = ba^{-1}$. Hence $b - a = b + (-a)$ and $b/a = ba^{-1}$.

The reader may easily prove that the operations of addition, subtraction, multiplication, and division in an arbitrary field possess all the basic properties which these operations possess in the field of real numbers. In particular, in any field the two members of an equation can be multiplied or divided by the same nonzero number; every term can be transported from one member to the other reversing its sign, etc. Consider, for instance, the property which relates subtraction and multiplication.

Problem 200. Prove that in any field $(a - b)c = ac - bc$ for any three elements a, b, c .

If K is a field then it is possible, as for the field of real numbers, to consider the polynomials with coefficients in the field K , or, in other words, the polynomials over K .

Definition. An expression like the following (n being a natural number)

$$a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n = 0, \quad (2.1)$$

where $a_0, a_1, \dots, a_n$ are elements of the field K , and $a_0 \neq 0$, is called a polynomial of degree n in one variable x over K .

If a is an element of the field K , the expression a is itself considered as a polynomial over K , and if $a \neq 0$ it represents a polynomial of degree zero, whereas if $a = 0$ the degree of this polynomial is considered to be undefined.

The elements $a_0, a_1, \dots, a_n$ are called the coefficients of the polynomial (2.1), and a_0 the leading coefficient.

Two polynomials in one variable x are considered to be equal if and only if the coefficients of the terms of the same degree in both polynomials coincide. Let

$$P(x) = a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n.$$

If in the second member of this equation one replaces x with an element a of the field K and one carries out the calculations indicated, i.e., the operations of addition and multiplication in the field K , one obtains as a result some element b of the field K . One thus writes $P(a) = b$. If $P(a) = 0$, where 0 is the zero element of the field K , one says that a is a root of the equation $P(x) = 0$; one also says that a is a root of the polynomial $P(x)$.

The polynomials over any field can be added, subtracted, and multiplied.

The sum of two polynomials $P(x)$ and $Q(x)$ is a polynomial $R(x)$ in which the coefficient of x^k ($k = 0, 1, 2, \dots$) is equal to the sum (in the field K) of the coefficients of x^k in the polynomials $P(x)$ and $Q(x)$. In the same way one defines the difference of two polynomials. It is evident that the degree of the sum or of the difference of two polynomials is not higher than the maximum of the degrees of the given polynomials.

To calculate the product of the polynomials $P(x)$ and $Q(x)$ one must multiply every monomial ax^k of the polynomial $P(x)$ by every monomial bx^l of the polynomial $Q(x)$ according to the rule $ax^k \cdot bx^l = abx^{k+l}$, where ab is the product in K , and $k + l$ is the usual sum of integers. Afterwards one must sum all the obtained expressions, collecting the monomials where the variable x has the same degree, and replacing the sum $d_1x^r + d_2x^r + \dots + d_sx^r$ by the expression $(d_1 + d_2 + \dots + d_s)x^r$. If

$$P(x) = a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n,$$

$$Q(x) = b_0x^m + b_1x^{m-1} + \dots + b_{m-1}x + b_m,$$

then¹

$$\begin{aligned} P(x) \cdot Q(x) &= a_0b_0x^{n+m} + (a_0b_1 + a_1b_0)x^{n+m-1} \\ &\quad + (a_0b_2 + a_1b_1 + a_2b_0)x^{n+m-2} + \dots + a_nb_m. \end{aligned}$$

Since $a_0 \neq 0$ and $b_0 \neq 0$ (cf. Problem 197), the degree of the product $P(x) \cdot Q(x)$ is equal to $n + m$, i.e., the degree of the product of two nonzero polynomials is equal to the sum of the degrees of the given polynomials.

Taking into account that the operations of addition and multiplication of the elements of the field K possess

the commutative, associative, and distributive properties, it is not difficult to verify that the introduced operations of addition and multiplication of polynomials also possess all these properties.

If

$$P(x) + Q(x) = R_1(x), \quad P(x) - Q(x) = R_2(x),$$

$$P(x) \cdot Q(x) = R_3(x),$$

and a is any element of the field K , one obtains

$$P(a) + Q(a) = R_1(a), \quad P(a) - Q(a) = R_2(a),$$

$$P(a) \cdot Q(a) = R_3(a).$$

The polynomials over an arbitrary field K can be divided by one another with a remainder. Dividing the polynomial $P(x)$ by the polynomial $Q(x)$ with a remainder means finding the polynomials $S(x)$ (quotient) and $R(x)$ (remainder) such that

$$P(x) = S(x) \cdot Q(x) + R(x);$$

moreover, either the degree of the polynomial $R(x)$ must be lower than the degree of the polynomial $Q(x)$, or one must have $R(x) = 0$.

Let $P(x)$ and $Q(x)$ be any two polynomials over the field K and $Q(x) \neq 0$. We show that it is possible to divide the polynomial $P(x)$ by the polynomial $Q(x)$ with a remainder.

Let

$$P(x) = a_0x^n + a_1x^{n-1} + \dots + a_n,$$

$$Q(x) = b_0x^m + b_1x^{m-1} + \dots + b_m.$$

If $n < m$ we put $S(x) = 0$ and $R(x) = P(x)$, and we obtain the quotient and the remainder required. If $n \geq m$ then consider the polynomial

$$P(x) - \frac{a_0}{b_0}x^{n-m}Q(x) = R_1(x).$$

The polynomial $R_1(x)$ contains no monomial in x^n because either its degree is not higher than $n - 1$, or $R_1(x) = 0$. If

$$R_1(x) = c_0x^k + c_1x^{k-1} + \dots + c_k$$

and $k \geq m$, then consider the polynomial

$$R_1(x) - \frac{c_0}{b_0}x^{k-m}Q(x) = R_2(x),$$

etc. Since the degree of the polynomial obtained is strictly lower than the degree of the preceding polynomial, this procedure must end, i.e., at some step we obtain

$$R_{s-1}(x) - \frac{d_0}{b_0}x^{l-m}Q(x) = R_s(x),$$

where either the degree of $R_s(x)$ is lower than the degree of $Q(x)$ or $R_s(x) = 0$. We thus have

$$\begin{aligned} P(x) &= \frac{a_0}{b_0} x^{n-m} Q(x) + R_1(x) \\ &= \frac{a_0}{b_0} x^{n-m} Q(x) + \frac{c_0}{b_0} x^{k-m} Q(x) + R_2(x) \\ &\vdots \\ &= \frac{a_0}{b_0} x^{n-m} Q(x) + \frac{c_0}{b_0} x^{k-m} Q(x) + \cdots + \frac{d_0}{b_0} x^{l-m} Q(x) + R_s(x) \\ &= \left(\frac{a_0}{b_0} x^{n-m} + \frac{c_0}{b_0} x^{k-m} + \cdots + \frac{d_0}{b_0} x^{l-m} \right) Q(x) + R_s(x). \end{aligned}$$

Consequently the expression within brackets is the quotient of the division of the polynomial $P(x)$ by the polynomial $Q(x)$ and $R_s(x)$ is the remainder. The procedure of division of two polynomials described here is called the Euclidean algorithm.

Problem 201. Let

$$P(x) = S_1(x) \cdot Q(x) + R_1(x),$$

$$P(x) = S_2(x) \cdot Q(x) + R_2(x),$$

where the degrees of $R_1(x)$ and $R_2(x)$ are lower than the degree of $Q(x)$ (possibly $R_1(x) = 0$ or $R_2(x) = 0$).

Prove that

$$S_1(x) = S_2(x), \quad R_1(x) = R_2(x).$$

2.2 The Field of Complex Numbers

From the solution of Problem 194 it follows that there exist fields smaller than the field of real numbers; for example, the field of rational numbers. We now construct a field which is bigger than the field of real numbers: the field of complex numbers.

Consider all possible pairs of real numbers, i.e., pairs of the type (a, b) , where a and b are two arbitrary real numbers. We will say that $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$. In the set of all these pairs we define two binary operations, addition and multiplication, in the following way:

$$(a, b) + (c, d) = (a + c, b + d), \tag{2.2}$$

$$(a, b) \cdot (c, d) = (ac - bd, ad + bc). \tag{2.3}$$

Here within brackets in the second members of the equations the operations are the usual operations on real numbers. For example, we obtain

$$(\sqrt{2}, 3) + (\sqrt{2}, -1) = (2\sqrt{2}, 2),$$

$$(0, 1) \cdot (0, 1) = (-1, 0).$$

Definition. The set of all pairs of real numbers with the operations of addition and multiplication defined by (2.2) and (2.3) is called the set of complex numbers.

From this definition it is clear that in the complex numbers there is nothing “supernatural”: the complex numbers are nothing but pairs of real numbers. However, a question may arise: is it correct to call such objects

numbers? We will answer this question at the end of this section. Another question, which perhaps the reader may put, is the reason why the operations of addition and multiplication of complex numbers are defined exactly in this manner and, in particular, why the operation of multiplication is so strange. We will answer this question in §2.3.

First, we clarify the remarkable properties of the set of complex numbers which we have defined.

Problem 202. Prove that the complex numbers form a commutative group under addition. Which complex number is the unit element (zero) of this group?

In the sequel it will be convenient to denote complex numbers by a single letter, for example by z (or w).

Problem 203. Prove that the operation of multiplication of complex numbers is commutative and associative, i.e., that $z_1 \cdot z_2 = z_2 \cdot z_1$ and $(z_1 \cdot z_2) \cdot z_3 = z_1 \cdot (z_2 \cdot z_3)$.

It is easy to verify that

$$(a, b) \cdot (1, 0) = (1, 0) \cdot (a, b) = (a, b)$$

for every complex number (a, b) . The complex number $(1, 0)$ is therefore the unit element in the set of complex numbers under multiplication.

Problem 204. Let z be an arbitrary complex number and suppose that $z \neq (0, 0)$. Prove that there exists a complex number z^{-1} such that

$$z \cdot z^{-1} = z^{-1} \cdot z = (1, 0).$$

The results of Problems 203 and 204 show that the nonzero complex numbers form a commutative group under multiplication.

Problem 205. Prove that the operations of addition and multiplication of complex numbers possess the distributive property, i.e., that

$$(z_1 + z_2) \cdot z_3 = z_1 \cdot z_3 + z_2 \cdot z_3$$

for any complex numbers z_1, z_2, z_3 .

From the results of Problems 202–205 it follows that the complex numbers with the operations of addition and multiplication defined by (2.2) and (2.3) form a field. This field is the field of complex numbers.

For complex numbers of type $(a, 0)$, where a is any real number, formulae (2.2) and (2.3) give

$$(a, 0) + (b, 0) = (a + b, 0),$$

$$(a, 0) \cdot (b, 0) = (a \cdot b, 0).$$

Consequently, if one associates to every complex number of type $(a, 0)$ the real number a , then to the operations on numbers of type $(a, 0)$ there correspond the usual operations on real numbers. Therefore we simply identify the complex number $(a, 0)$ with the real number a and we say that the field of complex numbers contains the field of real numbers².

The complex number $(0, 1)$ is not real (under our definition) and we will denote it by i : $i = (0, 1)$. Since the field of complex numbers contains all real numbers and the number i , it also contains all numbers of the

form $b \cdot i$ and $a + b \cdot i$, where a and b are any two real numbers and the operations of addition and multiplication are extended to operations on complex numbers.

Problem 206. Let (a, b) be a complex number. Prove that

$$(a, b) = a + b \cdot i.$$

From the result of Problem 206 we obviously obtain that $a + bi = c + di$ if and only if $a = c$ and $b = d$.

As a consequence, every complex number can be represented in a unique way in the form $a + bi$, where a and b are two real numbers. If $z = a + bi$ then, following tradition, a is called the real part of the complex number, bi the imaginary part, and b the coefficient of the imaginary part.

The representation of a complex number z in the form $z = a + bi$ is called the algebraic representation of z .

For complex numbers in algebraic representation formulae (2.2) and (2.3) read:

$$\begin{aligned}(a + bi) + (c + di) &= (a + c) + (b + d)i, \\ (a + bi) \cdot (c + di) &= (ac - bd) + (ad + bc)i.\end{aligned}$$

Problem 207. Solve the equation (i.e., find the formula for the difference)

$$(a + bi) + (x + yi) = (c + di).$$

Problem 208. Solve the equation (i.e., find the formula for the quotient)

$$(a + bi) \cdot (x + yi) = (c + di), \quad \text{where} \quad a + bi \neq 0.$$

It is easy to verify that

$$i \cdot i = (0, 1) \cdot (0, 1) = (-1, 0) = -1,$$

i.e., $i^2 = -1$. Hence in the field of complex numbers, square roots of negative numbers are well defined.

Problem 209. Calculate (a) i^3 ; (b) i^4 ; (c) i^n .

Problem 210. Find all complex numbers $z = x + yi$ such that: (a) $z^2 = 1$; (b) $z^2 = -1$; (c) $z^2 = a^2$; (d) $z^2 = -a^2$ (a is a real number).

Definition. The complex number $a - bi$ is called the conjugate of $z = a + bi$ and it is denoted by $\bar{z}$. It is easy to verify that

$$z + \bar{z} = 2a, \quad z \cdot \bar{z} = a^2 + b^2.$$

Problem 211. Let z_1 and z_2 be two arbitrary complex numbers. Prove that: (a) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$; (b) $\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$; (c) $\overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2$; (d) $\overline{z_1 / z_2} = \bar{z}_1 / \bar{z}_2$.

Problem 212. Let

$$P(z) = a_0 z^n + a_1 z^{n-1} + \dots + a_{n-1} z + a_n,$$

where z is a complex number and all the a_i are real. Prove that $\overline{P(z)} = P(\bar{z})$.

The passage to the complex numbers is a successive step in the series: natural numbers—integer numbers—rational numbers—real numbers—complex numbers. The reader may feel that up to the real numbers one deals with numbers, whereas the complex numbers are objects of another nature. Of course, one may use whatever terminology one wishes, but the complex numbers must, in fact, be considered as numbers.

The first objection against this is that complex numbers are not numbers, but pairs of numbers. Recall, however, that in a similar way one introduces rational numbers. A rational number is a class of equivalent fractions, and a fraction is a pair of integer numbers of the form m/n (where $n \neq 0$); in this way the operations on rational numbers are simply operations on pairs of integer numbers. Another objection might be that a number is an object which allows us to measure something. If we think that numbers are entities by which one can measure everything, then one must exclude from the set of these entities, for example, the negative numbers because there are no segments of length -3 cm, and a train cannot go for -4 hours. If, on the contrary, one thinks that numbers are objects by which it is possible (or convenient) to measure at least one quantity, then complex numbers are similar to the other numbers: with them one describes very well, for example, the potential and the resistance of alternating currents in electric circuits, which are extensively utilized in electrotechnology. Complex numbers are also successfully employed in hydrodynamics and aerodynamics.

So the passage from real to complex numbers is as natural as, for example, the passage from integer to rational numbers.

2.3 Uniqueness of the Field of Complex Numbers

Consider now the question of whether complex numbers could be defined otherwise.

In other words, the question that we answer in this section is the following: we want to obtain a field which is an extension of the field of real numbers; does there exist more than one field which is an extension of the field of real numbers?

Definition. We call an isomorphic mapping (or simply an isomorphism) of one field onto another a bijective mapping ϕ which is an isomorphism with respect both to addition and to multiplication, i.e., such that $\phi(a+b) = \phi(a) + \phi(b)$ and $\phi(ab) = \phi(a)\phi(b)$. Two fields between which one can define an isomorphism are said to be isomorphic.

If in a field one considers exclusively the operations of addition and multiplication, then isomorphic fields all have identical properties. As a consequence, as in the case of groups, isomorphic fields cannot be distinguished.

As we have seen in the preceding section, in the field of complex numbers there is only one element i such that $i^2 = -1$. The following problem shows that on adding this element to the field of real numbers one necessarily obtains the field of complex numbers.

Problem 213. Let M be a field containing the field of real numbers and a certain element i_0 such that $i_0^2 = -1$. Prove that M contains a field M' isomorphic to the field of complex numbers.

We will say that a field is the minimal field with the required properties if it possesses these properties and it does not contain other fields with the same properties.

The result of Problem 213 can then be formulated in this way: the minimal field which contains the field of real numbers and an element i_0 such that $i_0^2 = -1$ is the field of complex numbers. This result proves in a certain sense the uniqueness of the field of complex numbers. However, another, much stronger, result holds. Indeed, suppose one renounces the requirement that the field M contains an element i_0 such that $i_0^2 = -1$, and poses the problem of finding all fields that are minimal extensions of the field of real numbers. It turns out that there are only two such extensions (up to isomorphism): one of them is the field of complex numbers. Prove this statement.

Suppose the field M contains all real numbers and that the operations on them coincide with the usual operations on real numbers. Suppose, moreover, that M contains an element j , different from all real numbers. Thus for every set of n real numbers $a_1, a_2, \dots, a_n$, there exists in M an element equal to

$$j^n + a_1 j^{n-1} + a_2 j^{n-2} + \dots + a_n. \quad (2.6)$$

We call n the degree of expression (2.6).

There are two possible cases:

- (a) a certain expression of the form (2.6) is equal to 0 for $n \geq 1$;
- (b) there are no expressions of the form (2.6) equal to 0 for $n \geq 1$.

Suppose first that we are in case (a).

Definition. A polynomial with coefficients in a certain field K is said to be reducible over K if it can be represented as a product of two polynomials of lower degree with coefficients in K . In the opposite case it is said to be irreducible³ over K .

For example, the polynomials $x^3 - 1$ and $x^2 - x - 1$ are reducible over the field of real numbers, because

$$x^3 - 1 = (x - 1)(x^2 + x + 1)$$

and

$$x^2 - x - 1 = \left(x - \frac{1 + \sqrt{5}}{2}\right) \left(x - \frac{1 - \sqrt{5}}{2}\right),$$

whereas the polynomials $x^2 + 1$ and $x^2 + x + 1$ are irreducible over the field of real numbers. It is evident that polynomials of the first degree over any field are irreducible.

Problem 214. Let us choose, amongst all expressions of type (2.6), the expression of minimal degree n ($n \geq 1$) that vanishes in M : the corresponding equation is

$$j^n + a_1 j^{n-1} + a_2 j^{n-2} + \dots + a_n = 0.$$

Prove that the polynomial

$$x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n$$

is not reducible over the field of real numbers.

In the sequel we will prove (cf. Problem 272) that every polynomial with real coefficients of degree higher than 2 is reducible over the field of real numbers. Hence in Problem 214, n cannot be higher than 2. But since $n \neq 1$ (otherwise we should have $j + a = 0$, and j would be equal to the real number $-a$), we obtain that $n = 2$.

Consequently, in case (a) there exist two real numbers p and q in M which satisfy

$$j^2 + pj + q = 0$$

and for which the polynomial $x^2 + px + q$ is irreducible over the field of real numbers.

Problem 215. Prove that in case (a) the field M contains an element i_0 such that $i_0^2 = -1$.

From the results of Problems 215 and 213 it follows that in case (a) the field M contains a field M' isomorphic to the field of complex numbers. Therefore, if the field M is a minimal extension of the field of real numbers, then the field M must coincide with M' . As a consequence, in case (a) any field which represents a minimal extension of the field of real numbers coincides (i.e., is isomorphic) with the field of complex numbers. So in case (a) there is only one field (up to isomorphism) which is a minimal extension of the field of real numbers, namely the field of complex numbers.

Problem 216. Find all fields that are minimal extensions of the field of real numbers in case (b).

2.4 Geometrical Descriptions of the Complex Numbers

Consider on the plane a system of orthogonal coordinates (x, y) and let us associate to every complex number $a + ib$ the point of the plane with coordinates (a, b) . We obtain a bijective correspondence between all complex numbers and all points of the plane. This is the first geometrical representation of the complex numbers.

Problem 217. Which complex numbers correspond to the points shown in Figure 13?

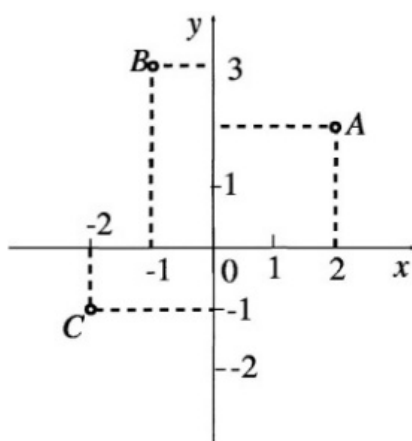


图 2.1

Problem 218. Let the complex numbers be represented by the points of the plane. What is the geometrical meaning of the mapping ϕ if for every complex number z : (a) $\phi(z) = -z$; (b) $\phi(z) = 2z$; (c) $\phi(z) = \bar{z}$ ($\bar{z}$ is

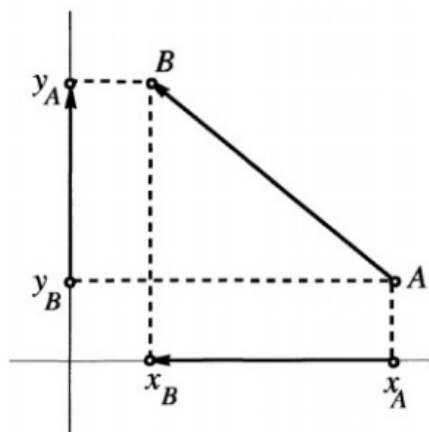


图 2.2

the conjugate of z)?

Let $A(x_A, y_A)$ and $B(x_B, y_B)$ be two points of the plane (Figure 14). The segment AB directed from A to B is called the vector $\overrightarrow{AB}$. The coordinates of the vector $\overrightarrow{AB}$ are by definition calculated in the following way:

$$x_{\overrightarrow{AB}} = x_B - x_A, \quad y_{\overrightarrow{AB}} = y_B - y_A.$$

Two vectors are considered equal if they are parallel and have the same direction and the same length.

Problem 219. Prove that two vectors are equal if and only if their coordinates are equal.

The set of equal vectors is considered to be a unique vector, characterized by its coordinates, which is called a free vector. Putting into correspondence every complex number $a + bi$ with the free vector having coordinates (a, b) , we obtain the second geometrical representation of the complex numbers.

Problem 220. Let u, v , and w be the free vectors corresponding to the complex numbers z_1, z_2 , and z_3 . Prove that $z_1 + z_2 = z_3$ if and only if $u + v = w$, where the vectors are added according to the parallelogram rule.

Problem 221. Prove the following relation between the two geometrical representations of the complex numbers: if z_A, z_B , and $z_{\overrightarrow{AB}}$ are the complex numbers corresponding to the points A, B , and to vector $\overrightarrow{AB}$, then

$$z_{\overrightarrow{AB}} = z_B - z_A.$$

By definition, two equal vectors have equal lengths. This length is additionally assumed to be the length of the free vector corresponding to a given set of equal vectors.

Definition. One calls the modulus of the complex number z (denoted by $|z|$) the length of the corresponding free vector⁴.

Problem 222. Let $z = a + bi$. Prove that

$$|z|^2 = a^2 + b^2 = z \cdot \bar{z},$$

where $\bar{z}$ is the conjugate of z .

Problem 223. Prove the inequalities

$$|z_1 + z_2| \leq |z_1| + |z_2|, \quad |z_1 - z_2| \geq ||z_1| - |z_2||,$$

where z_1, z_2 are arbitrary complex numbers. In which case does equality hold?

Problem 224. Prove by means of the complex numbers that in any parallelogram the sum of the squares of the lengths of the diagonals is equal to the sum of the squares of the lengths of all the sides.

2.5 The Trigonometric Form of the Complex Numbers

Recall that the angle between the rays OA and OB is defined as the angle by which one has to turn counterclockwise the ray OA around O in order to take it over to the ray OB (if the rotation is clockwise, the angle has the sign “minus”). Thus the angle is not defined uniquely, but up to rotations by $2\pi k$, where k is any integer.

Definition. Let point O be the origin of the coordinates, and suppose that the vector OA with coordinates (a, b) corresponds to the complex number $z = a + bi$. One calls the argument of the complex number z (denoted by $\arg z$) the angle between the positive direction of the axis Ox and the ray OA (Figure 15). If $z = 0$, then $\arg z$ is not defined.

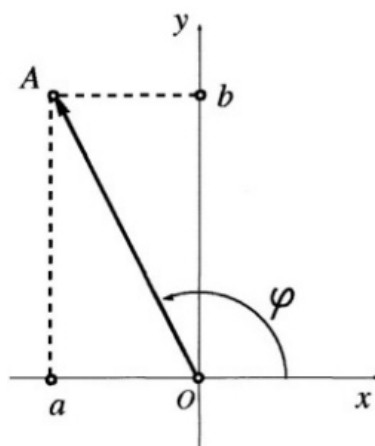


图 2.3

Since for a given number $z \neq 0$ the angle is not uniquely defined, by the expression $\arg z$ we mean a multi-valued function taking infinitely many values, between which the differences are equal to multiples of 2π .

The expression $\arg z = \varphi$ will mean that one of the values of the argument is equal to φ .

Let $z = a + bi \neq 0$ and $|z| = r$. The vector $\overrightarrow{OA}$ with coordinates (a, b) corresponds to the complex number $a + bi$, and its length is therefore equal to r . Let $\arg z = \varphi$. Thus, by the definition of the trigonometric functions (see Figure 15),

$$\cos \varphi = \frac{a}{r}, \quad \sin \varphi = \frac{b}{r},$$

and therefore

$$z = a + bi = r \cos \varphi + i r \sin \varphi = r(\cos \varphi + i \sin \varphi),$$

where $r = |z|$ and $\varphi = \arg z$. Thus we have obtained the trigonometric representation of the complex number z .

For example, if $z = -1 + \sqrt{3}i$, then $|z| = \sqrt{1+3} = 2$ (cf. Problem 222) and $\cos \varphi = -1/2$, $\sin \varphi = \sqrt{3}/2$. We can assume that $\varphi = 2\pi/3$; thus

$$z = -1 + \sqrt{3}i = 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right).$$

Problem 225. Represent in trigonometric form the following complex numbers: (a) $1+i$; (b) $-\sqrt{3}-i$; (c) $3i$; (d) -5 ; (e) $1+2i$.

Problem 226. Let $z_1 = r_1(\cos \varphi_1 + i \sin \varphi_1)$ and $z_2 = r_2(\cos \varphi_2 + i \sin \varphi_2)$. Prove that

$$\begin{aligned} z_1 \cdot z_2 &= r_1 r_2 (\cos(\varphi_1 + \varphi_2) + i \sin(\varphi_1 + \varphi_2)), \\ \frac{z_1}{z_2} &= \frac{r_1}{r_2} (\cos(\varphi_1 - \varphi_2) + i \sin(\varphi_1 - \varphi_2)), \quad z_2 \neq 0. \end{aligned}$$

Thus, as a result of multiplication, the moduli of the complex numbers are multiplied and their arguments are added; as a result of division, the moduli are divided and the arguments are subtracted.

Problem 227. Prove the De Moivre formula⁵

$$[r(\cos \varphi + i \sin \varphi)]^n = r^n (\cos n\varphi + i \sin n\varphi)$$

for every integer $n > 0$.

Problem 228. Calculate $(1 - \sqrt{3}i)^{100} / 2^{100}$.

Problem 229. Let $z = r(\cos \varphi + i \sin \varphi)$ be a given complex number and n a natural number. Find all complex numbers w satisfying the equation

$$w^n = z. \quad (2.7)$$

Definition. The expression $\sqrt[n]{z}$ (the n th root of z) denotes a multi-valued function which puts into correspondence with every complex number $z \neq 0$ all the n roots of equation (2.7). For $z = 0$ one has $\sqrt[n]{0} = 0$.

Problem 230. Find all the values of the roots:

- (a) $\sqrt{-1}$;
- (b) $\sqrt[3]{8}$;
- (c) $\sqrt[4]{\cos 100^\circ + i \sin 100^\circ}$;
- (d) $\sqrt[3]{1+i}$.

It will be convenient in the sequel to adopt the notation

$$\epsilon_n = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}.$$

Problem 231. Prove that all the values of $\sqrt[n]{1}$ are $1, \epsilon_n, \epsilon_n^2, \dots, \epsilon_n^{n-1}$.

Remark. Since $\epsilon_n^n = 1$, the set of elements $1, \epsilon_n, \epsilon_n^2, \dots, \epsilon_n^{n-1}$ is a cyclic group under multiplication.

Problem 232. Let z_1 be one of the values of $\sqrt[n]{z_0}$. Find all the values of $\sqrt[n]{z_0}$.

We shall use the representation of complex numbers by the points of the plane, i.e., the complex number $a + bi$ will correspond to the point of the plane having coordinates (a, b) . So instead of the point corresponding to the complex number z we shall say simply point z .

Problem 233. Let the complex numbers be represented by the points of the plane. What is the geometrical meaning of the following expressions: (a) $|z|$; (b) $\arg z$; (c) $|z_1 - z_2|$; (d) $\arg(z_1/z_2)$?

Problem 234. Find the position on the plane of the points z satisfying the following conditions (where z_0, z_1, z_2 are given complex numbers and R is a given real number): (a) $|z| = 1$; (b) $|z| = R$; (c) $|z - z_0| = R$; (d) $|z - z_0| \leq R$; (e) $|z - z_1| = |z - z_2|$; (f) $\arg z = \pi$; (g) $\arg z = 9\pi/4$; (h) $\arg z = \varphi$.

Problem 235. How are all the values of $\sqrt[3]{z}$ distributed on the plane, z being a given complex number?

2.6 Continuity

In the sequel an important role will be played by the notion of continuity and, in particular, by that of a continuous curve. If the reader does not know the precise definition of such entities he understands intuitively, however, what a continuous curve is, as well as what a continuous function of one real variable is (at the intuitive level one may say that it is a function whose graph is a continuous curve). But if the function of one real variable is sufficiently complicated (for example, $f(x) = (x^3 - 2x)/(x^2 - \sin x + 1)$), then saying whether it is continuous or not, using only the intuitive idea of continuity, is rather difficult. Hence we give the rigorous definition of continuity and, by means of it, we will prove some basic properties of continuous functions. We give the definition of continuity for functions of one real variable as well as for functions of one complex variable.

The graph of a function of one real argument can be discontinuous at some points, and at some points it can have some breaks. It is therefore natural to consider first the notion of continuity of a function at a given point rather than the general definition of continuity.

If we try to define more precisely our intuitive idea of continuity of a function $f(x)$ at a given point x_0 , we obtain that continuity means the following: under small changes of the argument near the point x_0 the value of the function changes a little with respect to the value $f(x_0)$. Moreover, it is possible to obtain a variation of the function's value about $f(x_0)$ as small as we want by choosing a sufficiently small interval of the variation of the argument around x_0 . One can formulate this more rigorously in the following way.

Definition. Let f be a function of one real or complex variable z . One says that the function $f(z)$ is continuous at z_0 if for every arbitrary real number $\varepsilon > 0$ one can choose a real number $\delta > 0$ (which depends on z_0 and on ε) such that for all numbers z satisfying the condition $|z - z_0| < \delta$ the inequality⁶ $|f(z) - f(z_0)| < \varepsilon$ holds.

Example 15. Prove that the function with complex argument $f(z) = 2z$ is continuous at any point z_0 . Suppose a point z_0 and a real number $\varepsilon > 0$ be given. We have to choose a real number $\delta > 0$ such that for all numbers z which satisfy the condition $|z - z_0| < \delta$ the inequality $|f(z) - f(z_0)| = |2z - 2z_0| < \varepsilon$ is satisfied. It is

not difficult to see that one can choose $\delta = \varepsilon/2$ (independently of the point z_0). Indeed, from the condition $|z - z_0| < \delta$ it follows that

$$|2z - 2z_0| = |2(z - z_0)| = |2| \cdot |z - z_0| < 2\delta = \varepsilon,$$

i.e., $|2z - 2z_0| < \varepsilon$. As a consequence the function $f(z) = 2z$ is continuous at any point z_0 . In particular, it is continuous for all real values of the argument z . Therefore, if one restricts the function to real values of the argument, one obtains that the function of the real variable $f(x) = 2x$ is continuous for all real values x . **Problem 236.** Let a be a complex number (or, as a particular case, real). Prove that the complex (or real) function $f(z) \equiv a$ is continuous for all values of the argument.

Problem 237. Prove that the function of a complex argument $f(z) = z$ and the function of a real argument $f(x) = x$ are continuous for all values of their argument.

Problem 238. Prove that the function of a complex argument $f(z) = z^2$ is continuous for all values of z .

定义 2.1

Let $f(z)$ and $g(z)$ be two functions of a complex (or real) argument. One calls the sum of the functions $f(z)$ and $g(z)$ the function $h(z)$ of a complex (or real) argument which satisfies at every point z_0 the equation

$$h(z_0) = f(z_0) + g(z_0).$$

If the value $f(z_0)$ or the value $g(z_0)$ is not defined, then the value $h(z_0)$ is also not defined. In the same way one defines the difference, the product, and the quotient of two functions. 

Problem 239. Let $f(z)$ and $g(z)$ be functions of a complex or real argument, continuous at z_0 . Prove that at z_0 the following functions are continuous:

- (a) $h(z) = f(z) + g(z)$;
- (b) $h(z) = f(z) - g(z)$;
- (c) $h(z) = f(z) \cdot g(z)$.

From the result of Problem 239(c) we obtain, in particular, that if a function $f(z)$ is continuous at a point z_0 and n is an integer, then the function $[f(z)]^n$ is also continuous at the point z_0 .

Problem 240. Let $f(z)$ and $g(z)$ be two functions of a complex or real argument, continuous at z_0 , and suppose that $g(z_0) \neq 0$. Prove that at z_0 the following functions are continuous:

- (a) $h(z) = 1/g(z)$;
- (b) $h(z) = f(z)/g(z)$.

定义 2.2

Let $f(z)$ and $g(z)$ be two functions of a complex or real argument. One calls the composition of the functions $f(z)$ and $g(z)$ the function $h(z)$ which satisfies at every point z_0 the equation $h(z_0) = f(g(z_0))$.

If the value $g(z_0)$ is not defined, or the function $f(z)$ is not defined at the point $g(z_0)$, then the value $h(z_0)$ is also not defined. 

Problem 241. Let $f(z)$ and $g(z)$ be two functions of a complex or real argument. Let $g(z_0) = z_1$, and let the functions $f(z)$ and $g(z)$ be continuous at the points z_1 and z_0 , respectively. Prove that the function $h(z) = f(g(z))$ is continuous at the point z_0 .

From the results of Problems 239–241 it follows, in particular, that any expression obtained from any functions of one complex (or real) argument, continuous for all values of the argument, by means of the operations of addition, subtraction, multiplication, division, elevation to an integer power, and composition, represents a continuous function at all points at which the denominator does not vanish.

For example, from the results of Problems 236 and 237, we obtain that the functions $f(z) = z^n$, $f(z) = az^n$, and, more generally,

$$f(z) = a_0 z^n + a_1 z^{n-1} + \dots + a_n,$$

are continuous functions of z for all complex numbers $a, a_0, a_1, \dots, a_n$.

Problem 242. Prove that the functions of a real argument $f(x) = \sin x$ and $f(x) = \cos x$ are continuous for all values of x .

Problem 243. Consider for all real values $x \geq 0$ the function $f(x) = \sqrt[n]{x}$, where n is a non-zero integer and a non-negative value is chosen for $\sqrt[n]{x}$. Prove that this function is continuous for every $x > 0$.

During the study of continuity one encounters some statements which intuitively seem evident; however, their exact proofs involve serious technical difficulties and require a definition of the real numbers stricter than that learnt at school, as well as the study of the foundations of set theory and topology.

An example of such statements can be represented by the following proposition. If a function $f(x)$ of a real argument is continuous in some interval and in this interval takes only integer values, then it takes only one value in the whole interval. Indeed, it seems intuitively evident that, as the point x moves in the interval, the value of the function $f(x)$ must change continuously and cannot “jump” from one integer value to another one. Proving this proposition in an exact way is, however, rather difficult.

In the present exposition we rely on the “intuitive evidence” of some propositions related to continuity without giving demonstrations of them. In particular, we adopt, without proof, some propositions which we have formulated in the form of examples.

2.7 Continuous curves

Suppose that the parameter t takes real values in the interval $0 \leq t \leq 1$, and that each of these values is in correspondence with some complex number

$$z(t) = x(t) + iy(t).$$

The plane on which the values of z are represented will be called simply the z -plane. If the functions $x(t)$ and $y(t)$ are continuous for $0 \leq t \leq 1$, then as t varies from 0 to 1, the point $z(t)$ describes a continuous curve in the z -plane. We provide this curve with an orientation, assuming that $z_0 = z(0)$ and $z_1 = z(1)$ are the initial and the final points, respectively. The function $z(t)$ is called the parametric equation of this curve.

例题 2.1 Let $z(t) = t + it^2$. Thus $x(t) = t$, $y(t) = t^2$, and $y(t) = x^2(t)$ for every t , i.e., the point $z(t)$ lies on the parabola $y = x^2$ for every t . As t varies from 0 to 1, $x(t)$ also varies from 0 to 1 and the point $z(t)$ runs along the parabola $y = x^2$ from the point $z_0 = 0$ to the point $z_1 = 1 + i$ (Figure 2.4).

Problem 244. Trace on the z -plane the curves given by the following parametric equations:

- (a) $z(t) = 2t$;
- (b) $z(t) = it$;

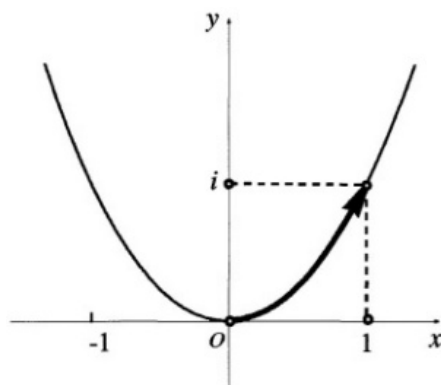


图 2.4: The curve $z(t) = t + it^2$.

- (c) $z(t) = it^2$;
- (d) $z(t) = t - it$;
- (e) $z(t) = t^2 + it$;
- (f) $z(t) = R(\cos 2\pi t + i \sin 2\pi t)$;
- (g) $z(t) = R(\cos 4\pi t + i \sin 4\pi t)$;
- (h) $z(t) = R(\cos \pi t + i \sin \pi t)$;
- (i)

$$z(t) = \begin{cases} \cos 2\pi t + i \sin 2\pi t, & \text{for } 0 \leq t \leq \frac{1}{2}, \\ 4t - 3, & \text{for } \frac{1}{2} < t \leq 1. \end{cases}$$

Problem 245. Write a parametric equation for the segment joining the points $z_0 = a_0 + b_0i$ and $z_1 = a_1 + b_1i$.

注 In the following problems the parametric equations have some indices. These numbers have to be viewed only as labels, but all curves lie in the same z -plane.

Problem 246. By means of which geometrical transformations of the curve C_1 with equation $z_1(t)$ can we obtain the curve C_2 with equation $z_2(t)$ if:

- (a) $z_2(t) = z_1(t) + z_0$, where z_0 is a given complex number;
- (b) $z_2(t) = a \cdot z_1(t)$, where a is a positive real number;
- (c) $z_2(t) = z_0 \cdot z_1(t)$, where $|z_0| = 1$;
- (d) $z_2(t) = z_0 \cdot z_1(t)$, where z_0 is an arbitrary complex number?

Problem 247. Let $z_1(t)$ be the parametric equation of curve C . What is the curve described by the equation $z_2(t)$ if $z_2(t) = z_1(1 - t)$?

Problem 248. Let $z_1(t)$ and $z_2(t)$ be the parametric equations of curves C_1 and C_2 , and suppose that $z_1(1) = z_2(0)$. What is the curve described by the equation $z_3(t)$ if

$$z_3(t) = \begin{cases} z_1(2t), & \text{for } 0 \leq t \leq \frac{1}{2}, \\ z_2(2t - 1), & \text{for } \frac{1}{2} < t \leq 1? \end{cases}$$

Problem 249. Let $z(t) = \cos \pi t + i \sin \pi t$ (Figure 2.5). Find all values of $\arg z(t)$ as a function of t .

Problem 250. Let $z(t) = \cos \pi t + i \sin \pi t$. How should we choose one of the values of $\arg z(t)$ for every t in such a way that the values of $\arg z(t)$ vary continuously as t varies from 0 to 1, with the condition that $\arg z(0)$ is equal to: (a) 0; (b) 2π ; (c) -4π ; (d) $2\pi k$, where k is a given integer?

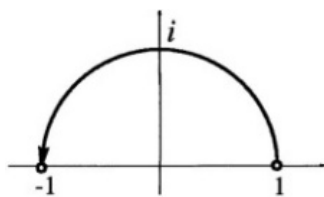


图 2.5: The curve in Problems 249 and 250.

定理 2.1

Assume that a continuous curve C with a parametric equation $z(t)$ does not cross the origin of coordinates and that at the initial point of the curve C the argument is φ_0 . Thus it is possible to choose one of the values of the argument for all points of the curve C so that along the entire curve the argument of $z(t)$ changes continuously, starting from φ_0 .

In other words, one can choose for every t one of the values $\varphi(t)$ of $\arg z(t)$ in order that the function $\varphi(t)$ be continuous for $0 \leq t \leq 1$ and $\varphi(0) = \varphi_0$. (See note⁷.)



Problem 251. Let $\varphi(t)$ and $\varphi'(t)$ be two functions describing the continuous variation of $\arg z(t)$ along the curve C . Prove that $\varphi(t) - \varphi'(t) = 2\pi k$, where k is a given integer which does not depend on t .

Problem 252. Prove that if one chooses a value $\varphi(0) = \varphi_0$, then the function $\varphi(t)$, which describes the continuous variation of $\arg z(t)$ along the curve C , is uniquely defined.

Problem 253. Let $\varphi(t)$ be a function describing the continuous variation of $\arg z(t)$. Prove that the function $\psi(t) = \varphi(t) - \varphi(0)$ is uniquely defined by the function $z(t)$ and does not depend on the choice of $\varphi(0)$.

From the statement of Problem 253 it follows, in particular for $t = 1$, that for a continuous curve C not passing through the point $z = 0$ the quantity $\varphi(1) - \varphi(0)$ is uniquely defined by the continuity condition of $\varphi(t)$.

定义 2.3

The difference $\varphi(1) - \varphi(0)$ is called the variation of the argument along the curve C .



Problem 254. Find the variation of the argument along the curves with the following parametric equations: (a) $z(t) = \cos \pi t + i \sin \pi t$; (b) $z(t) = \cos 2\pi t + i \sin 2\pi t$; (c) $z(t) = \cos 4\pi t + i \sin 4\pi t$; (d) $z(t) = (1 - t) + it$.

Problem 255. What is the variation of the argument along the curves shown in Figure 2.6?



图 2.6: Curves for Problem 255.

注 If a curve C is closed, i.e., $z(1) = z(0)$, then the quantity $\varphi(1) - \varphi(0)$ is equal to $2\pi k$, where k is an integer.

定义 2.4

If for a continuous closed curve C not passing through a point $z = 0$ the variation of the argument is equal to $2\pi k$, then we say that the curve C turns k times around the point $z = 0$.



Problem 256. How many times do the following curves turn around the point $z = 0$: (a) $z(t) = 2 \cos 2\pi t + 2i \sin 2\pi t$ (Figure 19); (b) $z(t) = \frac{1}{2} \cos 4\pi t - \frac{1}{2}i \sin 4\pi t$ (Figure 20); (c) the curve in Figure 21; (d) the curve in Figure 22?

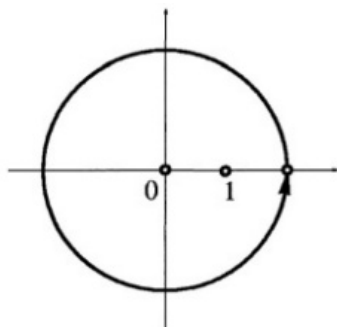


FIGURE 19

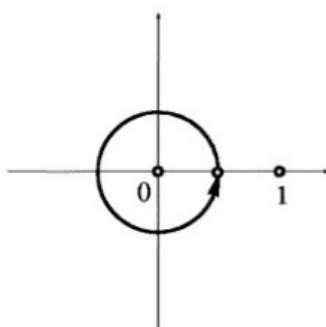


FIGURE 20

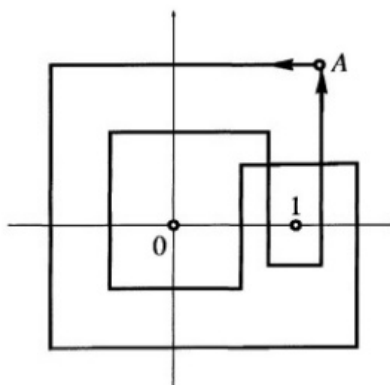
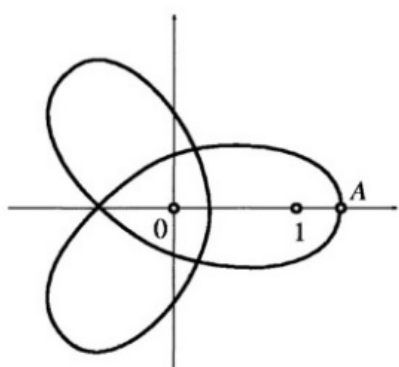


图 2.7: Figures 19–22: curves for Problem 256.

Problem 257. Prove that the number of turns of a closed continuous curve around the point $z = 0$ does not depend on the choice of the initial point, and depends only on the orientation of the curve.

Problem 258. Suppose that a curve C with equation $z_1(t)$ turns k times around the point $z = 0$. How many times does the curve with the equation $z_2(t)$ turn around the point $z = 0$ if: (a) $z_2(t) = 2 \cdot z_1(t)$; (b) $z_2(t) = -z_1(t)$; (c) $z_2(t) = z_0 \cdot z_1(t)$, where $z_0 \neq 0$; (d) $z_2(t) = \overline{z_1(t)}$, where $\bar{z}$ is the conjugate of z ?

定义 2.5

Suppose that a continuous curve C with the equation $z_1(t)$ does not pass through the point $z = z_0$. We say that the curve C turns k times around the point $z = z_0$ if the curve with equation $z_2(t) = z_1(t) - z_0$ turns k times around the point $z = 0$ (Figure 2.8).



Consequently, to define the number of turns of a curve around the point $z = z_0$, we have to look at the rotation of the vector $z_1(t) - z_0$, i.e., the vector joining points z_0 and $z_1(t)$ (cf. 221).

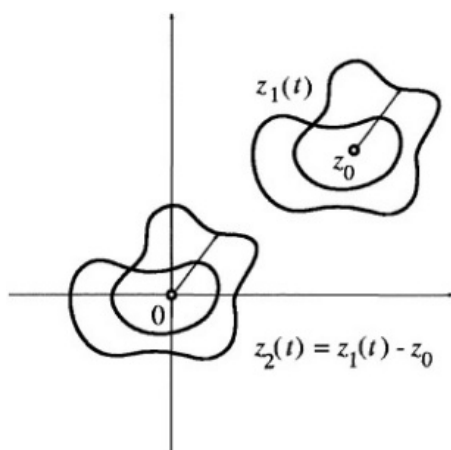


图 2.8: Turns of a curve around the point z_0 .

Problem 259. How many times do the curves described in Problem 256 turn around the point $z = 1$?

Problem 260. Let $z_1(t)$ and $z_2(t)$ be the equations of two curves C_1 and C_2 not passing through the point $z = 0$. Let the variations of the argument along these curves be equal, respectively, to φ_1 and φ_2 . What is the variation of the argument along the curve C with equation $z(t)$ if: (a) $z(t) = z_1(t) \cdot z_2(t)$; (b) $z(t) = z_1(t)/z_2(t)$?

2.8 Images of curves: the basic theorem of the algebra of complex numbers

Consider two planes of complex numbers: the z -plane and the w -plane, and a function $w = f(z)$, which puts into correspondence with every value z a uniquely defined value w . If on the z -plane there is a continuous curve C having equation $z(t)$, then by the function $w = f(z)$ every point of this curve is sent to a point of the w -plane. Hence, if the function $f(z)$ is continuous, we also obtain on the w -plane a continuous curve having equation $w_0(t) = f(z(t))$. We shall denote this curve by $f(C)$.

Problem 261. What is the curve $f(C)$ if $w = f(z) = z^2$ and the curve C is

- (a) a quarter of a circle: $z(t) = R(\cos(\pi t/2) + i \sin(\pi t/2))$;
- (b) a semicircle: $z(t) = R(\cos \pi t + i \sin \pi t)$;
- (c) a circle: $z(t) = R(\cos 2\pi t + i \sin 2\pi t)$?

Problem 262. Let the variation of the argument along a curve C be equal to φ . What is the variation of the argument along the curve $f(C)$ if: (a) $f(z) = z^2$; (b) $f(z) = z^3$; (c) $f(z) = z^n$, where n is an arbitrary non-zero integer?

Problem 263. Suppose that the curve C turns k times around the point $z = z_0$. How many times does the curve $f(C)$ turn around the point $w = 0$ if $f(z) = (z - z_0)^n$?

Problem 264. Suppose that a curve C turns around the points $z = 0$, $z = 1$, $z = i$, and $z = -i$, respectively, k_1 , k_2 , k_3 , and k_4 times. How many times does the curve $f(C)$ turn around the point $w = 0$ if: (a) $f(z) = z^2 - z$; (b) $f(z) = z^2 + 1$; (c) $f(z) = (z + iz)^4$; (d) $f(z) = z^3 - z^2 + z - 1$?

Consider the equation

$$a_0 z^n + a_1 z^{n-1} + \dots + a_{n-1} z + a_n = 0,$$

where all the a_i are arbitrary complex numbers, $n \geq 1$, and $a_0 \neq 0$. Our first aim is to show that this equation has at least one complex root. If $a_n = 0$, then the equation possesses the root $z = 0$. In the sequel, therefore,

we will assume that $a_n \neq 0$.

Let us denote by A the maximum amongst the numbers $|a_0|, |a_1|, \dots, |a_n|$. Since $a_0 \neq 0$, $A > 0$. Choose two real numbers R_1 and R_2 with the following conditions: let R_1 be sufficiently small to satisfy $R_1 \leq 1$ and $R_1 < |a_n|/(10An)$; let R_2 be sufficiently large to satisfy $R_2 \geq 1$ and $R_2 > 10An/|a_0|$.

Problem 265. Let $|z| = R_1$. Prove that

$$|a_0 z^n + \dots + a_{n-1} z| < \frac{|a_n|}{10}.$$

Problem 266. Let $|z| = R_2$. Prove that

$$\left| \frac{a_1}{z} + \dots + \frac{a_n}{z^n} \right| < \frac{|a_0|}{10}.$$

Let us denote by C_R the curve with equation $z(t) = R(\cos 2\pi t + i \sin 2\pi t)$ (i.e., the circle with radius R , oriented counterclockwise). Since the curve C_R is closed ($z(1) = z(0)$), the curve $f(C_R)$, where $f(z) = a_0 z^n + \dots + a_n$, is closed as well ($f(z(1)) = f(z(0))$). Let $\nu(R)$ be the number of turns of the curve $f(C_R)$ around the point $w = 0$ (if $f(C_R)$ does not pass through the point $w = 0$).

Problem 267. Calculate $\nu(R_1)$ and $\nu(R_2)$.

Let us now increase the radius R from R_1 to R_2 . The curve $f(C_R)$ will consequently be deformed from $f(C_{R_1})$ to $f(C_{R_2})$. If for a value R^* the curve $f(C_{R^*})$ does not pass through the point $w = 0$, by a sufficiently small variation of R near R^* the curve $f(C_R)$ will turn out to be deformed by too small an amount for the number of its turns around the point $w = 0$ to change: the function $\nu(R)$ is indeed continuous at the value R^* . If the curves $f(C_R)$ avoid the point $w = 0$ for all values of R between R_1 and R_2 , then $\nu(R)$ is a continuous function for all $R_1 \leq R \leq R_2$. Since the function $\nu(R)$ takes only integer values, it can be continuous only if for all $R_1 \leq R \leq R_2$ it takes a unique value. But, solving Problem 267, we have obtained that $\nu(R_1) = 0$ and $\nu(R_2) = n$. Therefore the claim that none of the curves $f(C_R)$ passes through the point $w = 0$ for all $R_1 \leq R \leq R_2$ is untrue. We thus have $f(z) = 0$ for a certain z . In this way we have proved the following theorem⁸.

定理 2.2 (The fundamental theorem of the algebra of complex numbers⁹)

The equation

$$a_0 z^n + a_1 z^{n-1} + \dots + a_{n-1} z + a_n = 0,$$

where all the a_i are arbitrary complex numbers, $n \geq 1$, and $a_0 \neq 0$, has at least one complex root. 

Problem 268. Prove Bézout's theorem¹⁰: If z_0 is a root of the equation $a_0 z^n + \dots + a_{n-1} z + a_n = 0$, then the polynomial $a_0 z^n + \dots + a_{n-1} z + a_n$ is divisible by the binomial $z - z_0$ without remainder.

Problem 269. Prove that the polynomial $a_0 z^n + \dots + a_{n-1} z + a_n$, where $a_0 \neq 0$, can be represented in the form

$$a_0 z^n + \dots + a_{n-1} z + a_n = a_0 (z - z_1)(z - z_2) \cdots (z - z_n).$$

注 Suppose that the polynomial $P(z)$ decomposes into factors:

$$P(z) = a_0 (z - z_1)(z - z_2) \cdots (z - z_n).$$

The left member of the equation is equal to 0 if and only if at least one of the factors inside brackets is equal to 0 (cf. 195, 197). Hence the roots of the equation $P(z) = 0$ are the numbers $z_1, z_2, \dots, z_n$ and them alone.

Problem 270. Let z_0 be a root of the equation

$$a_0 z^n + a_1 z^{n-1} + \dots + a_{n-1} z + a_n = 0,$$

where the a_i are real numbers. Prove that the number $\overline{z_0}$, the conjugate of z_0 , is also a root of that equation.

Problem 271. Suppose that the equation with real coefficients

$$a_0 z^n + a_1 z^{n-1} + \dots + a_{n-1} z + a_n = 0$$

has a complex root z_0 which is not a real number. Prove that the polynomial

$$a_0 z^n + a_1 z^{n-1} + \dots + a_{n-1} z + a_n$$

is divisible by a polynomial of second degree with real coefficients.

Problem 272. Prove that every polynomial with real coefficients can be written in the form of a product of polynomials of first and second degree with real coefficients.

注 From the result of Problem 272 it follows that the sole irreducible polynomials (cf. §2.3) over the field of real numbers are the polynomials of first degree and those of second degree with no real roots. We had used this property in §2.3 of this chapter. Over the field of complex numbers, according to the result of Problem 269, only polynomials of first degree are irreducible.

定义 2.6

Let z_0 be a root of the equation

$$a_0 z^n + a_1 z^{n-1} + \dots + a_{n-1} z + a_n = 0.$$

One says that z_0 is a root of order k if the polynomial $a_0 z^n + \dots + a_{n-1} z + a_n$ is divisible by $(z - z_0)^k$ and not by $(z - z_0)^{k+1}$.



Problem 273. What is the order of the roots $z = 1$ and $z = -1$ in the equation

$$z^5 - z^4 - 2z^3 + 2z^2 + z - 1 = 0?$$

定义 2.7

One calls the derivative of the polynomial

$$P(z) = a_0 z^n + a_1 z^{n-1} + \dots + a_k z^{n-k} + \dots + a_{n-1} z + a_n$$

the polynomial

$$P'(z) = a_0 n z^{n-1} + a_1 (n-1) z^{n-2} + \dots + a_k (n-k) z^{n-k-1} + \dots + a_{n-1}.$$

The derivative is usually denoted by the symbol ' (prime).



Problem 274. Let $P(z)$ and $Q(z)$ be two polynomials. Prove that: (a) $(P(z) + Q(z))' = P'(z) + Q'(z)$; (b) $(c \cdot P(z))' = c \cdot P'(z)$, where c is an arbitrary complex constant; (c) $(P(z)Q(z))' = P'(z)Q(z) + P(z)Q'(z)$.

Problem 275. Let $P(z) = (z - z_0)^n$, where $n \geq 1$ is an integer. Prove that $P'(z) = n(z - z_0)^{n-1}$.

Problem 276. Prove that if the equation $P(z) = 0$ has a root z_0 of order $k > 1$, then the equation $P'(z) = 0$ has a root z_0 of order $k - 1$, and that if the equation $P(z) = 0$ has a root z_0 of first order, then $P'(z_0) \neq 0$.

2.9 The Riemann surface of the function $w = \sqrt{z}$

We had considered single-valued functions for which there corresponds a unique value of the function to every value of the independent variable. In what follows we will deal mainly with multi-valued functions, for which there correspond distinct values of the function to a value of the independent variable¹¹. We will explain the reason for our interest in such functions. In fact, the final aim of our study is the proof of the Abel theorem, according to which a function expressing the roots of the general equation of fifth degree cannot be represented by radicals. But this function is multi-valued, because an equation of fifth degree has, in general, for given coefficients, five roots. Also the functions which are represented by radicals are multi-valued.

The principal idea of the demonstration of the Abel theorem is the following. We put into correspondence with a multi-valued function of a complex variable a certain group, the so-called monodromy group.

The monodromy group of the function expressing the roots of the general equation of fifth degree in terms of a parameter z cannot coincide with any monodromy group of functions representable by radicals, and therefore this function cannot be represented by radicals.

In order to introduce the notion of the monodromy group, we consider first another notion very important in the theory of functions of one complex variable—the notion of the Riemann¹² surface of a function. We begin with the construction of the Riemann surface for the simplest example of a multi-valued function, the function $w = \sqrt{z}$.

We already know that the function $w = \sqrt{z}$ takes the single value $w = 0$ for $z = 0$ and two values for all $z \neq 0$ (cf. 229). Moreover, if w_0 is one of the values of $\sqrt{z_0}$, then the other value of $\sqrt{z_0}$ is $-w_0$.

Problem 277. Find all values of: (a) $\sqrt{1}$; (b) $\sqrt{-1}$; (c) $\sqrt{i}$; (d) $\sqrt{1 + i\sqrt{3}}$ (here $\sqrt{3}$ is the positive value of the root).

Let us cut the z -plane along the negative side of the real axis from 0 to $-\infty$, and for every z not belonging to the cut let us choose the value $w = \sqrt{z}$ which lies on the right half of the w -plane. In this way we obtain a continuous single-valued function over the whole z -plane, except the cut. This function, which we denote by ${}_1\sqrt{z}$, defines a continuous and single-valued mapping of the z -plane, except the cut, onto the right half of the w -plane (Figure 2.9).

注 If we choose $\arg z$ in such a way that $-\pi < \arg z < \pi$, then for the function ${}_1\sqrt{z}$ we obtain

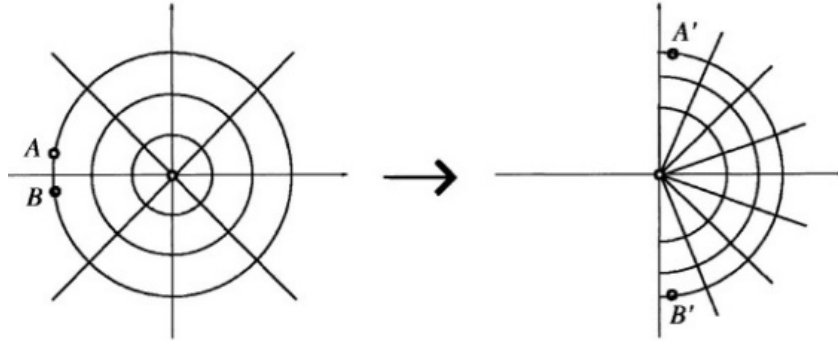
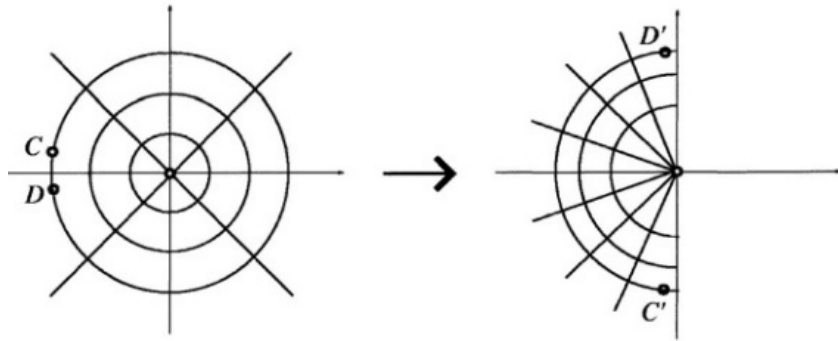
$$\arg({}_1\sqrt{z}) = \frac{1}{2} \arg z$$

(cf. 229). Therefore, under the mapping $w = {}_1\sqrt{z}$, the z -plane shrinks like a fan whose radii are shortened as its opening angle is halved.

If we now choose, for every z not lying on the cut, the value of $w = \sqrt{z}$ which lies on the left half of the w -plane, we obtain another function, still single-valued and continuous over the whole z -plane except the cut. This function, which we denote by ${}_2\sqrt{z}$, defines a continuous single-valued mapping of the z -plane except the cut onto the left half of the w -plane (Figure 2.10). Here ${}_2\sqrt{z} = -{}_1\sqrt{z}$.

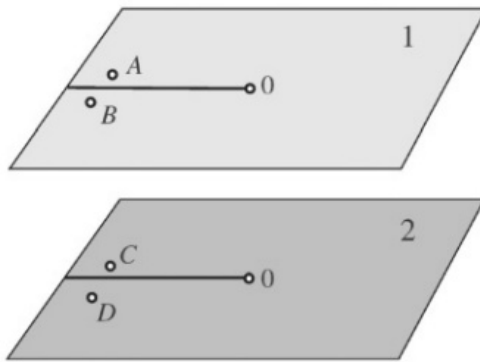
The functions ${}_1\sqrt{z}$ and ${}_2\sqrt{z}$ so defined are called the continuous single-valued branches of the function $w = \sqrt{z}$ (for the given cut).

Consider now two copies of the z -plane, which we shall call sheets, and cut every sheet along the negative side of the real axis from 0 to $-\infty$ (Figure 2.11).

图 2.9: The branch ${}_1\sqrt{z}$.图 2.10: The branch ${}_2\sqrt{z}$.

Let us take the function ${}_1\sqrt{z}$ on the first sheet and the function ${}_2\sqrt{z}$ on the second sheet. Thus we can see the functions ${}_1\sqrt{z}$ and ${}_2\sqrt{z}$ as a unique single-valued function, defined no longer on the z -plane but on a more complex surface consisting of two distinct sheets.

So if a point z moves continuously on the first (or on the second) sheet, not crossing the cut, the single-valued function that we have defined varies continuously. But if the point z , moving, for example, on the first sheet, traverses the cut, then continuity is lost. This follows from the close points A and B on the z -plane being sent by the mapping ${}_1\sqrt{z}$, respectively, to points A' and B' , far from each other (cf. Figure 2.9).

图 2.11: Two cut sheets of the z -plane.

On the other hand, it is easy to see in Figures 2.9 and 2.10 that the image of the point A under the mapping $w = {}_1\sqrt{z}$ (the point A') is close to the image of the point D under the mapping $w = {}_2\sqrt{z}$ (the point D').

Consequently, if, traversing the cut, the point z moves from the upper side of one sheet to the lower side of the other sheet, the single-valued function we have defined varies continuously. To guarantee that the point

z moves as requested, we consider the upper side of the cut on the first sheet joined to the lower side of the cut on the second sheet, and the lower side of the cut on the first sheet joined to the upper side of the cut on the second sheet (Figure 2.12). Furthermore, when joining the sheets we shall add between them the real negative axis from the point 0 to $-\infty$. During the first joining, for the points z which lie on this half-axis we choose the values of $w = \sqrt{z}$ lying on the positive side of the imaginary axis, and during the second joining we choose the values of $w = \sqrt{z}$ which lie on the negative side of the imaginary axis.

By means of the joining explained above we have transformed the two-valued function $w = \sqrt{z}$ into another function which is single-valued and continuous, no longer on the z -plane but on a new surface. This surface is called the Riemann surface of the function $w = \sqrt{z}$.

Attempts to do the joining without intersections (and without reversal of the plane) are made in vain. We assume that Figure 2.12 represents an image of the Riemann surface of the function $w = \sqrt{z}$, using the further convention that the self-intersection along the negative side of the real axis is only apparent. In order to understand this, consider the following example. In Figure 7 (§1.12) we see the image of a cube. Whereas in this image some edges intersect, we know that such intersections are only apparent, and this knowledge allows us to avoid mistaken interpretations.

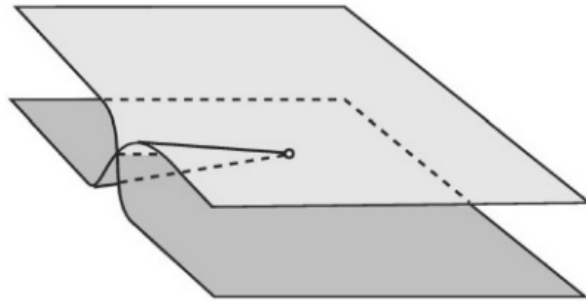


图 2.12: The Riemann surface of $w = \sqrt{z}$.

The Riemann surface of a multi-valued function $w(z)$ can be constructed in a way analogous to that used for the Riemann surface of the function $w = \sqrt{z}$. To do this we have first to separate the continuous single-valued branches of the function $w(z)$, excluding the points z which belong to the cuts. Afterwards we have to join the branches obtained, choosing the values on the cuts in such a way as to obtain a continuous single-valued function on the whole surface. The surface obtained is called the Riemann surface of the multi-valued function¹³ $w(z)$.

It remains, therefore, to explain how to separate the continuous single-valued branches of a function $w(z)$ and how to join them. To understand this, look again more attentively at the Riemann surface of the multi-valued function $w = \sqrt{z}$.

Let $w(z)$ be a multi-valued function, and fix one of the values w_0 of the function $w(z)$ at a certain point z_0 . Let $w'(z)$ be a continuous single-valued branch of the function $w(z)$ defined on some region of the z -plane (for example, on the whole plane except some cuts), and such that $w'(z_0) = w_0$. Suppose, moreover, that there exists a continuous curve C , connecting z_0 to a point z_1 , lying entirely in the region of the plane considered. Thus, while the point z moves continuously along the curve C from z_0 to z_1 , the function $w'(z)$ varies continuously from $w'(z_0)$ to $w'(z_1)$.

One can also use this property conversely, i.e., for the definition of the function $w'(z)$.

Indeed, suppose at a certain point z_0 one of the values w_0 of the function $w(z)$ is chosen. Let C be a

continuous curve beginning at z_0 and ending at a certain point z_1 . Moving along the curve C , we choose for every point z lying on C one of the values of the function $w(z)$ in such a way that these values vary continuously while we move along the curve C , starting from the value w_0 . Thus, when we arrive at the point z_1 , the value $w_1 = w(z_1)$ is completely defined. We say that w_1 is the value of $w(z_1)$ defined by continuity along the curve C under the condition $w(z_0) = w_0$. If the values of the function $w(z)$ chosen for all points of the curve C are represented on the w -plane, then we obtain a continuous curve beginning at the point w_0 and ending at the point w_1 . This curve is one of the continuous images of the curve C under the mapping $w = w(z)$.

Problem 278. For the function $w(z) = \sqrt{z}$, let us choose $w(1) = \sqrt{1} = 1$. Define by continuity $w(-1) = \sqrt{-1}$ along the following curves: (a) the upper semicircle of radius 1 with centre at the origin of coordinates; (b) the lower semicircle (Figure 2.13).

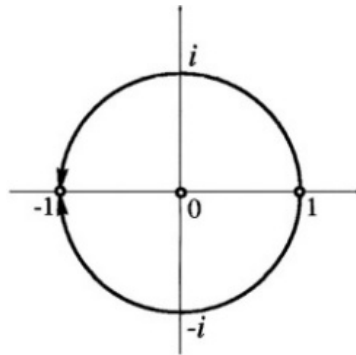


图 2.13: The curves in Problem 278.

In fact, using for a function the definition by continuity along a certain curve, we may encounter some difficulties. Consider the following example.

Problem 279. Find all continuous images $w_0(t)$ of a curve C with parametric equation $z(t) = 2t - 1$ (Figure 2.14) under the mapping $w = \sqrt{z}$, beginning: (a) at the point i ; (b) at the point $-i$.

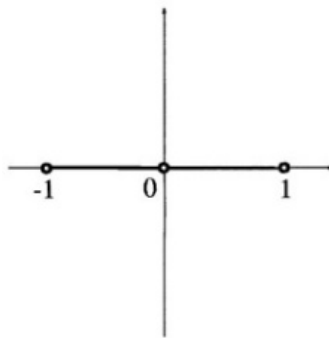


图 2.14: The curve in Problem 279.

From the solution of Problem 279 we obtain that, even by fixing the image of the initial point of the curve C , the continuous image of the curve C under the mapping $w = \sqrt{z}$ may be defined non-uniquely. The uniqueness is lost when the curve C passes through the point $z = 0$. In fact, for the function $w = \sqrt{z}$, the uniqueness is lost only in this case, because only in this case do the two images of the point $z(t)$ merge into one point.

To avoid the non-uniqueness in the definition of the images of curves under the mapping $w = \sqrt{z}$, we may exclude the point $z = 0$ and forbid any curve to pass through this point. This restriction, however, does not always allow the continuous single-valued branches of the function $\sqrt{z}$ to be separated.

Indeed, if we fix at a point z_0 one of the values $w_0 = w(z_0)$ and we try to define $w(z_1)$ at a certain point z_1 by continuity along two distinct curves joining z_0 and z_1 , we may obtain different values of $w(z_1)$ (for example, see Problem 278). Observe now how we can avoid the non-uniqueness of the obtained value.

Problem 280. Suppose the variation of the argument of $z(t)$ along a curve C is equal to φ . Find the variation of the argument of $w_0(t)$ along an arbitrary continuous image of the curve C under the mapping $w(z) = \sqrt{z}$.

Problem 281. Let $w(z) = \sqrt{z}$ and choose $w(1) = \sqrt{1} = -1$. Define the value of $w(i) = \sqrt{i}$ by continuity along: (a) the segment joining the points $z = 1$ and $z = i$; (b) the curve with parametric equation $z(t) = \cos \frac{3}{2}\pi t - i \sin \frac{3}{2}\pi t$; (c) the curve with parametric equation $z(t) = \cos \frac{5}{2}\pi t + i \sin \frac{5}{2}\pi t$.

Problem 282. Let $w(z) = \sqrt{z}$ and choose at the initial point of a curve C , $w(1) = \sqrt{1} = 1$. Define by continuity along the curve C the value of $w(1) = \sqrt{1}$ at the end point if the curve C has the equation: (a) $z(t) = \cos 2\pi t + i \sin 2\pi t$; (b) $z(t) = \cos 4\pi t - i \sin 4\pi t$; (c) $z(t) = 2 - \cos 2\pi t - i \sin 2\pi t$.

Problem 283. Let C be a closed curve on the z -plane (i.e., $z(0) = z(1)$). Prove that the value of the function $\sqrt{z}$ at the end point of the curve C , defined by continuity, coincides with the value at the initial point if and only if the curve C wraps around the point $z = 0$ an even number of times.

In the sequel it will be convenient to use the following notation.

定义 2.8

Let C be a continuous curve with a parametric equation $z(t)$. We shall denote by C^{-1} the curve geometrically identical to C but oriented in the opposite direction; its equation is (cf. 247) $z_1(t) = z(1 - t)$.

定义 2.9

Suppose that the initial point of a curve C_2 coincides with the final point of a curve C_1 . We will denote by $C_1 C_2$ the curve obtained by joining the end point of C_1 to the initial point of C_2 (cf. 248).

Problem 284. Let C_1 and C_2 be two curves joining the points z_0 and z_1 , and let one of the values $\sqrt{z_0} = w_0$ be chosen. Prove that the values of $\sqrt{z_1}$ defined by continuity along the curves C_1 and C_2 are equal if and only if the curve $C_1^{-1} C_2$ (Figure 2.15) turns around the point $z = 0$ an even number of times.

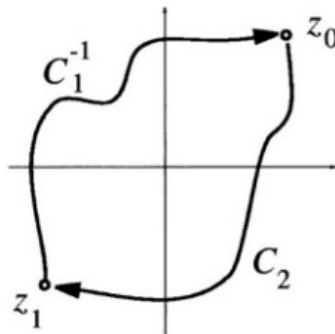


图 2.15: The curve $C_1^{-1} C_2$ in Problem 284.

From the statement of the last problem it follows that, in particular, if the curve $C_1^{-1} C_2$ turns zero times around the point $z = 0$, then the values of the function $\sqrt{z}$ at the final points of the curves C_1 and C_2 will coincide if the values at the initial point coincide.

Thus, to separate the single-valued continuous branches of the function $\sqrt{z}$, it suffices to take the curves C_1 and C_2 in such a way that the curve $C_1^{-1} C_2$ does not turn around the point $z = 0$. For this it suffices to make

a cut from the point $z = 0$ to infinity, avoiding intersection with the curve. We operated exactly in this way, drawing the cut from the point $z = 0$ to $-\infty$ along the negative side of the real axis.

If, after having made the cut, one fixes at a certain point z_0 one of the values $w'_0 = {}_1\sqrt{z_0}$, and if the value at every point z_1 is defined by continuity along an arbitrary curve C joining z_0 and z_1 and not passing through the cut, then at every point of the plane (except those on the cut) a certain single-valued continuous branch ${}_1\sqrt{z}$ of the function $\sqrt{z}$ is defined. If at the point z_0 one fixes the other value $w''_0 = {}_2\sqrt{z_0}$, then one defines the other branch ${}_2\sqrt{z}$ of the function $\sqrt{z}$.

Problem 285. Prove that ${}_1\sqrt{z} \neq {}_2\sqrt{z}$ for every point z outside the cut.

Problem 286. Fix the value $w' = {}_1\sqrt{z'}$ at a certain point z' and define the values of the function $\sqrt{z}$ at the other points of the z -plane (except the cut) by continuity along curves starting from z' and not intersecting the cut. Prove that the continuous single-valued branch so obtained coincides with the function ${}_1\sqrt{z}$ (defined by the value at the point z').

It follows from the result of Problem 286 that, choosing as initial point different points of the z -plane, one obtains the same splitting of the Riemann surface into single-valued continuous branches. Therefore, this splitting depends only on the way in which we have made the cut.

Problem 287. Suppose that points z_0 and z_1 do not lie on the cut and that the curve C , joining them, traverses the cut once (Figure 2.16). Choose a value $w_0 = \sqrt{z_0}$ and by continuity along C define the value $w_1 = \sqrt{z_1}$. Prove that the values w_0 and w_1 correspond to different branches of $\sqrt{z}$.

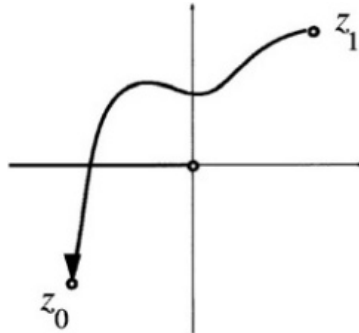


图 2.16: A curve traversing the cut once.

In this way, traversing the cut, one moves from one branch to the other, i.e., the branches join each other exactly as we have put them together in joining the sheets (Figure 2.12). One obtains in this way the Riemann surface of the function $\sqrt{z}$.

We say that a certain property is satisfied by any turn around a point z_0 if it is satisfied by a simple turn counterclockwise along all circles centred on z_0 and with sufficiently small radii¹⁴.

Problem 288. Prove that by a turn around a point z_0 one remains on the same sheet of the Riemann surface of the function $\sqrt{z}$ if $z_0 \neq 0$, and one moves onto the other sheet if $z_0 = 0$.

The following notion is very important in the sequel.

定义 2.10

Points around which one may turn and move from one sheet to another (i.e., changing the value of the function) are called the branch points of the given multi-valued function.



The Riemann surface of the function $\sqrt{z}$ can be represented by a scheme (Figure 2.17). This scheme shows that the Riemann surface of the function $w = \sqrt{z}$ has two sheets, that the point $z = 0$ is a branch point, and that by turning around the point $z = 0$ one moves from one of the sheets to the other. Moreover, the arrow between the two sheets in correspondence with the point $z = 0$ indicates the passage from one sheet to the other not only by a turn around the point $z = 0$, but also by crossing a point of the cut joining the point $z = 0$ to infinity. We have seen that this relation between branch points and cuts coming from these points is not arbitrary.

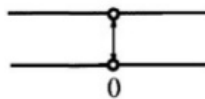


图 2.17: Scheme of the Riemann surface of $w = \sqrt{z}$.

In the sequel, instead of the Riemann surfaces of multi-valued functions, we will represent their schemes.

2.10 The Riemann surfaces of more complicated functions

Consider the multi-valued function $\sqrt[3]{z}$.

Problem 289. Let the variation of the argument along the curve $z(t)$ be equal to φ , and let $w_0(t)$ be the continuous image of the curve $z(t)$ under the mapping $w = \sqrt[3]{z}$. Find the variation of the argument along the curve $w_0(t)$.

Problem 290. Find the branch points of the function $\sqrt[3]{z}$.

Problem 291. Let us make a cut from the point $z = 0$ to $-\infty$ along the negative side of the real axis and assume, moreover, the single-valued continuous branches of the function $w = \sqrt[3]{z}$ to be given by the conditions

$$f_1(1) = 1,$$

$$f_2(1) = \cos(2\pi/3) + i\sin(2\pi/3) = -1/2 + i\sqrt{3}/2,$$

$$f_3(1) = \cos(4\pi/3) + i\sin(4\pi/3) = -1/2 - i\sqrt{3}/2.$$

Find: (a) $f_1(i)$; (b) $f_2(i)$; (c) $f_1(8)$; (d) $f_3(8)$; (e) $f_3(-i)$.

Problem 292. Draw the Riemann surface (and its scheme) of the function $w = \sqrt[3]{z}$.

Problem 293. Let C be a continuous curve with parametric equation $z(t)$, and let w_0 be one of the values of $\sqrt[n]{z_0}$. Prove that there exists at least one continuous image of the curve C under the mapping $w(z) = \sqrt[n]{z}$, starting at the point w_0 .

Problem 294. Suppose the variation of the argument along a curve $z(t)$ is equal to φ and $w_0(t)$ is the continuous image of the curve $z(t)$ under the mapping $w(z) = \sqrt[n]{z}$. Find the variation of the argument along the curve $w_0(t)$.

Problem 295. Find the branch points of the function $\sqrt[n]{z}$.

In §2.5 we have used the notation

$$\epsilon_n = \cos(2\pi/n) + i\sin(2\pi/n)$$

and we have considered some properties of this complex number.

Problem 296. Suppose that a curve $z(t)$ does not pass through the point $z = 0$ and that $w_0(t)$ is one of the

continuous images of the curve $z(t)$ under the mapping $w = \sqrt[n]{z}$. Find all the continuous images of the curve $z(t)$ under the mapping $w = \sqrt[n]{z}$.

Suppose that two curves C_1 and C_2 join a point z_0 to a different point z_1 . Exactly as in the case of the function $\sqrt{z}$ (cf. 284), one proves that if the curve $C_1^{-1}C_2$ does not turn around the point $z = 0$, then the function $\sqrt[n]{z}$ is uniquely defined by continuity along the curves C_1 and C_2 . In this way, if, as for the function $\sqrt{z}$, we make a cut from the point $z = 0$ to infinity, the image of the function $w = \sqrt[n]{z}$ turns out to be decomposed into single-valued continuous branches.

Problem 297. Make a cut from the point $z = 0$ to infinity, not passing through the point $z = 1$, and define the single-valued continuous branches of the function $\sqrt[n]{z}$ by the conditions $f_i(1) = \epsilon_n^i$, where i takes the integer values from 0 to $n - 1$. How are the branches $f_i(z)$ expressed in terms of $f_0(z)$?

Problem 298. Draw the scheme of the Riemann surface of the function $\sqrt[n]{z}$.

Problem 299. Find the branch points and draw the scheme of the Riemann surface for the function $\sqrt{z-1}$.

Problem 300. Find the branch points and draw the scheme of the Riemann surface of the function $\sqrt[n]{z+i}$.

When a multi-valued function has several branch points, in order to separate the single-valued continuous branches we make cuts from every branch point to infinity along lines which do not intersect each other.

In this way the scheme of the Riemann surface of a given function may depend essentially on the choice of the cuts from the branch points to infinity (some examples of such cases are given later in Problems 327 and 328). Whenever we are in this situation, and only in this case, we will indicate how to make the cuts.

The schemes of the Riemann surfaces drawn by the reader solving the above proposed problems may differ from the schemes given in Solutions, owing to different possibilities in numbering the sheets. Yet different schemes must coincide under a suitable relabelling.

301. Let $f(z)$ be a continuous single-valued function and C be a continuous curve on the z plane, starting at the point z_0 . Let w_0 be one of the values of $\sqrt[n]{f(z_0)}$. Prove that there exists at least one continuous image of the curve C under the mapping $w = \sqrt[n]{f(z)}$ starting at the point w_0 .

From the result of Problem 301 it follows that it is possible to define the function $\sqrt[n]{f(z)}$ by continuity along an arbitrary curve not passing through the points at which the uniqueness of the continuous images is lost.

302. Let $f(z)$ be a continuous single-valued function and $w_0(z)$ one of the single-valued continuous branches (according to the corresponding cuts) of the function $w = \sqrt[n]{f(z)}$. Find all the continuous single-valued branches (according to the same cuts) of the function $w(z)$.

303. Find all the branch points and draw the schemes of the Riemann surface for the functions: a) $\sqrt{z(z-i)}$; b) $\sqrt{z^2+1}$.

304. Draw the schemes of the Riemann surfaces of the following functions: a) $\sqrt[3]{z^2-1}$; b) $\sqrt[3]{(z-1)^2z}$; c) $\sqrt[3]{(z^2+1)^2}$.

305. Separate the single-valued continuous branches and draw the scheme of the Riemann surface of the function $\sqrt{z^2}$.

Remark. From the result of Problem 305 we obtain that the point $z = 0$ is not a branch point of the function $\sqrt{z^2}$. However, the images of the curves passing through the point $z = 0$ are not uniquely defined. For example, the continuous images of the broken line AOB (Figure 33) under the mapping $w = \sqrt{z^2}$ are the broken lines COD, COF, EOD, and EOF (Figure 33). Passing through the point $z = 0$ one may either remain on the same sheet (the lines COD and EOF) or move onto the other one (the lines COF and EOD). The Riemann surface of the function $w(z) = \sqrt{z^2}$ is shown in Figure 34.

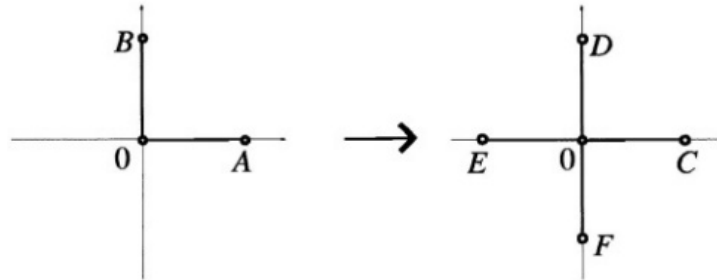


FIGURE 33

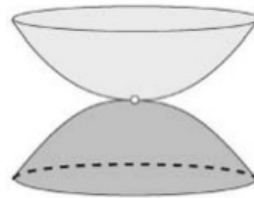


FIGURE 34

Definition. Points where the uniqueness of the continuous images of the curves is lost but that are not branch points are called the *non-uniqueness points* of the given function.

When building the Riemann surfaces one should draw no cuts from the points of non-uniqueness to infinity: in drawing any curve these points must always be avoided.

306. Draw the schemes of the Riemann surfaces of the following functions: a) $\sqrt[4]{z^2 + 2}$; b) $\sqrt[4]{z^2}$; c) $\sqrt[4]{(z-1)^2(z+1)^3}$; d) $\sqrt[4]{(z^2-1)^3(z+1)^3}$; e) $\sqrt[4]{z(z^3-1)}$.

Later we shall also consider functions which are not defined at some points. These points may, however, be branch points.

307. Draw the scheme of the Riemann surface of the function $\sqrt{1/z}$.

308. Draw the schemes of the Riemann surfaces of the following functions: a) $\sqrt{1/(z-i)}$; b) $\sqrt[3]{(z-1)/(z+1)}$; c) $\sqrt[4]{(z+i)^2/(z(z-1)^3)}$.

Solving the problems of this section we have always found that, after having made the cuts from all the branch points to infinity, the function considered turned out to be decomposed into continuous single-valued branches which join to each other in a way defined by the cuts. This property is possessed by a quite wide class of functions. In particular, it is possessed by all functions we have considered, for instance, by all functions

representable by radicals (Section 2.11) and by algebraic functions (Section 2.14) (these two classes are special cases of a wider class of functions, called analytic, which possess this property as well).

The proof of this statement lies outside the purposes of this book. We have therefore to accept this proposition without proof, knowing that the proof does exist¹⁵. The reader can thus pass directly to Section 2.11.

However, a sense of dissatisfaction may arise in the reader about something. Although we cannot destroy it completely, we prove that the property formulated above follows from another property, the so-called monodromy property, which turns out to be more evident.

We know that after having decomposed the function $w(z)$ into continuous single-valued branches (in a certain region of the z plane), the function $w(z)$ is defined by continuity in the same way along two arbitrary curves C_1 and C_2 lying in this region and joining two arbitrary points z_0 and z_1 . The monodromy property is related to this condition.

Suppose a multi-valued function $w(z)$ be such that if one fixes one of its values w_0 at an arbitrary point z_0 then the value of the function can be defined by continuity (possibly in a unique way) along an arbitrary curve beginning at the point z_0 (and not passing through the points at which $w(z)$ is not defined). We say that the function $w(z)$ possesses the monodromy property if it satisfies the following condition.

Monodromy property. Suppose that two continuous curves C_1 and C_2 on the z plane join two points z_0 and z_1 , passing neither through the branch points nor through the points of non-uniqueness of the function $w(z)$. Furthermore, suppose that the curve C_1 can be transformed, varying continuously, into the curve C_2 , in such a way that none of the curves during the deformation passes through the branch points, and that the ends of these curves are fixed (see Figure 35; a and b are branch points). Thus the value $w(z_1)$ is uniquely defined by continuity along the curves C_1 and C_2 (when a value $w_0 = w(z_0)$ is chosen).

309. Suppose a function $w(z)$ possesses the monodromy property. On the z plane make the cuts, not intersecting each other, from the branch points of $w(z)$ to infinity. Prove that in this way the function $w(z)$ is decomposed into continuous single-valued branches.

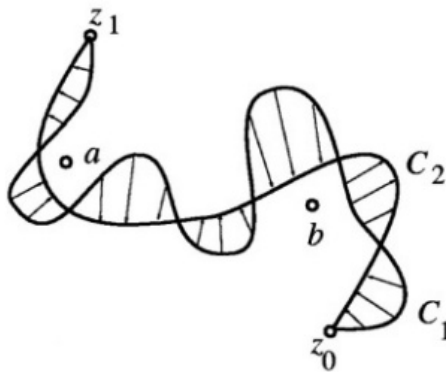


FIGURE 35

310. Suppose that in the conditions of the preceding problem the cuts do not pass through the non-uniqueness points of the function $w(z)$ and that $w(z)$ has a finite number of branch points. Prove that on traversing the

cut (in a defined direction) one moves from a given branch of the function $w(z)$ to another, a well-defined one, independently of the actual point at which the cut is crossed.

Remark 1. During a turn around a branch point one traverses once the cut joining this point to infinity. Consequently by virtue of the result of Problem 310 the passages between two different branches traversing the cut at an arbitrary point coincide with the passages obtained by a turn (with the corresponding orientation) around the branch point from which the cut has been drawn, and they thus coincide with the passages indicated by the arrows in correspondence with this point in the scheme of the Riemann surface.

Remark 2. From the results of Problems 309 and 310 it follows that if a multi-valued function $w(z)$ possesses the monodromy property then one can build its Riemann surface. In order to understand the structure of this surface it thus suffices to find the branch points of the function $w(z)$ and to define the passages between the branches corresponding to the turns around these points.

All functions which we shall consider in the sequel possess the monodromy property. We do not prove exactly this claim, because for this it would be needed to possess the precise notion of the analytic function. We give, however, the idea of the proof of the statement that a function $w(z)$ possesses the monodromy property if it is “sufficiently good.” What this means will be clarified during the exposition of the proof’s arguments.

Suppose the conditions required by the monodromy property be satisfied. Let C'_1 and C'_2 be the continuous images of the curves C_1 and C_2 under the mapping $w(z)$, with $w_0 = w(z_0)$ as the initial point. We have to prove that the curves C'_1 and C'_2 end at the same point.

Suppose first that all the curves obtained deforming C_1 into C_2 pass neither through the branch points, nor through the non-uniqueness points of the function. Let C be one of these curves. Thus there exists a unique image C' of the curve C under the mapping $w(z)$, beginning at the point $w_0 = w(z_0)$. If the function $w(z)$ is “sufficiently good”¹⁶ then during the continuous deformation of the curve C from C_1 to C_2 the curve C' is continuously deformed from C'_1 to C'_2 . The end point of the curve C' is continuously displaced as well. But the curve C ends at the point z_1 , thus the final point of the curve C' must coincide with one of the images of z_1 . If the function $w(z)$ takes at every z (in particular at z_1) only a finite number of values (and we consider only functions of this type), then the final point of the curve C' cannot jump from an image of the point z_1 to another, because this should destroy the continuity of the deformation. Hence the final points of the curves C' and, in particular, of C'_1 and of C'_2 do coincide.

Consider now what happens when the curve C passes through a non-uniqueness point (when it is not a branch point) of the function $w(z)$. Consider only the particular case in which the curve varies only in a neighbourhood of one non-uniqueness point, say a (Figure 36). If at the point z_0 one has chosen a value $w_0 = w(z_0)$ then one defines by continuity the value of $w(z)$ at the point A . Afterwards the values of $w(z)$ at the point E are uniquely defined by continuity along the curves ADE and ABE , because otherwise, making a turn along the curve $EDABE$, the value of the function $w(z)$ should change and the point a should be a branch point of the function $w(z)$. Afterwards, as we had uniquely defined the value of $w(z)$ at the point E along the two curves, we also define by continuity along the curve Ez_1 the value of $w(z)$ at the point z_1 .

So, the “obscure” point in our exposition remains the claim that all functions which we shall consider are “sufficiently good.”

The reader either accepts this proposition or will apply himself to the study of analytic functions¹⁷.

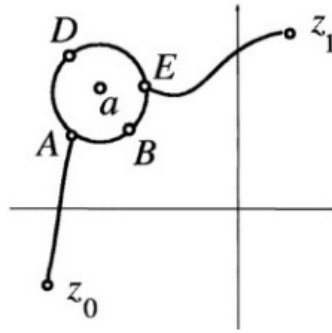


FIGURE 36

2.11 Functions representable by radicals

Definition. Let $f(z)$ and $g(z)$ be two multi-valued functions. By $f(z) + g(z)$ we will denote the multi-valued function whose values at a point z_0 are obtained by adding each value $f(z_0)$ to each value of $g(z_0)$. Similarly one defines the functions $f(z) - g(z)$, $f(z) \cdot g(z)$ and $f(z)/g(z)$.

By $[f(z)]^n$, where n is an arbitrary non-zero integer, we will denote a function whose values at the point z_0 are obtained raising to power n each value $f(z_0)$.

By $\sqrt[n]{f(z)}$, where n is a non-zero integer, we will denote the function whose values at a point z_0 are obtained extracting all roots of order n of each value $f(z_0)$.

311. Find all values of: a) $\sqrt[3]{-8} + \sqrt{2i}$; b) $(1 - \sqrt{-2i})/\sqrt{-4}$; c) $\sqrt{i + \sqrt{-1}}$; d) $\sqrt[4]{(1 + i)^2}$; e) $(\sqrt{i} + \sqrt{i})^2$.

Definition. We will say that a function $h(z)$ is *representable by radicals* if it can be written in terms of the function $f(z) = z$ and of constant functions ($g(z) = a$, a being any complex number) by means of the operations of addition, subtraction, multiplication, division, raising to an integer power and extraction of a root of integer order.

For example, the function

$$h(z) = \left(\sqrt[3]{\sqrt{z}} + 3z^2 - \frac{3}{\sqrt{z}} \right)^4$$

is representable by radicals. We have already seen other functions that are representable by radicals.

312. Let $h(z)$ be a function representable by radicals and let C be a continuous curve on the z plane, beginning at a point z_0 and not passing through the points at which $h(z)$ is not defined. Prove that if w_0 is one of the values $h(z_0)$ then there exists at least one continuous image of the curve C under the mapping $w = h(z)$, beginning at the point w_0 . (We suppose that the parametric equation $w(t) = a$, where a is a given complex number, describes a curve degenerated to a point.)

From the result of Problem 312 one obtains that an arbitrary function representable by radicals can be defined by continuity along an arbitrary continuous curve C , not passing through the points at which $h(z)$ is not

defined. Moreover, if the curve C passes neither through the branch points nor those of non-uniqueness of the function $h(z)$, then the function $h(z)$ is uniquely defined by continuity along the curve C .

We had already remarked in the preceding section that functions representable by radicals are “sufficiently good”¹⁸, i.e., they possess the monodromy property. Hence for every function representable by radicals one can build the Riemann surface¹⁹ (cf. Problems 309 and 310). Let us analyze the structure of such Riemann surfaces.

In this section we shall consider only functions representable by radicals.

313. Let $h(z) = f(z) + g(z)$. Eliminate from the plane all points of non-uniqueness of the function $h(z)$ and make the cuts not intersect each other, starting from all branch points of $f(z)$ and of $g(z)$ and going to infinity. Let $f_1(z), \dots, f_n(z)$ and $g_1(z), \dots, g_m(z)$ be the continuous single-valued branches of the functions $f(z)$ and $g(z)$ defined on the plane with the cuts. Find the continuous single-valued branches of the function $h(z)$.

If, turning once around the point z_0 , one moves from the branch $f_{i_1}(z)$ to the branch $f_{i_2}(z)$ and from the branch $g_{j_1}(z)$ to the branch $g_{j_2}(z)$, then evidently one moves from the branch $h_{i_1, j_1}(z) = f_{i_1}(z) + g_{j_1}(z)$ to the branch $h_{i_2, j_2}(z) = f_{i_2}(z) + g_{j_2}(z)$. This result indicates to us the formal method for drawing the scheme of the Riemann surface of the function $h(z) = f(z) + g(z)$ when we already have the schemes (with the same cuts) of the Riemann surfaces of the functions $f(z)$ and $g(z)$. In correspondence with every pair of branches $f_i(z)$ and $g_j(z)$ we take a sheet on which we consider defined the branch $h_{i,j}(z) = f_i(z) + g_j(z)$. If in the schemes of the Riemann surfaces of the functions $f(z)$ and $g(z)$ at the point z_0 arrows indicate, respectively, the passage from the branch $f_{i_1}(z)$ to the branch $f_{i_2}(z)$ and the passage from the branch $g_{j_1}(z)$ to the branch $g_{j_2}(z)$, then in the scheme of the Riemann surface of the function $h(z)$ we indicate by an arrow over the point z_0 the passage from the branch $h_{i_1, j_1}(z)$ to the branch $h_{i_2, j_2}(z)$.

314. Draw the schemes of the Riemann surfaces of the following functions: a) $\sqrt{z} + \sqrt{z-1}$; b) $\sqrt[3]{z^2-1} + \sqrt{1/z}$; c) $\sqrt{z} + \sqrt[3]{z}$; d) $\sqrt{z^2-1} + \sqrt[4]{z-1}$.

The formal method described above for building the scheme of the Riemann surface of the function $h(z) = f(z) + g(z)$ does not always give the correct result, because it may happen that some of the branches $h_{i,j}(z)$ coincide.

For simplicity we shall suppose that the cuts do not pass through the non-uniqueness points of the function $h(z)$. In this case, traversing a cut, by virtue of the uniqueness, we shall move from one set of sheets, corresponding to the same branch of the function $h(z)$, to a new set of sheets, all corresponding to a different branch. As a consequence, if we join the sheets corresponding to the same branches of the function $h(z)$, i.e., if we substitute every set of sheets with a single sheet, then the passages between the sheets so obtained are uniquely defined by any turn round an arbitrary branch point z_0 .

315. Find all values of $f(1)$ if: a) $f(z) = \sqrt{z} + \sqrt{z}$; b) $f(z) = \sqrt{z} + \sqrt[4]{z^2}$; c) $f(z) = \sqrt[3]{z} + \sqrt[3]{z}$.

316. Draw the schemes of the Riemann surfaces using the formal method and the correct schemes for the following functions: a) $f(z) = \sqrt{z} + \sqrt{z}$; b) $f(z) = \sqrt{z} + \sqrt[4]{z^2}$; c) $f(z) = \sqrt[3]{z} + \sqrt[3]{z}$.

We obtain finally that to draw the scheme of the Riemann surface of the function $h(z) = f(z) + g(z)$, starting from the schemes of the Riemann surfaces of the functions $f(z)$ and $g(z)$ (with the same cuts), it suffices

to build the scheme using the formal method described above and afterwards identify the sheets corresponding to equal values.

It is easy to see that one can apply this procedure to build the schemes of the Riemann surface for the functions $h(z) = f(z) - g(z)$; $h(z) = f(z) \cdot g(z)$; $h(z) = f(z)/g(z)$.

317. Draw the schemes of the Riemann surfaces of the functions: a) $i\sqrt{z} - \sqrt[4]{z^2}$; b) $\sqrt{z-1} \cdot \sqrt[4]{z}$; c) $\sqrt{z^2-1}/\sqrt[4]{z+1}$; d) $(\sqrt{z} + \sqrt{z})/\sqrt[3]{z(z-1)}$.

318. Let $f_1(z), f_2(z), \dots, f_m(z)$ be the single-valued continuous branches of the function $f(z)$. Find all single-valued continuous branches, having the same cuts, of the function $h(z) = [f(z)]^n$, where n is a non-zero integer.

From the result of the last problem it follows that the scheme of the Riemann surface of the function $h(z) = [f(z)]^n$ coincides with the scheme of the Riemann surface of the function $f(z)$ if all branches $h_i(z) = [f_i(z)]^n$ are distinct. But this is not always true. If one obtains some equal branches, then traversing the cuts one moves, by virtue of the uniqueness, from equal branches to equal branches.

We thus obtain that to draw the scheme of the Riemann surface of the function $h(z) = [f(z)]^n$ it suffices to consider, in the scheme of the Riemann surface of the function $f(z)$, the branches $h_i(z) = [f_i(z)]^n$ instead of the branches $f_i(z)$. If this produces equal branches it suffices to identify the corresponding sheets.

319. Build the schemes of the Riemann surfaces of the following functions: a) $(\sqrt[4]{z})^2$; b) $(\sqrt{z} + \sqrt{z})^2$; c) $(\sqrt{z} \cdot \sqrt[3]{z-1})^3$.

Let us now analyse how the Riemann surface of the function $\sqrt[n]{f(z)}$ is related to the scheme of the Riemann surface of the function $f(z)$.

320. Which points can be branch points of the function $\sqrt[n]{f(z)}$?

On the z plane make the cuts between the branch points of the function $f(z)$ and infinity in such a way that these cuts do not pass through the points at which some value of $f(z)$ vanishes, and separate the single-valued continuous branches of the function $f(z)$. Let $f_1(z), f_2(z), \dots, f_m(z)$ be these branches. Make the cuts between the points at which one of the values of $f(z)$ is equal to 0 and infinity. Let $g(z)$ be one of the single-valued continuous branches of the function $\sqrt[n]{f(z)}$ with these cuts.

321. Prove that the function $[g(z)]^n$ coincides with one of the functions $f_i(z)$ everywhere, except on the cuts.

It follows from the result of the preceding problem that every branch of the function $\sqrt[n]{f(z)}$ corresponds to some branch of the function $f(z)$.

322. Let $g(z)$ be a single-valued continuous branch of the function $\sqrt[n]{f(z)}$, corresponding to the branch $f_i(z)$ of the function $f(z)$. Find all single-valued continuous branches of the function $\sqrt[n]{f(z)}$, corresponding to the branch $f_i(z)$.

From the result of the last problem we obtain that to every branch $f_i(z)$ of the function $f(z)$ there corresponds a “bunch” consisting of n branches of the function $\sqrt[n]{f(z)}$. Enumerate the branches of this “bunch” by $f_{i,0}(z), f_{i,1}(z), \dots, f_{i,n-1}(z)$ in such a way that for every k the equation $f_{i,k}(z) = f_{i,0}(z) \cdot \epsilon_n^k$ be satisfied.

Let z_0 be a branch point of the function $f(z)$ and suppose that turning once around the point z_0 one moves from the branch $f_i(z)$ to the branch $f_j(z)$. Thus evidently for the function $\sqrt[n]{f(z)}$ we obtain that on turning around the point z_0 one moves from all branches of the bunch which corresponds to the branch $f_i(z)$ to all branches of the bunch which corresponds to the branch $f_j(z)$.

323. Let C be a curve on the z plane with a parametric equation $z(t)$ and let $w_0(t)$ be the parametric equation of the continuous image of the curve C on the w plane under the mapping $w = \sqrt[n]{f(z)}$. Prove that the curve with the equation $w_k(t) = w_0(t) \cdot \epsilon_n^k$ is also a continuous image of the curve C under the mapping $w = \sqrt[n]{f(z)}$.

324. Suppose that a curve C on the z plane avoids the branch points and the non-uniqueness points of the function $\sqrt[n]{f(z)}$. Prove that if on moving along the curve C one moves from the branch $f_{i,s}(z)$ to the branch $f_{j,r}(z)$, then one moves from the branch $f_{i,s+k}(z)$ to the branch $f_{j,r+k}(z)$, where the sums $s+k$ and $r+k$ are calculated modulo n (cf. Problem 40).

In this way, to define where one arrives, starting from a branch of a given bunch, on turning around a given branch point of the function $\sqrt[n]{f(z)}$, it suffices to define where one arrives from one of the branches of that bunch; for the other branches the passages turn out to be automatically defined by virtue of the result of Problem 324.

325. Draw the scheme of the Riemann surface of the function $f(z) = \sqrt{\sqrt{z} - 1}$.

326. Draw the schemes of the Riemann surfaces of the following functions: a) $\sqrt[3]{\sqrt{z} - 2}$; b) $\sqrt{\sqrt[3]{z} - 1}$.

327. Draw the scheme of the Riemann surface of the function $f(z) = \sqrt{z^2 + 1} - 2$ with different cuts, respectively shown: a) in Figure 37; b) in Figure 38. In both cases say whether the points z at which $f(z) = 0$ lie on the same sheet or on different sheets.

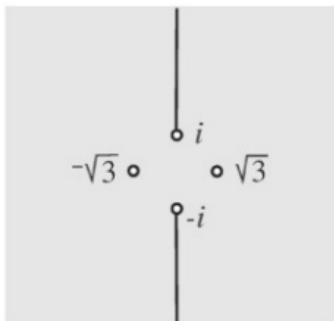


FIGURE 37

(a) FIGURE 37

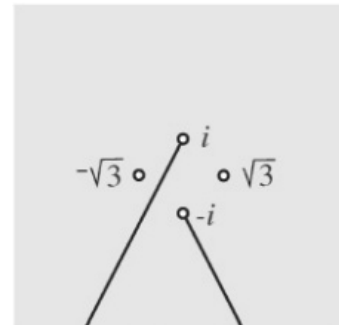


FIGURE 38

(b) FIGURE 38

328. Draw the scheme of the Riemann surface of the function $h(z) = \sqrt{\sqrt{z^2 + 1} - 2}$ with the cuts shown:
a) in Figure 39; b) in Figure 40.

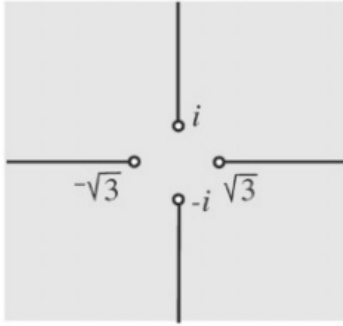
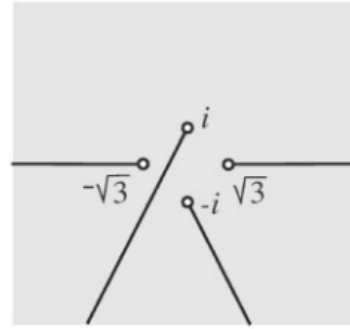


FIGURE 39

(a) FIGURE 39



(b) FIGURE 40

We formulate once more the results of this section which will be useful in the sequel.

Theorem 8. To build the schemes of the Riemann surfaces of the functions $h(z) = f(z) + g(z)$, $h(z) = f(z) - g(z)$, $h(z) = f(z) \cdot g(z)$, $h(z) = f(z)/g(z)$ starting from the schemes of the Riemann surfaces of the functions $f(z)$ and $g(z)$, with the same cuts, it suffices to do the following:

- put into correspondence with every pair of branches, $f_i(z)$ and $g_j(z)$, a sheet on which the branch $h_{i,j}(z)$, equal respectively to $f_i(z) + g_j(z)$, $f_i(z) - g_j(z)$, $f_i(z) \cdot g_j(z)$, $f_i(z)/g_j(z)$, is defined;
- if by turning around the point z_0 one moves from the branch $f_{i_1}(z)$ to the branch $f_{i_2}(z)$ and from the branch $g_{j_1}(z)$ to the branch $g_{j_2}(z)$, then for the function $h(z)$ by the same turn one moves from the branch $h_{i_1,j_1}(z)$ to the branch $h_{i_2,j_2}(z)$;
- identify the sheets on which the branches $h_{i,j}$ coincide.

Theorem 9. To build the scheme of the Riemann surface of the function $h(z) = [f(z)]^n$ starting from the scheme of the Riemann surface of the function $f(z)$, defined by the same cuts, it suffices to do the following:

- in the scheme of the Riemann surface of the function $f(z)$ consider, instead of the branches $f_i(z)$, the branches $h_i(z) = [f_i(z)]^n$;
- identify the sheets on which the branches $h_i(z)$ coincide.

Theorem 10. To build the scheme of the Riemann surface of the function $h(z) = \sqrt[n]{f(z)}$ starting from the scheme of the Riemann surface of the function $f(z)$, defined by the same cuts, it suffices to do the following:

- replace every sheet of the scheme of the Riemann surface of the function $f(z)$ by a “pack” of n sheets;
- turning around an arbitrary branch point of the function $h(z)$ one moves from all sheets of one pack to all sheets of a different pack;
- these passages from one pack of sheets to another correspond to the passages between the sheets of the Riemann surface of the function $f(z)$;
- if the branches in the bunches are enumerated in such a way that $f_{i,k}(z) = f_{i,0}(z)\epsilon_n^k$, then by moving from one bunch to another the sheets of the corresponding packs are not mixed, but they permute cyclically (cf. Problem 324).

2.12 Monodromy groups of multi-valued functions

We now associate with every scheme of the Riemann surface a group of permutations.

329. Suppose that a curve C on the z plane avoids the branch points and the non-uniqueness points of the function $w(z)$. Prove that moving along the curve C , starting from distinct sheets of the scheme of the Riemann surface of the function $w(z)$, one arrives at distinct sheets.

So by virtue of the result of Problem 249, to each turn (counterclockwise) around any branch point of the function $w(z)$ there corresponds a permutation of the sheets of the scheme of the Riemann surface of the function $w(z)$.

330. Consider the schemes of the Riemann surfaces of the functions of Problem 314, drawn as in the solution of this problem (see the chapter “Hints, Solutions, and Answers”) and consider in these schemes the sheets enumerated from bottom to top by the numbers $1, 2, 3, \dots$. For every function write the permutations of the sheets corresponding to one turn around every branch point.

331. Let $g_1, g_2, \dots, g_s$ be some elements of a group G . Consider all elements of G which can be obtained from $g_1, \dots, g_s$ by means of the iterated operations consisting of multiplying and of taking of the inverse of an element. Prove that the set of the elements obtained is a subgroup of the group G .

Definition. The group obtained in Problem 331 is called the *subgroup generated by the elements* $g_1, \dots, g_s$.

Definition. Let $g_1, \dots, g_s$ be the permutations of the sheets of the scheme of a Riemann surface corresponding to the turns (counterclockwise) around all the branch points. We call the subgroup generated by the elements $g_1, \dots, g_s$ the *permutation group of the sheets* of the given scheme of the Riemann surface, for brevity the *permutation group of the given scheme*.

Remark 1. If the number of sheets in a scheme is finite (and we consider only schemes of such a type) then to define the permutation group of this scheme it suffices to use the operation of composition, without using any permutation's inversion. Indeed, in this case every permutations of the sheets g has a finite order k : $g^k = e$, and therefore

$$g^{-1} = g^{k-1} = \underbrace{g \cdot g \cdot \dots \cdot g}_{k-1}.$$

Remark 2. The permutation group of the scheme we have considered is defined, as usual, up to isomorphism. The numbering of the sheets will be not important, because for different numberings one obtains different, but isomorphic, subgroups of S_n .

332. Which of the groups you already know are isomorphic to the permutation groups of the schemes of the following functions: a) $\sqrt{z}$; b) $\sqrt[3]{z}$; c) $\sqrt[n]{z}$; d) $\sqrt[3]{z^2 - 1}$ (cf. Problem 304); e) $\sqrt[4]{(z-1)^2(z+1)^3}$ (cf. Problem 306)?

333. To which of the groups you already know are the permutation groups of the schemes of the functions considered in Problems: 1) 314; 2) 317; 3) 319 isomorphic?

334. For the two schemes of the function $h(z) = \sqrt{\sqrt{z^2 + 1} - 2}$, built in the solution of Problem 328, describe the permutation groups.

Suppose that the point z_0 is neither a branch point nor a non-uniqueness point of the multi-valued function $w(z)$, and that $w_1, w_2, \dots, w_n$ are all values of the function $w(z)$ at the point z_0 . Consider a continuous curve C starting and ending at the point z_0 and not passing through any branch point and any non-uniqueness point of the function $w(z)$. Take a value $w_i = w(z_0)$ and define by continuity along the curve C a new value $w_j = w(z_0)$. Starting from distinct values w_i we obtain different values w_j (otherwise along the curve C the uniqueness would be lost). Hence to the curve C there corresponds a certain permutation of the values $w_1, w_2, \dots, w_n$. Thus if to the curve C there corresponds the permutation g , to the curve C^{-1} there corresponds the permutation g^{-1} , and if to the curves C_1 and C_2 (both with ending points at z_0) there correspond the permutations g_1 and g_2 , then to the curve $C_1 C_2$ there corresponds the permutation $g_2 g_1$ (recall that the permutations in a product are kept from right to left).

In this way if we consider all possible curves starting and ending at z_0 , then the corresponding permutations will form a group, the group of permutations of the values $w(z_0)$.

335. Let G_1 be the permutation group of the values $w(z_0)$ and G_2 the permutation group of some scheme of the function $w(z)$. Prove that the groups G_1 and G_2 are isomorphic.

Notice that in the definition of the permutation group of the values $w(z_0)$ one does not use any scheme of the Riemann surface of the function $w(z)$. From the result of Problem 335 it then follows that the permutation group of the values $w(z_0)$ for an arbitrary point z_0 and the permutation group of an arbitrary scheme of the Riemann surface of the function $w(z)$ are isomorphic. Consequently the permutation group of the values $w(z_0)$ for all points z_0 and the permutation group of all the schemes of the Riemann surfaces of the function $w(z)$ are isomorphic, i.e., they represent one unique group. This group is called the *monodromy group* of the multi-valued function $w(z)$.

2.13 Monodromy groups of functions representable by radicals

In this section we prove one of the main theorems of this book.

Theorem 11. If the multi-valued function $h(z)$ is representable by radicals its monodromy group is soluble (cf. Section 1.14).

The proof of Theorem 11 consists in the solutions of the following problems.

336. Let $h(z) = f(z) + g(z)$, or $h(z) = f(z) - g(z)$, or $h(z) = f(z) \cdot g(z)$, or $h(z) = f(z)/g(z)$, and suppose that we have built the scheme of the Riemann surface of the function $h(z)$ from the schemes of the Riemann surfaces of the functions $f(z)$ and $g(z)$ by the formal method (cf. Theorem 8(a), Section 2.11). Prove

that if F and G are the permutation groups of the initial schemes, then the permutation group of the scheme built is isomorphic to a subgroup of the direct product $G \times F$ (cf. Chapter 1, Section 1.7).

337. Let H_1 be the permutation group of the scheme built by the formal method under the hypotheses of the preceding problem, and let H_2 be the group of permutations of the real scheme of the Riemann surface of the function $h(z)$. Prove that there exists a surjective homomorphism (cf. Section 1.7) of the group H_1 onto the group H_2 .

338. Suppose the monodromy groups of the functions $f(z)$ and $g(z)$ be soluble. Prove that the monodromy groups of the functions $h(z) = f(z) + g(z)$, $h(z) = f(z) - g(z)$, $h(z) = f(z) \cdot g(z)$, $h(z) = f(z)/g(z)$ are soluble as well.

339. Suppose the monodromy group of the function $f(z)$ be soluble. Prove that the monodromy group of the function $h(z) = [f(z)]^n$ is also soluble.

340. Let H be the permutation group of a scheme of the function $h(z) = \sqrt[n]{f(z)}$, and F the permutation group of a scheme of the function $f(z)$, made with the same cuts. Define a surjective homomorphism of the group H onto the group F .

341. Prove that the kernel of the homomorphism (cf. Section 1.13) defined by the solution of the preceding problem is commutative.

342. Suppose the monodromy group F of the function $f(z)$ be soluble. Prove that the monodromy group H of the function $h(z) = \sqrt[n]{f(z)}$ is also soluble.

The functions $h(z) = a$ and $h(z) = z$ are continuous single-valued functions on the entire z plane. Their Riemann surfaces thus consist of a single sheet and therefore the corresponding monodromy groups consist of a single element e and are therefore soluble. As a consequence, taking into account the definition of the functions representable by radicals (Section 2.11) and the results of Problems 338, 339, 342, one obtains the proposition of Theorem 11.

Remark 1. This remark is for that reader who knows the theory of analytic functions. Theorem 11 holds for a wider class of functions. For example, to define a function $h(z)$ one can be allowed to use, besides of constant functions, the identity function, the functions expressed by arithmetic operations and radicals, also all analytic single-valued functions (for example, e^z , $\sin z$, etc.), the multi-valued function $\ln z$ and some others. In this case the monodromy group of the function $h(z)$ will be soluble, though it is not necessarily finite.

2.14 The Abel theorem

Consider the equation

$$3w^5 - 25w^3 + 60w - z = 0. \quad (2.8)$$

We consider z as a parameter and for every complex value of z we look for all complex roots w of this equation. By virtue of the result of Problem 269 the given equation for every z has 5 roots (taking into account the multiplicities).

343. Which values of w can be multiple roots (of order higher than 1, cf. Section 2.8) of the equation

$$3w^5 - 25w^3 + 60w - z = 0?$$

For which values of z are these roots multiples?

It follows from the solution of the preceding problem that for $z = \pm 38$ and $z = \pm 16$ equation (2.8) has four distinct roots, and for the other values of z it has 5 distinct roots. Let us study the function $w(z)$.

First we prove that for small variations of the parameter z the roots of equation (2.8) vary only slightly. This property is more precisely expressed by the following problem.

344. Let z_0 be an arbitrary complex number and w_0 be one of the roots of equation (2.8) for $z = z_0$. Consider a disc of radius r arbitrarily small with its centre at w_0 . Prove that there exists a real number $\rho > 0$ such that if $|z'_0 - z_0| < \rho$ then in the disc considered there exists at least one root of equation (2.8) for $z = z'_0$ also.

Suppose the function $w(z)$ express the roots of equation (2.8) in terms of the parameter z and w_0 be one of the values $w(z_0)$. It follows from the result of Problem 344 that if z changes continuously along a curve, starting at the point z_0 , then one can choose one of the values $w(z)$ in such a way that the point w , too, moves continuously along a curve starting from the point w_0 . In other words, the function $w(z)$ can be defined by continuity along an arbitrary curve C . Therefore if the curve C avoids the branch points and the non-uniqueness points of the function $w(z)$, the function $w(z)$ is uniquely defined by continuity along the curve C .

345. Prove that points different from $z = \pm 38$ and $z = \pm 16$ can be neither branch points nor non-uniqueness points of a function expressing the roots of equation (2.8) in terms of the parameter z .

Let $w(z)$ be the function expressing the roots of equation (2.8) in terms of the parameter z . The function $w(z)$, being an algebraic function¹, is “sufficiently good” (cf. Section 2.10), i.e., it possesses the monodromy property. One can therefore build for the function $w(z)$ the Riemann surface (cf. Problems 309 and 310). This Riemann surface evidently has 5 sheets.

By virtue of the result of Problem 345 the only possible branch points of the function $w(z)$ are the points $z = \pm 38$ and $z = \pm 16$, but it is not yet clear whether this is really the case.

346. Suppose it is known that the point $z_0 = +38$ (or $z_0 = -38$, or $z_0 = \pm 16$) is a branch point of the function $w(z)$ expressing the roots of equation (2.8) in terms of the parameter z . How do the sheets of the Riemann surface of the function $w(z)$ at the point z_0 (more precisely, along the cuts joining the point z_0 to infinity; cf. Remark 2, Section 2.10) join?

347. Let $w(z)$ be a function expressing the roots of equation (2.8) in terms of the parameter z . Moreover, let z_0 and z_1 be two arbitrary points different from $z = \pm 38$ and $z = \pm 16$, and w_0 and w_1 be two arbitrary images

¹ $a_0(z)w^n + a_1(z)w^{n-1} + \dots + a_n(z) = 0$

of these points under the mapping $w(z)$. Prove that it is possible to draw a continuous curve joining the points z_0 and z_1 , not passing through the points $z = \pm 38$ and $z = \pm 16$ and such that its continuous image, starting from the point w_0 , ends at the point w_1 .

348. Prove that all four points $z = \pm 38$ and $z = \pm 16$ are branch points of the function $w(z)$. How can we represent the scheme of the Riemann surface of the function $w(z)$? Draw all different possible schemes (we consider two schemes different if they cannot be obtained one from another by a permutation of the sheets and of the branch points).

349. Find the monodromy group of the function $w(z)$ expressing the roots of the equation

$$3w^5 - 25w^3 + 60w - z = 0$$

in terms of the parameter z .

350. Prove that the function $w(z)$, expressing the roots of the equation

$$3w^5 - 25w^3 + 60w - z = 0$$

in terms of the parameter z is not representable by radicals.

351. Prove that the algebraic general equation of fifth degree

$$a_0w^5 + a_1w^4 + a_2w^3 + a_3w^2 + a_4w + a_5 = 0$$

(where $a_0, a_1, a_2, a_3, a_4, a_5$ are complex parameters, $a_0 \neq 0$) is not solvable by radicals, i.e., that there are no formulae expressing the roots of this equation in terms of the coefficients by means of the operations of addition, subtraction, multiplication, division, elevation to an integer power and extraction of a root of integer order.

352. Consider the equation

$$(3w^5 - 25w^3 + 60w - z)w^{n-5} = 0$$

and prove that for $n > 5$ the general algebraic equation of degree n is not solvable by radicals.

The results of Problems 351 and 352 contain the proof of the theorem which is the subject of the present book. We have indeed proved the following.

Abel's theorem. For $n \geq 5$ the general algebraic equation of degree n

$$a_0w^n + a_1w^{n-1} + \dots + a_{n-1}w + a_n = 0$$

is not solvable by radicals.

Remark 1. In the introduction we deduced the Cardano formulae for the solution of the general algebraic equation of third degree. The roots of the equation are not given by all values expressed by these formulae, but only by those which satisfy some supplementary conditions. One may therefore pose the question of whether it is possible, also for the general equation of degree n ($n \geq 5$), to build by radicals a formula such that the roots

of the equation are only a part of the values that are expressed by this formula. This is not possible even for equation (2.8).

Indeed, if the values of the function $w(z)$, expressing the roots of equation (2.8) in terms of the parameter z , are only a part of the values of a function $w_1(z)$, represented by radicals, then the Riemann surface of the function $w(z)$ is a separate part of the Riemann surface of the function $w_1(z)$. If G is the monodromy group of the function $w_1(z)$ then to every permutation of the group G there corresponds a permutation of the five sheets of the function $w(z)$. This mapping is a homomorphism of the group G into the group S_5 . Since the group S_5 is not soluble then the group G is also not soluble (cf. Problem 163). On the other hand, the group G must be soluble, being the monodromy group of a function representable by radicals. We have thus obtained a contradiction.

Remark 2. From Remark 1 in Section 2.13 it follows that the Abel theorem also holds if one is allowed to use, besides radicals, some other functions, for example all analytic functions (such as e^z , $\sin z$, etc.), the function $\ln z$, and some others.

Remark 3. Consider equation (2.8) only in the domain of real numbers. Suppose that the function $y(x)$ expresses the real roots of the equation

$$3y^5 - 25y^3 + 60y - x = 0$$

in terms of the real parameter x . Is the function $y(x)$ representable by radicals? The answer is ‘no.’ To the reader who knows the theory of analytic functions we say that this follows from the theorem of the analytic continuation. Indeed, the function $w(z)$ expressing the roots of equation (2.8) in terms of the parameter z is analytic. If the function $y(x)$ were representable by radicals, then the corresponding formula, considered in the domain of complex numbers, should give, by virtue of the theorem of the analytic continuation, the function $w(z)$; i.e., the function $w(z)$ would be representable by radicals.

Hence the Abel theorem remains true also if one considers only the real roots of the general equation of degree n ($n \geq 5$) for all possible real values of the coefficients. Moreover, by virtue of Remark 2 the theorem holds also if one is allowed to use, besides radicals, some other functions, for example all functions with an analytic single-valued continuation (e^x , $\sin x$, etc.), the function $\ln x$, and some others.

Remark 4. The class of algebraic functions (cf. note 20) is sufficiently rich and interesting. In particular, one can prove that all functions representable by radicals are algebraic. We have proved that every function representable by radicals possesses a soluble monodromy group (Theorem 11). It turns out that if the analysis is restricted to algebraic functions then the converse also holds: if the monodromy group of an algebraic function is soluble then this function is representable by radicals. An algebraic function is thus representable by radicals if and only if its monodromy group is soluble.

第三章 Hints, Solutions, and Answers

3.1 Problems of Chapter 1

1. Answer. In the cases 1(a), 1(c), 2(c), and 3(a).
2. See Table 3.

表 3.1: Table 3

	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

3. See Table 4.

表 3.2: Table 4

	a	b	c	d	f
e	e	a	b	c	d
a	a	b	e	f	c
b	b	e	a	d	f
c	c	d	f	e	a
d	d	f	c	b	c
f	f	c	d	a	b

4. See Table 5.

表 3.3: Table 5

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

5. See Table 6.
6. See Table 7, where e and a are rotations of the rhombus about its centre by 0° and 180° , respectively, and b and c are reflections of the rhombus with respect to its diagonals.
7. See Table 7, where e and a are rotations of the rectangle about its centre by 0° and 180° , respectively, and b and c are reflections of the rectangle with respect to the straight lines passing through the middle points of its opposite sides.

表 3.4: Table 6

	e	a	b	c	d	f	g	h
e	e	a	b	c	d	f	g	h
a	a	e	c	b	f	d	h	g
b	b	c	a	e	g	h	f	d
c	c	b	e	a	h	g	d	f
d	d	f	h	g	e	a	c	b
f	f	d	g	h	a	e	b	c
g	g	h	d	f	b	c	e	a
h	h	g	f	d	c	b	a	e

表 3.5: Table 7

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

8. No, because there are no capitals in the world whose names begin with the letter X.
9. Answer. (a) ϕ is not a mapping onto because, for example, an integer n such that $\phi(n) = 5$ does not exist; that is, 5 has no pre-image. (b) ϕ is a mapping onto, but not one-to-one, because every positive integer under $\phi(n) = |n|$ has two pre-images (for example, 5 and -5). (c) ϕ is bijective, because 0, 1, 2, $\dots$ map to 0, 2, 4, $\dots$, while $-1, -2, -3, \dots$ map to 1, 3, 5, $\dots$.
10. Answer. $e^{-1} = e, a^{-1} = b, b^{-1} = a, c^{-1} = c, d^{-1} = d, f^{-1} = f$.
11. Answer. $g^{-1}(x) = x/2$.
12. Suppose that the given group of transformations contains g . By definition it also contains g^{-1} and $gg^{-1} = e$.
13. By the definition of e and multiplication, $(eg)(A) = e(g(A)) = g(A)$ and $(ge)(A) = g(e(A)) = g(A)$ for every element A . Hence $eg = g$ and $ge = g$.
- 14.
- $$((g_1g_2)g_3)(A) = g_1(g_2(g_3(A))), \quad (g_1(g_2g_3))(A) = g_1(g_2(g_3(A))).$$
15. (1) No. The only possible unit is 1, and an element 0^{-1} satisfying $0x = x0 = 1$ does not exist. (2) Yes.
16. Yes.

17. (a) No. Among the natural numbers there is no element x such that $n + x = x + n = n$ for every natural n . (If zero is included, only 0 has an opposite.) (b) No. The only possible unit is 1, and no element except 1 has an inverse.

18. Let e_1 and e_2 be unit elements. Then $e_1a = a$ and $ae_2 = a$ for every a , so $e_1e_2 = e_2$ and $e_1e_2 = e_1$. Hence $e_1 = e_2$.

19. If a has inverse elements a_1 and a_2 , then $(a_1a)a_2 = ea_2 = a_2$ and $a_1(aa_2) = a_1e = a_1$. Therefore $a_1 = a_2$.

20. $ee = e$; and $a^{-1}a = aa^{-1} = e$.

21. We prove this by induction. For $n = 3$ there are only $(a_1a_2)a_3$ and $a_1(a_2a_3)$, equal by associativity. Assume the assertion for $3 \leq n < k$. An arbitrary well-arranged expression A with k factors splits at its last operation as $A = (A_1)(A_2)$, where A_1, A_2 contain l and $k - l$ factors. By induction, A_1 and A_2 equal the corresponding left-associated products, so A equals

$$(\cdots((a_1a_2)a_3) \cdots a_l)(\cdots((a_{l+1}a_{l+2})a_{l+3}) \cdots a_k).$$

Put $a = (\cdots((a_1a_2)a_3) \cdots a_{l-1})a_l$ and $b = (\cdots((a_{l+1}a_{l+2})a_{l+3}) \cdots a_{k-2})a_{k-1}$. Then A expresses $a \cdot (b \cdot a_k) = (a \cdot b) \cdot a_k$. Since $a \cdot b$ has $k - 1$ factors, induction gives

$$ab = (\cdots((a_1a_2)a_3) \cdots a_{k-2})a_{k-1},$$

and hence $A = (\cdots((a_1a_2)a_3) \cdots a_{k-1})a_k$.

22. Answer. (1) Yes; (2) yes; (3) no; (4) yes; (5) yes.

23. (1) $(ab)(a^{-1}b^{-1}) = a(bb^{-1})a^{-1} = e$ and $(b^{-1}a^{-1})(ab) = b^{-1}(a^{-1}a)b = e$. (2) Induction gives $(a_1 \cdots a_n)^{-1} = a_n^{-1} \cdots a_1^{-1}$.

24. If $ax = b$, left multiplication by a^{-1} gives $x = a^{-1}b$, which is indeed a solution. Similarly $ya = b$ has the unique solution $y = ba^{-1}$.

25. Since $aa = e$ for every a , $(bc)(bc) = e$. Multiplying on the left by b and on the right by c gives $bbcbcc = bec$. As $bb = e$ and $cc = e$, $(ec)(be) = (be)c$, hence $cb = bc$. Thus G is commutative.

26. We prove $a^m(a^{-1})^m = (a^{-1})^ma^m = e$. Indeed,

$$a^m(a^{-1})^m = a^{m-1}aa^{-1}(a^{-1})^{m-1} = a^{m-1}(a^{-1})^{m-1} = e,$$

and the other equality is analogous (or follows by induction).

- 27.** For $m, n > 0$, $a^m a^n = a^{m+n}$. If $m, n < 0$, writing $m = -k, n = -l$ gives $a^m a^n = (a^{-1})^{k+l} = a^{m+n}$. If $m > 0, n < 0, m + n \geq 0$, then $a^m a^n = a^{m+n} a^{-n} (a^{-n})^{-1} = a^{m+n}$. If $m > 0, n < 0, m + n < 0$, then $a^m a^n = a^m (a^{-m} a^{m+n}) = a^{m+n}$. The case $m < 0, n > 0$ is analogous; cases with a zero exponent are immediate.
- 28.** If $n > 0$, $(a^m)^n = a^{mn}$. If $n < 0, m > 0$, write $n = -l$ and obtain $(a^m)^n = ((a^m)^l)^{-1} = a^{-ml} = a^{mn}$. If $m, n < 0$, write $m = -k, n = -l$ and obtain $(a^m)^n = ((a^k)^{-1})^{-l} = (a^k)^l = a^{mn}$. Zero cases are immediate.
- 29.** In the triangle symmetry group, a, b have order 3 and c, d order 2; in the square, b, c have order 4 and a, d, f, g, h order 2; in the rhombus all elements except e have order 2.
- 30.** (1) If $a^k = a^l, 0 \leq k \leq n-1, k > l$, then $a^{k-l} = e$, contradicting that a has order n . (2) Write $m = nt + r, 0 \leq r \leq n-1$. Then $a^m = (a^n)^t a^r = a^r$.
- 31.** Hint. The generator is a rotation of order $2\pi/n$.
- 32.** In the triangle rotation group the generators are a (rotation 120°) and b (rotation 240°); in the square rotation group they are a (rotation 90°) and c (rotation 270°).
- 33.** Let $m = nd + r, 0 \leq r \leq n-1$. Then $a^m = a^r$, and $a^r = e$ iff $r = 0$. Thus $a^m = e$ iff $m = nd$.
- 34.** $(a^m)^p = a^{mp} = (a^p)^m = e$, so the order of a^m divides the prime p ; the assertion follows.
- 35.** Let $d = (m, n)$. Since $(a^m)^{n/d} = a^{mn/d} = e$, the order divides n/d . If $(a^m)^k = e$, then mk is divisible by n , so $(m/d)k$ is divisible by n/d . Coprimality of m/d and n/d gives $n/d \mid k$. Hence the order is n/d .
- 36.** Let a be counterclockwise rotation by $2\pi/12$. The generator a^m has order 12 precisely when m is relatively prime to 12, namely $m = 1, 5, 7, 11$. The generators rotate by $\pi/6, 5\pi/6, 7\pi/6, 11\pi/6$.
- 37.** If $a^k = a^l$ with $k > l$, then $a^{k-l} = e$, contradicting infinite order.
- 38.** Under addition, a^m denotes m summands and a^{-1} denotes $-a$; the generators are 1 and -1 .
- 39.** (a) See Table 8; (b) see Table 9; (c) see Table 10.

表 3.6: Table 8

	0	1
0	0	1
1	1	0

表 3.7: Table 10

	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

40. All group properties hold for the remainders modulo n : associativity follows because both sides have the remainder of $a + b + c$; the unit is 0; for $m \neq 0$ the inverse is $n - m$ and the inverse of 0 is 0. The group is cyclic with generator 1, since the least $k > 0$ for which $1 + \cdots + 1 = 0 \pmod{n}$ is n .

41. From $a^m a^r = a^k$ we obtain $a^{(m+r)-k} = e$, so $(m+r) - k$ is divisible by n and $m+r \equiv k \pmod{n}$.

42. The groups (1) and (4) are isomorphic via $\phi(e) = 0, \phi(a) = 2, \phi(b) = 1, \phi(c) = 3$; groups (2) and (3) are isomorphic via $\phi(e) = e, \phi(a) = a, \phi(b) = b, \phi(c) = c$.

43. Since ϕ is bijective, for $c, d \in G_2$ choose unique $a, b \in G_1$ with $\phi(a) = c, \phi(b) = d$. Then $\phi^{-1}(cd) = ab = \phi^{-1}(c)\phi^{-1}(d)$, so ϕ^{-1} is an isomorphism.

44. The composition $\phi_2\phi_1 : G_1 \rightarrow G_3$ is bijective and

$$(\phi_2\phi_1)(ab) = \phi_2(\phi_1(a))\phi_2(\phi_1(b)),$$

so it is an isomorphism.

45. Hint. Use Problem 41. The isomorphism is $\phi(a^m) = m$.

46. Hint. Use Problem 27. The isomorphism is $\phi(a^m) = m$.

47. Let $\phi(e_G) = x$. Then $x^2 = x$; multiplying by x^{-1} gives $x = e_F$.

48. $\phi(g)\phi(g^{-1}) = \phi(gg^{-1}) = \phi(e_G) = e_F$, hence $\phi(g^{-1}) = [\phi(g)]^{-1}$.

49. For nonzero n , $\phi(g^n) = h^n$. If n_1, n_2 are the orders of g, h , then $g^{n_2} = e_G$ and $h^{n_1} = e_F$, so $n_1 \leq n_2$ and $n_2 \leq n_1$. Thus $n_1 = n_2$.

50. Answer. (a) The sole group is $\mathbb{Z}_2$. If a and e are its elements, the only undetermined product is a^2 , which must be e . (b) The sole group is $\mathbb{Z}_3$. For elements e, a, b , one obtains $ab = ba = e, a^2 = b, b^2 = a$, and $a^3 = e$.

51. The rotations of the square are not isomorphic to the symmetries of the rhombus: the former has an element of order 4, the latter does not.

52. The map $\phi(x) = 2^x$ bijects the real numbers under addition with the positive reals under multiplication, since $\phi(x+y) = \phi(x)\phi(y)$.

- 53.** For $y \in G$, $\phi_a(a^{-1}y) = y$, so ϕ_a is surjective. If $\phi_a(x) = \phi_a(y)$, then $ax = ay$ and $x = y$; hence it is bijective.
- 54.** $(\phi_a\phi_b)(x) = \phi_a(bx) = abx = \phi_{ab}(x)$ and $(\phi_a)^{-1} = \phi_{a^{-1}}$. Thus these transformations form a group.
- 55.** Define $\psi(a) = \phi_a$. Since $\phi_a(e) = a$, ψ is bijective, and $\psi(ab) = \phi_{ab} = \phi_a\phi_b = \psi(a)\psi(b)$; it is an isomorphism.
- 56.** (a) If e_G, e_H are units of G, H , then $e_H e_H = e_H$ and $e_G e_H = e_H$, so $e_H = e_G$. (b) For $a \in H$, uniqueness of inverses gives $a_H^{-1} = a_G^{-1}$.
- 57.** Necessity follows from Problem 56 and the subgroup definition. Conversely, closure, inclusion of e_G , and inclusion of inverses imply all group axioms for H , so H is a subgroup.
- 58.** Answer. For the triangle: $\{e, a, b\}$, $\{e, a\}$, $\{e, d\}$, $\{e, f\}$, $\{e\}$, and the whole group. For the square: $\{e, a, b, c\}$, $\{e, a\}$, $\{e, d\}$, $\{e, f\}$, $\{e, g\}$, $\{e, h\}$, $\{e, a, d, f\}$, $\{e, a, g, h\}$, $\{e\}$, and the whole group.
- 59.** For cyclic groups of orders 5, 8, and 15, respectively, the subgroups are: (a) $\{e\}$ and $\mathbb{Z}_5$; (b) $\{e\}$, $\{e, a^4\} \cong \mathbb{Z}_2$, $\{e, a^2, a^4, a^6\} \cong \mathbb{Z}_4$, and $\mathbb{Z}_8$; (c) $\{e\}$, $\{e, a^5, a^{10}\} \cong \mathbb{Z}_3$, $\{e, a^3, a^6, a^9, a^{12}\} \cong \mathbb{Z}_5$, and $\mathbb{Z}_{15}$.
- 60.** Let $H \neq \{e\}$ and let a^d have minimum positive exponent in H . Then every a^{kd} lies in H . For $a^m \in H$, write $m = td + r$, $0 \leq r < d$. Since $a^r = a^m a^{-td} \in H$, minimality gives $r = 0$, so $d \mid m$. As $a^n = e \in H$, $d \mid n$; this gives the required subgroup description.
- 61.** The solution is the same as that of Problem 60.
- 62.** An element of infinite order generates an infinite cyclic subgroup with infinitely many subgroups. If all element orders are finite, successively choose elements outside the previously generated finite cyclic subgroups; the procedure cannot terminate, yielding infinitely many subgroups.
- 63.** If $H = \bigcap_{i=1}^m H_i$, products and inverses of elements of H lie in every H_i , and e lies in every H_i . Hence H is a subgroup by Problem 57.
- 64.** Answer. (a) Yes; (b) yes; (c) no; (d) no.
- 65.** Vertex A can be sent to any of 4 vertices, B to any of the remaining 3, and C to any of the remaining 2. Thus $4 \cdot 3 \cdot 2 = 24$.
- 66.** (a) All symmetries fixing D :

$$e = \begin{pmatrix} A & B & C & D \\ A & B & C & D \end{pmatrix}, \quad d = \begin{pmatrix} A & B & C & D \\ B & C & D & A \end{pmatrix}, \quad d^2 = \begin{pmatrix} A & B & C & D \\ C & D & A & B \end{pmatrix}, \quad d^3 = \begin{pmatrix} A & B & C & D \\ D & A & B & C \end{pmatrix}.$$

67. A transformation preserving tetrahedron orientation preserves the position of every face relative to the opposite vertex; one changing orientation changes all such positions. Products of two preserving transformations or two reversing transformations preserve orientation, while a preserving and a reversing transformation compose to a reversing one. The identity and inverses preserve orientation, so the orientation-preserving symmetries form a subgroup. There are $4 \cdot 3 = 12$ of them.

68. Answer. (a) The subgroup of rotations about an axis through the midpoints of two opposite edges; (b) the subgroup of rotations about the axis through D perpendicular to the plane of triangle ABC .

69. For $(g_i, h_i) \in G \times H$,

$$\begin{aligned} ((g_1, h_1)(g_2, h_2))(g_3, h_3) &= ((g_1g_2)g_3, (h_1h_2)h_3) \\ &= (g_1(g_2g_3), h_1(h_2h_3)) \\ &= (g_1, h_1)((g_2, h_2)(g_3, h_3)). \end{aligned}$$

Moreover $(e_G, e_H)(g, h) = (g, h) = (g, h)(e_G, e_H)$ and (g^{-1}, h^{-1}) is the inverse of (g, h) . Thus $G \times H$ is a group.

70. Answer. nk .

71. Hint. The map $\phi((g, h)) = (h, g)$ is an isomorphism between $G \times H$ and $H \times G$.

72. Elements (g, e_H) form a subgroup isomorphic to G , and elements (e_G, h) form a subgroup isomorphic to H .

73. We have $(g_1, h_1)(g_2, h_2) = (g_1g_2, h_1h_2) = (g_2g_1, h_2h_1) = (g_2, h_2)(g_1, h_1)$.

74. If $(g_i, h_i) \in G_1 \times H_1$, closure gives $(g_1g_2, h_1h_2) \in G_1 \times H_1$; (e_G, e_H) belongs to it; and inverses (g^{-1}, h^{-1}) belong to it. Hence it is a subgroup of $G \times H$.

75. No. Let $G = \{e_1, c\}$ and $H = \{e_2, d\}$ be cyclic groups of order two. Then $\{(e_1, e_2), (c, d)\}$ is a subgroup of $G \times H$ not representable in the requested form.

76. For rhombus vertices A, B, C, D , let $G = \{e_1, g\}$ permute A, C and $H = \{e_2, h\}$ permute B, D . The map

$$\phi(e) = (e_1, e_2), \quad \phi(a) = (g, h), \quad \phi(b) = (e_1, h), \quad \phi(c) = (g, e_2)$$

is an isomorphism onto $G \times H \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.

77. (1) For $a = (g, h) \in \mathbb{Z}_2 \times \mathbb{Z}_3$, $a^2 = (e_1, h^2)$, $a^3 = (g, e_2)$, $a^4 = (e_1, h)$, $a^5 = (g, h^2)$, and $a^6 = (e_1, e_2)$. Hence $\mathbb{Z}_2 \times \mathbb{Z}_3 \cong \mathbb{Z}_6$. (2) Every $(g, h) \in \mathbb{Z}_2 \times \mathbb{Z}_4$ satisfies $(g, h)^4 = (e_1, e_2)$, so the group has no element of order 8 and is not $\mathbb{Z}_8$.

- 78.** Let g, h generate $\mathbb{Z}_m, \mathbb{Z}_n$ and let r be the order of (g, h) . Then $r \leq mn$ and $m, n \mid r$. If $(m, n) = 1$, $r = mn$ and $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$. Otherwise, with $k = \text{lcm}(m, n) < mn$, every (g, h) satisfies $(g, h)^k = (e_1, e_2)$, so no element has order mn and the groups are not isomorphic.
- 79.** Answer (see §1.1, Examples 1 and 2): (a) $\{e, a, b\}$; (b) $\{e, c\}, \{a, f\}, \{b, d\}$.
- 80.** Since $x = xe$ and $e \in H$, $x \in xH$.
- 81.** If $y = xh_1$ with $h_1 \in H$, then $x = yh_1^{-1}$. For $h \in H$, $yh = x(h_1h) \in xH$ and $xh = y(h_1^{-1}h) \in yH$. Thus $xH = yH$.
- 82.** If $z \in xH \cap yH$, then $xH = zH = yH$ by Problem 81.
- 83.** Hint. The order of an element equals the order of the cyclic subgroup it generates; apply Lagrange's theorem.
- 84.** Hint. If a group has prime order p , every element other than e has order p (Problem 83).
- 85.** By Lagrange's theorem, the only subgroups are $\{e\}$ and the whole group.
- 86.** Hint. Use Problems 84 and 45.
- 87.** If G has order m and $n = md$, take $G \times \mathbb{Z}_d$.
- 88.** It is possible: the rotation group of a tetrahedron has 12 elements but no subgroup of order 6. The group contains e , eight rotations by 120° or 240° about face altitudes, and three rotations by 180° about axes through midpoints of opposite edges. A hypothetical subgroup of order 6 containing a nontrivial altitude rotation must contain both rotations about that altitude. A further element moving the corresponding vertex yields, by conjugation, nontrivial rotations fixing another vertex, and then rotations about all four altitudes. This gives at least 9 elements, a contradiction.
- 89.** (a) Left and right partitions coincide: $\{e, a, b\}$ and $\{c, d, f\}$. (b) Left partition: $\{e, c\}, \{a, f\}, \{b, d\}$; right partition: $\{e, c\}, \{a, d\}, \{b, f\}$.
- 90.** (a) The two partitions coincide: $\{e, a\}, \{b, c\}, \{d, f\}, \{g, h\}$. (b) Left partition: $\{e, d\}, \{b, g\}, \{a, f\}, \{c, h\}$; right partition: $\{e, d\}, \{b, h\}, \{a, f\}, \{c, g\}$.
- 91.** The two partitions coincide and contain three cosets: numbers $3k$, numbers $3k + 1$, and numbers $3k + 2$, where $k = 0, \pm 1, \pm 2, \dots$

92. (a) Two groups: $\mathbb{Z}_4$ and $\mathbb{Z}_2 \times \mathbb{Z}_2$. (b) Two groups: $\mathbb{Z}_6$ and the symmetry group of the equilateral triangle. (c) Five groups: $\mathbb{Z}_8$, $\mathbb{Z}_4 \times \mathbb{Z}_2$, $(\mathbb{Z}_2 \times \mathbb{Z}_2) \times \mathbb{Z}_2$, the symmetry group of the square, and the quaternion group with elements $\pm 1, \pm i, \pm j, \pm k$ and multiplication table 11. **2)** The orders of the elements a , b , and c are equal to 2,

表 3.8: Table 11

	1	-1	i	-i	j	-j	k	-k
1	1	-1	i	-i	j	-j	k	-k
-1	-1	1	-i	i	-j	j	-k	k
i	i	-i	-1	1	k	-k	-j	j
-i	-i	i	1	-1	-k	k	j	-j
j	j	-j	-k	k	-1	1	i	-i
-j	-j	j	k	-k	1	-1	-i	i
k	k	-k	j	-j	-i	i	-1	1
-k	-k	k	-j	j	i	-i	1	-1

i.e., $a^2 = b^2 = c^2 = e$. We now seek the element corresponding to the product ab . It is impossible that $ab = e$ (otherwise $ab = a^2$ and $b = a$), that $ab = a$ (otherwise $b = e$), or that $ab = b$ (otherwise $a = e$). Therefore only $ab = c$ remains. Similarly, one obtains $ba = c$, $ac = ca = b$, and $bc = cb = a$. The multiplication table is complete, and we obtain (see 6) the group of symmetries of the rhombus, isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$ (see 76).

b) The elements of the group can have order 1, 2, 3, or 6 (see 83). We consider several cases.

1) Suppose that there is an element of order 6. The given group is then the cyclic group $\mathbb{Z}_6$.

2) Suppose that all elements are of order 2. Then the group is commutative (see 25), and if a and b are elements of the original group, the elements $\{e, a, b, ab\}$ form a subgroup. This is impossible by Lagrange's theorem, so this case must be excluded.

3) All elements are of order 2 or 3, and there is an element of order 3. Let a be an element of order 3 and let c be an element that is not a power of a . Then $\{c, ca, ca^2\}$ is a left coset of the subgroup $\{e, a, a^2\}$, so the six elements e, a, a^2, c, ca, ca^2 are all distinct (see 82). We prove that there is only one way to define a multiplication table on this set of six elements. First we prove that $c^2 = e$. Indeed, it is impossible that $c^2 = ca^k$ (otherwise $c = a^k$). If $c^2 = a$ (or $c^2 = a^2$), then $c^3 = cc^2 = ca \neq e$ (or $c^3 = ca^2 \neq e$), contrary to the assumption that all elements have order 2 or 3. Thus $c^2 = e$. Since c is an arbitrary element outside $\{e, a, a^2\}$, we also have $(ca)^2 = e$ and $(ca^2)^2 = e$. The product of every pair of the six elements is now determined. Indeed,

$$a^k a^l = a^{k+l}, \quad (ca^k) a^l = ca^{k+l}, \quad (ca^k)(ca^l) = (ca^k)(ca^k) a^{l-k} = a^{l-k},$$

and

$$a^k (ca^l) = c(ca^k)(ca^l) = ca^{l-k}.$$

Consequently, there is only one possible multiplication table that gives a group. Hence there is only one group of six elements whose element orders are 1, 2, and 3. We know this group: it is the group of symmetries of the equilateral triangle.

c) The elements of the group can have order 1, 2, 4, or 8 (see 83). Consider the following cases.

1) Suppose that there is an element of order 8. The given group is then the cyclic group $\mathbb{Z}_8$.

2) Suppose that every element other than the identity has order 2. The given group is then commutative (see 25). Let a and b be two distinct elements of order 2. Then $\{e, a, b, ab\}$ is a subgroup. If c does not belong to

this subgroup, then $\{c, ac, bc, abc\}$ is a right coset of it, and the eight elements

$$e, a, b, ab, c, ac, bc, abc$$

are all distinct. Their products are determined because the group is commutative and $a^2 = b^2 = c^2 = e$; for example, $(ac)(abc) = a^2bc^2 = b$. Therefore, if every nonidentity element has order 2, there is only one possible group. It is $(\mathbb{Z}_2 \times \mathbb{Z}_2) \times \mathbb{Z}_2$.

3) Suppose that a has order 4 and that among the elements which are not powers of a there is an element b of order 2, i.e., $b^2 = e$. Then

$$\{e, a, a^2, a^3\}, \quad \{b, ba, ba^2, ba^3\}$$

are the two left cosets of $\{e, a, a^2, a^3\}$, and their eight elements are all distinct. We seek the element equal to ab . It is impossible that $ab = a^k$ (otherwise $b = a^{k-1}$), or that $ab = b$ (otherwise $a = e$). If $ab = ba^2$, then $ab^2 = ba^2b$ and, since $b^2 = e$, $a = ba^2b$. This gives the contradiction

$$a^2 = (ba^2b)(ba^2b) = ba^2a^2b = bb = e.$$

Thus either $ab = ba$ or $ab = ba^3$.

α) If $ab = ba$, the multiplication table is uniquely determined. Indeed,

$$a^k a^l = a^{k+l}, \quad a^k (ba^l) = ba^{k+l}, \quad (ba^k) a^l = ba^{k+l}, \quad (ba^k)(ba^l) = a^{k+l}.$$

Thus there can be only one group, and it exists: it is $\mathbb{Z}_4 \times \mathbb{Z}_2$. If e_1, g are the identity and a generator of $\mathbb{Z}_4$, and e_2, h are the identity and a generator of $\mathbb{Z}_2$, then taking $a = (g, e_2)$ and $b = (e_1, h)$ gives all the properties above.

β) If $ab = ba^3$, the multiplication table is again uniquely determined. Indeed,

$$\begin{aligned} a^k a^l &= a^{k+l}, & (ba^k) a^l &= ba^{k+l}, & a^k b &= ba^{3k}, \\ a^k (ba^l) &= ba^{3k+l}, & (ba^k)(ba^l) &= a^{3k+l}. \end{aligned}$$

Thus again there can be only one group, and it exists: it is the group of symmetries of the square. It suffices to take a to be rotation by 90° and b to be reflection in a diagonal.

4) Suppose that there is an element a of order 4 and that all elements other than e, a, a^2, a^3 also have order 4. Let b be any element outside $\{e, a, a^2, a^3\}$. Then

$$e, a, a^2, a^3, b, ba, ba^2, ba^3$$

are all distinct. We determine which of these elements is b^2 . It is impossible that $b^2 = ba^k$ (otherwise $b = a^k$), or that $b^2 = e$ (because b has order 4). If $b^2 = a$ or $b^2 = a^3$, then $b^4 = a^2 \neq e$, a contradiction. Hence $b^2 = a^2$. Since b is arbitrary outside $\{e, a, a^2, a^3\}$, we also have

$$(ba)^2 = (ba^2)^2 = (ba^3)^2 = a^2.$$

From $baba = a^2 = b^2$ it follows that $aba = b$, $aba^4 = ba^3$, and $ab = ba^3$. The multiplication table is now uniquely determined. Indeed,

$$\begin{aligned} a^k a^l &= a^{k+l}, & (ba^k) a^l &= ba^{k+l}, & a^k b &= ba^{3k}, \\ a^k (ba^l) &= ba^{3k+l}, & (ba^k)(ba^l) &= b^2 a^{3k+l} = a^{3k+l+2}. \end{aligned}$$

Thus there can be only one group in this case. One verifies that this multiplication table does define a group,

called the quaternion group. It is better to denote its elements as follows: in place of

$$e, a, a^2, a^3, b, ba, ba^2, ba^3$$

write, respectively,

$$1, i, -1, -i, j, -j, k, -k.$$

Multiplication by 1 and -1 , and the rules for signs, are the same as in ordinary algebra. Moreover,

$$\begin{aligned} i^2 = j^2 = k^2 = -1, \quad ij = k, \quad ji = -k, \\ jk = i, \quad kj = -i, \quad ki = j, \quad ik = -j. \end{aligned}$$

The multiplication table of the quaternion group is shown in Table 11.

93. Consider the vertex named A in the new notation. Its old name was $g^{-1}(A)$. Under the given transformation this vertex is sent to the vertex whose old name is $hg^{-1}(A)$ and whose new name is $ghg^{-1}(A)$. Similarly, in the new notation, B is sent to $ghg^{-1}(B)$ and C to $ghg^{-1}(C)$. Hence the permutation corresponding to this transformation in the new notation is ghg^{-1} .

94. $ghg^{-1} = h_1$ if and only if $h = g^{-1}h_1g$. Hence every element h_1 has under ϕ_g one and only one preimage. It follows that $\phi_g(h) = ghg^{-1}$ is a bijection of the group onto itself. Moreover,

$$\begin{aligned} \phi_g(h_1h_2) &= g(h_1h_2)g^{-1} = gh_1(g^{-1}g)h_2g^{-1} \\ &= (gh_1g^{-1})(gh_2g^{-1}) = \phi_g(h_1)\phi_g(h_2). \end{aligned}$$

Thus ϕ_g is an isomorphism.

95. *Answer.* The reflections in all the altitudes.

96. *Answer.* The rotations through 120° and 240° .

97. *Answer.* Divide the elements of the group of symmetries of the tetrahedron into the following classes:

1. e ;
2. all nonidentity rotations about the altitudes;
3. all rotations through 180° about axes through the midpoints of opposite edges;
4. all reflections in planes through a vertex and the midpoint of the opposite edge;
5. all transformations generated by a cyclic permutation of the vertices, for example

$$\begin{pmatrix} A & B & C & D \\ B & C & D & A \end{pmatrix}.$$

Two elements can be transformed into one another by an inner automorphism of the group of symmetries of the tetrahedron if and only if they belong to the same class.

In the group of rotations of the tetrahedron, classes 4 and 5 are absent, whereas class 2 splits into two subclasses: (2a) all clockwise rotations through 120° about the altitudes, viewed from the vertex toward the base; and (2b) all rotations through 240° .

Solution. The classes above are characterized as follows: elements of class 2 have order 3 and preserve orientation; elements of class 3 have order 2 and preserve orientation; elements of class 4 have order 2 and

reverse orientation; and elements of class 5 have order 4 and reverse orientation. Since an inner automorphism is an isomorphism (see 94), two elements of different orders cannot be transformed into one another (see 49). Moreover, h and ghg^{-1} either both reverse orientation or both preserve it. Thus elements in distinct classes cannot be transformed into one another by an inner automorphism.

Let h_1 and h_2 be rotations through 180° about two axes through the midpoints of opposite edges, and let g be a rotation sending the first axis onto the second. Then gh_1g^{-1} sends the second axis into itself without reversing it. Moreover, $gh_1g^{-1} \neq e$, since otherwise $h_1 = g^{-1}eg = e$. Hence $gh_1g^{-1} = h_2$. Therefore any two elements of class 3 can be transformed into one another by an inner automorphism in the rotation group, and hence also in the full symmetry group, of the tetrahedron.

Let h_1 and h_2 be reflections of the tetrahedron in two planes of symmetry, and let g be a rotation sending the first plane onto the second. Then, as above, $gh_1g^{-1} = h_2$.

If $g_1hg_1^{-1} = h_1$ and $g_2hg_2^{-1} = h_2$, then $h = g_1^{-1}h_1g_1$, and therefore

$$(g_2g_1^{-1})h_1(g_2g_1^{-1})^{-1} = h_2.$$

Thus, if h can be transformed into both h_1 and h_2 , then h_1 and h_2 can be transformed into one another. It therefore suffices to show that one element of each class can be sent to all other elements of that class.

Let

$$a = \begin{pmatrix} A & B & C & D \\ B & C & D & A \end{pmatrix}$$

be an element of class 5, and define rotations g_i ($i = 1, \dots, 5$) by

$$\begin{aligned} g_1 &= \begin{pmatrix} A & B & C & D \\ A & C & D & B \end{pmatrix}, & g_2 &= \begin{pmatrix} A & B & C & D \\ A & D & B & C \end{pmatrix}, \\ g_3 &= \begin{pmatrix} A & B & C & D \\ C & B & D & A \end{pmatrix}, & g_4 &= \begin{pmatrix} A & B & C & D \\ B & D & C & A \end{pmatrix}, \\ g_5 &= \begin{pmatrix} A & B & C & D \\ B & A & D & C \end{pmatrix}. \end{aligned}$$

Then, as one verifies, the elements $g_ia g_i^{-1}$ ($i = 1, \dots, 5$), together with a , form all of class 5.

Let

$$b = \begin{pmatrix} A & B & C & D \\ A & C & D & B \end{pmatrix}$$

be rotation through 120° about the altitude from A . We prove that in the full symmetry group of the tetrahedron it can be carried by inner automorphisms to every other rotation about an altitude. By symmetry it is enough to show that b can be transformed into the other rotation about the same altitude,

$$b^2 = \begin{pmatrix} A & B & C & D \\ A & D & B & C \end{pmatrix},$$

and into an arbitrary rotation about a different altitude, for example

$$c = \begin{pmatrix} A & B & C & D \\ D & B & A & C \end{pmatrix}.$$

Let

$$g_1 = \begin{pmatrix} A & B & C & D \\ A & B & D & C \end{pmatrix}, \quad g_2 = \begin{pmatrix} A & B & C & D \\ B & C & A & D \end{pmatrix}.$$

Then $g_1 b g_1^{-1} = b^2$ and $g_2 b g_2^{-1} = c$.

If g is restricted to rotations of the tetrahedron, however, a counterclockwise rotation through 120° about one altitude (viewed from the vertex toward the base) cannot be transformed into the rotation through 240° about that altitude. Indeed, rotations through 120° about altitudes can be transformed only into rotations through 120° , and similarly for rotations through 240° .

98. $\phi_b(ab) = b(ab)b^{-1} = ba$. Since ϕ_b is an isomorphism (see 94), ab and ba have the same order (see 49).

99. *Hint.* For every $a \in N$ and $g \in G$, one has $gag^{-1} = agg^{-1} = a \in N$.

100. *Answer.* Yes. Verify that for every element g of the group of symmetries of the square, $geg^{-1} = e$ and $gag^{-1} = a$.

101. Suppose first that the left and right coset partitions coincide. Let a be an arbitrary element of N and g an arbitrary element of G . Since gN and Ng have the element g in common, they must coincide. Hence $ga \in gN$ also belongs to Ng , so there is a $b \in N$ such that $ga = bg$. Therefore $gag^{-1} \in N$, and N is a normal subgroup of G .

Conversely, let N be a normal subgroup of G . We prove that $gN = Ng$ for every $g \in G$. Let ga be an arbitrary element of gN . Then $gag^{-1} = b$ for some $b \in N$, so $ga = bg$ and hence $gN \subseteq Ng$. Now let cg be an arbitrary element of Ng . Then $(g^{-1})c(g^{-1})^{-1} = d$ for some $d \in N$. Hence $cg = gd$, and therefore $Ng \subseteq gN$. Thus $gN = Ng$.

102. *Hint.* Both the left and right partitions contain two classes: one is the given subgroup and the other contains all remaining elements. See also Theorem 2 (§1.10).

103. Let $N_1, N_2, \dots, N_s$ be the given normal subgroups of G , and let N be their intersection. If $a \in N$, then a belongs to every N_i . Hence, for every $g \in G$, gag^{-1} belongs to every N_i , and therefore to N . Thus N is a normal subgroup of G .

104. Let g be arbitrary in G . Since $eg = ge$, e belongs to the center. If a belongs to the center, then $ag = ga$. Multiplying on the left by a^{-1} gives $ga^{-1} = a^{-1}g$, so a^{-1} also belongs to the center. If a and b belong to the center, then

$$g(ab) = (ga)b = a(gb) = (ab)g,$$

so ab also belongs to the center. By Problem 57, the center is a subgroup.

For any a in the center and $g \in G$, $gag^{-1} = agg^{-1} = a$, which also belongs to the center. The center is therefore a normal subgroup.

105. Let $h_1 \in N_1$, $h_2 \in N_2$, $g_1 \in G_1$, and $g_2 \in G_2$. Then $g_1 h_1 g_1^{-1} \in N_1$ and $g_2 h_2 g_2^{-1} \in N_2$. Consequently,

$$\begin{aligned} & (g_1, g_2)(h_1, h_2)(g_1^{-1}, g_2^{-1}) \\ &= (g_1 h_1, g_2 h_2)(g_1^{-1}, g_2^{-1}) = (g_1 h_1 g_1^{-1}, g_2 h_2 g_2^{-1}) \in N_1 \times N_2. \end{aligned}$$

Thus $N_1 \times N_2$ is a normal subgroup of $G_1 \times G_2$.

106. Since $x_1 \in x_1 N$ and x_2 belongs by hypothesis to the same coset, there is an $h_1 \in N$ such that $x_2 = x_1 h_1$. Similarly, there is an $h_2 \in N$ such that $y_2 = y_1 h_2$. Since N is normal, $N y_1 = y_1 N$, so there is an $h_3 \in N$ such that $h_1 y_1 = y_1 h_3$. Therefore

$$x_2 y_2 = x_1 h_1 y_1 h_2 = x_1 y_1 h_3 h_2.$$

Since $h_3 h_2 \in N$, the elements $x_1 y_1$ and $x_2 y_2$ belong to the same coset $x_1 y_1 N$.

107. Let a, b, c be arbitrary elements of T_1, T_2, T_3 , respectively. By the definition of coset multiplication, $(T_1 T_2) T_3$ and $T_1 (T_2 T_3)$ are the cosets containing $(ab)c$ and $a(bc)$, respectively. Since $(ab)c = a(bc)$,

$$(T_1 T_2) T_3 = T_1 (T_2 T_3).$$

108. *Hint.* Take e as a representative of the class E .

109. *Hint.* Let a be an arbitrary element of the class T . Take as T^{-1} the coset containing a^{-1} .

110. It is easy to verify (see Table 2, §1.11) that $A^2 = B^2 = C^2 = E$. Hence this quotient group is isomorphic to the group of symmetries of the rhombus.

111. We list only the normal subgroups different from $\{e\}$ and from the whole group.

a) See 58(1), 95, 96, and 102. The normal subgroup is the group of rotations of the triangle; the corresponding quotient group is isomorphic to $\mathbb{Z}_2$.

b) See 99, 74, and 75. Let $\{e_1, c\} \times \{e_2, d\}$ be the given group. The normal subgroups are

$$\begin{aligned} & \{(e_1, e_2), (c, e_2)\}, \\ & \{(e_1, e_2), (e_1, d)\}, \\ & \{(e_1, e_2), (c, d)\}. \end{aligned}$$

The corresponding quotient groups are isomorphic to $\mathbb{Z}_2$.

c) For the notation see Examples 3 and 4 (§1.1). If a normal subgroup of the symmetry group of the square contains b or c , then it contains the entire rotation subgroup. This gives the normal subgroup $\{e, a, b, c\}$ (see 102), with quotient group $\mathbb{Z}_2$.

We have $bdb^{-1} = f$ and $bfb^{-1} = d$. Thus, if one of d, f belongs to a normal subgroup, then so does the other. Since $df = a$, the element a also belongs to it. This gives the normal subgroup $\{e, a, d, f\}$ (see 102), with quotient group $\mathbb{Z}_2$.

Since $bgb^{-1} = h$, $bhb^{-1} = g$, and $hg = a$, the same argument gives the normal subgroup $\{e, a, g, h\}$, with quotient group $\mathbb{Z}_2$.

If the normal subgroup contains none of b, c, d, f, g, h , then it is $\{e, a\}$, and the corresponding quotient group is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$ (see 100 and 110).

Thus the normal subgroups are

$$\{e, a, b, c\}, \quad \{e, a, d, f\}, \quad \{e, a, g, h\}, \quad \{e, a\}.$$

In cases 1–3 the quotient group is isomorphic to $\mathbb{Z}_2$; in case 4 it is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$.

d) Let $\{1, -1, i, -i, j, -j, k, -k\}$ be the given group. If h is any element other than 1 and -1 , then $h^2 = -1$. Hence every nontrivial normal subgroup contains -1 . This first gives the normal subgroup $\{1, -1\}$. Its coset partition is shown in Table 12.

表 3.9: Cosets of $\{1, -1\}$ in the quaternion group

$\{1, -1\}$	$\{i, -i\}$	$\{j, -j\}$	$\{k, -k\}$
E	A	B	C

Since $i^2 = j^2 = k^2 = -1$, one has $A^2 = B^2 = C^2 = E$, and the quotient group is therefore isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$. Since -1 belongs to every nontrivial subgroup, the two elements $i, -i$ either both belong or both do not belong to a normal subgroup. The same holds for $j, -j$ and for $k, -k$. By Lagrange's theorem, a nontrivial proper normal subgroup of the quaternion group can contain only two or four elements. We therefore obtain only three more normal subgroups (see 102):

$$\{1, -1, i, -i\}, \quad \{1, -1, j, -j\}, \quad \{1, -1, k, -k\}.$$

The corresponding quotient groups are isomorphic to $\mathbb{Z}_2$.

Thus the normal subgroups are

$$\begin{aligned} &\{1, -1\}, && \{1, -1, i, -i\}, \\ &\{1, -1, j, -j\}, && \{1, -1, k, -k\}. \end{aligned}$$

The quotient group in case 1 is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$, and in cases 2–4 it is isomorphic to $\mathbb{Z}_2$.

112. a) See 99 and 60. Let $n = dk$. The partition of the cyclic group

$$\mathbb{Z}_n = \{e, a, a^2, \dots, a^{n-1}\}$$

by the subgroup

$$\{e, a^d, a^{2d}, \dots, a^{(k-1)d}\}$$

is represented in Table 13, where l takes all values from 0 to $k - 1$.

表 3.10: Cosets in $\mathbb{Z}_n$

a^{dl}	a^{dl+1}	a^{dl+2}	$\dots$	$a^{dl+(d-1)}$
E	A_1	A_2	$\dots$	A_{d-1}

The element a belongs to A_1 , and the least positive integer m such that $a^m \in E$ is d . Hence the order of A_1 in the quotient group is d , so the quotient group is isomorphic to $\mathbb{Z}_d$.

b) See 99 and 61. Table 13 also represents the partition of

$$\mathbb{Z} = \{\dots, a^{-2}, a^{-1}, e, a, a^2, \dots\}$$

by the subgroup

$$\{\dots, a^{-2d}, a^{-d}, e, a^d, a^{2d}, \dots\},$$

where $l = 0, \pm 1, \pm 2, \dots$. As in part (a), the quotient group is isomorphic to $\mathbb{Z}_d$.

113. See 97. A set of rotations is a normal subgroup of the rotation group of the tetrahedron if and only if it is a union of some of the classes described in the solution of Problem 97 (for the rotation group) and is itself a subgroup. If a normal subgroup contains one rotation through 120° or 240° about an altitude, then it also contains the other rotation about that altitude and hence all rotations about all altitudes. If

$$a = \begin{pmatrix} A & B & C & D \\ A & C & D & B \end{pmatrix}, \quad b = \begin{pmatrix} A & B & C & D \\ C & B & D & A \end{pmatrix}$$

are rotations about the altitudes from A and B , respectively, then

$$ab = \begin{pmatrix} A & B & C & D \\ D & C & B & A \end{pmatrix}$$

is rotation through 180° about the axis through the midpoints of AD and BC . Hence the normal subgroup contains every rotation through 180° about an axis through the midpoints of opposite edges, and therefore it is the entire rotation group of the tetrahedron.

Thus the rotation group of the tetrahedron has only one nontrivial proper normal subgroup: it consists of the identity and the three rotations through 180° about the axes through the midpoints of opposite edges. The quotient group has three elements and is therefore isomorphic to $\mathbb{Z}_3$ (see 50).

114. Let (g_1, g_2) be arbitrary in $G_1 \times G_2$ and (g_3, e_2) arbitrary in $G_1 \times \{e_2\}$. Then

$$\begin{aligned} (g_1, g_2)(g_3, e_2)(g_1, g_2)^{-1} \\ = (g_1 g_3, g_2)(g_1^{-1}, g_2^{-1}) = (g_1 g_3 g_1^{-1}, e_2) \in G_1 \times \{e_2\}. \end{aligned}$$

Thus $G_1 \times \{e_2\}$ is a normal subgroup of $G_1 \times G_2$.

Let (g_1, a) be arbitrary in $G_1 \times G_2$. Multiplying it, for example on the right, by all elements of $G_1 \times \{e_2\}$ produces all elements of the form (g, a) , where g runs through G_1 . Denote this coset by T_a . The cosets of $G_1 \times \{e_2\}$ are precisely the T_a , where a runs through G_2 . Since (e_1, a) , (e_1, b) , and (e_1, ab) belong to T_a , T_b , and T_{ab} , respectively, and

$$(e_1, a)(e_1, b) = (e_1, ab),$$

we have $T_a T_b = T_{ab}$. The bijection ϕ from G_2 to the quotient group defined by $\phi(a) = T_a$ is an isomorphism because

$$\phi(ab) = T_{ab} = T_a T_b = \phi(a)\phi(b).$$

Thus $(G_1 \times G_2)/(G_1 \times \{e_2\})$ is isomorphic to G_2 .

115. See 57. The commutant clearly has property 1 of Problem 57. Also $eee^{-1}e^{-1} = e$, so e belongs to the commutant. If $k = aba^{-1}b^{-1}$ is a commutator, then

$$k^{-1} = (aba^{-1}b^{-1})^{-1} = bab^{-1}a^{-1}$$

(see 23), which is also a commutator. By definition, every element a of the commutant can be written as

$$a = k_1 k_2 \cdots k_n,$$

where each k_i is a commutator. Therefore

$$a^{-1} = k_n^{-1} \cdots k_2^{-1} k_1^{-1}$$

also belongs to the commutant. Thus the commutant is a subgroup.

116. If g is arbitrary and $k = aba^{-1}b^{-1}$ is a commutator, then gkg^{-1} is a commutator, since

$$\begin{aligned} gkg^{-1} &= gaba^{-1}b^{-1}g^{-1} \\ &= ga(g^{-1}g)b(g^{-1}g)a^{-1}(g^{-1}g)b^{-1}g^{-1} \\ &= (gag^{-1})(gbg^{-1})(ga^{-1}g^{-1})(gbg^{-1})^{-1}. \end{aligned}$$

If a is any element of the commutant, write $a = k_1 k_2 \cdots k_n$ with each k_i a commutator. Then

$$gag^{-1} = (gk_1g^{-1})(gk_2g^{-1}) \cdots (gk_ng^{-1})$$

is a product of commutators and hence belongs to the commutant. Since g is arbitrary, the commutant is a normal subgroup.

117. *Hint.* Show that $aba^{-1}b^{-1} = e$ if and only if $ab = ba$.

118. a) The symmetry group of the triangle is not commutative, so its commutant is not $\{e\}$. If g is a transformation of the triangle, then g and g^{-1} either both reverse the triangle or both preserve it. Thus $g_1 g_2 g_1^{-1} g_2^{-1}$ contains zero, two, or four orientation-reversing factors and never reverses the triangle; it is a rotation. The commutant therefore contains only rotations. Since it is nontrivial and is a subgroup, it coincides with the subgroup of all rotations (see 58).

b) As in part (a), the commutant is nontrivial and contains only rotations of the square. If g is a transformation of the square, then g and g^{-1} either both interchange the diagonals or both fix them. Therefore every commutator fixes the diagonals. Being a rotation, it is either e or the central symmetry a . Thus the commutant is contained in $\{e, a\}$; since it is nontrivial, it equals $\{e, a\}$.

c) The elements 1 and -1 commute with every element of the quaternion group. Hence, if one of g_1, g_2 is 1 or -1 , then $g_1 g_2 g_1^{-1} g_2^{-1} = 1$. If $g \neq \pm 1$, then $g(-g) = -g^2 = 1$, so $g^{-1} = -g$. Therefore, for $g_1, g_2 \neq \pm 1$,

$$g_1 g_2 g_1^{-1} g_2^{-1} = g_1 g_2 (-g_1) (-g_2) = (g_1 g_2)^2.$$

The square of every quaternion-group element is 1 or -1 . Hence the commutant is contained in $\{1, -1\}$. Since the group is noncommutative, the commutant is not $\{1\}$, and therefore it is $\{1, -1\}$.

119. The argument used in Problems 118(a) and 118(b) shows that the commutant in the symmetry group of a regular n -gon contains only rotations.

Let n be odd, let a be reflection in the axis l (Figure 41), and let b be counterclockwise rotation through $\pi - \pi/n$, sending A to B . Then $aba^{-1}b^{-1}$ is the rotation sending B to C , namely counterclockwise rotation through $2\pi/n$. Since the commutant is a subgroup, it contains all rotations through multiples of $2\pi/n$. Since it

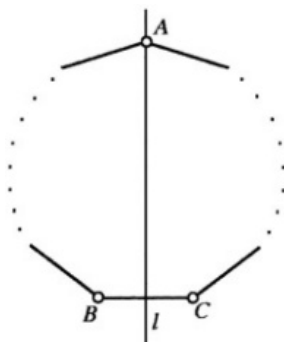


Figure 41

contains only rotations of the regular n -gon, for odd n it is the entire rotation subgroup, isomorphic to $\mathbb{Z}_n$ (see 31).

Now let $n = 2k$. Inscribe a regular k -gon by joining the even-numbered vertices, and a second regular k -gon by joining the odd-numbered vertices. If g is any transformation of the n -gon, then g and g^{-1} either both interchange these two k -gons or both fix them. Therefore every commutator sends each k -gon into itself. Consequently the commutant can contain only rotations through multiples of $2\pi/k$. Let c be rotation of the regular n -gon through $2\pi/n$, and let d be reflection in the axis m (Figure 42).

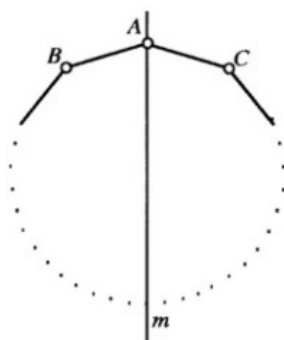


Figure 42

Then $cdc^{-1}d^{-1}$ is the rotation sending C to B , namely counterclockwise rotation through

$$\frac{4\pi}{n} = \frac{2\pi}{k}.$$

Hence the commutant contains every rotation through a multiple of $2\pi/k$, and no others. It is the subgroup of plane rotations sending each regular k -gon into itself, and it is isomorphic to $\mathbb{Z}_k = \mathbb{Z}_{n/2}$.

120. Let k, l, m be the axes through the midpoints K, L, M , respectively, of the three pairs of opposite edges of the tetrahedron. Joining K, L, M gives the equilateral triangle KLM . Under any rotation of the tetrahedron, either all three axes are fixed or none is fixed. Associate a permutation of the axes k, l, m with the corresponding permutation of the vertices K, L, M of the triangle. Every rotation of the tetrahedron then induces a rotation of the triangle KLM . Hence every commutator in the rotation group of the tetrahedron induces a commutator in the rotation group of KLM . The latter group is commutative, so every such induced commutator is e . Thus every commutator in the rotation group of the tetrahedron fixes each of k, l, m .

The commutant can therefore contain only the identity and the rotations through 180° about axes through the midpoints of opposite edges. Since the rotation group of the tetrahedron is noncommutative, its commutant

is nontrivial. The commutant is a normal subgroup, so by 113 it is exactly the subgroup consisting of the identity and these three rotations through 180° .

121. See the solution of Problem 113.

122. A symmetry g of the tetrahedron and its inverse g^{-1} either both reverse orientation or both preserve it (see the solution of Problem 67). Thus every commutator $g_1 g_2 g_1^{-1} g_2^{-1}$ preserves orientation, and the commutant in the full symmetry group contains only rotations. If

$$a = \begin{pmatrix} A & B & C & D \\ A & C & D & B \end{pmatrix}, \quad b = \begin{pmatrix} A & B & C & D \\ A & B & D & C \end{pmatrix},$$

then

$$aba^{-1}b^{-1} = \begin{pmatrix} A & B & C & D \\ A & D & B & C \end{pmatrix}$$

is a rotation about the altitude from A . Since the commutant is a normal subgroup (see 116 and 121), it coincides with the rotation subgroup.

123. *Answer.* There are 24 rotations of the cube:

1. the identity;
2. nine rotations through 90° , 180° , or 270° about axes through centers of opposite faces;
3. six rotations through 180° about axes through midpoints of opposite edges;
4. eight rotations through 120° or 240° about axes through opposite vertices.

124. Joining the centers of adjacent faces of a cube produces an octahedron. Thus every rotation of the cube determines a rotation of the octahedron, and vice versa. Composition is preserved, so this correspondence is an isomorphism between the rotation groups of the cube and the octahedron.

125. Fix the position of the cube, and regard two colorings as different if at least one face has a different color. There are

$$6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 = 720$$

colorings. Each coloring has 24 distinct images under rotations of the cube (see 123), so there are $720/24 = 30$ ways to color the cube.

Only four rotations send a matchbox to itself: the identity and the three rotations through 180° about axes through centers of opposite faces. Hence there are $720/4 = 180$ ways to color a matchbox with six colors.

126. *Answer.* The symmetry group of the rhombus and the group $\mathbb{Z}_2 \times \mathbb{Z}_2$.

127. *Hint.* a) See 57. b) Use the fact that g and g^{-1} either both interchange the tetrahedra or both fix them.

128. A rotation g of the cube and its inverse g^{-1} either both interchange the tetrahedra ACB_1D_1 and A_1C_1BD (see Figure 8), or both fix them. Thus every commutator sends each tetrahedron into itself. Hence each element of the commutant of the rotation group of the cube induces a rotation of the tetrahedron ACB_1D_1 .

Let a be rotation of the cube through 90° about the axis through the centers of the faces $ABCD$ and $A_1B_1C_1D_1$, in the direction that sends B to A . Let b be rotation through 120° about the axis through A_1 and C_1 , in the direction that sends A to D_1 . Then $aba^{-1}b^{-1}$ fixes A and sends A_1 to D ; thus it is a nonidentity rotation about the axis through A and C_1 . It is also a rotation of the tetrahedron ACB_1D_1 about the altitude from A . It is now easy to show (see 121) that the commutant of the cube's rotation group contains every rotation that fixes the tetrahedron. Since it contains only such rotations, it is isomorphic to the rotation group of the tetrahedron.

129. Let A and B be arbitrary cosets, represented by a and b . Since $aba^{-1}b^{-1}$ belongs to the commutant,

$$ABA^{-1}B^{-1} = E,$$

and hence $AB = BA$.

130. Let a, b be arbitrary elements of the group, and let A, B be their cosets. Since $AB = BA$, one has $ABA^{-1}B^{-1} = E$. Therefore $aba^{-1}b^{-1}$ belongs to the normal subgroup N . Thus every commutator belongs to N , and hence the commutant is contained in N .

131. Let h_1, h_2 be arbitrary elements of N , and let g be arbitrary in G . Since N is normal, gh_1g^{-1} and gh_2g^{-1} belong to N . Moreover,

$$\begin{aligned} g(h_1h_2h_1^{-1}h_2^{-1})g^{-1} \\ = (gh_1g^{-1})(gh_2g^{-1})(gh_1g^{-1})^{-1}(gh_2g^{-1})^{-1}, \end{aligned}$$

which is a commutator in N , hence belongs to $K(N)$. Therefore $K(N)$ is a normal subgroup of G .

132. Let f_1, f_2 be arbitrary elements of F . Since ϕ is a surjective homomorphism from G onto F , there are $g_1, g_2 \in G$ with $\phi(g_1) = f_1$ and $\phi(g_2) = f_2$. Hence

$$f_1f_2 = \phi(g_1)\phi(g_2) = \phi(g_1g_2) = \phi(g_2g_1) = \phi(g_2)\phi(g_1) = f_2f_1.$$

Thus F is commutative.

The converse is false; see Example 12 in §1.13.

133. Let $\phi(e_G) = x$. Then

$$x^2 = \phi(e_G)\phi(e_G) = \phi(e_Ge_G) = \phi(e_G) = x.$$

Therefore $x = e_F$.

134.

$$\phi(a)\phi(a^{-1}) = \phi(aa^{-1}) = \phi(e_G) = e_F$$

by Problem 133. Hence $\phi(a^{-1}) = [\phi(a)]^{-1}$.

135. Let $a, b \in G$. Then

$$\begin{aligned}(\phi_2\phi_1)(ab) &= \phi_2(\phi_1(ab)) = \phi_2(\phi_1(a)\phi_1(b)) \\ &= \phi_2(\phi_1(a))\phi_2(\phi_1(b)) = ((\phi_2\phi_1)(a))((\phi_2\phi_1)(b)).\end{aligned}$$

136. If $\phi(a) = A$ and $\phi(b) = B$, then, by the definition of coset multiplication,

$$\phi(a)\phi(b) = AB = \phi(ab).$$

137. See 57. If $a, b \in \ker \phi$, then $\phi(ab) = \phi(a)\phi(b) = e_F e_F = e_F$, so $ab \in \ker \phi$. Also $\phi(e_G) = e_F$ by 133, so $e_G \in \ker \phi$. Finally, if $\phi(a) = e_F$, then by 134,

$$\phi(a^{-1}) = [\phi(a)]^{-1} = e_F^{-1} = e_F.$$

Hence $a \in \ker \phi$ implies $a^{-1} \in \ker \phi$, and $\ker \phi$ is a subgroup.

138. Let a be arbitrary in $\ker \phi$ and g arbitrary in G . Then $\phi(a) = e_F$ and

$$\phi(gag^{-1}) = \phi(g)\phi(a)\phi(g^{-1}) = \phi(g)e_F[\phi(g)]^{-1} = e_F$$

by 134. Thus $gag^{-1} \in \ker \phi$, and $\ker \phi$ is a normal subgroup of G .

139. Suppose g_1 and g_2 belong to the same coset $g \ker \phi$. Then there are $r_1, r_2 \in \ker \phi$ such that $g_1 = gr_1$ and $g_2 = gr_2$. Therefore

$$\phi(g_1) = \phi(g)e_F = \phi(g_2).$$

Conversely, suppose $\phi(g_1) = \phi(g_2)$. Then, by 134,

$$\phi(g_1^{-1}g_2) = [\phi(g_1)]^{-1}\phi(g_2) = e_F.$$

Hence $g_1^{-1}g_2 = r$ for some $r \in \ker \phi$, so $g_2 = g_1r$. Thus g_1 and g_2 belong to the same coset $g_1 \ker \phi$.

140. Let f be arbitrary in F . Since ϕ is surjective, there is an $a \in G$ such that $\phi(a) = f$. Let A be the coset containing a . By definition, $\psi(A) = \phi(a) = f$.

141. Suppose $\psi(A) = \psi(B)$, and let a, b be representatives of A, B , respectively. Then $\phi(a) = \psi(A) = \psi(B) = \phi(b)$. By 139, $A = B$.

142. Let A, B be arbitrary cosets, represented by a, b . The element ab belongs to AB . By the definition of ψ and because ϕ is a homomorphism,

$$\psi(AB) = \phi(ab) = \phi(a)\phi(b) = \psi(A)\psi(B).$$

Since ψ is bijective (see 141), it is an isomorphism.

143. Let k, l, m be the three axes through the midpoints of opposite edges of the tetrahedron. Every symmetry of the tetrahedron permutes these axes, giving a mapping ϕ_1 from the symmetry group of the tetrahedron to the group of permutations of k, l, m . This mapping is onto: every permutation is induced by a suitable symmetry

of the tetrahedron. For any symmetries g_1, g_2 , the permutation induced by $g_1 g_2$ is the composition of those induced by g_1 and g_2 , so

$$\phi_1(g_1 g_2) = \phi_1(g_1) \phi_1(g_2).$$

Thus ϕ_1 is a homomorphism.

Associate each permutation of the axes k, l, m in the natural way with a symmetry of an equilateral triangle KLM . This gives an isomorphism ϕ_2 from the permutation group of the three axes to the symmetry group of KLM .

The composition $\phi_2 \phi_1$ is a surjective homomorphism (see 135) from the symmetry group of the tetrahedron onto the symmetry group of KLM . Its kernel consists of the symmetries that send each of k, l, m to itself: the identity and the rotations through 180° about these axes. By Problem 138 and Theorem 3, these symmetries form a normal subgroup of the symmetry group of the tetrahedron, and the corresponding quotient group is isomorphic to the symmetry group of the triangle.

144. Let k, l, m be the axes through the centers of opposite faces of the cube. As in the solution of Problem 143, define a homomorphism from the rotation group of the cube onto the symmetry group of an equilateral triangle KLM . Its kernel consists of the rotations that send each of k, l, m to itself: the identity and the rotations through 180° about these three axes. By Problem 138 and Theorem 3, these four rotations form a normal subgroup of the rotation group of the cube, and the corresponding quotient group is isomorphic to the symmetry group of the triangle.

145. Let r_α denote counterclockwise rotation of the plane about O through the angle α . The mapping

$$\phi(r_\alpha) = r_{n\alpha}$$

is a homomorphism from the group R to itself, because

$$\phi(r_\alpha r_\beta) = \phi(r_{\alpha+\beta}) = r_{n(\alpha+\beta)} = r_{n\alpha} r_{n\beta} = \phi(r_\alpha) \phi(r_\beta).$$

and for every rotation r_α there exists a rotation $r_{\alpha/n}$ such that $\phi(r_{\alpha/n}) = r_\alpha$. The kernel of the homomorphism ϕ contains all rotations such that $n\alpha = 2\pi k$, i.e., $\alpha = 2\pi k/n$. They are precisely the rotations of the plane that send the regular n -gon onto itself. From the result of Problem 138 and Theorem 3 we obtain the proposition that we had to prove.

146. Let ϕ_1 and ϕ_2 be the natural homomorphisms of the groups G_1 and G_2 onto the quotient groups G_1/N_1 and G_2/N_2 , respectively. Let ϕ be the surjective mapping of the group $G_1 \times G_2$ onto the group $(G_1/N_1) \times (G_2/N_2)$ such that $\phi((g_1, g_2)) = (\phi_1(g_1), \phi_2(g_2))$. This mapping is a homomorphism; indeed,

$$\begin{aligned} \phi((g_1, g_2) \cdot (g_3, g_4)) &= \phi((g_1 g_3, g_2 g_4)) \\ &= (\phi_1(g_1 g_3), \phi_2(g_2 g_4)) \\ &= (\phi_1(g_1) \phi_1(g_3), \phi_2(g_2) \phi_2(g_4)) \\ &= (\phi_1(g_1), \phi_2(g_2)) \cdot (\phi_1(g_3), \phi_2(g_4)) \\ &= \phi((g_1, g_2)) \cdot \phi((g_3, g_4)). \end{aligned}$$

The kernel of the homomorphism ϕ contains all pairs (g_1, g_2) such that $\phi((g_1, g_2)) = (E_1, E_2)$, where E_1

and E_2 are the unit elements in the quotient groups G_1/N_1 and G_2/N_2 , respectively. Since $\phi((g_1, g_2)) = (\phi_1(g_1), \phi_2(g_2))$, the kernel of ϕ is the subgroup $N_1 \times N_2$. From the result of Problem 138 and Theorem 3 we obtain the statement which we had to prove.

147. Yes, it is possible. For example, the group $\mathbb{Z}_4 = \{e, a, a^2, a^3\}$ and the group $\mathbb{Z}_2 \times \mathbb{Z}_2$ contain the normal subgroups $\{e, a^2\}$ and $\mathbb{Z}_2 \times \{e_2\}$, respectively, which are isomorphic to $\mathbb{Z}_2$. The corresponding quotient groups, too, are isomorphic.

148. Yes, it is possible. For example, consider the group $\mathbb{Z}_4 \times \mathbb{Z}_2$, where $\mathbb{Z}_4 = \{e_1, a, a^2, a^3\}$. The group $\mathbb{Z}_4 \times \mathbb{Z}_2$ contains two isomorphic normal subgroups, $\{e_1\} \times \mathbb{Z}_2$ and $\{e_1, a^2\} \times \{e_2\}$, and the corresponding quotient groups are, respectively,

$$\frac{\mathbb{Z}_4 \times \mathbb{Z}_2}{\{e_1\} \times \mathbb{Z}_2} \cong \frac{\mathbb{Z}_4}{\{e_1\}} \times \frac{\mathbb{Z}_2}{\mathbb{Z}_2} \cong \mathbb{Z}_4$$

and

$$\frac{\mathbb{Z}_4 \times \mathbb{Z}_2}{\{e_1, a^2\} \times \{e_2\}} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$$

(see 146).

149. Yes, it is possible. An example of an infinite group of this type is given in Problem 145, where $R/\mathbb{Z}_n \cong R$ and, evidently, $R/\{e\} \cong R$.

An example of a finite group is given by the group $\mathbb{Z}_4 \times \mathbb{Z}_2$, which contains two normal subgroups of the type $\mathbb{Z}_4 \times \{e_2\}$ and $\mathbb{Z}_2 \times \mathbb{Z}_2$, whose corresponding quotient groups are isomorphic to $\mathbb{Z}_2$ (see 146).

150. See 57. Suppose that f_1 and f_2 belong to $\phi(H)$. This means that in H there exist two elements h_1 and h_2 such that $\phi(h_1) = f_1$ and $\phi(h_2) = f_2$. Thus $h_1 h_2$ belongs to H , and

$$\phi(h_1 h_2) = \phi(h_1) \phi(h_2) = f_1 f_2,$$

since ϕ is a homomorphism. It follows that $f_1 f_2$ also belongs to $\phi(H)$.

2) Since e_G belongs to H and $\phi(e_G) = e_F$ (see 133), e_F belongs to the image $\phi(H)$ of the subgroup H .

3) Suppose that f belongs to $\phi(H)$. This means that in H there exists an element h such that $\phi(h) = f$. Thus h^{-1} belongs to H , and $\phi(h^{-1}) = [\phi(h)]^{-1} = f^{-1}$ (see 134). Hence f^{-1} also belongs to $\phi(H)$.

151. See 57. 1) Suppose that g_1 and g_2 belong to $\phi^{-1}(H)$. This means that the elements $\phi(g_1) = h_1$ and $\phi(g_2) = h_2$ belong to H . Thus $h_1 h_2$ also belongs to H , and $\phi(g_1 g_2) = \phi(g_1) \phi(g_2) = h_1 h_2$. Hence $g_1 g_2$ belongs to $\phi^{-1}(H)$.

2) Since $\phi(e_G) = e_F$ (see 133) and e_F belongs to H , e_G belongs to $\phi^{-1}(H)$.

3) Suppose that g belongs to $\phi^{-1}(H)$. This means that the element $\phi(g) = h_1$ belongs to H . Thus h_1^{-1} also belongs to H , and $\phi(g^{-1}) = [\phi(g)]^{-1} = h_1^{-1}$ (see 134). Hence g^{-1} belongs to $\phi^{-1}(H)$.

152. Let a be an arbitrary element of $\phi^{-1}(N)$. This means that $\phi(a) = h$ belongs to N . If g is an arbitrary element of the group G and $\phi(g) = f$, then $\phi(g^{-1}) = [\phi(g)]^{-1} = f^{-1}$ (see 134). Thus

$$\phi(g a g^{-1}) = \phi(g) \phi(a) \phi(g^{-1}) = f h f^{-1}$$

belongs to N , because N is a normal subgroup of F . Therefore $g a g^{-1}$ belongs to $\phi^{-1}(N)$, and $\phi^{-1}(N)$ is a normal subgroup of G .

153. If g_1 and g_2 are two arbitrary elements of G , with $\phi(g_1) = f_1$ and $\phi(g_2) = f_2$, then (see 134) $\phi(g_1^{-1}) = f_1^{-1}$ and $\phi(g_2^{-1}) = f_2^{-1}$. It follows that

$$\phi(g_1 g_2 g_1^{-1} g_2^{-1}) = f_1 f_2 f_1^{-1} f_2^{-1},$$

i.e., the image of every commutator of G is a commutator in F . Every element of the commutant K_1 can be written in the form $k_1 k_2 \cdots k_n$, where each k_i is a commutator. The element

$$\phi(k_1 k_2 \cdots k_n) = \phi(k_1) \phi(k_2) \cdots \phi(k_n)$$

is a product of commutators in F , and therefore belongs to the commutant K_2 . It follows also that K_1 is contained in $\phi^{-1}(K_2)$.

154. Let a be an arbitrary element of $\phi(N)$. This means that in N there exists an element h such that $\phi(h) = a$. Let f be an arbitrary element of F . Since ϕ is a surjective homomorphism, there exists in G an element g such that $\phi(g) = f$. Thus $\phi(g^{-1}) = f^{-1}$ (see 134), and $\phi(ghg^{-1}) = f a f^{-1}$. Since N is a normal subgroup of G , the element $f a f^{-1}$ belongs to $\phi(N)$, and $\phi(N)$ is thus a normal subgroup of F .

155. Let $f_1 f_2 f_1^{-1} f_2^{-1}$ be an arbitrary commutator in F . Since ϕ is a surjective homomorphism, there exist in G two elements g_1 and g_2 such that $\phi(g_1) = f_1$ and $\phi(g_2) = f_2$. Thus (see 134) $\phi(g_1^{-1}) = f_1^{-1}$, $\phi(g_2^{-1}) = f_2^{-1}$, and

$$\phi(g_1 g_2 g_1^{-1} g_2^{-1}) = f_1 f_2 f_1^{-1} f_2^{-1}.$$

Since $g_1 g_2 g_1^{-1} g_2^{-1}$ belongs to K_1 , the commutator $f_1 f_2 f_1^{-1} f_2^{-1}$ belongs to $\phi(K_1)$. But since $\phi(K_1)$ is a subgroup of F (see 150) and contains all commutators, $\phi(K_1)$ contains the entire commutant K_2 . On the other hand, $\phi(K_1)$ is contained in K_2 (see 153). Hence $\phi(K_1) = K_2$.

The equality $K_1 = \phi^{-1}(K_2)$ does not hold in general. For example, the mapping ϕ of the group $\mathbb{Z}_2 = \{e_1, a\}$ into the group $\{e_2\}$ consisting of a sole element, such that $\phi(e_1) = e_2$ and $\phi(a) = e_2$, is a surjective homomorphism. One has $K_1 = \{e_1\}$, $K_2 = \{e_2\}$, and $\phi^{-1}(K_2) = \{e_1, a\} \neq K_1$.

156. *Answer.* a) Yes: the group $\mathbb{Z}_n$ is commutative; b) yes (see 118); c) yes (see 118); d) yes (see 118); e) yes (see 120); f) yes (see 122 and (e)); g) yes (see 118 and (e)).

157. A given face can be sent onto any one of the 12 faces in 5 distinct ways.

Answer. 60.

158. *Answer.* 1) 1; 2) 24; 3) 20; 4) 15.

159. Let l_1 and l_2 be two axes of the same type (i.e., two axes either through the centres of opposite faces, through opposite vertices, or through the centres of opposite edges). There exists a rotation g of the dodecahedron which sends l_1 to l_2 . If a is any nontrivial rotation about l_1 , then the rotation $g a g^{-1}$ sends (verify) the axis l_2 to itself, without reversing it. The element $g a g^{-1}$ is not the identity; otherwise $a = g^{-1} e g = e$ would be the identity rotation about l_1 . Hence, if a normal subgroup in the group of rotations contains one rotation about one axis, then it contains at least one rotation about every one of the axes of the same type. A subgroup of rotations of the dodecahedron about one axis has order 5, 3, or 2, according to the type of axis. Since 5, 3, and 2 are prime numbers, any element different from e is a generator of this subgroup (see 34), i.e., it generates the entire subgroup. Hence, if a normal subgroup of the group of rotations of the dodecahedron contains one rotation about some axis, then it contains all the rotations about all the axes of that type.

160. From the result of Problem 159 it follows that a normal subgroup in the group of rotations of the dodecahedron must consist of some of the classes 1–4, necessarily including class 1. The order of the normal subgroup must be a divisor of the order of the group of rotations of the dodecahedron (i.e., 60) (see Lagrange's theorem, §1.8). It is easy to verify (see 158) that this is possible only if the normal subgroup contains only class 1 or contains all the classes.

161. Since the group G is not commutative, the commutant $K(G) \neq \{e\}$. Since $K(G)$ is a normal subgroup of G (see 116), it follows from the hypothesis of the problem that $K(G) = G$. Therefore, in the sequence $G, K(G), K_2(G) = K(K(G)), \dots$, all groups coincide with G , and this sequence cannot end with the unit group. Hence G is not soluble.

162. Let the group G be soluble. Thus there exists an integer n such that

$$K_n(G) = \underbrace{K \circ K \circ \dots \circ K}_{n \text{ times}}(G)$$

is the unit group. If H is a subgroup of G , then $K(H)$ is contained in $K(G)$, $K_2(H)$ is contained in $K_2(G)$, and so on. Since $K_n(H)$ is contained in $K_n(G)$, and $K_n(G)$ is the unit group, $K_n(H)$ is also the unit subgroup. It follows that H is soluble.

163. Denote by $K(G)$ the commutant in the group G , and put

$$K_r(G) = \underbrace{K \circ K \circ \dots \circ K}_{r \text{ times}}(G).$$

Since ϕ is a surjective homomorphism of G onto F , $\phi(K(G)) = K(F)$ (see 155), $\phi(K_2(G)) = K_2(F)$, and in general $\phi(K_r(G)) = K_r(F)$. Since G is soluble, for a certain n the subgroup $K_n(G)$ is the unit group. But $\phi(K_n(G)) = K_n(F)$, and thus $K_n(F)$ is also the unit group. Hence F is soluble.

164. Answer. For example, $\phi : G \rightarrow \{e\}$, where G is the group of rotations of the tetrahedron.

165. Hint. Consider the natural homomorphism of the group G onto the quotient group G/N (§1.13). Then use the result of Problem 163.

166. Let $K(G)$ be the commutant in the group G , and let

$$K_r(G) = \underbrace{K \circ K \circ \dots \circ K}_{r \text{ times}}(G).$$

Consider the natural homomorphism ϕ (see §1.13) of G onto the quotient group G/N . Thus (see 155) $\phi(K(G)) = K(G/N)$, $\phi(K_2(G)) = K_2(G/N)$, and in general $\phi(K_r(G)) = K_r(G/N)$. Since by hypothesis G/N is soluble, for a certain n the subgroup $K_n(G/N)$ is the unit group, i.e., $K_n(G/N) = \{E\}$. Since $\phi(K_n(G)) = K_n(G/N) = \{E\}$, the subgroup $K_n(G)$ is contained in the normal subgroup N . The group N being soluble, for a certain s the subgroup $K_s(N)$ is the unit group. Since $K_n(G)$ is contained in N , the subgroup $K_{n+s}(G)$ is contained in $K_s(N)$, and is thus the unit group. It follows that G is soluble.

167. From Problem 146 we obtain

$$(G \times F)/(G \times \{e_2\}) \cong \{e\} \times F.$$

Since the groups $G \times \{e_2\}$ and $\{e\} \times F$ are isomorphic to the groups G and F , respectively, they are soluble. Therefore the group $G \times F$ (see 166) is also soluble.

168. Since G is soluble, in the sequence of commutants $K(G), K_2(G), \dots$ we have, for a certain n , $K_n(G) = \{e\}$. Consider the sequence

$$G, K(G), K_2(G), \dots, K_n(G).$$

This is the sequence we seek, because every group after the first is the commutant of the preceding group, and is therefore a normal subgroup (see 116) of it. Moreover, all the quotient groups $K_i(G)/K_{i+1}(G)$, as well as $G/K(G)$, are commutative (see 129); the group $K_n(G) = \{e\}$, too, is commutative.

169. Since by hypothesis the quotient group G_{i-1}/G_i is commutative, the commutant $K(G_{i-1})$ is contained in G_i (see 130). It follows that $K_2(G_{i-1})$ is contained in $K(G_i)$, and in general $K_r(G_{i-1})$ is contained in $K_{r-1}(G_i)$ for every $r \geq 2$ and $1 \leq i \leq n$. The subgroup $K_{n+1}(G_0)$ is contained in $K_n(G_1)$, which is contained in $K_{n-1}(G_2)$, and so on, up to $K(G_n)$. It follows that $K_{n+1}(G)$ is contained in $K(G_n)$. But $K(G_n) = \{e\}$, because G_n is by hypothesis commutative. Hence $K_{n+1}(G) = \{e\}$, i.e., G is soluble.

We can obtain the result of Problem 169 also by induction, going from G_n to G_{n-1} , then to G_{n-2} , and so on, using the result of Problem 166.

170. By hypothesis the group G is soluble. This means that for a certain n the subgroup $K_n(G)$ is the unit group, and therefore $K_{n-1}(G)$ is commutative. Since $K_{n-1}(G)$ is a normal subgroup of G (see 131), we may consider the quotient group $G_1 = G/K_{n-1}(G)$. We prove that $K_{n-2}(G_1)$ is commutative. Consider the natural homomorphism (see §1.13) $\phi : G \rightarrow G_1$ with kernel $K_{n-1}(G)$. Since ϕ is surjective, $\phi(K(G)) = K(G_1)$ (see 155). Thus $\phi(K_2(G)) = K_2(G_1)$, and so on. Consequently,

$$K_{n-1}(G_1) = \phi(K_{n-1}(G)) = \{e_{G_1}\},$$

because $K_{n-1}(G)$ is the kernel of ϕ . Since $K_{n-1}(G_1) = K(K_{n-2}(G_1)) = \{e_{G_1}\}$, the subgroup $K_{n-2}(G_1)$ is commutative (see 117). Denote by G_2 the quotient group $G_1/K_{n-2}(G_1)$. As before, we can prove that the subgroup $K_{n-3}(G_2)$ is commutative. Put $G_3 = G_2/K_{n-3}(G_2)$, and so on. The group $G_{n-1} = G_{n-2}/K(G_{n-2})$ is commutative (see 129). The sequence of groups

$$G, G_1, G_2, \dots, G_{n-1},$$

with normal subgroups

$$K_{n-1}(G), K_{n-2}(G_1), \dots, K(G_{n-2}),$$

is the requested sequence.

171. Let $G_0, G_1, \dots, G_n$ be a sequence of groups with the properties mentioned in Problem 170. We will prove that all the groups of the sequence, in particular G_0 , are soluble. We shall use induction from n to 0. The group G_n is soluble, because by hypothesis it is commutative and therefore $K(G_n) = \{e\}$. Suppose that we have already proved that G_i is soluble. We will prove that G_{i-1} , too, is soluble. The group G_{i-1} contains by hypothesis a normal, commutative, and thus soluble, subgroup N_{i-1} , for which the quotient group $G_{i-1}/N_{i-1} \cong G_i$ is soluble by the induction hypothesis. Hence G_{i-1} is also soluble by virtue of the result of Problem 166. In conclusion, all the groups in the sequence $G_0, G_1, \dots, G_n$ are soluble, and in particular G_0 is soluble.

172. Any permutation of degree n can be written in the form

$$\begin{pmatrix} 1 & 2 & \dots & n \\ i_1 & i_2 & \dots & i_n \end{pmatrix},$$

where the i_m 's are all distinct and take values from 1 to n . In particular, i_1 may take any one of the n values. Then i_2 can take any one of the remaining $n - 1$ values, and so on. It follows that the number of distinct permutations of degree n is

$$n(n-1) \cdots 2 \cdot 1 = n!.$$

173. If

$$a = \begin{pmatrix} 1 & 2 & 3 & 4 & \dots & n \\ 2 & 3 & 1 & 4 & \dots & n \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 1 & 2 & 3 & 4 & \dots & n \\ 1 & 3 & 2 & 4 & \dots & n \end{pmatrix},$$

then

$$ab = \begin{pmatrix} 1 & 2 & 3 & 4 & \dots & n \\ 2 & 1 & 3 & 4 & \dots & n \end{pmatrix} \quad \text{and} \quad ba = \begin{pmatrix} 1 & 2 & 3 & 4 & \dots & n \\ 3 & 2 & 1 & 4 & \dots & n \end{pmatrix},$$

i.e., $ab \neq ba$. (Recall that in the product ab , the permutation b is carried out first, and a afterward.)

174. Suppose that the element i_1 is sent to i_2 , i_2 to i_3 , and so on. Let i_r be the first returning element. If we suppose that $i_r = i_k$, where $2 \leq k \leq r-1$, then two distinct elements, i_{k-1} and i_{r-1} , are sent by the permutation to the same element, which is a contradiction. It follows that $i_r = i_1$, and one obtains a cycle. Starting with an arbitrary element not belonging to this cycle, one constructs the second cycle, and so on. It is easy to see that every permutation is the product of independent cycles.

Suppose now that a given permutation is the product of independent cycles. If one of the cycles sends the element i_1 to i_2 , and the elements i_1 and i_2 do not appear in other cycles, then all products of the cycles send i_1 to i_2 . Hence the element i_2 , which follows i_1 in the cycle containing i_1 , is uniquely defined by the given permutation. Therefore all cycles are uniquely defined. Note that if the cycles are not independent, the decomposition into cycles may not be unique. For example,

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = (12)(23) \quad \text{and} \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = (13)(12).$$

175. Hint. Verify the equality

$$(i_1 i_2 \dots i_m) = (i_1 i_m)(i_1 i_{m-1}) \cdots (i_1 i_3)(i_1 i_2).$$

176. Hint. Let $i < j$. Verify the equality

$$(i, j) = (i, i+1)(i+1, i+2) \cdots (j-2, j-1)(j-1, j) \\ \cdot (j-2, j-1)(j-3, j-2) \cdots (i+1, i+2)(i, i+1).$$

177. The pairs corresponding to inversions are

$$(3, 2), (3, 1), (2, 1), (5, 4), (5, 1), (4, 1).$$

Answer. 6.

178. If the numbers i and j are interchanged and $k_1, k_2, \dots, k_s$ are the numbers between i and j , the property

of being or not being an inversion changes into the opposite for the pairs

$$(i, j), \quad (i, k_r), \quad (k_r, j), \quad r = 1, 2, \dots, s,$$

i.e., for $2s + 1$ pairs. Since $2s + 1$ is odd, the parity of the number of inversions changes.

179. Answer. The permutation is even (6 inversions).

180. Since

$$\begin{pmatrix} \dots & r & \dots & s & \dots \\ \dots & i_r & \dots & i_s & \dots \end{pmatrix} (r, s) = \begin{pmatrix} \dots & r & \dots & s & \dots \\ \dots & i_s & \dots & i_r & \dots \end{pmatrix},$$

the lower row of the product is obtained from the lower row of the initial permutation by interchanging i_r and i_s . By virtue of the result of Problem 178, the permutation obtained and the initial permutation have different parities.

181. Every permutation splits into a product of transpositions (see 174, 175). Suppose a permutation a is decomposed into a product of m transpositions: $a = a_1 a_2 \cdots a_m$. We can write $a = e a_1 a_2 \cdots a_m$, where e is the identity permutation. Since e is even and is multiplied m times by a transposition, it follows (see 180) that if m is even, a is even, and if m is odd, a is odd.

182. See the hint to Solution 175.

Answer. a) even; b) odd; c) even for m odd, odd for m even.

183. Hint. Decompose the given permutations into products of transpositions (see 181). Count the number of transpositions in these products.

184. If the permutations had different parities, then (see 183) the permutation $aa^{-1} = e$ would be odd, which is not true.

185. For example,

$$(1\ 2\ 3)(2\ 3\ 4) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & \dots & n \\ 2 & 1 & 4 & 3 & 5 & \dots & n \end{pmatrix},$$

whereas

$$(2\ 3\ 4)(1\ 2\ 3) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & \dots & n \\ 3 & 4 & 1 & 2 & 5 & \dots & n \end{pmatrix}.$$

186. If a is an even permutation, then gag^{-1} is also even, independently of the parity of g . Hence A_n is a normal subgroup of S_n . Let b be an odd permutation. We prove that the coset bA_n contains all the odd permutations. Let c be an odd permutation. Thus $b^{-1}c$ is even. It follows that $c = b(b^{-1}c)$ belongs to the coset bA_n . Therefore S_n is decomposed by A_n into two cosets: that of all even permutations and that of all odd permutations.

187. Answer. Since S_n is decomposed by A_n into two cosets having the same number of elements, the number of elements of A_n is (see 172)

$$\frac{n!}{2} = \frac{1 \cdot 2 \cdots n}{2}.$$

188. The group S_2 contains 2 elements and is therefore isomorphic to the commutative group $\mathbb{Z}_2$. The groups S_3 and S_4 are isomorphic to the group of symmetries of the triangle and to the group of symmetries of the tetrahedron, respectively. Both these groups are soluble (see 156).

189. Enumerate the vertices of the dodecahedron as shown in Figure 43. The tetrahedra are, for example, those with vertices chosen in the following way¹:

$$(1, 8, 14, 16), \quad (2, 9, 15, 17), \quad (3, 10, 11, 18), \quad (4, 6, 12, 19), \quad (5, 7, 13, 20).$$

The edges of the cubes are the diagonals of the dodecahedron faces. Every pair of opposite vertices of the dodecahedron is a pair of two opposite vertices of two Kepler cubes. Each cube has only one pair of vertices in common with any one of the others. (Two Kepler cubes—black and white—having two opposite vertices in common are shown in the figure.) In each cube one can inscribe two tetrahedra (see Problems 126 and 127). Since each tetrahedron is defined by four vertices of the cube, and any two Kepler cubes have only 2 vertices in common, all tetrahedra inscribed in the Kepler cubes are distinct. So there are in all 10 tetrahedra, two for every vertex of the dodecahedron. Any two such tetrahedra either have no vertices in common, or they have only one vertex in common. Indeed, if two vertices belonged to two tetrahedra, the edge of such tetrahedra joining them would be the diagonal of a face of two different cubes, but we know that any two cubes have in common only opposite vertices. There are two possible choices of 5 tetrahedra without common vertices inside the dodecahedron: indeed, when we choose one of them, we have to

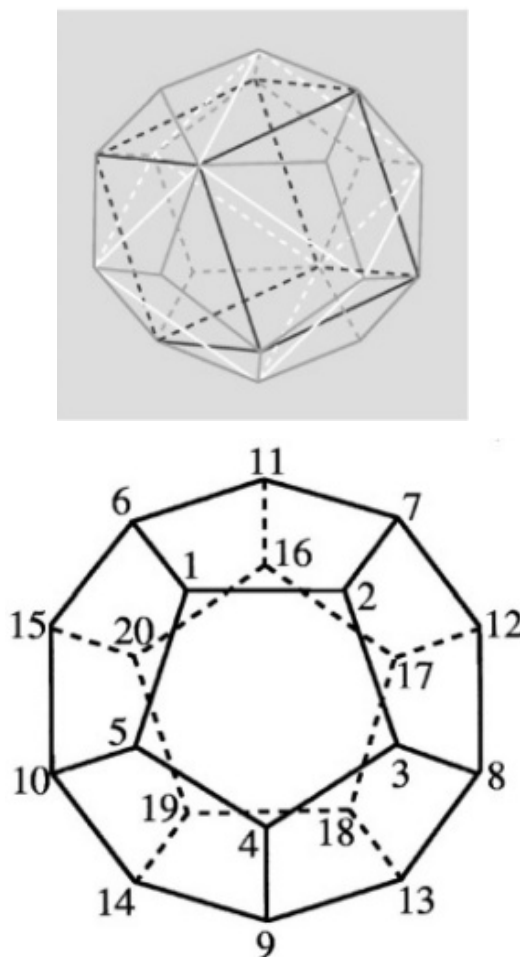


图 3.1

190. The permutations of degree 5 may be represented as products of independent cycles only in the following

¹The inscription of the five Kepler cubes inside the dodecahedron helps us to find the five tetrahedra.

ways:

$$\begin{array}{lll} \text{a) } (i_1 i_2 i_3 i_4 i_5), & \text{b) } (i_1 i_2 i_3), & \text{c) } (i_1 i_2)(i_3 i_4), \\ \text{d) } (i_1 i_2 i_3 i_4), & \text{e) } (i_1 i_2 i_3)(i_4 i_5), & \text{f) } (i_1 i_2). \end{array}$$

Applying the results of Problems 182 and 183, we obtain that in cases (a), (b), and (c) the permutations are even, whereas in cases (d), (e), and (f) they are odd.

191. Suppose that the normal subgroup N in A_5 contains a permutation h of type (a) (see 190). Without loss of generality, suppose that $h = (1\ 2\ 3\ 4\ 5)$. We prove that any permutation $(i_1\ i_2\ i_3\ i_4\ i_5)$ of type (a) belongs to N . If the row i_1, i_2, i_3, i_4, i_5 has an even number of inversions, then the permutation

$$g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ i_1 & i_2 & i_3 & i_4 & i_5 \end{pmatrix}$$

is even. Hence, by the definition of a normal subgroup, N contains $ghg^{-1} = (i_1\ i_2\ i_3\ i_4\ i_5)$. If the row i_1, i_2, i_3, i_4, i_5 has an odd number of inversions, then the row i_1, i_4, i_2, i_5, i_3 has an even number of inversions, because the order of elements is reversed in three pairs. In this case the permutation

$$g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ i_1 & i_4 & i_2 & i_5 & i_3 \end{pmatrix}$$

is even. Hence N contains $ghg^{-1} = (i_1\ i_4\ i_2\ i_5\ i_3)$, and together with it the permutation $(ghg^{-1})^2 = (i_1\ i_2\ i_3\ i_4\ i_5)$.

Suppose now that N contains a permutation $h = (1\ 2\ 3)$ of type (b) (see 190), and let $(i_1\ i_2\ i_3)$ be arbitrary. Choose i_4 and i_5 , different from i_1, i_2, i_3 , from the set $\{1, 2, 3, 4, 5\}$. Either the row i_1, i_2, i_3, i_4, i_5 or the row i_1, i_2, i_3, i_5, i_4 has an even number of inversions. Consequently, one of the permutations

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ i_1 & i_2 & i_3 & i_4 & i_5 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ i_1 & i_2 & i_3 & i_5 & i_4 \end{pmatrix}$$

is even. Denoting this permutation by g , we obtain in either case that N contains $ghg^{-1} = (i_1\ i_2\ i_3)$.

If the normal subgroup contains a permutation of type (c) (see 190), for example $h = (1\ 2)(3\ 4)$, then it also contains every permutation $(i_1\ i_2)(i_3\ i_4)$ of type (c). Indeed, one of the permutations

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ i_1 & i_2 & i_3 & i_4 & i_5 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ i_2 & i_1 & i_3 & i_4 & i_5 \end{pmatrix}$$

is even. Let this permutation be denoted by g . Thus N contains $ghg^{-1} = (i_1\ i_2)(i_3\ i_4)$.

192. We count the number of permutations of each of types (a), (b), and (c) (see 190).

a) There are $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ sequences of the five numbers 1, 2, 3, 4, 5. Since every permutation of type (a) can be written in five equivalent ways, depending on the choice of the first element, the number of permutations of type (a) is $120/5 = 24$.

b) By the same reasoning as in (a), the number of permutations of type (b) is $5 \cdot 4 \cdot 3/3 = 20$.

c) A permutation of type (c) can be written in 8 different ways: one can choose i_1 in 4 ways and afterward i_3 in 2 ways. Hence the number of permutations of type (c) is $5 \cdot 4 \cdot 3 \cdot 2/8 = 15$.

Every normal subgroup N contains the unit element. Moreover, from the result of Problem 191 it follows that a normal subgroup of A_5 either contains all the permutations of a given type (see 190) or contains none of them. The order of a normal subgroup must divide the order of A_5 , which is 60. But adding to the number 1 any selection of the numbers 24, 20, and 15 gives a divisor of 60 only in two cases: when nothing is added and

when all three numbers are added. The first case corresponds to the unit subgroup, and the second to the whole group A_5 .

193. Such a subgroup is, for example, the subgroup containing all permutations of the form

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & \dots & n \\ i_1 & i_2 & i_3 & i_4 & i_5 & 6 & \dots & n \end{pmatrix}$$

with an even number of inversions in the row i_1, i_2, i_3, i_4, i_5 .

3.2 Problems of Chapter 2

194. a) *Answer.* No, because the integer numbers do not form a group under addition (cf. 17).

b) *Answer.* No, because the integer numbers without zero do not form a group under multiplication: all numbers except 1 and -1 have no inverse.

c) *Answer.* Yes. Use the result of Problem 57.

Solution. Since the real numbers form a commutative group under addition, and the nonzero real numbers also form a commutative group under multiplication, it suffices to verify that the rational numbers form a subgroup of the real numbers under addition, and that the nonzero rational numbers form a subgroup under multiplication. This follows easily from Problem 57. Indeed: 1) if a and b are rational, then $a + b$ and ab are rational as well; 2) 0 and 1 are rational; 3) if a is rational, then $-a$ and $1/a$ (for $a \neq 0$) are also rational. The distributive law is obviously satisfied. Consequently, the rational numbers form a field.

d) *Answer.* Yes. Use the result of Problem 57.

Solution. If $r_1 + r_2\sqrt{2} = 0$, where r_1 and r_2 are rational and $r_2 \neq 0$, then $\sqrt{2} = -r_1/r_2$, which is impossible because $\sqrt{2}$ is not rational. Hence if $r_1 + r_2\sqrt{2} = 0$, with r_1, r_2 rational, then $r_1 = r_2 = 0$. All numbers of the form $r_1 + r_2\sqrt{2}$, for different pairs (r_1, r_2) , are different, because if $r_1 + r_2\sqrt{2} = r_3 + r_4\sqrt{2}$, then $(r_1 - r_3) + (r_2 - r_4)\sqrt{2} = 0$, and therefore $r_1 = r_3$ and $r_2 = r_4$.

Let us now prove that all numbers of the form $r_1 + r_2\sqrt{2}$, where r_1 and r_2 are rational, form a field. We prove that these numbers form a subgroup under addition in the real numbers, and that the nonzero ones also form a subgroup under multiplication. Using Problem 57, we obtain:

1) if $a = r_1 + r_2\sqrt{2}$ and $b = r_3 + r_4\sqrt{2}$, then

$$a + b = (r_1 + r_3) + (r_2 + r_4)\sqrt{2}$$

and

$$ab = (r_1r_3 + 2r_2r_4) + (r_1r_4 + r_2r_3)\sqrt{2};$$

2) 0 and 1 belong to the set considered, because $0 = 0 + 0\sqrt{2}$ and $1 = 1 + 0\sqrt{2}$;

3) if $a = r_1 + r_2\sqrt{2}$, then $-a = (-r_1) + (-r_2)\sqrt{2}$, and, for $r_1 + r_2\sqrt{2} \neq 0$,

$$\begin{aligned} a^{-1} &= \frac{1}{r_1 + r_2\sqrt{2}} = \frac{r_1 - r_2\sqrt{2}}{(r_1 + r_2\sqrt{2})(r_1 - r_2\sqrt{2})} \\ &= \frac{r_1 - r_2\sqrt{2}}{r_1^2 - 2r_2^2} = \frac{r_1}{r_1^2 - 2r_2^2} - \frac{r_2}{r_1^2 - 2r_2^2}\sqrt{2}. \end{aligned}$$

Since the distributive law is satisfied by all real numbers, the set considered is a field.

195. We have

$$a0 + a0 = a(0 + 0) = a0.$$

Since a field is a group under addition, one can add $-a0$ to both sides of the equation. One obtains $a0 = 0$. Since multiplication in the field is commutative, one also has $0a = 0$.

196. 1) We have

$$ab + (-a)b = (a + (-a))b = 0b = 0$$

(cf. 195). It follows that $(-a)b = -(ab)$. In the same way one proves that $a(-b) = -ab$.

2) By part (1) and Problem 20,

$$(-a)(-b) = -(a(-b)) = -(-(ab)) = ab.$$

197. If $ab = 0$ and $a \neq 0$, then there exists an element a^{-1} , the inverse of a . Thus $a^{-1}ab = a^{-1}0 = 0$ (cf. 195). But $a^{-1}ab = 1 \cdot b = b$. Hence $b = 0$.

198. *Answer.* See Table 14.

表 3.11

$\cdot$	0	1
0	0	0
1	0	1

$\cdot$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

$\cdot$	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

199. Let n be a non-prime number, i.e., $n = n_1n_2$, where $n_1 < n$ and $n_2 < n$. Thus modulo n we have $n_1n_2 = 0$, but $n_1 \neq 0$ and $n_2 \neq 0$. Since this is not possible in a field (cf. 197), for non-prime n the remainders with the operations modulo n do not form a field.

Observe now the following property. Let x and y be two integers, and let r_1 and r_2 be the remainders of their division by n , i.e., $x = k_1n + r_1$ and $y = k_2n + r_2$. Thus

$$x + y = (k_1 + k_2)n + (r_1 + r_2)$$

and

$$xy = (k_1k_2n + k_1r_2 + k_2r_1)n + r_1r_2.$$

We obtain that $x + y$ and $r_1 + r_2$, as well as xy and r_1r_2 , give the same remainder when divided by n . In other words, we obtain the same result whether we first take the remainders of x and y upon division by n and afterward take their sum or product modulo n , or first take the sum or product of x and y as usual and then take its remainder upon division by n . Thus, to calculate an expression with operations modulo n , one may perform the calculation as usual with integers and take the remainder modulo n only at the end.

Under addition modulo n , the remainders form a commutative group (see 40). Since for all integers a, b, c the numbers $(a + b)c$ and $ac + bc$ are equal, and consequently their remainders upon division by n are equal, the equality $(a + b)c = ac + bc$ also holds modulo n ; thus distributivity is satisfied. In the same way one verifies the associativity and commutativity of multiplication modulo n : $(ab)c = a(bc)$ and $ab = ba$. The unit element

for multiplication is 1. It remains to prove that for prime n , every remainder $a \neq 0$ has an inverse, i.e., that there exists a remainder x such that $ax = 1$ modulo n .

Let $0 < a < n$. Consider the numbers $a \cdot 0, a \cdot 1, \dots, a \cdot (n - 1)$ under ordinary multiplication. The difference between two of these numbers, $ak - al = a(k - l)$, is not divisible by n , because n is prime, $a < n$, and $0 < |k - l| < n$. As a consequence, these n numbers give distinct remainders upon division by n , and thus give all possible remainders. This means that one of them has remainder 1, i.e., for a certain remainder x , one has $ax = 1$ modulo n . Hence, for prime n , all the properties of fields are satisfied.

200. Since $a - b = a + (-b)$, it follows that

$$(a - b)c = (a + (-b))c = ac + (-b)c = ac + (-bc) = ac - bc$$

(cf. 196).

201. Subtract from

$$P(x) = S_1(x)Q(x) + R_1(x)$$

the equality

$$P(x) = S_2(x)Q(x) + R_2(x).$$

One obtains

$$0 = (S_1(x) - S_2(x))Q(x) + (R_1(x) - R_2(x)),$$

so

$$(S_1(x) - S_2(x))Q(x) = R_2(x) - R_1(x). \quad (3.1)$$

If $S_1(x) - S_2(x) \neq 0$, the degree of the polynomial on the left-hand side of (3.1) is not lower than the degree of $Q(x)$. But the degree of the polynomial on the right-hand side is strictly lower than the degree of $Q(x)$. This contradiction shows that $S_1(x) - S_2(x) = 0$, and consequently $R_2(x) - R_1(x) = 0$, i.e., $S_1(x) = S_2(x)$ and $R_1(x) = R_2(x)$.

202. This group is the direct product of the group of real numbers under addition with itself (cf. 69 and 73). The unit element (zero) is the pair $(0, 0)$.

203. Let $z_1 = (a, b)$, $z_2 = (c, d)$, and $z_3 = (e, f)$. Thus

$$z_1 z_2 = (ac - bd, ad + bc) \quad \text{and} \quad z_2 z_1 = (ca - db, cb + da).$$

But $ac - bd = ca - db$ and $ad + bc = cb + da$. Hence $z_1 z_2 = z_2 z_1$. Moreover,

$$\begin{aligned} (z_1 z_2) z_3 &= (ac - bd, ad + bc)(e, f) \\ &= (ace - bde - adf - bcf, acf - bdf + ade + bce), \end{aligned}$$

whereas

$$\begin{aligned} z_1(z_2 z_3) &= (a, b)(ce - df, cf + de) \\ &= (ace - adf - bcf - bde, acf + ade + bce - bdf). \end{aligned}$$

Therefore $(z_1 z_2) z_3 = z_1(z_2 z_3)$.

204. Let $z = (a, b) \neq (0, 0)$, and let the required complex number be $z^{-1} = (x, y)$. Thus

$$zz^{-1} = z^{-1}z = (ax - by, ay + bx).$$

In order that $zz^{-1} = (1, 0)$, the following equations must be satisfied:

$$\begin{cases} ax - by = 1, \\ bx + ay = 0. \end{cases}$$

This system has exactly one solution,

$$x = \frac{a}{a^2 + b^2}, \quad y = -\frac{b}{a^2 + b^2},$$

where $a^2 + b^2 \neq 0$, because $(a, b) \neq (0, 0)$. Hence

$$(a, b)^{-1} = \left(\frac{a}{a^2 + b^2}, -\frac{b}{a^2 + b^2} \right).$$

205. Let $z_1 = (a, b)$, $z_2 = (c, d)$, and $z_3 = (e, f)$. Thus

$$(z_1 + z_2)z_3 = (a + c, b + d)(e, f) = (ae + ce - bf - df, af + cf + be + de)$$

and

$$\begin{aligned} z_1z_3 + z_2z_3 &= (ae - bf, af + be) + (ce - df, cf + de) \\ &= (ae - bf + ce - df, af + be + cf + de), \end{aligned}$$

from which it follows that $(z_1 + z_2)z_3 = z_1z_3 + z_2z_3$.

206. We have

$$a + bi = (a, 0) + (b, 0)(0, 1) = (a, 0) + (0, b) = (a, b).$$

207. Answer.

$$(c + di) - (a + bi) = (c - a) + (d - b)i.$$

208.

$$\begin{aligned} x + yi &= \frac{c + di}{a + bi} = \frac{(c + di)(a - bi)}{(a + bi)(a - bi)} \\ &= \frac{(ac + bd) + (ad - bc)i}{a^2 + b^2} \\ &= \frac{ac + bd}{a^2 + b^2} + \frac{ad - bc}{a^2 + b^2}i, \end{aligned}$$

since $a + bi \neq 0$, and hence $a^2 + b^2 \neq 0$.

209. Answer. a) $i^3 = -i$; b) $i^4 = 1$; c)

$$i^n = \begin{cases} 1, & \text{if } n = 4k, \\ i, & \text{if } n = 4k + 1, \\ -1, & \text{if } n = 4k + 2, \\ -i, & \text{if } n = 4k + 3. \end{cases}$$

210. Hint. The equation

$$(x + yi)^2 = (x^2 - y^2) + 2xyi = b,$$

where b is real, is equivalent to the system

$$\begin{cases} x^2 - y^2 = b, \\ 2xy = 0. \end{cases}$$

Answer: a) $z = \pm 1$; b) $z = \pm i$; c) $z = \pm a$; d) $z = \pm ai$.

211. Let $z_1 = a + bi$ and $z_2 = c + di$. Thus:

a)

$$\begin{aligned} \overline{z_1 + z_2} &= \overline{(a + c) + (b + d)i} \\ &= (a + c) - (b + d)i \\ &= (a - bi) + (c - di) = \overline{z_1} + \overline{z_2}. \end{aligned}$$

b) $(z_1 - z_2) + z_2 = z_1$. By part (a),

$$\overline{z_1} = \overline{(z_1 - z_2) + z_2} = \overline{z_1 - z_2} + \overline{z_2}.$$

It follows that $\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$.

c)

$$\begin{aligned} \overline{z_1 z_2} &= \overline{(a + bi)(c + di)} \\ &= \overline{(ac - bd) + (bc + ad)i} \\ &= (ac - bd) - (bc + ad)i \\ &= (a - bi)(c - di) = \overline{z_1} \overline{z_2}. \end{aligned}$$

d)

$$\overline{z_1} = \overline{(z_1/z_2)z_2} = \overline{z_1/z_2} \overline{z_2}$$

by part (c). Hence

$$\overline{z_1/z_2} = \frac{\overline{z_1}}{\overline{z_2}}.$$

212. From Problem 211 we obtain

$$\begin{aligned} \overline{P(z)} &= \overline{a_0 z^n + a_1 z^{n-1} + \cdots + a_{n-1} z + a_n} \\ &= \overline{a_0} \overline{z}^n + \overline{a_1} \overline{z}^{n-1} + \cdots + \overline{a_{n-1}} \overline{z} + \overline{a_n} \\ &= a_0 \overline{z}^n + a_1 \overline{z}^{n-1} + \cdots + a_{n-1} \overline{z} + a_n \\ &= P(\overline{z}), \end{aligned}$$

because $\overline{a_i} = a_i$.

213. Since the field M contains all real numbers and the element i_0 , it contains all elements of the form $a + bi_0$, where a, b are arbitrary real numbers. Let M' denote the set of all elements of M represented in this form. By the commutative, associative, and distributive properties of addition and multiplication in M , we obtain for elements of M' :

$$(a + bi_0) + (c + di_0) = (a + c) + (b + d)i_0, \quad (3.2)$$

and

$$\begin{aligned} (a + bi_0)(c + di_0) &= ac + adi_0 + bci_0 + bdi_0^2 \\ &= (ac - bd) + (ad + bc)i_0, \end{aligned} \quad (3.3)$$

since $i_0^2 = -1$.

Let C be the field of complex numbers. Consider the mapping ϕ of C into M such that

$$\phi(a + bi) = a + bi_0.$$

Comparing (3.2) and (3.3) with (2.4) and (2.5) of §2.2, we see that ϕ is a homomorphism of C into M with respect to addition and multiplication. Since $\phi(C) = M'$, the set M' is a subgroup (cf. 150) of M with respect to addition and with respect to multiplication. The operations of addition and multiplication possess the properties of commutativity, associativity, and distributivity in M , so these properties also hold in M' . Hence M' is a field.

If $a + bi_0 = c + di_0$, then $(b - d)i_0 = c - a$. If $b - d \neq 0$, then $i_0 = (c - a)/(b - d)$ is a real number, but this cannot be true because the square of a real number is never -1 . Consequently $b - d = 0$, and thus also $c - a = 0$. Hence $b = d$ and $a = c$. Therefore the elements $a + bi_0$ are different for different pairs (a, b) . It follows that ϕ is a bijection of C onto M' . Since ϕ is a homomorphism, it is an isomorphism of C onto M' ; thus M' is isomorphic to the field of complex numbers.

214. Suppose the polynomial considered is reducible, i.e.,

$$x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n = (b_0x^m + b_1x^{m-1} + \cdots + b_m) \cdot (c_0x^k + c_1x^{k-1} + \cdots + c_k),$$

where $m < n$, $k < n$, all the b_i 's and c_i 's are real, and $b_0 \neq 0$, $c_0 \neq 0$. Put j in place of x in this equation. Since M is a field, one can eliminate the brackets, carrying out the product as usual. We obtain

$$j^n + a_1j^{n-1} + a_2j^{n-2} + \cdots + a_n = (b_0j^m + b_1j^{m-1} + \cdots + b_m) \cdot (c_0j^k + c_1j^{k-1} + \cdots + c_k).$$

By hypothesis,

$$j^n + a_1j^{n-1} + a_2j^{n-2} + \cdots + a_n = 0,$$

and thus (cf. 197) at least one of the two polynomials in brackets is zero. By dividing this polynomial by its leading coefficient, b_0 or c_0 , we obtain a vanishing expression of type (2.6), of degree lower than n . This contradicts the choice of n as the lowest degree for which an expression of type (2.6) vanishes. Therefore the polynomial considered is not reducible.

215. As we proved, in case (a) the element j of the field M satisfies

$$j^2 + pj + q = 0,$$

where p and q are real numbers, and $x^2 + px + q$ is not reducible over the field of real numbers. We have

$$x^2 + px + q = \left(x^2 + px + \frac{p^2}{4}\right) - \left(\frac{p^2}{4} - q\right) = \left(x + \frac{p}{2}\right)^2 - \left(\frac{p^2}{4} - q\right).$$

If $p^2/4 - q \geq 0$, then $p^2/4 - q = a^2$ for some real number a . Thus

$$x^2 + px + q = \left(x + \frac{p}{2}\right)^2 - a^2 = \left(x + \frac{p}{2} + a\right)\left(x + \frac{p}{2} - a\right),$$

i.e., the polynomial $x^2 + px + q$ would be reducible over the real numbers. It follows that $p^2/4 - q < 0$, i.e., $p^2/4 - q = -b^2$ for a certain nonzero real number b . Since $j^2 + pj + q = 0$ in M , we have

$$j^2 + pj + \frac{p^2}{4} = \frac{p^2}{4} - q \quad \text{and} \quad \left(j + \frac{p}{2}\right)^2 = \frac{p^2}{4} - q = -b^2.$$

Consequently,

$$\left(\frac{j}{b} + \frac{p}{2b}\right)^2 = \frac{-b^2}{b^2} = -1.$$

Hence the element

$$i_0 = \frac{j}{b} + \frac{p}{2b},$$

which belongs to M , is the element sought.

216. Answer. The sole field satisfying the required properties is, up to isomorphism, the field whose elements are fractions having polynomials in j as numerators and denominators, with the usual operations on these fractions.

217. Answer. A: $2 + 2i$, B: $-1 + 3i$, C: $-2 - i$.

218. Answer. a) The symmetry with respect to the origin of the coordinates (or, equivalently, the rotation by 180° around the origin of the coordinates). b) The dilation of the plane by 2 (fixing the origin of the coordinates). c) The reflection with respect to the x -axis.

219. Hint. Use the property of the triangles $A_1B_1C_1$ and $A_2B_2C_2$ being equal (Figure 44).

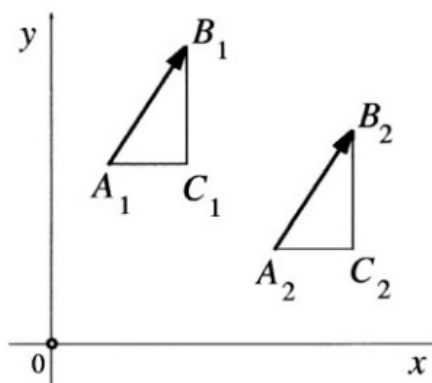


Figure 44

220. By hypothesis $z_1 = x_u + iy_u$, $z_2 = x_v + iy_v$, $z_3 = x_w + iy_w$. Consequently the equation $z_3 = z_1 + z_2$ is equivalent to the two equations

$$\begin{cases} x_w = x_u + x_v, \\ y_w = y_u + y_v. \end{cases} \quad (3.4)$$

On the other hand, if $w = u + v$, then (cf. Figure 45)

$$x_w = (\text{by definition}) x_C - x_A = (x_C - x_D) + (x_D - x_A) = x_u + x_v,$$

and similarly $y_w = y_u + y_v$. Thus the equation $w = u + v$ is equivalent to equations (3.4).

221. By definition, $x_{\overrightarrow{AB}} = x_B - x_A$ and $y_{\overrightarrow{AB}} = y_B - y_A$.

222. The equation $|z|^2 = a^2 + b^2$ comes from Pythagoras' theorem (Figure 46). The equations

$$z\bar{z} = (a + bi)(a - bi) = a^2 + b^2$$

are easily verified.

223. See Figure 47. **Hint.** The inequalities stated in the problem follow from the property that in a triangle the length of any side is smaller than the sum and longer than the difference of the lengths of the other two sides.

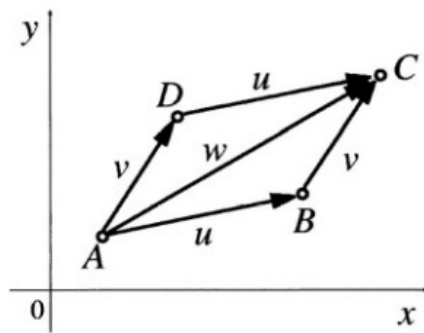


Figure 45

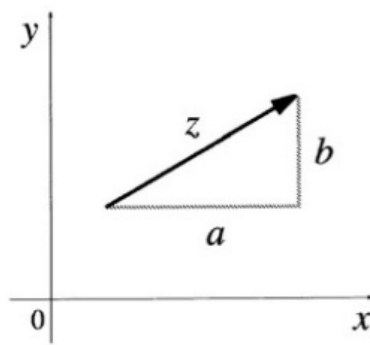


Figure 46

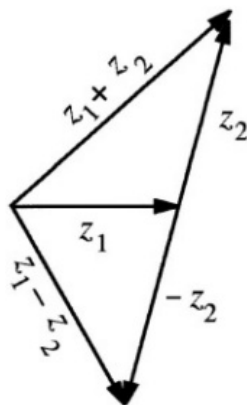


Figure 47

224. See Figure 48.

$$\begin{aligned} |z_1 + z_2|^2 + |z_1 - z_2|^2 &= (\text{cf. 222}) (z_1 + z_2)(\overline{z_1 + z_2}) + (z_1 - z_2)(\overline{z_1 - z_2}) \\ &= 2z_1\overline{z_1} + 2z_2\overline{z_2} = 2|z_1|^2 + 2|z_2|^2. \end{aligned}$$

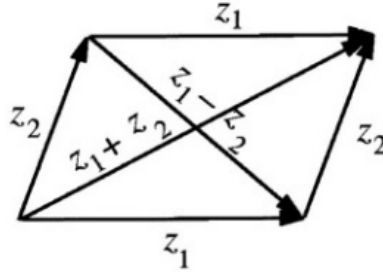


Figure 48

225. Answer.

- a) $\sqrt{2}(\cos(\pi/4) + i \sin(\pi/4))$;
- b) $2(\cos(7\pi/6) + i \sin(7\pi/6))$;
- c) $3(\cos(\pi/2) + i \sin(\pi/2))$;
- d) $5(\cos \pi + i \sin \pi)$;
- e) $\sqrt{5}(\cos \varphi + i \sin \varphi)$, where $\varphi = \arctan 2$.

226. We have

$$\begin{aligned} z_1 z_2 &= r_1 r_2 [(\cos \varphi_1 \cos \varphi_2 - \sin \varphi_1 \sin \varphi_2) + i(\sin \varphi_1 \cos \varphi_2 + \cos \varphi_1 \sin \varphi_2)] \\ &= r_1 r_2 (\cos(\varphi_1 + \varphi_2) + i \sin(\varphi_1 + \varphi_2)), \end{aligned}$$

by the trigonometric formulae for the sum. The second equation in the problem is equivalent to

$$z_1 = z_2 \frac{r_1}{r_2} (\cos(\varphi_1 - \varphi_2) + i \sin(\varphi_1 - \varphi_2)),$$

which follows directly from the first equality.

227. Hint. Use the result of Problem 226.

228. We have

$$1 - \sqrt{3}i = 2 \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) = 2(\cos(-\pi/3) + i \sin(-\pi/3)).$$

Thus, by Problem 227,

$$\begin{aligned} (1 - \sqrt{3}i)^{100} &= 2^{100} (\cos(-100\pi/3) + i \sin(-100\pi/3)) \\ &= 2^{100} (\cos(2\pi/3) + i \sin(2\pi/3)). \end{aligned}$$

It follows that

$$\frac{(1 - \sqrt{3}i)^{100}}{2^{100}} = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i.$$

Answer. $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$.

229. If $z = 0$, then also $w = 0$ (cf. 197). If $z \neq 0$, then also $w \neq 0$. In this case let

$$w = \rho(\cos \psi + i \sin \psi).$$

Thus, by de Moivre's formula (cf. 227),

$$w^n = \rho^n (\cos n\psi + i \sin n\psi) = r (\cos \varphi + i \sin \varphi).$$

It follows that $\rho^n = r$ and $n\psi = \varphi + 2\pi k$, where k is any integer. Therefore

$$w = \sqrt[n]{r} \left(\cos \frac{\varphi + 2\pi k}{n} + i \sin \frac{\varphi + 2\pi k}{n} \right). \quad (3.5)$$

If $k_1 - k_2 = ln$, where l is an integer, then the quantities $(\varphi + 2\pi k_1)/n$ and $(\varphi + 2\pi k_2)/n$ differ by $2\pi l$, and the values of w expressed by formula (3.5) coincide. Formula (3.5) thus gives n distinct values of w , obtained by assigning to k the values $0, 1, \dots, n-1$.

230. Hint. Put the expressions under the root into trigonometric form, then use formula (3.5) from the solution of Problem 229.

Answer.

- a) $+i, -i$;
- b) $2, -1 + \sqrt{3}i, -1 - \sqrt{3}i$ (here $\sqrt{3}$ is the positive real root);
- c) $\cos(25^\circ + 90^\circ n) + i \sin(25^\circ + 90^\circ n)$, where $n = 0, 1, 2, 3$;
- d) $\sqrt[6]{2}(\cos(15^\circ + 120^\circ n) + i \sin(15^\circ + 120^\circ n))$, where $n = 0, 1, 2$ (here $\sqrt[6]{2}$ is the positive real root).

231. Hint. See formula (3.5) from the solution of Problem 229; $1 = 1(\cos 0 + i \sin 0)$; $\epsilon_n^k = \cos(2\pi k/n) + i \sin(2\pi k/n)$ (cf. 227).

232. Hint. If $z_1^n = z_0$ and $z_2^n = z_0$, then

$$\left(\frac{z_2}{z_1} \right)^n = \frac{z_2^n}{z_1^n} = 1.$$

Answer. $z_1, z_1 \epsilon_n, z_1 \epsilon_n^2, \dots, z_1 \epsilon_n^{n-1}$ (cf. 231).

233. By virtue of the result of 221, $z_A = z_{\overrightarrow{OA}}$, where O is the origin of the coordinates. **Answer.** a) The distance of the point z from the origin of the coordinates; b) the angle between the positive side of the axis Ox and the ray Oz ; c) the distance between the points z_1 and z_2 (because $z_B - z_A = z_{\overrightarrow{AB}}$, cf. 221); d) the angle between the rays Oz_2 and Oz_1 (cf. 226).

234. See 233. **Answer.** a), b) The circle of radius 1 (and R) with centre at the origin of the coordinates; c) the circle of radius R and centre at the point z_0 ; d) the disc of radius R with centre z_0 , together with its bounding circle; e) the straight line perpendicular to the segment joining z_1 and z_2 , and passing through its midpoint; f) the negative side of the real axis; g) the bisector of the first quadrant of the plane; h) the ray defining the angle φ with the positive side of the axis Ox .

235. See 229 and 232. **Answer.** At the vertices of a regular n -gon with centre at the origin of the coordinates.

236. Let a point z_0 and an arbitrary real number $\varepsilon > 0$ be given. Choose $\delta = 1$, independently of z_0 and ε . Thus, for every z satisfying $|z - z_0| < 1$,

$$|f(z) - f(z_0)| = |a - a| = 0 < \varepsilon.$$

Consequently the function $f(z) \equiv a$ is continuous for every value of z . (For δ , one may take any positive real number.)

237. Let a point z_0 and an arbitrary real number $\varepsilon > 0$ be given. Choose $\delta = \varepsilon$, independently of z_0 . Thus, for every z satisfying $|z - z_0| < \delta$,

$$|f(z) - f(z_0)| = |z - z_0| < \delta = \varepsilon.$$

Consequently the function of a complex argument $f(z) = z$ is continuous for every value of the argument. Considering only real values of z , we obtain that the function of a real argument $f(x) = x$ is continuous for all values of the argument.

238. Let a point z_0 and an arbitrary real number $\varepsilon > 0$ be chosen. If δ is an arbitrary positive real number and $|z - z_0| < \delta$, then

$$\begin{aligned} |f(z) - f(z_0)| &= |z^2 - z_0^2| = |(z - z_0)(z + z_0)| \\ &= |z - z_0| |z + z_0| \leq \delta |z + z_0|. \end{aligned}$$

But

$$|z + z_0| = |(z - z_0) + 2z_0| \leq |z - z_0| + 2|z_0| < \delta + 2|z_0|.$$

Therefore

$$|f(z) - f(z_0)| < \delta(\delta + 2|z_0|).$$

Now choose δ satisfying

$$\delta(\delta + 2|z_0|) \leq \varepsilon.$$

If $z_0 = 0$, put $\delta = \sqrt{\varepsilon}$, taking the positive root. If $z_0 \neq 0$, choose as δ the smaller of $|z_0|$ and $\varepsilon/(3|z_0|)$. Then

$$\delta(\delta + 2|z_0|) \leq \frac{\varepsilon}{3|z_0|}(|z_0| + 2|z_0|) = \varepsilon.$$

Hence $|z - z_0| < \delta$ implies $|f(z) - f(z_0)| < \varepsilon$. Therefore $f(z) = z^2$ is continuous for all values of its argument.

239. a) Let an arbitrary real number $\varepsilon > 0$ be given. We have

$$\begin{aligned} |h(z) - h(z_0)| &= |(f(z) + g(z)) - (f(z_0) + g(z_0))| \\ &= |(f(z) - f(z_0)) + (g(z) - g(z_0))| \\ &\leq |f(z) - f(z_0)| + |g(z) - g(z_0)| \quad (\text{cf. 223}). \end{aligned}$$

By continuity of f at z_0 , there is $\delta_1 > 0$ such that $|z - z_0| < \delta_1$ implies $|f(z) - f(z_0)| < \varepsilon/2$. Similarly, there is $\delta_2 > 0$ such that $|z - z_0| < \delta_2$ implies $|g(z) - g(z_0)| < \varepsilon/2$. Take $\delta = \min\{\delta_1, \delta_2\}$. Then $|z - z_0| < \delta$ implies

$$|h(z) - h(z_0)| \leq |f(z) - f(z_0)| + |g(z) - g(z_0)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence $h(z) = f(z) + g(z)$ is continuous at z_0 .

b) If $h(z) = f(z) - g(z)$, then

$$\begin{aligned} |h(z) - h(z_0)| &= |(f(z) - g(z)) - (f(z_0) - g(z_0))| \\ &= |(f(z) - f(z_0)) - (g(z) - g(z_0))| \\ &\leq |f(z) - f(z_0)| + |g(z) - g(z_0)|. \end{aligned}$$

This is the same inequality as in part (a), so the proof proceeds in the same way.

c) Let $\varepsilon > 0$. We have

$$\begin{aligned} |h(z) - h(z_0)| &= |f(z)g(z) - f(z_0)g(z_0)| \\ &= |f(z)g(z) - f(z)g(z_0) + f(z)g(z_0) - f(z_0)g(z_0)| \\ &\leq |f(z)| |g(z) - g(z_0)| + |g(z_0)| |f(z) - f(z_0)|. \end{aligned}$$

We choose $\delta > 0$ so that, whenever $|z - z_0| < \delta$, both terms in the last sum are smaller than $\varepsilon/2$.

1) If $f(z_0) \neq 0$, let $\varepsilon_1 = |f(z_0)|$. By continuity of f , there is $\delta' > 0$ such that $|z - z_0| < \delta'$ implies $|f(z) - f(z_0)| < |f(z_0)|$. Thus

$$|f(z)| \leq |f(z) - f(z_0)| + |f(z_0)| < 2|f(z_0)|.$$

Let $\varepsilon_2 = \varepsilon/(4|f(z_0)|)$. By continuity of g , there is $\delta'' > 0$ such that $|z - z_0| < \delta''$ implies

$$|g(z) - g(z_0)| < \frac{\varepsilon}{4|f(z_0)|}.$$

Choose $\delta_1 = \min\{\delta', \delta''\}$. Then

$$|f(z)| |g(z) - g(z_0)| < \frac{\varepsilon}{2}.$$

If $f(z_0) = 0$, continuity gives $\delta' > 0$ such that $|z - z_0| < \delta'$ implies $|f(z)| < 1$, and it gives $\delta'' > 0$ such that $|z - z_0| < \delta''$ implies $|g(z) - g(z_0)| < \varepsilon/2$. Again taking $\delta_1 = \min\{\delta', \delta''\}$, we obtain

$$|f(z)| |g(z) - g(z_0)| < \frac{\varepsilon}{2}.$$

2) If $g(z_0) \neq 0$, let $\varepsilon_3 = \varepsilon/(2|g(z_0)|)$. There is $\delta_2 > 0$ such that $|z - z_0| < \delta_2$ implies

$$|f(z) - f(z_0)| < \frac{\varepsilon}{2|g(z_0)|},$$

and consequently

$$|g(z_0)| |f(z) - f(z_0)| < \frac{\varepsilon}{2}.$$

If $g(z_0) = 0$, any positive δ_2 will do, since

$$|g(z_0)| |f(z) - f(z_0)| = 0 < \frac{\varepsilon}{2}.$$

Now choose $\delta = \min\{\delta_1, \delta_2\}$. Then

$$|h(z) - h(z_0)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence $h(z) = f(z)g(z)$ is continuous at z_0 .

240. a) Let an arbitrary real number $\varepsilon > 0$ be given. We have

$$\begin{aligned} |h(z) - h(z_0)| &= \left| \frac{1}{g(z)} - \frac{1}{g(z_0)} \right| = \left| \frac{g(z_0) - g(z)}{g(z)g(z_0)} \right| \\ &= \frac{|g(z) - g(z_0)|}{|g(z)||g(z_0)|}. \end{aligned}$$

Let $\varepsilon_1 = |g(z_0)|/2 > 0$. By continuity of g , there is $\delta_1 > 0$ such that $|z - z_0| < \delta_1$ implies $|g(z) - g(z_0)| < |g(z_0)|/2$. Therefore

$$\begin{aligned} |g(z)| &= |g(z_0) - (g(z_0) - g(z))| \\ &\geq |g(z_0)| - |g(z_0) - g(z)| > \frac{|g(z_0)|}{2}. \end{aligned}$$

Let $\varepsilon_2 = \varepsilon|g(z_0)|^2/2 > 0$. There is $\delta_2 > 0$ such that $|z - z_0| < \delta_2$ implies

$$|g(z) - g(z_0)| < \frac{\varepsilon|g(z_0)|^2}{2}.$$

Choose $\delta = \min\{\delta_1, \delta_2\}$. Then

$$|h(z) - h(z_0)| = \frac{|g(z) - g(z_0)|}{|g(z)||g(z_0)|} < \frac{\varepsilon|g(z_0)|^2}{2} \frac{2}{|g(z_0)|^2} = \varepsilon.$$

Thus $h(z) = 1/g(z)$ is continuous at z_0 .

b) Since f and g are continuous at z_0 , the function $1/g$ is continuous there by part (a), and consequently

$$h(z) = \frac{f(z)}{g(z)} = f(z) \frac{1}{g(z)}$$

is continuous there as well (cf. 239(c)).

241. Let an arbitrary real number $\varepsilon > 0$ be given. Since f is continuous at z_1 , there exists $\delta_1 > 0$ such that

$$|u - z_1| < \delta_1 \implies |f(u) - f(z_1)| < \varepsilon.$$

Since g is continuous at z_0 , there exists $\delta > 0$ such that

$$|z - z_0| < \delta \implies |g(z) - g(z_0)| < \delta_1,$$

that is, $|g(z) - z_1| < \delta_1$. Hence

$$|f(g(z)) - f(z_1)| < \varepsilon,$$

or $|f(g(z)) - f(g(z_0))| < \varepsilon$. Consequently $h(z) = f(g(z))$ is continuous at z_0 .

242. Let a point x_0 and a real number $\varepsilon > 0$ be given. For $f(x) = \sin x$,

$$\begin{aligned} |f(x) - f(x_0)| &= |\sin x - \sin x_0| \\ &= 2 \left| \sin \frac{x - x_0}{2} \right| \left| \cos \frac{x + x_0}{2} \right| \\ &\leq 2 \left| \sin \frac{x - x_0}{2} \right|. \end{aligned}$$

For $f(x) = \cos x$,

$$\begin{aligned} |f(x) - f(x_0)| &= |\cos x - \cos x_0| \\ &= 2 \left| \sin \frac{x - x_0}{2} \right| \left| \sin \frac{x + x_0}{2} \right| \\ &\leq 2 \left| \sin \frac{x - x_0}{2} \right|. \end{aligned}$$

Thus in both cases

$$|f(x) - f(x_0)| \leq 2 \left| \sin \frac{x - x_0}{2} \right|.$$

Choose $\delta > 0$ so that $|x - x_0| < \delta$ implies

$$\left| \sin \frac{x - x_0}{2} \right| < \frac{\varepsilon}{2}.$$

If $\varepsilon/2 > 1$, this inequality holds for all x , and δ can be chosen arbitrarily. If $\varepsilon/2 \leq 1$, consider in the (x, y) -plane a circle of radius 1 with centre at the origin and the lines $y = \varepsilon/2$ and $y = -\varepsilon/2$ (Figure 49). For the angles α and β shown in the figure,

$$\sin \alpha = \frac{\varepsilon}{2}, \quad \sin \beta = -\frac{\varepsilon}{2},$$

so $\beta = -\alpha$.

Choose $\delta = 2\alpha$. Then $|x - x_0| < \delta$ implies

$$-\alpha < \frac{x - x_0}{2} < \alpha,$$

and therefore

$$-\frac{\varepsilon}{2} < \sin \frac{x - x_0}{2} < \frac{\varepsilon}{2}.$$

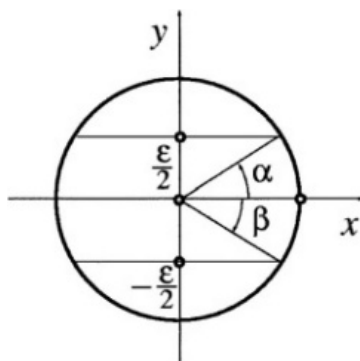


Figure 49

Thus

$$|f(x) - f(x_0)| \leq 2 \left| \sin \frac{x - x_0}{2} \right| < \varepsilon.$$

Consequently $f(x) = \sin x$ and $f(x) = \cos x$ are continuous for all real values of x .

243. Let a point x_0 and an arbitrary positive real number ε be given. We must choose $\delta > 0$ such that every x satisfying $|x - x_0| < \delta$ and $x \geq 0$ also satisfies

$$|\sqrt[n]{x} - \sqrt[n]{x_0}| < \varepsilon. \quad (3.6)$$

This is equivalent to

$$\sqrt[n]{x_0} - \varepsilon < \sqrt[n]{x} < \sqrt[n]{x_0} + \varepsilon. \quad (3.7)$$

Since $\sqrt[n]{x}$ is strictly increasing for $x \geq 0$, if $\sqrt[n]{x_0} - \varepsilon \geq 0$, inequalities (3.7) are equivalent to

$$(\sqrt[n]{x_0} - \varepsilon)^n < x < (\sqrt[n]{x_0} + \varepsilon)^n. \quad (3.8)$$

Equivalently,

$$(\sqrt[n]{x_0} - \varepsilon)^n - x_0 < x - x_0 < (\sqrt[n]{x_0} + \varepsilon)^n - x_0.$$

Taking δ to be the smaller of

$$x_0 - (\sqrt[n]{x_0} - \varepsilon)^n \quad \text{and} \quad (\sqrt[n]{x_0} + \varepsilon)^n - x_0$$

makes (3.8), and hence (3.6), follow from $|x - x_0| < \delta$.

If $\sqrt[n]{x_0} - \varepsilon < 0$, the left-hand inequality in (3.7) is automatic for $x \geq 0$, and (3.7) is equivalent to

$$x < (\sqrt[n]{x_0} + \varepsilon)^n. \quad (3.9)$$

In this case it suffices to take

$$\delta = (\sqrt[n]{x_0} + \varepsilon)^n - x_0.$$

244. Answer. a) See Figure 50. **Hint.** $y(t) \equiv 0$; b), c) see Figure 50. **Hint.** $x(t) \equiv 0$; d) see Figure 50: $x(t) = t$, $y(t) = -t$, $y = -x$; e) see Figure 51: $x(t) = t^2$, $y(t) = t$, $x = y^2$; f), g) one turn in case (f) and two turns in case (g), both counterclockwise, along a circle of radius R , with initial point $z = R + 0i$ (Figure 52). **Hint.** In case (f), $|z(t)| = R$, $\arg z(t) = 2\pi t$, and in case (g), $\arg z(t) = 4\pi t$; h) the semicircle of radius R (Figure 53); i) see Figure 54.

245. See Figure 55. By similarity,

$$t = \frac{x - a_0}{a_1 - a_0} = \frac{y - b_0}{b_1 - b_0}.$$

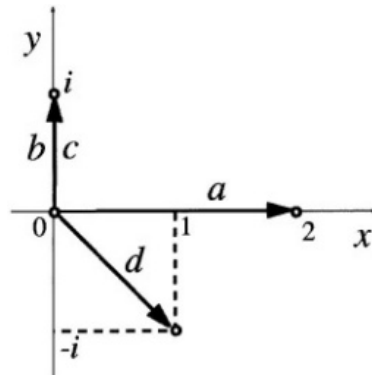


Figure 50

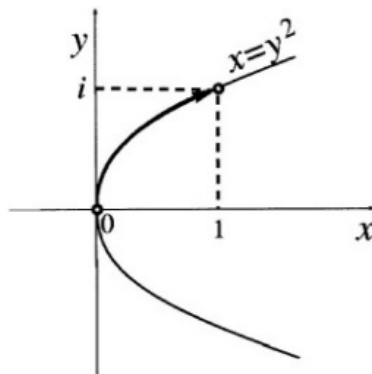


Figure 51

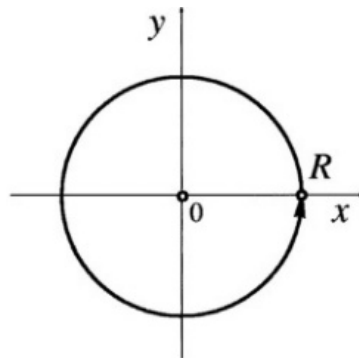


Figure 52

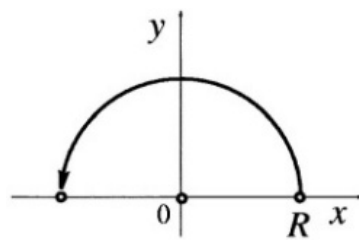


Figure 53

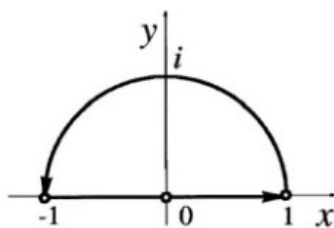


Figure 54

When z moves along the segment from $z_0 = a_0 + b_0i$ to $z_1 = a_1 + b_1i$, the parameter varies from 0 to 1. Thus one obtains

$$x = a_0 + (a_1 - a_0)t, \quad y = b_0 + (b_1 - b_0)t,$$

and

$$z(t) = (a_0 + (a_1 - a_0)t) + i(b_0 + (b_1 - b_0)t).$$

It is easy to verify that this formula describes the segment for any positions of z_0 and z_1 , including $a_0 = a_1$ or $b_0 = b_1$.

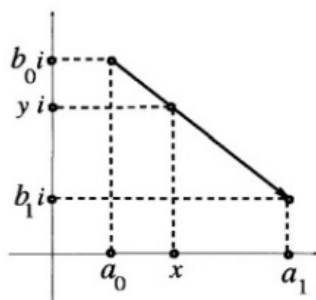


Figure 55

246. Since $z_A + z_{\overrightarrow{AB}} = z_B$ (cf. 221), case (a) corresponds to a displacement of the curve by the vector represented by z_0 (Figure 56).

If $z_2 = z_0 z_1$, $\arg z_0 = \varphi_0$, and $\arg z_1 = \varphi_1$, then (cf. 226)

$$|z_2| = |z_0| |z_1|, \quad \arg z_2 = \varphi_0 + \varphi_1.$$

Therefore case (b) corresponds to a dilation of C_1 by a factor a , fixing the origin (Figure 57); case (c) corresponds to a rotation of C_1 through the angle $\varphi_0 = \arg z_0$ about the origin (Figure 58); and case (d) corresponds to a dilation by $|z_0|$ together with a rotation through $\varphi_0 = \arg z_0$ (Figure 59).

247. When t varies continuously from 0 to 1, $1 - t$ varies continuously from 1 to 0. Thus $z_2(t) = z_1(1 - t)$ describes the same geometrical curve as C , but with the opposite orientation.

248. When t varies continuously from 0 to $1/2$, $2t$ varies continuously from 0 to 1. When t varies continuously from $1/2$ to 1, $2t - 1$ varies continuously from 0 to 1. Hence the function $z_3(t)$ given in the problem describes the curve obtained by first drawing C_1 and then C_2 . The condition $z_1(1) = z_2(0)$ guarantees continuity of the resulting curve.

249. Answer. $\pi t + 2\pi k$, where $k = 0, \pm 1, \pm 2, \dots$

250. Answer. a) πt ; b) $2\pi + \pi t$; c) $-4\pi + \pi t$; d) $2\pi k + \pi t$.

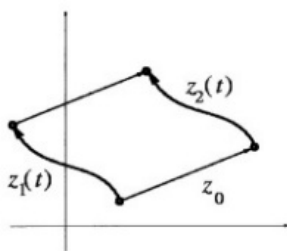


FIGURE 56

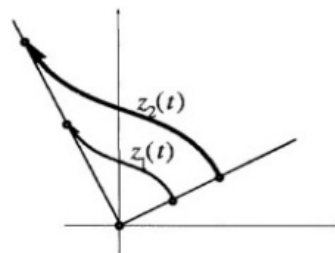
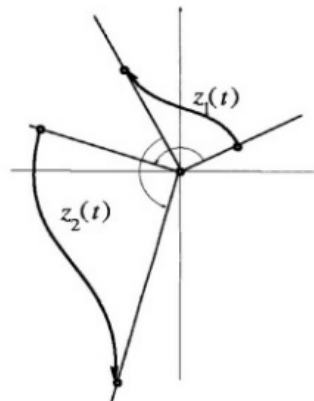
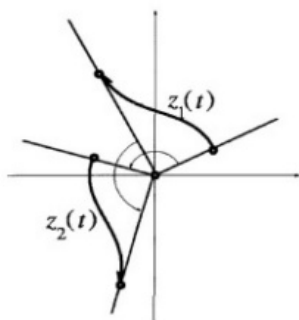


FIGURE 57



Figures 56–59

251. For every t , we have $\varphi(t) - \varphi'(t) = 2\pi k$, where initially the integer k might appear to depend on t . Write

$$\varphi(t) - \varphi'(t) = 2\pi k(t).$$

Then

$$k(t) = \frac{\varphi(t) - \varphi'(t)}{2\pi}.$$

Since φ and φ' are continuous for $0 \leq t \leq 1$, so is $k(t)$ (cf. 239, 240). But $k(t)$ takes only integer values and therefore must be constant. Hence

$$\varphi(t) - \varphi'(t) = 2\pi k$$

for a fixed integer k .

252. Let $\varphi(t)$ and $\varphi'(t)$ describe continuous variations of $\arg z(t)$, with $\varphi(0) = \varphi'(0) = \varphi_0$. By 251,

$$\varphi(t) - \varphi'(t) = 2\pi k$$

for a fixed integer k . But $\varphi(0) - \varphi'(0) = 0$, so $k = 0$. Consequently $\varphi(t) \equiv \varphi'(t)$.

253. Let $\varphi(t)$ and $\varphi'(t)$ describe two continuous variations of $\arg z(t)$. By 251,

$$\varphi(t) - \varphi'(t) = 2\pi k$$

for a fixed integer k . In particular, $\varphi(0) - \varphi'(0) = 2\pi k$. Therefore

$$\varphi(t) - \varphi(0) = \varphi'(t) - \varphi'(0).$$

254. a) One can take $\varphi(t) = \pi t$. **Answer.** $\varphi(1) - \varphi(0) = \pi$ (see Figure 53); b) $\varphi(t) = 2\pi t$. **Answer.** 2π (see Figure 52); c) $\varphi(t) = 4\pi t$. **Answer.** 4π ; d) **Answer.** $\pi/2$ (see Figure 60).

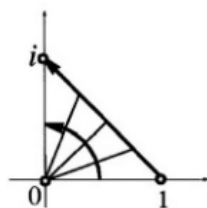


Figure 60

255. Answer. a) $3\pi/2$; b) $-\pi/2$.

256. Answer. a) 1; b) -2 ; c) 2; d) 0.

257. Suppose that on an oriented continuous closed curve C , point A is chosen as the initial point in one case and point B in another (Figure 61). If the variation of the argument along the segment AB , according to the orientation of C , is φ_1 , and that along the segment BA is φ_2 , then the variation along the entire curve is $\varphi_1 + \varphi_2$.

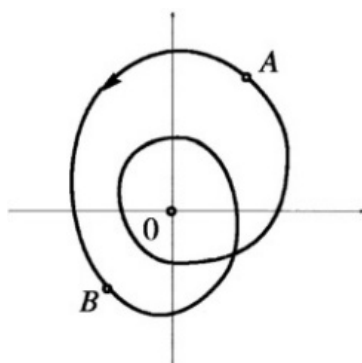


Figure 61

258. a), b), c) If $z_2(t) = z_0 z_1(t)$ and $\varphi_1(t)$ describes the continuous variation of $\arg z_1(t)$, then

$$\varphi_2(t) = \varphi_1(t) + \varphi_0, \quad \varphi_0 = \arg z_0,$$

describes the continuous variation of $\arg z_2(t)$ (cf. 226, 239). It follows that

$$\varphi_2(1) - \varphi_2(0) = \varphi_1(1) - \varphi_1(0).$$

Answer. k times. This result also depends on Problem 246.

d) If $\arg z = \varphi$, then $\arg \bar{z} = -\varphi$ (cf. 218(c)). Therefore, if $\varphi_1(t)$ describes the continuous variation of $\arg z_1(t)$, then $\varphi_2(t) = -\varphi_1(t)$ describes that of $\arg z_2(t)$. Hence

$$\varphi_2(1) - \varphi_2(0) = -(\varphi_1(1) - \varphi_1(0)).$$

Answer. $-k$ times.

259. Answer. a) 1; b) 0; c) 1; d) 2.

260. Solution. If $\varphi_1(t)$ and $\varphi_2(t)$ describe the continuous variations of the argument along C_1 and C_2 , then for a function $\varphi(t)$ describing the continuous variation along C , one can take

$$\varphi(t) = \varphi_1(t) + \varphi_2(t)$$

in case (a), and

$$\varphi(t) = \varphi_1(t) - \varphi_2(t)$$

in case (b) (cf. 226, 239). It follows that

$$\begin{aligned}\varphi(1) - \varphi(0) &= (\varphi_1(1) \pm \varphi_2(1)) - (\varphi_1(0) \pm \varphi_2(0)) \\ &= (\varphi_1(1) - \varphi_1(0)) \pm (\varphi_2(1) - \varphi_2(0)) \\ &= \varphi_1 \pm \varphi_2.\end{aligned}$$

Answer. a) $\varphi_1 + \varphi_2$; b) $\varphi_1 - \varphi_2$.

261. Hint. Let $w_0(t) = [z(t)]^2$ and use de Moivre's formula (cf. 227):

- a) $w_0(t) = R^2(\cos \pi t + i \sin \pi t)$, a semicircle of radius R^2 ;
- b) $w_0(t) = R^2(\cos 2\pi t + i \sin 2\pi t)$, a circle of radius R^2 ;
- c) $w_0(t) = R^2(\cos 4\pi t + i \sin 4\pi t)$, a circle of radius R^2 , covered twice.

262. Hint. Use the result of Problem 260(a). **Answer.** a) 2φ ; b) 3φ ; c) $n\varphi$.

263. Let $z(t)$ be a parametric equation of C , and let $z_1(t) = z(t) - z_0$. By hypothesis the variation of $\arg z_1(t)$ is $2\pi k$. The parametric equation of $f(C)$ is

$$w_0(t) = (z(t) - z_0)^n = [z_1(t)]^n.$$

The variation of the argument of $w_0(t)$ is therefore $2\pi kn$ (cf. 262(c)). Hence $f(C)$ turns around $w = 0$ exactly kn times. **Answer.** kn times.

264. By hypothesis the variations of the arguments of $z(t)$, $z(t) - 1$, $z(t) - i$, and $z(t) + i$ are $2\pi k_1$, $2\pi k_2$, $2\pi k_3$, and $2\pi k_4$, respectively. By Problem 260(a):

- a) $w_0(t) = z(t)(z(t) - 1)$. The variation of its argument is

$$2\pi k_1 + 2\pi k_2 = 2\pi(k_1 + k_2).$$

Answer. The curve $f(C)$ turns around $w = 0$ $k_1 + k_2$ times.

- b) $w_0(t) = (z(t) - i)(z(t) + i)$. **Answer.** $k_3 + k_4$ times.

- c) $w_0(t) = [z(t)]^4[z(t) + i]^4$. **Answer.** $4(k_1 + k_4)$ times.

- d) $w = (z - 1)(z^2 + 1) = (z - 1)(z + i)(z - i)$. **Answer.** $k_2 + k_3 + k_4$ times.

265.

$$\begin{aligned}|a_0 z^n + \cdots + a_{n-1} z| &\leq |a_0 z^n| + \cdots + |a_{n-1} z| && \text{(cf. 223(a))} \\ &= |a_0| |z|^n + \cdots + |a_{n-1}| |z| && \text{(cf. 226)} \\ &= |a_0| R_1^n + \cdots + |a_{n-1}| R_1 \\ &\leq |a_0| R_1 + \cdots + |a_{n-1}| R_1 && \text{(since } R_1 < 1) \\ &\leq n A R_1 && \text{(since } |a_i| \leq A) \\ &< n A \frac{|a_n|}{10nA} = \frac{|a_n|}{10} && \text{(since } R_1 < |a_n|/(10nA)).\end{aligned}$$

266.

$$\begin{aligned}
\left| \frac{a_1}{z} + \cdots + \frac{a_n}{z^n} \right| &\leq \left| \frac{a_1}{z} \right| + \cdots + \left| \frac{a_n}{z^n} \right| && \text{(cf. 223(a))} \\
&= \frac{|a_1|}{|z|} + \cdots + \frac{|a_n|}{|z|^n} && \text{(cf. 226)} \\
&= \frac{|a_1|}{R_2} + \cdots + \frac{|a_n|}{R_2^n} \\
&\leq \frac{|a_1|}{R_2} + \cdots + \frac{|a_n|}{R_2} && \text{(since } R_2 > 1) \\
&\leq \frac{nA}{R_2} && \text{(since } |a_i| \leq A) \\
&< \frac{nA|a_0|}{10nA} = \frac{|a_0|}{10} && \text{(since } R_2 > 10nA/|a_0|).
\end{aligned}$$

267. If $|z| = R_1$, then by 265,

$$|f(z) - a_n| < \frac{|a_n|}{10}.$$

Consequently $f(C_{R_1})$ lies entirely in the disc of radius $|a_n|/10$ centred at $w = a_n$ (Figure 62). It is evident that this curve does not turn around $w = 0$. Therefore $\nu(R_1) = 0$.

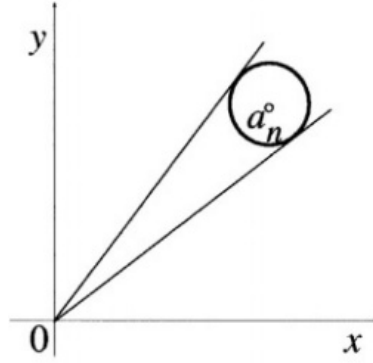


Figure 62

Let us now prove that $\nu(R_2) = n$. First we give a proof that is not strictly exact, but is appealing, called “the lady with her little dog.” From Problem 266, if $|z| = R_2$, then

$$\begin{aligned}
|f(z) - a_0 z^n| &= |a_1 z^{n-1} + \cdots + a_n| \\
&= \left| z^n \left(\frac{a_1}{z} + \cdots + \frac{a_n}{z^n} \right) \right| \\
&= |z|^n \left| \frac{a_1}{z} + \cdots + \frac{a_n}{z^n} \right| \\
&< \frac{|a_0| |z|^n}{10} = \frac{|a_0| R_2^n}{10}.
\end{aligned}$$

Put $R = |a_0| R_2^n$. Thus $|f(z) - a_0 z^n| < R/10$ on $|z| = R_2$. When z covers the circle of radius R_2 once, $w = a_0 z^n$, the “lady,” covers the circle of radius R exactly n times. Since $|f(z) - a_0 z^n| < R/10$, the point $w = f(z)$, the “little dog,” cannot be farther than $R/10$ from the lady. Thus, if the lady turns n times around $w = 0$ along the circle of radius R , her dog is also obliged to turn n times around $w = 0$ (Figure 63). It follows that $\nu(R_2) = n$.

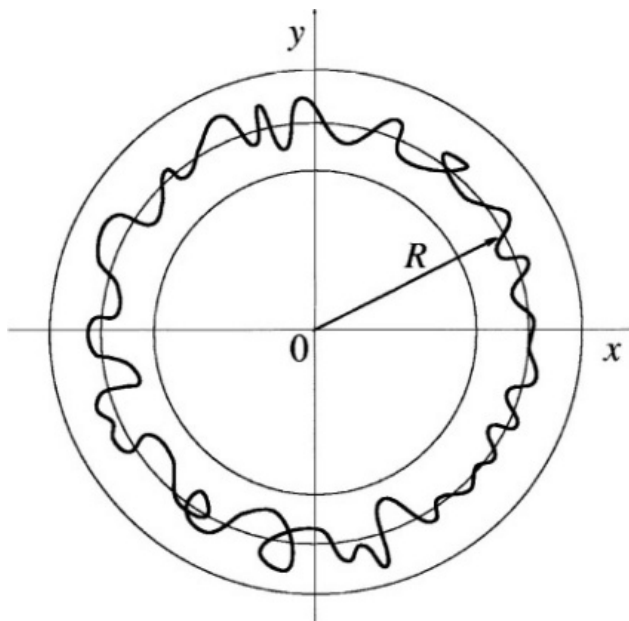


Figure 63

A more exact proof of $\nu(R_2) = n$ is as follows. If $|z| = R_2$, then by 266,

$$\left| \frac{f(z)}{z^n} - a_0 \right| < \frac{|a_0|}{10}.$$

As in the proof that $\nu(R_1) = 0$, when $z(t)$ covers C_{R_2} , the variation of the argument of $f(z)/z^n$ is zero. The variation of the argument of z is 2π , so that of z^n is $2\pi n$ (cf. 262). Since

$$f(z) = \frac{f(z)}{z^n} z^n,$$

the variation of the argument of $f(z)$ is $2\pi n$ (cf. 260). Thus $f(C_{R_2})$ turns n times around $w = 0$, and $\nu(R_2) = n$.

Answer. $\nu(R_1) = 0$ and $\nu(R_2) = n$.

268. Divide the polynomial

$$P(z) = a_0 z^n + \cdots + a_{n-1} z + a_n$$

by $z - z_0$ with remainder, for example by the Euclidean algorithm (cf. §2.1). The remainder is a complex number r , and the quotient is a polynomial $Q(z)$. Thus

$$P(z) = Q(z)(z - z_0) + r.$$

Substituting $z = z_0$, we obtain

$$P(z_0) = Q(z_0)(z_0 - z_0) + r = r.$$

But $P(z_0) = 0$, since z_0 is a root of $P(z) = 0$. Hence $r = 0$, and

$$P(z) = Q(z)(z - z_0).$$

269. Let

$$P(z) = a_0 z^n + \cdots + a_{n-1} z + a_n.$$

By the fundamental theorem of algebra, $P(z) = 0$ has a root z_1 . By Bézout's theorem (cf. 268),

$$P(z) = (z - z_1)Q(z),$$

where $Q(z) = a_0z^{n-1} + b_1z^{n-2} + \cdots + b_{n-1}$. If $n - 1 \geq 1$, then $Q(z) = 0$ has a root z_2 , so

$$Q(z) = (z - z_2)R(z)$$

and

$$P(z) = (z - z_1)(z - z_2)R(z).$$

Continuing this procedure, at the n th step the quotient is a complex constant, necessarily a_0 . This gives the required decomposition.

270. Let $P(z) = a_0z^n + \cdots + a_{n-1}z + a_n$. By 212,

$$\overline{P(z)} = P(\bar{z}).$$

By hypothesis $P(z_0) = 0$. It follows that

$$P(\bar{z}_0) = \overline{P(z_0)} = 0,$$

so $\bar{z}_0$ is also a root of $P(z) = 0$.

271. Let

$$P(z) = a_0z^n + \cdots + a_{n-1}z + a_n,$$

where all a_i are real. Since z_0 is a root, $\bar{z}_0$ is also a root (cf. 270). Because z_0 is not real, $z_0 \neq \bar{z}_0$. Thus the decomposition in 269 contains the factors $z - z_0$ and $z - \bar{z}_0$, and we may write

$$\begin{aligned} P(z) &= (z - z_0)(z - \bar{z}_0)Q(z) \\ &= (z^2 - (z_0 + \bar{z}_0)z + z_0\bar{z}_0)Q(z). \end{aligned}$$

Therefore P is divisible by a quadratic polynomial with real coefficients (cf. the solution of Problem 211).

272. Let $P(z)$ be the given polynomial. If its degree is greater than 2, then by the fundamental theorem of algebra, $P(z) = 0$ has a root z_0 . If z_0 is real, divide $P(z)$ by $z - z_0$, obtaining

$$P(z) = (z - z_0)Q(z)$$

(cf. 268). If z_0 is not real, then $P(z)$ is divisible by a quadratic polynomial with real coefficients (cf. 271).

In either case the quotient, which can be obtained by the Euclidean algorithm (cf. §2.1), has real coefficients. Apply the same procedure to the quotient. The procedure ends when the remaining quotient has degree one or two, giving the required decomposition.

273.

$$\begin{aligned} z^5 - z^4 - 2z^3 + 2z^2 + z - 1 &= (z - 1)(z^4 - 2z^2 + 1) \\ &= (z - 1)(z^2 - 1)^2 \\ &= (z - 1)^3(z + 1)^2. \end{aligned}$$

Answer. 1 is a root of order 3, and -1 is a root of order 2.

274. Compare the coefficients of terms of the same degree in the two sides of the given equalities. Let

$$\begin{aligned} P(z) &= a_0z^n + a_1z^{n-1} + \cdots + a_{n-1}z + a_n, \\ Q(z) &= b_0z^m + b_1z^{m-1} + \cdots + b_{m-1}z + b_m. \end{aligned}$$

Then

$$P'(z) = a_0 n z^{n-1} + a_1(n-1)z^{n-2} + \cdots + a_{n-1},$$

$$Q'(z) = b_0 m z^{m-1} + b_1(m-1)z^{m-2} + \cdots + b_{m-1}.$$

In case (a), for each k , the coefficients of z^k on the two sides are both

$$(a_{n-1-k} + b_{m-1-k})(k+1),$$

and in case (b), the coefficients of z^k on both sides are

$$ca_{n-1-k}(k+1).$$

c) For brevity, use the summation symbol $\sum$. The symbol $\sum_{i=0}^n$ means that the expression following it is evaluated for $i = 0, 1, \dots, n$ and the results are added; $\sum_{i+j=r}$ is interpreted similarly over all i, j such that $i + j = r$. Write

$$P(z) = \sum_{i=0}^n a_i z^{n-i}, \quad Q(z) = \sum_{j=0}^m b_j z^{m-j},$$

and

$$P'(z) = \sum_{i=0}^{n-1} a_i(n-i)z^{n-i-1}, \quad Q'(z) = \sum_{j=0}^{m-1} b_j(m-j)z^{m-j-1}.$$

Let k be an integer with $0 \leq k \leq n + m - 1$. Since

$$(n-i) + (m-j) = k+1 \iff i+j = n+m-k-1,$$

the coefficient of z^{k+1} in $P(z)Q(z)$ is

$$\sum_{i+j=n+m-k-1} a_i b_j.$$

Consequently the coefficient of z^k in $(P(z)Q(z))'$ is

$$\sum_{i+j=n+m-k-1} a_i b_j (k+1).$$

Similarly, the coefficient of z^k in $P'(z)Q(z)$ is

$$\sum_{i+j=n+m-k-1} a_i b_j (n-i),$$

and that in $P(z)Q'(z)$ is

$$\sum_{i+j=n+m-k-1} a_i b_j (m-j).$$

Hence the coefficient of z^k in $P'(z)Q(z) + P(z)Q'(z)$ is

$$\sum_{i+j=n+m-k-1} a_i b_j ((n-i) + (m-j)) = \sum_{i+j=n+m-k-1} a_i b_j (k+1),$$

which is exactly the coefficient of z^k in $(P(z)Q(z))'$.

275. For $n = 1$,

$$((z - z_0)^1)' = 1 = 1 \cdot (z - z_0)^0.$$

For $n = 2$,

$$((z - z_0)^2)' = (z^2 - 2z_0 z + z_0^2)' = 2z - 2z_0 = 2(z - z_0).$$

Suppose the assertion holds for $n = k$, so that

$$((z - z_0)^k)' = k(z - z_0)^{k-1}.$$

Then, using Problem 274(c),

$$\begin{aligned} ((z - z_0)^{k+1})' &= ((z - z_0)^k(z - z_0))' \\ &= ((z - z_0)^k)'(z - z_0) + (z - z_0)^k(z - z_0)' \\ &= k(z - z_0)^{k-1}(z - z_0) + (z - z_0)^k \\ &= (k + 1)(z - z_0)^k. \end{aligned}$$

Thus the assertion holds for every integer $n \geq 1$.

276. By hypothesis,

$$P(z) = (z - z_0)^k Q(z),$$

where $Q(z)$ is not divisible by $z - z_0$. By Problems 274(c) and 275,

$$\begin{aligned} P'(z) &= ((z - z_0)^k)'Q(z) + (z - z_0)^k Q'(z) \\ &= k(z - z_0)^{k-1}Q(z) + (z - z_0)^k Q'(z) \\ &= (z - z_0)^{k-1}(kQ(z) + (z - z_0)Q'(z)). \end{aligned}$$

The polynomial in the last parentheses is not divisible by $z - z_0$, since otherwise $Q(z)$ would be divisible by $z - z_0$. Consequently $P'(z)$ is divisible by $(z - z_0)^{k-1}$, but not by $(z - z_0)^k$.

277. Answer. a) ± 1 ; b) $\pm i$; c) $\pm(\sqrt{2}/2 + i\sqrt{2}/2)$; d) $\pm(\sqrt{6}/2 + i\sqrt{6}/2)$ (here $\sqrt{2}$ and $\sqrt{6}$ denote the positive square roots). **278.** The continuous image under the mapping $w = \sqrt{z}$ of the upper (respectively, lower) semicircle starting at the point $w = 1$ is the arc AB (respectively, the arc AC) (Figure 64). The curve AB ends at the point i , and the curve AC at the point $-i$.

Answer. a) $w(-1) = i$; b) $w(-1) = -i$.

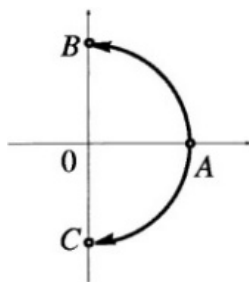


图 3.2

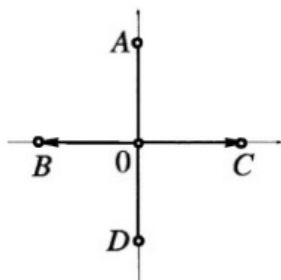


图 3.3

279. For $0 \leq t \leq \frac{1}{2}$, $w_0(t)$ may take two values: $w_0(t) = \pm i\sqrt{1-2t}$ (the value of the square root is considered positive). Also, for $\frac{1}{2} < t \leq 1$, $w_0(t)$ may take two values: $w_0(t) = \pm\sqrt{2t-1}$. For $t = \frac{1}{2}$, $w_0(t)$ takes a unique value: $w_0(\frac{1}{2}) = 0$.

Answer (see Figure 65). a) The continuous images are represented by the broken lines AOB and AOC; b) the continuous images are represented by the broken lines DOB and DOC.

280. Let C_1 be the continuous image of the curve C under the mapping $w = \sqrt{z}$, and let the variation of the argument along the curve C_1 be equal to φ_1 . Thus the curve C is the image of the curve C_1 under the mapping $z = w^2$, and (cf. 262(a)) $\varphi = 2\varphi_1$. Therefore $\varphi_1 = \varphi/2$.

Answer. $\varphi/2$.

281. Let $w = \sqrt{z}$ and let $w_0(t)$ be the continuous image of the given curve. Since

$$i = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2},$$

$\sqrt{i}$ (and $w_0(1)$) may take two values:

$$\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2}$$

and

$$\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} = -\frac{\sqrt{2}}{2} - i \frac{\sqrt{2}}{2}.$$

In accord with the condition $w_0(0) = -1$, one may take $\arg w_0(0) = \pi$.

a) The variation of the argument along the given segment is, evidently, $\pi/2$. Consequently (cf. 280), the variation of the argument along the curve $w_0(t)$ is equal to $\pi/4$, and the argument of $w_0(1)$ is equal to $\pi + \pi/4 = 5\pi/4$.

Answer. $\sqrt{i} = -\frac{\sqrt{2}}{2} - i \frac{\sqrt{2}}{2}$.

b) $z(t) = \cos(-3\pi t/2) + i \sin(-3\pi t/2)$. In order for $\arg z(t)$ to vary continuously, one may choose $\varphi(t) = -3\pi t/2$. The variation of the argument along the curve is thus $\varphi(1) - \varphi(0) = -3\pi/2$. Hence the variation of the argument along the curve $w_0(t)$ (cf. 280) is equal to $-3\pi/4$, and the argument of $w_0(1)$ is equal to $\pi - 3\pi/4 = \pi/4$.

Answer. $\sqrt{i} = \frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2}$.

c) $\varphi(t) = 5\pi t/2$. The variation of the argument along the given curve is $\varphi(1) - \varphi(0) = 5\pi/2$. The variation of the argument along the curve $w_0(t)$ is equal to $5\pi/4$. The argument of $w_0(1)$ is equal to $\pi + 5\pi/4 = 9\pi/4$.

Answer. $\sqrt{i} = \frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2}$.

282. Let $w = \sqrt{z}$, and let $w_0(t)$ be the continuous image of the given curve with $w_0(0) = 1$. We have to determine $w_0(1)$. We may take $\arg w_0(0) = 0$.

a) The variation of $\arg z(t)$ is equal to 2π . Therefore the variation of $\arg w_0(t)$ is equal to π (cf. 280), and $\arg w_0(1) = 0 + \pi = \pi$.

Answer. $\sqrt{1} = -1$.

b) $z(t) = \cos(-4\pi t) + i \sin(-4\pi t)$. The variation of $\arg z(t)$ is equal to -4π . The variation of $\arg w_0(t)$ is equal to -2π , and $\arg w_0(1) = 0 - 2\pi = -2\pi$.

Answer. $\sqrt{1} = 1$.

c) The given curve is a circle of unit radius whose centre is moved to the point $z = 2$ (cf. 246(a)). This curve does not turn at all around the point $z = 0$, and therefore the variation of $\arg z(t)$ vanishes. It follows that the variation of $\arg w_0(t)$ is also equal to zero.

Answer. $\sqrt{1} = 1$.

283. Let $w_0(t)$ be the continuous image of the curve C under the mapping $w = \sqrt{z}$. Since $z(1) = z(0)$, either $w_0(1) = w_0(0)$ or $w_0(1) = -w_0(0)$. In order to have $w_0(1) = w_0(0)$, it is necessary and sufficient that the variation of $\arg w_0(t)$ be equal to $2\pi k$, where k is any integer. To obtain this, the variation of $\arg z(t)$ must be equal to $4\pi k$ (cf. 280), i.e., the curve C must turn $2k$ times around the point $z = 0$.

284. Let L_1 and L_2 be the continuous images of the curves C_1 and C_2 under the mapping $w = \sqrt{z}$. If the curves L_1 and L_2 start from the same point w_0 (Figure 66), then the curve $L_1^{-1}L_2$ is a continuous image of the curve $C_1^{-1}C_2$. The ends of the curve $L_1^{-1}L_2$ (points A and B) coincide if and only if the curve $C_1^{-1}C_2$ turns around the point $z = 0$ an even number of times (cf. 283).

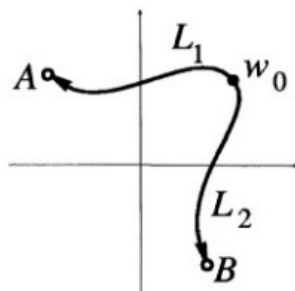


图 3.4

285. Let C be a curve not traversing the cut and joining the points z_0 and z_1 . Suppose that, by choosing two distinct values at the point z_0 and defining the function $\sqrt{z}$ by continuity along the curve C , we obtain the same value at the point z_1 . Consider the curve C^{-1} , i.e., the curve C oriented in the opposite direction. We obtain that the value at the initial point of C^{-1} is the same in both cases, but the values at the final point z_0 , defined by continuity, are different. This is impossible by virtue of the uniqueness of the continuous image, because the curve C does not pass through the point $z = 0$. It follows that the claim ${}_1\sqrt{z} = {}_2\sqrt{z}$ is false.

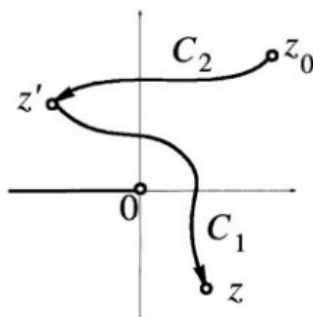


图 3.5

286. Let z be an arbitrary point outside the cut, and let C_1 be a continuous curve starting from z' and ending at z without crossing the cut. Let us draw another curve C_2 , not crossing the cut, going from the point z_0 to the point z' (Figure 67). By hypothesis we have chosen the value $w' = {}_1\sqrt{z'}$. This means that if we choose $\sqrt{z_0} = w'_0$ and

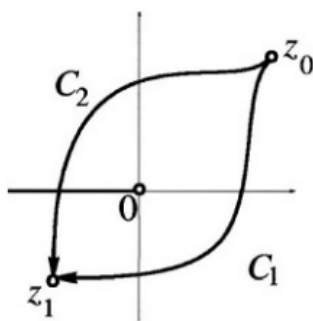


图 3.6

define $\sqrt{z'}$ by continuity along the curve C_2 , we obtain exactly w' . Thus the value of $\sqrt{z}$, defined by continuity along the curve C_1 according to the condition $\sqrt{z'} = w'$, coincides with the value of $\sqrt{z}$ defined by continuity along the curve C_2C_1 according to the condition $\sqrt{z_0} = w'_0$. Hence the value of $\sqrt{z}$ for every z outside the cut is equal to ${}_1\sqrt{z}$.

287. Let C_1 be a continuous curve joining the points z_0 and z_1 and not crossing the cut (Figure 68). Let $w'_1 = \sqrt{z_1}$ be the value of the function $\sqrt{z}$ defined by continuity along the curve C_1 according to the condition $\sqrt{z_0} = w_0$. Since the curve C_1 does not pass through the cut, the values w_0 and w'_1 correspond to the same branch of the function $\sqrt{z}$. The curve $C_1^{-1}C_2$ turns once around the point $z = 0$. The values w_1 and w'_1 are thus different (cf. 283). Since w_0 and w'_1 correspond to the same branch of the function $\sqrt{z}$, w_0 and w_1 correspond to different branches.

288. If $z_0 \neq 0$, then the circle with centre at the point z_0 and with a sufficiently small radius does not turn at all around the point $z = 0$. Therefore the variation of $\arg z(t)$ vanishes, and consequently the variation of $\arg w_0(t)$ is zero, i.e., the value of $\sqrt{z}$ does not change. The variation of $\arg z(t)$ along a circle with centre at the point $z = 0$ is equal to 2π . In this case the variation of $\arg w_0(t)$ is equal to π . The value of $\sqrt{z}$ during a turn around the point $z = 0$ thus changes into the opposite value.

289. The curve $z(t)$ is the image of the curve $w_0(t)$ under the mapping $z(w) = w^3$. Consequently, if φ_1 is the variation of the argument along the curve $w_0(t)$, then $\varphi = 3\varphi_1$ (cf. 262(b)), from which $\varphi_1 = \varphi/3$.

290. If $z_0 \neq 0$, then the circle with centre at the point z_0 and with a sufficiently small radius does not turn at all around the point $z = 0$. Consequently the variation of $\arg z(t)$ along this circle vanishes. Thus the variation of $\arg w_0(t)$ also vanishes (cf. 289), i.e., the value of the function $w = \sqrt[3]{z}$ does not vary. This means that none of the points $z \neq 0$ is a branch point.

The variation of $\arg z(t)$ along a circle with centre at the point $z = 0$ is equal to 2π . Thus the variation of $\arg w_0(t)$ is equal to $2\pi/3$. After a simple turn around the point $z = 0$, the value of the function $w = \sqrt[3]{z}$ is multiplied by

$$\epsilon_3 = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3},$$

i.e., the point $z = 0$ is a branch point of the function $\sqrt[3]{z}$.

Answer. $z = 0$.

291. Let $z(t)$ be a continuous curve, not crossing the cut and joining the point $z = 1$ to the given point. Let $w_0(t)$ be the continuous image of this curve under the mapping $w = \sqrt[3]{z}$, and let $\arg w_0 = \varphi_0$ be chosen. If

the variation of $\arg z(t)$ is equal to φ , then the variation of $\arg w_0(t)$ is equal to $\varphi/3$ (cf. 289). Consequently $\arg w_0(1) = \varphi_0 + \varphi/3$. One may choose $\varphi_0 = 0$ for the branch $f_1(z)$, $\varphi_0 = 2\pi/3$ for the branch $f_2(z)$, and $\varphi_0 = -2\pi/3$ for the branch $f_3(z)$.

a) **Answer.**

$$f_1(i) = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} = \frac{\sqrt{3}}{2} + \frac{i}{2}.$$

b) $\varphi_0 = 2\pi/3$, $\varphi = \pi/2$, and $\arg w_0(1) = 5\pi/6$.

Answer.

$$f_2(i) = \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} = -\frac{\sqrt{3}}{2} + \frac{i}{2}.$$

c) $\varphi_0 = 0$, $\varphi = 0$, and $\arg w_0(1) = 0$.

Answer. $f_1(8) = 2$.

d) $\varphi_0 = -2\pi/3$, $\varphi = 0$, and $\arg w_0(1) = -2\pi/3$.

Answer.

$$f_3(8) = 2 \left(\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3} \right) = -1 - i\sqrt{3}.$$

e) $\varphi_0 = -2\pi/3$, $\varphi = -\pi/2$, and $\arg w_0(1) = -5\pi/6$.

Answer.

$$f_3(-i) = \cos \frac{5\pi}{6} - i \sin \frac{5\pi}{6} = -\frac{\sqrt{3}}{2} - \frac{i}{2}.$$

292. As for the function $w = \sqrt[3]{z}$, one proves that after making the cut from the point $z = 0$ to infinity, for example along the negative side of the real axis, the function $\sqrt[3]{z}$ is decomposed into three single-valued continuous branches. During a simple counterclockwise turn around the point $z = 0$, $\arg w$ varies by $2\pi/3$, during a double turn by $4\pi/3$, and only after a triple turn around the point $z = 0$ does the value of the function $\sqrt[3]{z}$ return to its initial value. The scheme of the Riemann surface of the function $\sqrt[3]{z}$ thus has the form shown in Figure 69. The Riemann surface is represented in Figure 70.²

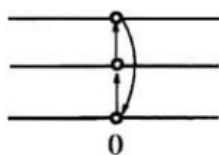


图 3.7

293. Suppose first that the curve C does not pass through the point $z = 0$. Let $\varphi(t)$ be a function which describes the continuous variation of $\arg z(t)$ (cf. Theorem 6, §2.7), and let $r(t) = |z(t)|$. Thus $\varphi(t)$ and $r(t)$ are continuous functions and

$$z(t) = r(t)(\cos \varphi(t) + i \sin \varphi(t)).$$

Let $\rho(t)$ be the positive real value of $\sqrt[n]{r(t)}$. Thus $\rho(t)$ is a continuous function (cf. 243), and the n continuous curves with parametric equations

$$w_k(t) = \rho(t) \left(\cos \left(\frac{\varphi(t)}{n} + \frac{2\pi k}{n} \right) + i \sin \left(\frac{\varphi(t)}{n} + \frac{2\pi k}{n} \right) \right), \quad k = 0, 1, \dots, n-1,$$

are the continuous images of the curve $z(t)$ under the mapping $w(z) = \sqrt[n]{z}$ (cf. 229). Since $w_k(0)$ on these

curves takes all values of $\sqrt[n]{z(0)}$, one of these curves begins at the point w_0 .

If the curve C passes through the point $z = 0$, then the points at which $z(t) = 0$ divide the curve C into segments. In this case we have, as before, a continuous image for every segment of the curve, and we take as the initial segment the image which starts from the point w_0 . If $z(t) = 0$, then $w(z(t)) = 0$ also. Hence the images obtained can be joined into one unique continuous curve, which is the required curve.

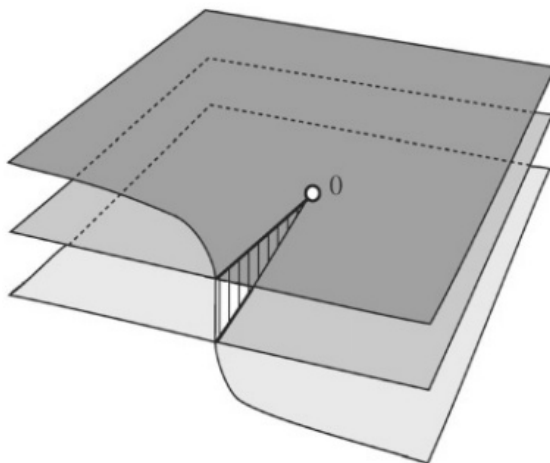


图 3.8

294. See solutions 280 and 289.

Answer. φ/n .

295. See solution 290.

Answer. $z = 0$.

296. $w_0(t)$, for every t , is one of the values of $\sqrt[n]{z(t)}$. All the values of $\sqrt[n]{z(t)}$ for a given t are

$$w_0(t), \quad w_0(t)\epsilon_n, \quad w_0(t)\epsilon_n^2, \quad \dots, \quad w_0(t)\epsilon_n^{n-1}$$

(cf. 232). Since the value of $\sqrt[n]{z}$ at the initial point of the curve $z(t)$ can be chosen in n different ways, we have exactly n continuous curves, images of the curve $z(t)$ under the mapping $w = \sqrt[n]{z}$ (uniqueness may be lost only at the point $z = 0$, but the curve $z(t)$ does not pass through this point).

Answer. The n continuous images are the curves

$$w_0(t), \quad w_1(t) = w_0(t)\epsilon_n, \quad w_2(t) = w_0(t)\epsilon_n^2, \quad \dots, \quad w_{n-1}(t) = w_0(t)\epsilon_n^{n-1}.$$

297. Let $z(t)$ be an arbitrary curve joining the point 1 to an arbitrary point without crossing the cut. From the solution of Problem 296, we obtain that if the value of the function at the initial point of this curve is multiplied by ϵ_n^k , then the value at the final point, defined by continuity, is multiplied by ϵ_n^k . Hence $f_i(z) = f_0(z)\epsilon_n^i$.

Answer. $f_i(z) = f_0(z)\epsilon_n^i$.

298. Solving Problem 297, we found that the n branches of the function $\sqrt[n]{z}$ are related by

$$f_i(z) = f_0(z)\epsilon_n^i, \quad i = 0, 1, \dots, n-1.$$

The unique branch point of the function $\sqrt[n]{z}$ is the point $z = 0$. During a simple turn around this point, the argument of the function $\sqrt[n]{z}$ changes by $2\pi/n$ (cf. 294), i.e., the value of the function is multiplied by ϵ_n .

Consequently the scheme of the Riemann surface of the function $\sqrt[n]{z}$ has the form shown in Figure 71 (the Riemann surface for $n = 6$ is shown in Figure 119).

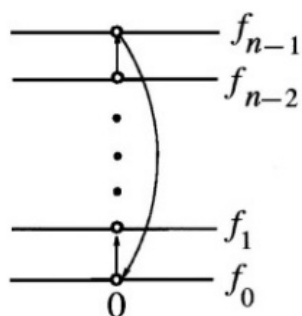


图 3.9

299. During a turn around the point $z = 1$, $\arg(z - 1)$ varies by 2π and it does not change during any turn around the other points (along sufficiently small circles). During a turn around the point $z = 1$, $\arg \sqrt{z - 1}$ thus changes by π , and it remains constant during any turn around the other points. Consequently the sole branch point is $z = 1$, turning around which the value of the function $\sqrt{z - 1}$ is multiplied by -1 . As for the function $\sqrt{z}$, one proves that after making an arbitrary cut from the point $z = 1$ to infinity, the image plane is decomposed into two single-valued continuous branches of the function $\sqrt{z - 1}$. The scheme of the Riemann surface of the function $\sqrt{z - 1}$ is shown in Figure 72.

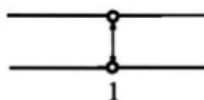


图 3.10

300. See solution 299. The sole branch point is the point $-i$ (because $z + i = z - (-i)$), turning around which the function $\sqrt[n]{z + i}$ is multiplied by ϵ_n . The scheme of the Riemann surface of the function $\sqrt[n]{z + i}$ is shown in Figure 73.

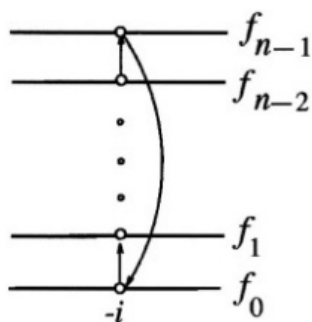


图 3.11

301. *Hint.* Consider the mapping $w = \sqrt[n]{f(z)}$ as the composition of two mappings: $\tau = f(z)$ and $w = \sqrt[n]{\tau}$ (cf. 293).

302. The mapping $w(z) = \sqrt[n]{f(z)}$ can be regarded as the composition of two mappings:

$$\tau = f(z) \quad \text{and} \quad w = \sqrt[n]{\tau}.$$

If C is a continuous curve in the z -plane, then in the τ -plane there is only one image $C' = f(C)$ of this curve.

Since $f(z)$ is a continuous function, C' is a continuous curve. If the curve C'' with equation $w_0(t)$ is one of the continuous images of the curve C' under the mapping $w(\tau) = \sqrt[n]{\tau}$, then the curves with equations

$$w_i(t) = w_0(t)\epsilon_n^i, \quad i = 1, 2, \dots, n-1,$$

are also continuous images of the curve C' under the mapping $w(\tau) = \sqrt[n]{\tau}$ (cf. 296), and consequently they are continuous images of the curve C under the mapping $w(z) = \sqrt[n]{f(z)}$. Thus, if the value of the function $w(z) = \sqrt[n]{f(z)}$ at the initial point is multiplied by ϵ_n^i , the value at the final point of the curve C , defined by continuity, is multiplied by ϵ_n^i . Therefore, if $w_0(z)$ is a continuous single-valued branch of the function $\sqrt[n]{f(z)}$, then all continuous single-valued branches are obtained by multiplying $w_0(z)$ by ϵ_n^i ($i = 1, 2, \dots, n-1$).

Answer.

$$w_0(z), \quad w_1(z) = w_0(z)\epsilon_n, \quad w_2(z) = w_0(z)\epsilon_n^2, \quad \dots, \quad w_{n-1}(z) = w_0(z)\epsilon_n^{n-1}.$$

303. a) During a turn around the point $z = 0$ or $z = i$, $\arg z(z-i)$ varies by 2π (cf. 260), and $\arg \sqrt{z(z-i)}$ varies by π (cf. 280), i.e., the value of the function $\sqrt{z(z-i)}$ is multiplied by -1 . To separate the single-valued continuous branches of the function $\sqrt{z(z-i)}$, it suffices to make two cuts, respectively from the point $z = 0$ and from the point $z = i$ to infinity (the proof is the same as for the function $w = \sqrt{z}$). The scheme of the function $\sqrt{z(z-i)}$ is shown in Figure 74 (the Riemann surface is shown in Figure 120a).

b) See Figure 75 (the Riemann surface is shown in Figure 120b). *Hint.* $z^2 + 1 = (z-i)(z+i)$.

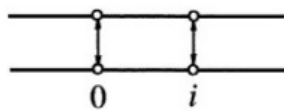


图 3.12

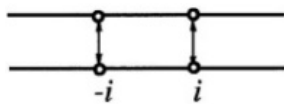


图 3.13

304. See 303.

a) Since $z^2 - 1 = (z-1)(z+1)$, during a turn around each of the points $z = 1$ and $z = -1$, the value of the function $\sqrt[3]{z^2 - 1}$ is multiplied by $\epsilon_3 = \cos(2\pi/3) + i\sin(2\pi/3)$. The scheme sought is shown in Figure 76 (the Riemann surface is shown in Figure 121).

b) After a turn around the point $z = 0$, the value of the function $\sqrt[3]{(z-1)^2 z}$ is multiplied by ϵ_3 . During a turn around the point $z = 1$, $\arg((z-1)^2 z)$ varies by 4π and $\arg \sqrt[3]{(z-1)^2 z}$ varies by $4\pi/3$, i.e., the value of the function $\sqrt[3]{(z-1)^2 z}$ is multiplied by ϵ_3^2 . The scheme sought is shown in Figure 77 (the Riemann surface is shown in Figure 122).

c) See Figure 78. *Hint.* $z^2 + 1 = (z-i)(z+i)$.

305. The single-valued continuous branches of the function $\sqrt{z^2}$, defined on the entire z -plane, are $w_0(z) = z$ and $w_1(z) = -z$. When one passes through the point $z = 0$, $\arg z^2$ changes by 4π and $\arg \sqrt{z^2}$ by 2π , i.e., the value of the function $\sqrt{z^2}$ does not change. The scheme sought consists of two disjoint sheets.

306. The problem is solved in the same way as Problem 304.

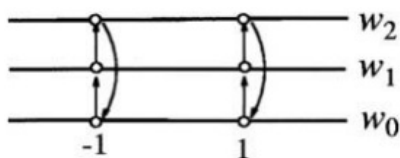


图 3.14

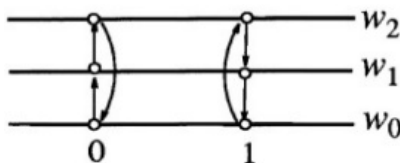


图 3.15

a) See Figure 79. *Hint.* $z^2 + 2 = (z - i\sqrt{2})(z + i\sqrt{2})$.

b) See Figure 80 (the Riemann surface is shown in Figure 123).

c) See Figure 81.

d) See Figure 82 (the Riemann surface is shown in Figure 124). *Hint.* $(z^2 - 1)^3(z + 1)^3 = (z - 1)^3(z + 1)^6$.

e) See Figure 83 (the Riemann surface is shown in Figure 125). *Hint.* $z^3 - 1 = (z - 1)(z - \epsilon_3)(z - \epsilon_3^2)$.

307. See Figure 84. *Hint.* During a turn around the point $z = 0$, $\arg z$ varies by 2π , $\arg(1/z)$ by -2π (cf. 260(b)), and $\arg \sqrt{1/z}$ by $-\pi$, i.e., the value of the function $\sqrt{1/z}$ is multiplied by -1 .

308. a) See Figure 85.

b) During a turn around the point $z = 1$, $\arg((z-1)/(z+1))$ varies by 2π , and during a turn around the point $z = -1$ by -2π (cf. 260). Consequently, around the point $z = 1$, the value of the function $\sqrt[3]{(z-1)/(z+1)}$ is multiplied by ϵ_3 , and around the point $z = -1$ by $\epsilon_3^{-1} = \epsilon_3^2$. The required scheme is shown in Figure 86.

c) See Figure 87 (the Riemann surface is shown in Figure 126).

309. Let a value $w_0 = w(z_0)$ be chosen at the point z_0 , and let z_1 be another point. If C_1 and C_2 are two arbitrary continuous curves joining z_0 and z_1 without crossing the cuts (Figure 88), then evidently the curve C_1 can be continuously deformed into the curve C_2 without passing through the branch points. Since the function $w(z)$ possesses the monodromy property, the values of $w(z_1)$ defined by continuity along the curves C_1 and C_2 coincide. Consequently the value $w(z_1)$ is uniquely defined by continuity along an arbitrary curve joining z_0 and z_1 without crossing the cut.

310. See Figure 89. Suppose that on moving along AB one moves from the i th branch to the j th branch. We want to know at which branch we arrive, starting from the i th branch and traversing the cut along CD.

Since the function $w(z)$ has a finite number of branch points, one can choose the curves AB and CD sufficiently short and the curves CA and BD sufficiently close to the cut so that there are no branch points of the function $w(z)$ in the region bounded by the curve CABDC. In this case one can evidently transform the

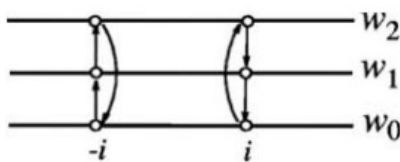


图 3.16

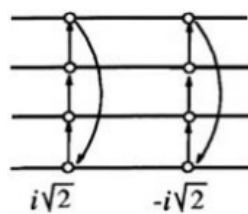


图 3.17

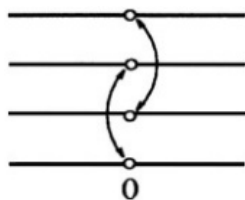


图 3.18

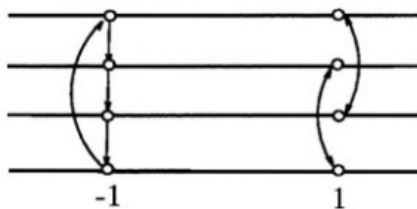


图 3.19

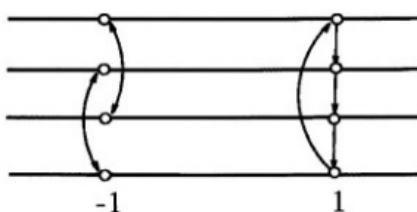


图 3.20

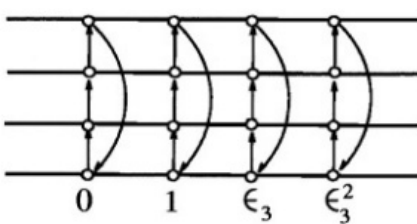


图 3.21

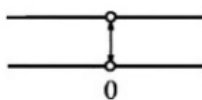


图 3.22

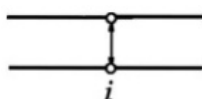


图 3.23

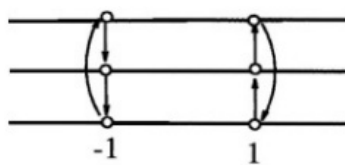


图 3.24

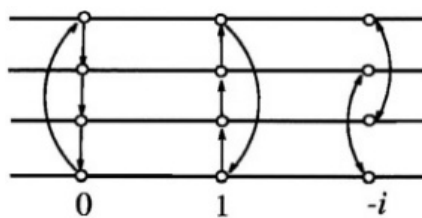


图 3.25

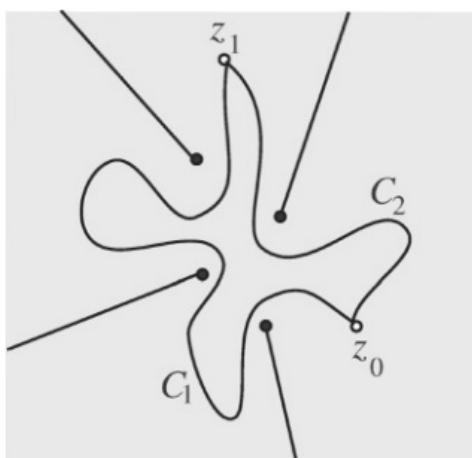


图 3.26

curve CABD into the curve CD without passing through any branch point. Since the function $w(z)$ possesses the monodromy property, the function $w(z)$ at the point D is uniquely defined by continuity along the curves CD and CABD. Starting from the i th branch and covering the curve CABD, moving first on the i th branch, one later passes to the j th branch and moves finally on it. In this way, along the curve CABD and therefore also along the curve CD, one moves from the i th branch to the j th branch, exactly as along the curve AB.

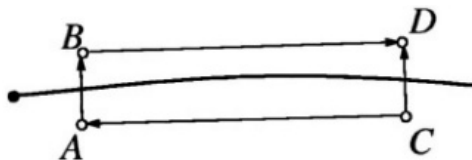


图 3.27

311. a) Hint. $\sqrt[3]{-8} = -2\epsilon_3^i$ ($i = 0, 1, 2$); $\sqrt{2i} = \pm(1 + i)$.

Answer. (Here $\sqrt{3}$ is the positive value of the square root.)

$$-1 + i, \quad -3 - i, \quad 2 + (\sqrt{3} + 1)i, \quad (\sqrt{3} - 1)i, \quad 2 + (1 - \sqrt{3})i, \quad -(\sqrt{3} + 1).$$

b) $\pm(1/2 + i)$, $\pm 1/2$; c) $\pm(1 + i)$, 0; d) $\pm(1 + i)$; e) $4i$, 0.

312. Hint. It suffices to prove that the functions $h(z) = z$ and $h(z) = a$ possess the properties stated by the problem, and that if the functions $f(z)$ and $g(z)$ possess these properties, then the functions

$$f(z) + g(z), \quad f(z) - g(z), \quad f(z)g(z), \quad \frac{f(z)}{g(z)}, \quad [f(z)]^n, \quad \sqrt[n]{f(z)}$$

(where n is an integer) also possess these properties.

Solution. 1) If $h(z) = z$, then $w_0 = h(z_0) = z_0$. The curve sought is the curve with parametric equation $w_0(t) = z(t)$, where $z(t)$ is the parametric equation of the curve C .

2) If $h(z) = a$, then $w_0 = a$ and the required curve is the curve with equation $w_0(t) = a$ (degenerate to a point).

3) Suppose that $h(z) = f(z) + g(z)$ and that the statement of the problem is true for the functions $f(z)$ and $g(z)$. By the definition of the sum of multivalued functions, we have $w_0 = w'_0 + w''_0$, where w'_0 is one of the values of $f(z_0)$ and w''_0 is one of the values of $g(z_0)$. Since the statement holds for the functions $f(z)$ and $g(z)$, there exist two continuous images $C' = f(C)$ and $C'' = g(C)$, starting respectively from the points w'_0 and w''_0 . If $w'(t)$ and $w''(t)$ are the parametric equations of the curves C' and C'' , then the function

$$w_0(t) = w'(t) + w''(t)$$

(which is continuous, being the sum of continuous functions) is the parametric equation of the required curve because

$$w_0(0) = w'(0) + w''(0) = w'_0 + w''_0 = w_0.$$

In an identical way one considers the cases in which $h(z) = f(z) - g(z)$, $h(z) = f(z)g(z)$, and $h(z) = f(z)/g(z)$. In the last case the continuous function sought is

$$w_0(t) = \frac{w'(t)}{w''(t)},$$

because by hypothesis the curve C does not pass through the points at which the function $h(z)$ is not defined, and consequently $w''(t) \neq 0$.

4) Suppose that $h(z) = \sqrt[n]{f(z)}$ and that the statement of the problem is true for $f(z)$. By the definition of the function $\sqrt[n]{f(z)}$, we have $w_0^n = \tau_0$, where τ_0 is one of the values of $f(z_0)$. The mapping $h(z) = \sqrt[n]{f(z)}$ can be considered as the composition of two mappings, $\tau = f(z)$ and $w = \sqrt[n]{\tau}$. Since the statement holds for the function $f(z)$, there exists at least one continuous image C' of the curve C under the mapping $\tau = f(z)$, beginning at the point τ_0 . By virtue of the result of Problem 293, there exists at least one continuous image C'' of the curve C' under the mapping $w = \sqrt[n]{\tau}$, beginning at the point w_0 . The curve C'' is the required curve.

313. At the point z_0 , chosen arbitrarily, the function $h(z)$ takes nm values:

$$h_{i,j}(z_0) = f_i(z_0) + g_j(z_0), \quad i = 1, \dots, n, \quad j = 1, \dots, m.$$

Since the sum of continuous functions is a continuous function, the single-valued continuous branches of the function $h(z)$ are the following nm functions:

$$h_{i,j}(z) = f_i(z) + g_j(z), \quad i = 1, \dots, n, \quad j = 1, \dots, m.$$

314. a) See Figure 90. *Hint.* Use the schemes of the Riemann surfaces of the functions $\sqrt{z}$ and $\sqrt{z-1}$ (cf. 288, 299).

b) See Figure 91. *Hint.* Cf. 304, 307.

c) See Figure 92. *Hint.* Cf. 288, 292.

d) See Figure 93 (the Riemann surface is shown in Figure 127). *Hint.* Draw first the schemes of the Riemann surfaces of the functions $\sqrt{z^2-1}$ and $\sqrt[4]{z-1}$.

315. Answer. a) Three values: 2, -2, 0.

b) Seven values: 2, 0, $1+i$, $-1+i$, -2 , $1-i$, $-1-i$.

c) Six values: 2, $2\epsilon_3$, $2\epsilon_3^2$, $1+\epsilon_3$, $1+\epsilon_3^2$, -1.

316. a) Let $f_0(z)$ and $f_1(z) = -f_0(z)$ be the single-valued continuous branches of the function $\sqrt{z}$. The scheme of the Riemann surface of the function $h(z) = \sqrt{z} + \sqrt{z}$, built by the formal method, is shown in Figure 94. The branches

$$h_{0,1}(z) = f_0(z) + f_1(z) \equiv 0, \quad h_{1,0}(z) = f_1(z) + f_0(z) \equiv 0$$

coincide. To obtain the correct scheme of the Riemann surface of the function $h(z) = \sqrt{z} + \sqrt{z}$, we therefore have to identify the branches $h_{0,1}$ and $h_{1,0}$. This scheme is shown in Figure 95.

b) Let $f_0(z)$ and $f_1(z) = -f_0(z)$ be the single-valued continuous branches of the function $\sqrt{z}$. Thus $[f_0(z)]^2 = z$ and $[f_0(z)]^4 = z^2$. Consequently $f_0(z)$ is one of the single-valued continuous branches of the function $\sqrt[4]{z^2}$. The branches of this function are

$$g_0(z) = f_0(z), \quad g_1(z) = if_0(z), \quad g_2(z) = -f_0(z), \quad g_3(z) = -if_0(z).$$

The scheme of the Riemann surface of the function $h(z) = \sqrt{z} + \sqrt[4]{z^2}$, built by the formal method, is shown in Figure 96. The correct scheme (Figure 97) is obtained by identifying the coincident branches $h_{0,2}(z) \equiv 0$ and $h_{1,0}(z) \equiv 0$.

c) Let $f_0(z)$ be one of the single-valued continuous branches of the function $\sqrt[3]{z}$. Thus the branches are

$$f_0(z), \quad f_1(z) = f_0(z)\epsilon_3, \quad f_2(z) = f_0(z)\epsilon_3^2.$$

The scheme of the Riemann surface of the function $h(z) = \sqrt[3]{z} + \sqrt[3]{z}$, built by the formal method, is shown in Figure 98. To obtain the correct scheme (Figure 99), one has to identify the following coincident branches:

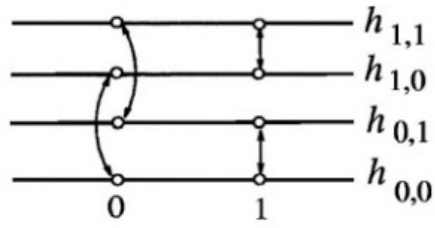


FIGURE 90

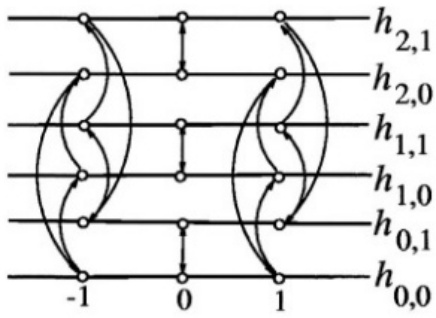


FIGURE 91

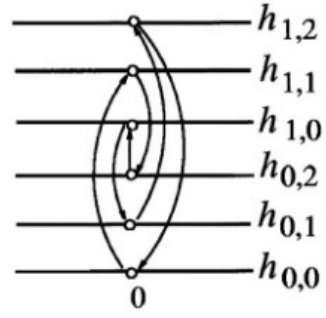


FIGURE 92

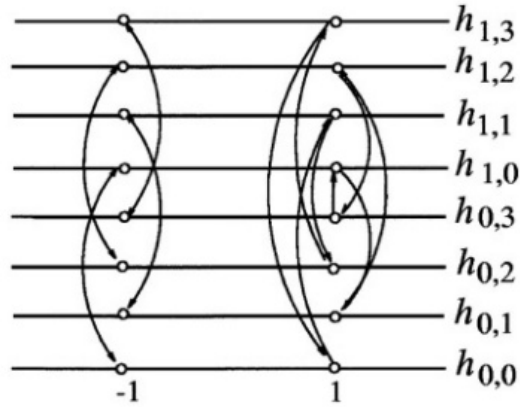


FIGURE 93

图 3.28: Schemes for Problem 314 (Figures 90–93).

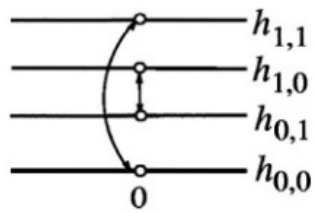


图 3.32

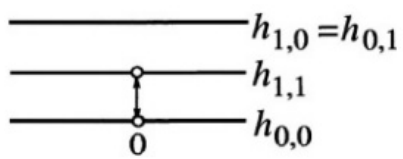


图 3.33

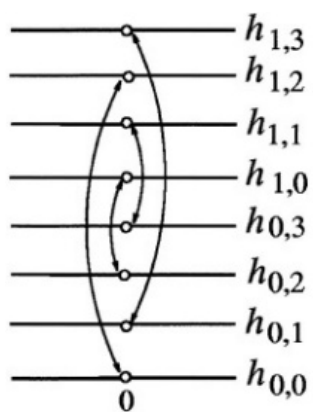


图 3.34

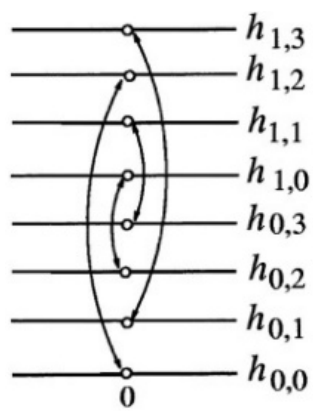


图 3.35

$h_{0,1}(z)$ and $h_{1,0}(z)$, $h_{0,2}(z)$ and $h_{2,0}(z)$, $h_{1,2}(z)$ and $h_{2,1}(z)$.

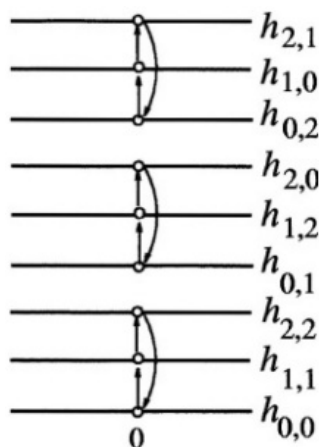


图 3.36

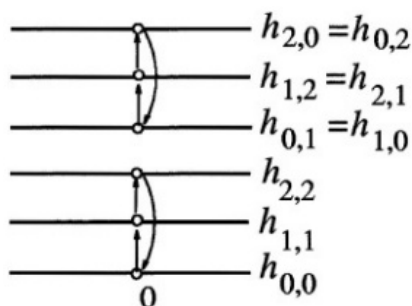


FIGURE 99

图 3.37

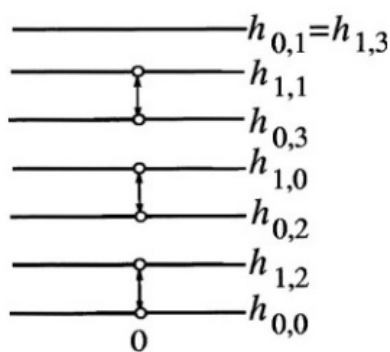


图 3.38

317. a) Let $f_0(z)$ and $f_1(z) = -f_0(z)$ be the single-valued continuous branches of the function $\sqrt{z}$. Thus $[f_0(z)]^2 = z$ and $[f_0(z)]^4 = z^2$. The single-valued continuous branches of the function $g(z) = \sqrt[4]{z^2}$ are therefore

$$g_0(z) = f_0(z), \quad g_1(z) = if_0(z), \quad g_2(z) = -f_0(z), \quad g_3(z) = -if_0(z).$$

One builds the scheme of the Riemann surface of the function $h(z) = i\sqrt{z} - \sqrt[4]{z^2}$ by the formal method and afterwards identifies the coincident branches $h_{0,1}(z) = h_{1,3}(z) \equiv 0$. The remaining branches are distinct; it suffices to calculate their values at the point $z = 1$. The correct scheme is shown in Figure 100.

b) Let $f_0(z)$ and $f_1(z) = -f_0(z)$ be the single-valued continuous branches of the function $\sqrt{z-1}$, and let

$$g_0(z), \quad g_1(z) = ig_0(z), \quad g_2(z) = -g_0(z), \quad g_3(z) = -ig_0(z)$$

be the single-valued continuous branches of the function $g(z) = \sqrt[4]{z}$. We build the scheme of the Riemann surface of the function $h(z) = \sqrt{z-1} \sqrt[4]{z}$ by the formal method (Figure 101) and identify the coincident branches:

$$h_{0,0}(z) = h_{1,2}(z), \quad h_{0,1}(z) = h_{1,3}(z),$$

$$h_{0,2}(z) = h_{1,0}(z), \quad h_{0,3}(z) = h_{1,1}(z).$$

The remaining branches are all distinct; it suffices to calculate their values at the point $z = 2$. The correct scheme is shown in Figure 102.

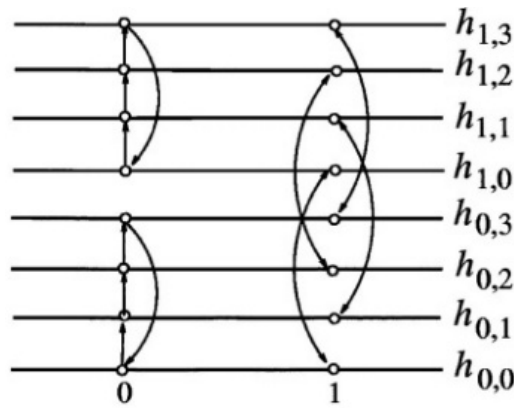


图 3.39

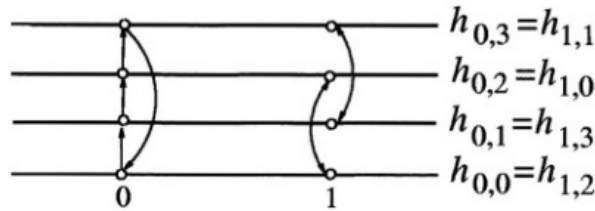


图 3.40

c) See Figure 103. The solution is similar to the solution of case (b).

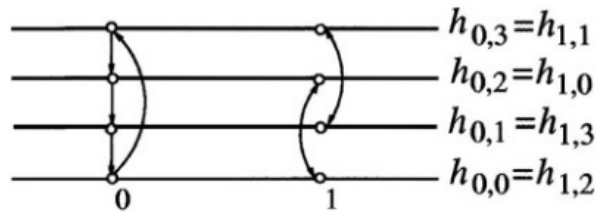


图 3.41

d) The function $f(z) = \sqrt{z} + \sqrt{z}$ has three single-valued continuous branches: $f_0(z) \equiv 0$, $f_1(z)$, and $f_2(z) = -f_1(z)$ (cf. solution 316(a)). The function $g(z) = \sqrt[3]{z(z-1)}$ also has three single-valued continuous branches:

$$g_0(z), \quad g_1(z) = \epsilon_3 g_0(z), \quad g_2(z) = \epsilon_3^2 g_0(z).$$

The branches $h_{0,0}(z)$, $h_{0,1}(z)$, and $h_{0,2}(z)$ coincide:

$$h_{0,0}(z) = h_{0,1}(z) = h_{0,2}(z) \equiv 0.$$

The remaining branches are all distinct; it suffices to calculate their values at the point $z = 2$. The correct scheme is shown in Figure 104 (the Riemann surface is shown in Figure 128).

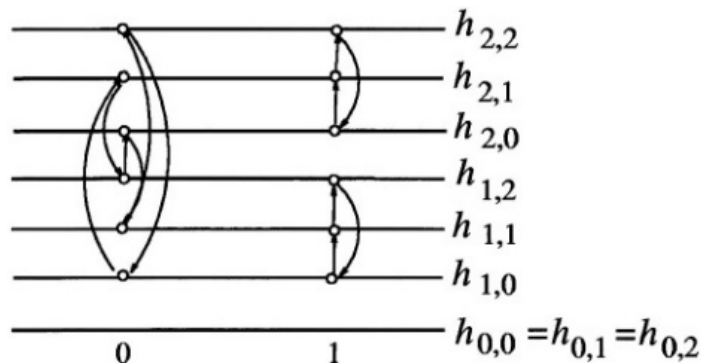


图 3.42

318. Answer. The single-valued continuous branches are the functions $h_i(z) = [f_i(z)]^n$, where $i = 1, 2, \dots, m$.

319. a) If

$$f_0(z), \quad f_1(z) = if_0(z), \quad f_2(z) = -f_0(z), \quad f_3(z) = -if_0(z)$$

are the single-valued continuous branches of the function $f(z) = \sqrt[4]{z}$, then $[f_0(z)]^2 = [f_2(z)]^2$ and $[f_1(z)]^2 = [f_3(z)]^2$. The correct scheme is shown in Figure 32 (§2.9).

b) Cf. 316(a). If $f_0 \equiv 0$, $f_1(z)$, and $f_2(z) = -f_1(z)$ are the single-valued continuous branches of the function $f(z) = \sqrt{z} + \sqrt{z}$, then $[f_1(z)]^2 = [f_2(z)]^2$. The correct scheme is shown in Figure 105.

c) If $f_0(z)$ is one of the single-valued continuous branches of the function $f(z) = \sqrt{z} \sqrt[3]{z-1}$, then the branches are

$$\begin{aligned} f_0(z), \quad f_1(z) = f_0(z)\epsilon_3, \quad f_2(z) = f_0(z)\epsilon_3^2, \\ f_3(z) = -f_0(z), \quad f_4(z) = -f_0(z)\epsilon_3, \quad f_5(z) = -f_0(z)\epsilon_3^2. \end{aligned}$$

Therefore

$$[f_0(z)]^3 = [f_1(z)]^3 = [f_2(z)]^3, \quad [f_3(z)]^3 = [f_4(z)]^3 = [f_5(z)]^3.$$

The required scheme is shown in Figure 32 (§2.9).

320. Suppose that the point z_0 is not a branch point of the function $f(z)$. Thus, after one turn around the point z_0 along a circle of sufficiently small radius, the value of $f(z)$ does not change.

Suppose all values of $f(z_0)$ are different from zero. Thus the continuous images under the mapping $w = f(z)$ of circles with centre at z_0 and with sufficiently small radii are closed continuous curves lying close to the point $w = f(z_0)$. Since all values of $f(z_0)$ are different from zero, all these image curves avoid the point $w = 0$ when the circles have sufficiently small radii. It follows that $\arg f(z)$ does not change. Thus the value of the function $\sqrt[n]{f(z)}$ does not change either. Hence the only possible branch points of the function $\sqrt[n]{f(z)}$ are the branch points of the function $f(z)$ and the points where one of the values of $f(z)$ vanishes.

Answer. The branch points of $f(z)$ and the points where one of the values of $f(z)$ vanishes.

321. Since $g(z)$ is a continuous function on the plane with the cuts described, $[g(z)]^n$ is also a continuous function. Since, for every z , $g(z)$ is one of the values of $\sqrt[n]{f(z)}$, $[g(z)]^n$ is, for every z , one of the values of the function $f(z)$. Consequently $[g(z)]^n$ is a single-valued continuous branch of the function $f(z)$ according to the chosen cuts.

322. Cf. 302.

Answer.

$$g(z), \quad g(z)\epsilon_n, \quad g(z)\epsilon_n^2, \quad \dots, \quad g(z)\epsilon_n^{n-1}.$$

323. Hint. Since $w_0(t)$ is a continuous curve, $w_k(t)$ is also a continuous curve; moreover,

$$[w_k(t)]^n = [w_0(t)]^n \epsilon_n^{kn} = [w_0(t)]^n,$$

and consequently $[w_k(t)]^n$ is equal to one of the values of $f(z(t))$.

324. Hint. From the result of Problem 323, it follows that if the value of the function $\sqrt[n]{f(z)}$ at the initial point of the curve C is multiplied by ϵ_n^k , then the value of the function $\sqrt[n]{f(z)}$ at the final point of the curve C is also multiplied by ϵ_n^k .

325. Let $f_0(z)$ and $f_1(z) = -f_0(z)$ be the single-valued continuous branches of the function $\sqrt{z}$ such that $f_0(1) = 1$ and $f_1(1) = -1$. Thus $f_0(z) - 1$ and $f_1(z) - 1$ are the single-valued continuous branches of the function $\sqrt{z} - 1$. Each one of these values corresponds to two branches of the function $\sqrt{\sqrt{z} - 1}$. If $\sqrt{z} - 1 = 0$, then $\sqrt{z} = 1$ and $z = 1$. Therefore, besides the point $z = 0$, only the point $z = 1$ can be a branch point. The branch point can lie only in the pack of two sheets corresponding to the branch $f_0(z) - 1$ (because one must have $\sqrt{1} = 1$). We have

$$f_0(z) - 1 = \frac{(f_0(z) - 1)(f_0(z) + 1)}{f_0(z) + 1} = \frac{[f_0(z)]^2 - 1}{f_0(z) + 1} = \frac{z - 1}{f_0(z) + 1}.$$

After one turn around the point $z = 1$, the argument of the denominator does not change, because $f_0(1) + 1 = 2 \neq 0$. The argument of the numerator varies by 2π . Thus $\arg(f_0(z) - 1)$ varies by 2π , $\arg \sqrt{f_0(z) - 1}$ varies by π , and therefore the value of $\sqrt{f_0(z) - 1}$ changes. After one turn around the point $z = 0$, the value of the function $\sqrt{z}$ changes; therefore one moves from the pack of two sheets corresponding to $f_0(z) - 1$ to the pack of two sheets corresponding to $f_1(z) - 1$, and vice versa. After a double turn around the point $z = 0$, the final value of the function $\sqrt{z}$ coincides with the initial value and $\arg(\sqrt{z} - 1)$ does not change (because $\sqrt{z} - 1 \neq 0$). After a double turn around the point $z = 0$, one returns to the first sheet. Combining the results obtained, we can build the scheme of the Riemann surface of the function $\sqrt{\sqrt{z} - 1}$, shown in Figure 106.

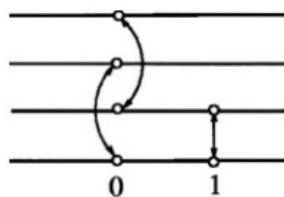


图 3.44

326. This problem is solved in the same way as Problem 325: a) cf. Figure 107; b) cf. Figure 108. *Hint.* If

$f_0(z)$ is a single-valued continuous branch of the function $\sqrt[3]{z}$ and $f_0(1) = 1$, then

$$f_0(z) - 1 = \frac{f_0^3(z) - 1}{f_0^2(z) + f_0(z) + 1} = \frac{z - 1}{f_0^2(z) + f_0(z) + 1},$$

where $f_0^2(1) + f_0(1) + 1 = 3 \neq 0$.

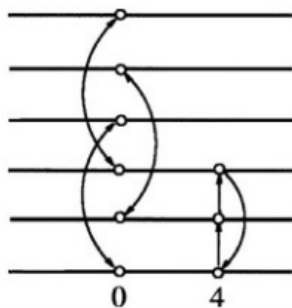


FIGURE 107

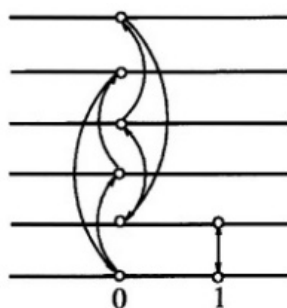


FIGURE 108

图 3.45: Schemes for Problem 326 (Figures 107 and 108).

327. Since $z^2 + 1 = (z - i)(z + i)$, the value of $\sqrt{z^2 + 1}$ changes as a consequence of one turn around the point $z = i$ and of one turn around the point $z = -i$, i.e., these points are the branch points of the function $\sqrt{z^2 + 1} - 2$. The scheme in both cases (a) and (b) is shown in Figure 75 (solution 303).

If $\sqrt{z^2 + 1} - 2 = 0$, then $z^2 + 1 = 4$ and $z = \pm\sqrt{3}$ ($\sqrt{3}$ is the positive value of the root). Let $f_0(\sqrt{3}) = 0$, i.e., in this case we have chosen $\sqrt{4} = 2$. We seek the value of $f_0(-\sqrt{3})$. Join the point $z = \sqrt{3}$ to the point $z = -\sqrt{3}$ by a continuous curve not crossing the cuts.

In case (a), one can take, for example, the segment joining the points $z = \sqrt{3}$ and $z = -\sqrt{3}$. It is easy to see that, on moving along this segment, $\arg(z + i) = \arg(z - (-i))$ increases by $2\pi/3$, whereas $\arg(z - i)$ decreases by $2\pi/3$. Therefore $\arg(z^2 + 1)$ does not change, and consequently the value of $\sqrt{z^2 + 1} - 2$ does not change. Thus, in case (a), $f_0(-\sqrt{3}) = f_0(\sqrt{3}) = 0$.

In case (b), on moving along an arbitrary curve joining the points $z = \sqrt{3}$ and $z = -\sqrt{3}$ and not crossing the cuts, $\arg(z + i)$ increases by $2\pi/3$, and $\arg(z - i)$ increases by $4\pi/3$. Thus $\arg(z^2 + 1)$ increases by 2π and $\arg \sqrt{z^2 + 1}$ increases by π : the value of $\sqrt{z^2 + 1}$ changes into the opposite value. Therefore, in case (b), $f_0(-\sqrt{3}) = -2 - 2 = -4 \neq 0$, whereas $f_1(-\sqrt{3}) = 2 - 2 = 0$.

328. Let $g_0(z)$ and $g_1(z)$ be the single-valued continuous branches of the function $g(z) = \sqrt{z^2 + 1}$ such that $g_0(z) - 2 = f_0(z)$ (cf. Solution 327) and $g_1(z) - 2 = f_1(z)$. In case (a), one has $g_0(\sqrt{3}) = g_0(-\sqrt{3}) = 2$ (cf. Solution 327), and

$$g_0(z) - 2 = \frac{g_0^2(z) - 4}{g_0(z) + 2} = \frac{z^2 - 3}{g_0(z) + 2} = \frac{(z - \sqrt{3})(z + \sqrt{3})}{g_0(z) + 2}.$$

After one turn around the points $z = \sqrt{3}$ and $z = -\sqrt{3}$, the argument of the denominator does not change because $g_0(\sqrt{3}) + 2 = g_0(-\sqrt{3}) + 2 = 4 \neq 0$, whereas the argument of the numerator increases by 2π . Therefore $\arg(g_0(z) - 2)$ increases by 2π and $\arg \sqrt{g_0(z) - 2}$ increases by π ; hence the value of $\sqrt{f_0(z)}$ changes. Consequently the branch points $z = \sqrt{3}$ and $z = -\sqrt{3}$ lie in the same pack of sheets. In the same way, one proves that in case (b) these branch points lie in different packs. It remains to calculate how the passages among the sheets obtained by turning around $z = i$ and $z = -i$ match each other.

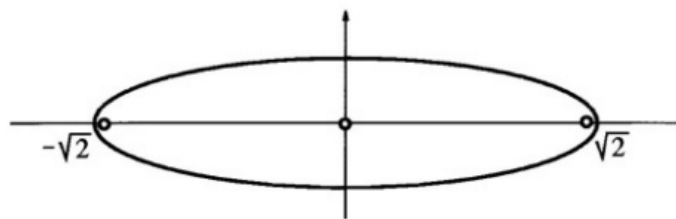


图 3.47: Continuous image of the circle C_R under $w = \sqrt{z^2 + 1}$.

The continuous image under the mapping $w = \sqrt{z^2 + 1}$ of the circle C_R of radius $R = 1.1$, with centre at $z = 0$, is the curve shown in Figure 3.47 (consider the mappings $w = z^2$, $w = z^2 + 1$, and $w = \sqrt{z^2 + 1}$). This curve does not turn around the point $w = 2$. Therefore, on turning along the circle C_R , neither $\arg(\sqrt{z^2 + 1} - 2)$ nor the value of

$$h(z) = \sqrt{\sqrt{z^2 + 1} - 2}$$

changes. Consequently, turning around $z = i$ and then around $z = -i$, we must return to the same sheet (cf. Remark 1 in § 2.10). The required schemes are shown in Figures 3.48 and 3.49.

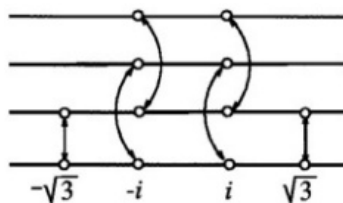


图 3.48

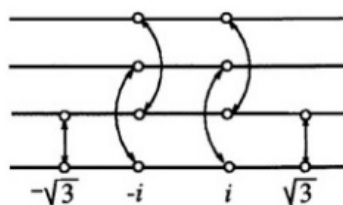


图 3.49

329. *Hint.* In the opposite case, on covering the inverse curve C^{-1} , one should lose uniqueness.

330. *Answer.*

$$\begin{aligned}
 \text{a) } z = 0 &: \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}, & z = 1 &: \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}; \\
 \text{b) } z = 0 &: \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 4 & 3 & 6 & 5 \end{pmatrix}, & z = 1, -1 &: \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 5 & 6 & 1 & 2 \end{pmatrix}; \\
 \text{c) } z = 0 &: \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 6 & 4 & 2 & 3 & 1 \end{pmatrix}; \\
 \text{d) } z = 1 &: \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 6 & 7 & 8 & 5 & 2 & 3 & 4 & 1 \end{pmatrix}, \\
 z = -1 &: \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 6 & 7 & 8 & 1 & 2 & 3 & 4 \end{pmatrix}.
 \end{aligned}$$

331. *Hint.* Use the result of Problem 57: $e = g \cdot g^{-1}$; conditions (1) and (3) are obviously satisfied.

332. *Answer.* a) The cyclic group $\mathbb{Z}_2$; b) the cyclic group $\mathbb{Z}_3$; c) the cyclic group $\mathbb{Z}_n$; d) $\mathbb{Z}_3$; e) $\mathbb{Z}_4$.

333. *Hint.* For the function $\sqrt{z} + \sqrt{z-1}$ given in Problem 314(a), a turn around $z = 0$ produces a permutation of the first indices in the branches $h_{i,j}(z)$, and a turn around $z = 1$ produces a permutation of the second indices. Hence the group sought is the direct product $\mathbb{Z}_2 \times \mathbb{Z}_2$ (cf. § 1.7).

For the function $\sqrt{z^2-1} + \sqrt[4]{z-1}$ given in Problem 314(d), let g_1 be the permutation of the sheets corresponding to one turn around $z = 1$, and let g_2 be the permutation corresponding to one turn around $z = -1$. Thus g_2 cyclically permutes the first indices of the branches $h_{i,j}(z)$, whereas $g_1 g_2^{-1}$ cyclically permutes the second indices. Since $g_1 = (g_1 g_2^{-1}) g_2$, the subgroup generated by g_1 and g_2 coincides with the subgroup generated by g_2 and $g_1 g_2^{-1}$. Therefore the required group is the direct product $\mathbb{Z}_2 \times \mathbb{Z}_4$.

The other cases of Problem 333 are solved similarly.

Answer. 1. a) $\mathbb{Z}_2 \times \mathbb{Z}_2$ (the direct product; cf. § 1.7); b) $\mathbb{Z}_3 \times \mathbb{Z}_2 \cong \mathbb{Z}_6$ (cf. $\mathbb{Z}_7$); c) $\mathbb{Z}_2 \times \mathbb{Z}_3 \cong \mathbb{Z}_6$; d) $\mathbb{Z}_2 \times \mathbb{Z}_4$. 2. a) $\mathbb{Z}_2$; b) $\mathbb{Z}_4$; c) $\mathbb{Z}_4$; d) $\mathbb{Z}_2 \times \mathbb{Z}_3 \cong \mathbb{Z}_6$. 3. a) $\mathbb{Z}_2$; b) $\{e\}$; c) $\mathbb{Z}_2$.

334. If the permutation g_1 exchanges the two packs of sheets, and the permutation g_2 changes the positions of the sheets in the same pack, then $g_1 g_2 g_1^{-1}$ changes the positions of the sheets in the other pack. Thus the permutation group of both schemes contains a permutation that exchanges the packs, a permutation that permutes the sheets in one pack, and a permutation that permutes the sheets in the other pack.

The group generated by these permutations contains only permutations for which every pack is sent either to itself or to the other pack, while the sheets in each pack are arbitrarily permuted. Number the branches of one bunch (or, equivalently, the sheets of one pack) by 1 and 3, and those of the other bunch by 2 and 4. Every permutation of the group so defined then corresponds to a symmetry of the square with vertices 1, 2, 3, 4, and, conversely, every symmetry of this square corresponds to a permutation in the permutation group of the scheme. Therefore, in both cases, the group is isomorphic to the group of symmetries of the square.

335. Let $w_1, w_2, \dots, w_n$ be all values of $w(z_0)$, and let $z_1, \dots, z_s$ be the branch points of the function $w(z)$. Number the sheets of the scheme of the Riemann surface of $w(z)$ so that, for every $i = 1, \dots, n$, the value $w_i = w(z_0)$ corresponds to the i th sheet. Then every permutation of the values w_i naturally corresponds to a permutation of the sheets. We prove that under this correspondence the groups G_1 and G_2 coincide.

Let a permutation g in G_1 be produced by a turn along a continuous curve C that starts and ends at z_0 . Suppose C crosses the cuts, according to which the Riemann surface was built, drawn from the points $z_{i_1}, z_{i_2}, \dots, z_{i_m}$. If the point z_j corresponds to a permutation g_j , then the curve C corresponds to a permutation of the sheets (together with a permutation of the values w_i) equal to

$$g_{i_m}^{\sigma_m} \cdots g_{i_2}^{\sigma_2} g_{i_1}^{\sigma_1},$$

where $\sigma_i = 1$ if the cut is traversed counterclockwise and $\sigma_i = -1$ if it is traversed clockwise (cf. Remark 1 in § 2.10). It follows that

$$g = g_{i_m}^{\sigma_m} \cdots g_{i_1}^{\sigma_1}$$

belongs to G_2 . Conversely, if $g = g_{i_m}^{\sigma_m} \cdots g_{i_1}^{\sigma_1}$ is an element of G_2 (where $\sigma_i = \pm 1$),³ one easily defines a curve C that produces the same permutation of the values w_i in G_1 . For example, Figure 3.50 shows a curve corresponding to the permutation $g_2^{-1} g_3 g_1^{-1}$.

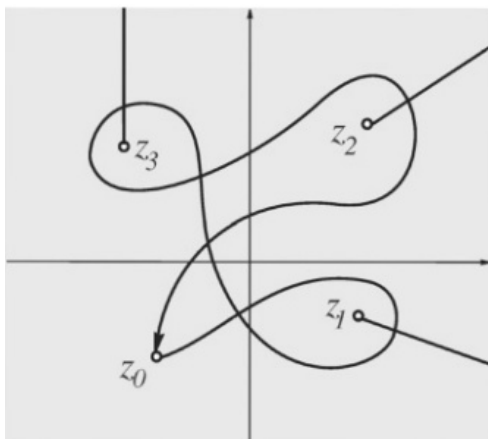


图 3.50

336. Let z_0 be a branch point of the function $h(z)$, and suppose that one turn around this point corresponds to the permutations d_1 and d_2 of the schemes of the functions $f(z)$ and $g(z)$. If z_0 is not a branch point for one of $f(z)$ or $g(z)$, the corresponding permutation d_1 or d_2 is the identity. If the branches $h_{i,j}(z)$ of $h(z)$ are numbered by two indices i, j as in Proposition (a) of Theorem 8 (§ 2.11), then, after a turn around z_0 , the first and second indices are independently permuted (Theorem 8, Proposition (b)). Moreover, the permutation of the first indices equals d_1 , and that of the second indices equals d_2 . Thus one turn around the branch point corresponds to a permutation of the sheets of the Riemann-surface scheme of $h(z)$ that can be regarded as a pair (d_1, d_2) . Since d_1 and d_2 are elements of the groups F and G , respectively, the pair (d_1, d_2) belongs to the direct product $F \times G$. The pairs corresponding to all branch points of $h(z)$ generate a certain subgroup of $F \times G$.

337. The scheme built by the formal method may contain sets of sheets on which the branches coincide. Theorem 8(c) says that the sheets in each such set must be identified to obtain the correct scheme of the Riemann surface of $h(z)$. By uniqueness, one turn around an arbitrary branch point moves us from the sheets of one set to the sheets of another set in the formally built scheme. Consequently, every permutation d of the sheets of that scheme corresponding to one turn around a branch point permutes the sets without destroying them. If d_1 and d_2 permute the sets without destroying them, then so does $d_1 d_2$. Therefore all permutations d_i in the group H_1 permute the sets without destroying them.

Associate with each permutation d_i a permutation d'_i of the sets. If the sheet permutations d_i and d_j correspond to the set permutations d'_i and d'_j , then $d_i d_j$ corresponds to $d'_i d'_j$. Thus the mapping from H_1 to the permutation group of the sets of sheets is a homomorphism.

Every set, including a set consisting of only one sheet, corresponds to a single sheet of the correct scheme of the Riemann surface of $h(z)$ (Theorem 8(c)), and the passages among the sheets of the original scheme are transformed exactly into passages among the sets of sheets. Hence the homomorphism just defined is a surjective homomorphism from H_1 onto H_2 .

338. Cf. Problems 336 and 337. By hypothesis the groups F and G (cf. Problem 336) are soluble. Hence $F \times G$ is also soluble (cf. Problem 167). Since H_1 , the permutation group of the sheets of the scheme built by the formal method, may be regarded as a subgroup of $F \times G$ (cf. Problem 336), H_1 is also soluble (cf. Problem 162). Since there is a surjective homomorphism from H_1 onto H_2 , the permutation group of the correct scheme for $h(z)$ (cf. Problem 337), it follows that H_2 is also soluble (cf. Problem 163).

339. *Hint.* Cf. Theorem 9, § 2.11. If F and H are the monodromy groups of the schemes of the functions $f(z)$ and $h(z)$, then, as in Problem 337, one proves the existence of a surjective homomorphism from F onto H . Afterwards one uses the result of Problem 163.

340. Every sheet of the scheme of the Riemann surface of $f(z)$ corresponds to a pack of n sheets in the scheme of the Riemann surface of $h(z) = \sqrt[n]{f(z)}$ (Proposition (a) of Theorem 10, § 2.11). The permutations of the scheme of $h(z)$ that correspond to turns around its branch points permute the packs without destroying them (Proposition (b) of Theorem 10). Thus every permutation in the group H permutes the packs without destroying them. To every permutation d in H there therefore corresponds a permutation d' of the packs. Moreover, if d_1 and d_2 correspond to d'_1 and d'_2 , respectively, then $d_1 d_2$ corresponds to $d'_1 d'_2$. This gives a homomorphism from H into the permutation group of the packs. The permutation of the packs obtained by a turn around an arbitrary point z_0 corresponds to the permutation of the sheets obtained by a turn around z_0 in the scheme of the Riemann surface of $f(z)$ (Proposition (c) of Theorem 10). Consequently, the group of permutations of the packs generated by H coincides with F (more precisely, it is isomorphic to F).

The homomorphism defined above is therefore a surjective homomorphism from H onto F .

341. The kernel of the homomorphism constructed in the solution of Problem 340 consists of those permutations in H that transform each pack into itself. Let d_1 and d_2 be two such permutations. If the sheets in the

packs are numbered by

$$f_{i,k}(z) = f_{i,0}(z)\epsilon_k,$$

then both d_1 and d_2 cyclically permute the sheets of every pack (cf. Proposition (d) of Theorem 10). Consider an arbitrary pack. If d_1 cyclically displaces its sheets by l places and d_2 displaces them by k places, then both d_1d_2 and d_2d_1 displace the sheets of the pack by $k+l$ places. Thus d_1d_2 and d_2d_1 permute the sheets identically in every pack, so $d_1d_2 = d_2d_1$.

342. If ϕ is the homomorphism defined in the solution of Problem 340 and $\ker \phi$ is its kernel, then the quotient group $H/\ker \phi$ is isomorphic to F (Theorem 3, § 1.13). Since $\ker \phi$ is commutative (cf. Problem 341) and F is soluble by hypothesis, H is soluble as well (cf. Problem 166).

343. Let

$$P_z(w) = 3w^5 - 25w^3 + 60w - z.$$

If w_0 is a multiple root of $P_z(w) = 0$, then w_0 is a root of $P'_z(w) = 0$, where the derivative is taken with respect to w (cf. Problem 276). We have

$$P'_z(w) = 15w^4 - 75w^2 + 60 = 15(w^4 - 5w^2 + 4) = 15(w-2)(w-1)(w+1)(w+2).$$

Since $P'_z(w) = 0$ has four roots of order one, namely $w_0 = -2, -1, 1, 2$, only these values can be multiple roots of $P_z(w) = 0$, and they can only have order two. Substituting these values into

$$3w^5 - 25w^3 + 60w - z = 0$$

shows that they are roots of order two when z takes the values

$$-16, -38, 38, 16,$$

respectively.

Answer. The roots of order two are $w_0 = -2$ for $z = -16$, $w_0 = -1$ for $z = -38$, $w_0 = 1$ for $z = 38$, and $w_0 = 2$ for $z = 16$.

344. Let $P_z(w) = 3w^5 - 25w^3 + 60w - z$. Set $z = z_0$ and consider the single-valued mapping from the w -plane to the complex τ -plane defined by $\tau = P_{z_0}(w)$. In the w -plane, let C be a circle of radius r with centre at w_0 (Figure 3.51), and let C' be its image under $\tau = P_{z_0}(w)$. Decompose the polynomial into monomials of first degree (cf. Problem 269):

$$P_{z_0}(w) = 3(w-w_1)(w-w_2)(w-w_3)(w-w_4)(w-w_5),$$

where all w_i are roots of $P_{z_0}(w) = 0$. During a counterclockwise turn along C , the argument of $(w-w_i)$ does not change if w_i lies outside the disc D bounded by C , and increases by 2π if w_i lies inside D . Therefore, on going counterclockwise along C , the argument of $P_{z_0}(w)$ increases by $2\pi m$, where m is the number of roots, counted with multiplicity, of $P_{z_0}(w) = 0$ inside D . Consequently the curve C' , the image of C under $\tau = P_{z_0}(w)$, turns m times around $\tau = 0$ (Figure 3.52).

By hypothesis, the point w_0 , the centre of C , is a root of $P_{z_0}(w) = 0$, so $m \geq 1$. Hence there is a value $\rho > 0$ such that the disc of radius ρ centred at $\tau = 0$ does not intersect C' (see Figure 3.52). Now consider

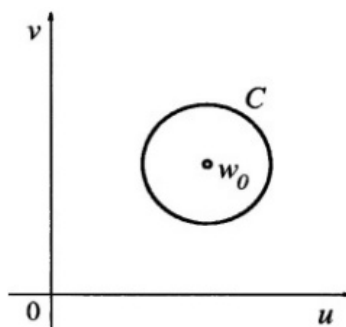


图 3.51

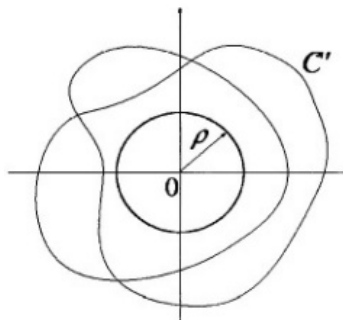


图 3.52

another complex number z'_0 and the mapping $\tau = P_{z'_0}(w)$. Let C'' be the image of C under this mapping. Since

$$\begin{aligned} P_{z'_0}(w) &= 3w^5 - 25w^3 + 60w - z'_0 \\ &= 3w^5 - 25w^3 + 60w - z_0 + (z_0 - z'_0) \\ &= P_{z_0}(w) + (z_0 - z'_0), \end{aligned}$$

the curve C'' is obtained from C' by displacement through the vector $z_0 - z'_0$ (cf. Problem 246). If the length of this vector is smaller than ρ , then C'' is displaced so little from C' that it turns around $\tau = 0$ the same number of times as C' . Equivalently, one may imagine that $\tau = 0$, rather than the curve, is displaced (see Figure 3.52). Since C' turns m times around $\tau = 0$, so does C'' . By the preceding reasoning, the disc D therefore contains $m \geq 1$ roots, counted with multiplicity, of $P_{z'_0}(w) = 0$.

345. Let z_0 be an arbitrary point different from $z = \pm 38$ and $z = \pm 16$. There are five different images of z_0 under the mapping $w(z)$; denote them by w_1, w_2, w_3, w_4, w_5 . If a continuous curve C starts at z_0 , then at least one continuous image of C under $w(z)$ starts at each $w_i, i = 1, \dots, 5$. If two continuous images of C started at the same w_i , the curve would have at least six continuous images. This is impossible because an equation of degree five cannot have more than five roots. Consequently z_0 is not a point of non-uniqueness of $w(z)$.

Consider five discs D_1, D_2, D_3, D_4, D_5 of radius r , centred at the points w_i . Choose r sufficiently small that the discs are disjoint. By Problem 344 there is a disc D_0 , centred at z_0 , such that every point z'_0 in D_0 has at least, and consequently exactly, one image in each of the discs $D_1, \dots, D_5$ in the w -plane. If a continuous curve C lies entirely in D_0 , all the images of its points lie in the discs $D_i, i = 1, \dots, 5$. Thus a continuous image of C under $w(z)$ cannot jump from one disc to another, and each image of C lies entirely in one of the discs D_i .

If C , lying entirely in D_0 , starts and ends at z'_0 , then the endpoints of its continuous image C'' are images

of z'_0 under $w(z)$. Since C'' lies entirely in one disc D_i , and that disc contains only one image of z'_0 , the curve C'' starts and ends at the same point. Thus, if C is a closed curve lying entirely in D_0 , the value of $w(z)$ at the final point of C , defined by continuity, coincides with its value at the initial point. In particular, this holds for every circle centred at z_0 with radius smaller than ρ . Consequently z_0 is not a branch point of $w(z)$.

346. By Problem 343, when $z = z_0 = 38$, equation (2.8) has four roots w_1, w_2, w_3, w_4 , of which one, say w_1 , has order two and the others are simple. Suppose z'_0 is close to z_0 . By Problem 344, near w_1 there are two images of z'_0 under $w(z)$, while near each of w_2, w_3 , and w_4 there is only one image. Let C be a circle of small radius centred at z_0 and starting and ending at z'_0 . As in Problem 345, the continuous images of C under $w(z)$ that start at w_2, w_3, w_4 end at their initial points, whereas a continuous image that starts at one of the two images of z'_0 near w_1 may end at the other image near w_1 . Consequently only two sheets can meet at z_0 , while there are no passages between the other three sheets.

347. Draw a continuous curve C' from w_0 to w_1 that does not cross the images under $w(z)$ of the points $z = \pm 38$ and $z = \pm 16$. This is possible because these four points have only finitely many images. Let C be the image of C' under

$$z(w) = 3w^5 - 25w^3 + 60w.$$

Since $z(w)$ and C' are continuous, so is C . The variables z and w are related under $z(w)$ by

$$3w^5 - 25w^3 + 60w - z = 0,$$

which is the same relation as that defining $w(z)$; hence C' is itself a continuous image of C under $w(z)$. Since C' does not pass through the images of $z = \pm 38$ and $z = \pm 16$, the curve C does not pass through these points. The initial and final points of C are $z(w_0) = z_0$ and $z(w_1) = z_1$. Thus C is the required curve.

348. By Problem 347, one can move from any sheet of the Riemann surface of $w(z)$ to any other along a curve that does not pass through $z = \pm 38$ or $z = \pm 16$. Moreover, every passage from one sheet to another while crossing a cut coincides with the passage indicated in the scheme at the branch point from which the cut starts (cf. Remark 1, § 2.10). Consequently the sheets must be connected at the branch points so as to obtain a connected scheme. Points other than $z = \pm 38$ and $z = \pm 16$ are not branch points (cf. Problem 345), and only two sheets can join at each of these four points (cf. Problem 346). Thus, to obtain a connected scheme, one arrow must correspond to each of these points; in particular, all four are branch points. All distinct connected schemes are shown in Figure 3.53. Every connected scheme for the Riemann surface of $w(z)$ agrees with one of these three after a permutation of the sheets and of the branch points; this does not assert that all three schemes can be realized.

349. We prove that the permutation group of each of the three schemes in Figure 3.53 contains all elementary transpositions (cf. § 1.15), namely $(1, 2)$, $(2, 3)$, $(3, 4)$, and $(4, 5)$. For the first scheme this is evident because these transpositions correspond to the branch points. In the second and third schemes, one branch point corresponds to $(1, 2)$. In both cases,

$$(2, 3) = (1, 2) \cdot (1, 3) \cdot (1, 2), \quad (3, 4) = (1, 3) \cdot (1, 4) \cdot (1, 3).$$

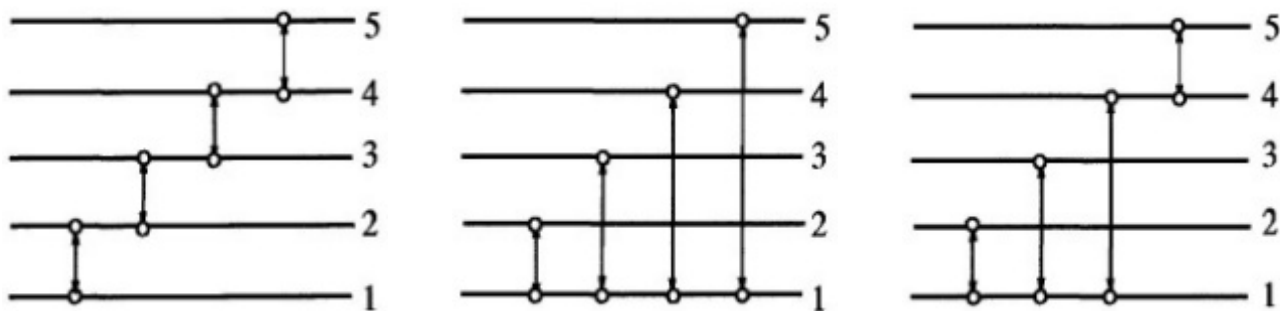


图 3.53: The distinct connected schemes.

For the second scheme,

$$(4, 5) = (1, 4) \cdot (1, 5) \cdot (1, 4),$$

while in the third it coincides with one of the branch-point transpositions. Thus in all cases the group contains every elementary transposition and hence, by Theorem 4 in § 1.15, is the full permutation group S_5 .

350. Problem 349 shows that the monodromy group of $w(z)$ is the full permutation group S_5 , which is not soluble (cf. Theorem 5, § 1.15). On the other hand, if $w(z)$ were representable by radicals, its monodromy group would be soluble (cf. Theorem 11, § 2.13). This contradiction shows that $w(z)$ is not representable by radicals.

351. *Hint.* If such a formula existed, then substituting $a_0 = 3$, $a_2 = -25$, $a_4 = 60$, $a_1 = a_3 = 0$, and $a_5 = z$ would show that the function $w(z)$ of Problem 350 is representable by radicals.

352. The function $w_1(z)$ expressing the roots of equation (2.9) in terms of the parameter z has a Riemann surface consisting of a separate sheet on which $w_1(z) \equiv 0$ and five sheets representing the scheme of the function $w(z)$ that expresses the roots of

$$3w^5 - 25w^3 + 60w - z = 0$$

in terms of z . Hence the monodromy group of $w_1(z)$ is the monodromy group of the scheme of $w(z)$, namely the full degree-five permutation group S_5 , which is not soluble (cf. Problem 349). If $w_1(z)$ were representable by radicals, however, its monodromy group would be soluble (cf. Theorem 11, § 2.13). This contradiction shows that $w_1(z)$ is not representable by radicals and that the general equation of degree n , for $n > 5$, is not solvable by radicals.

Drawings of Riemann surfaces

The Riemann surface of a complex function $w(z)$, as defined in this book, is a collection of different copies of the z -plane (the sheets), suitably joined to one another along cuts so that the function w , defined on these sheets, becomes single-valued and continuous.

Sometimes, however, the Riemann surface of $w = f(z)$ can be realized by a suitable projection of its graph in $\mathbb{C}^2$. For example, the Riemann surface of $\sqrt{z}$ in Figure 27 is homeomorphic to the surface shown in Figure 3.54. This is the graph of the real part of $\sqrt{z}$, i.e. the projection of the graph of this function onto the three-dimensional space with coordinates

$$(x = \Re z, y = \Im z, u = \Re w).$$

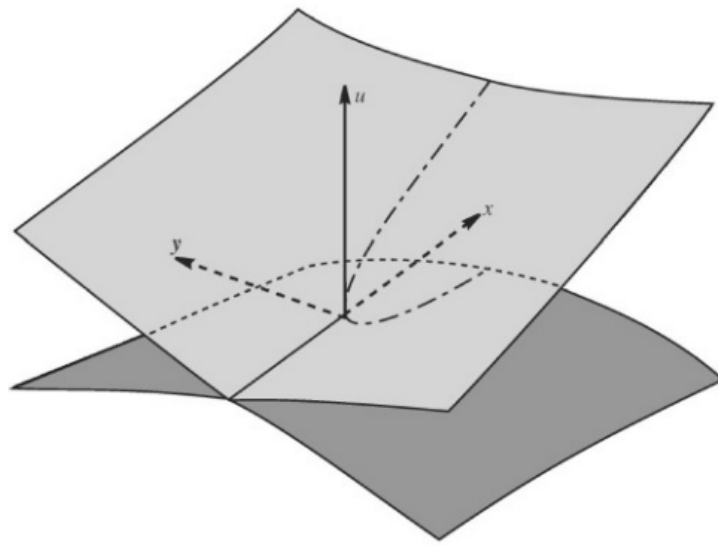


图 3.54: A realization of the Riemann surface of $\sqrt{z}$.

The drawings of Riemann surfaces in this appendix are not projections of graphs; they are “artificial” surfaces constructed by the method explained in § 2.10.

Here is how to “read” these drawings. The different sheets are joined so that passage from one sheet to another while traversing a cut is realized by a smooth curve on the surface. The drawing conventions are as follows:

- distinct grey colours indicate distinct sheets;
- ramification points are indicated by small white discs;
- cuts are black;
- a dashed vertical surface indicates a connection between two non-adjacent sheets (i.e. sheets with other sheets between them), and it always corresponds to a cut.

At every cut two sheets meet: along a smooth curve on the surface, one moves from one sheet to another (Figure 3.55).

Self-intersection lines of the surface that are not cuts are white. On crossing such a line transversally along a smooth curve on the surface, one remains on the same sheet (Figure 3.56).

The following Riemann surfaces are considered in Problems 298–317.

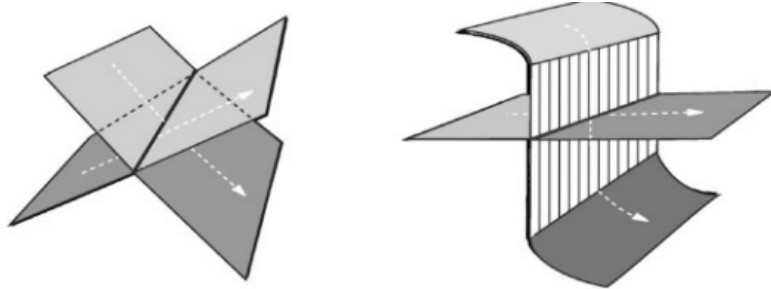


图 3.55

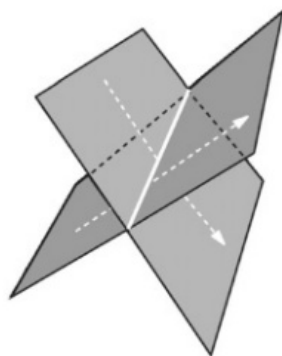


图 3.56

$$\sqrt[n]{z}, \quad n = 6$$

$$0$$

$$\text{a) } \sqrt{z(z-i)}, \quad \text{b) } \sqrt{z^2+1}$$

$$\sqrt[3]{z^2-1}$$

$$-1 \quad 1$$

$$\sqrt[3]{z(z-1)^2}$$

$$0 \quad 1$$

$$\sqrt[4]{(z^2-1)^3(z+1)^3}$$

$$-1 \quad 1$$

$$\sqrt[4]{z(z^3-1)}$$

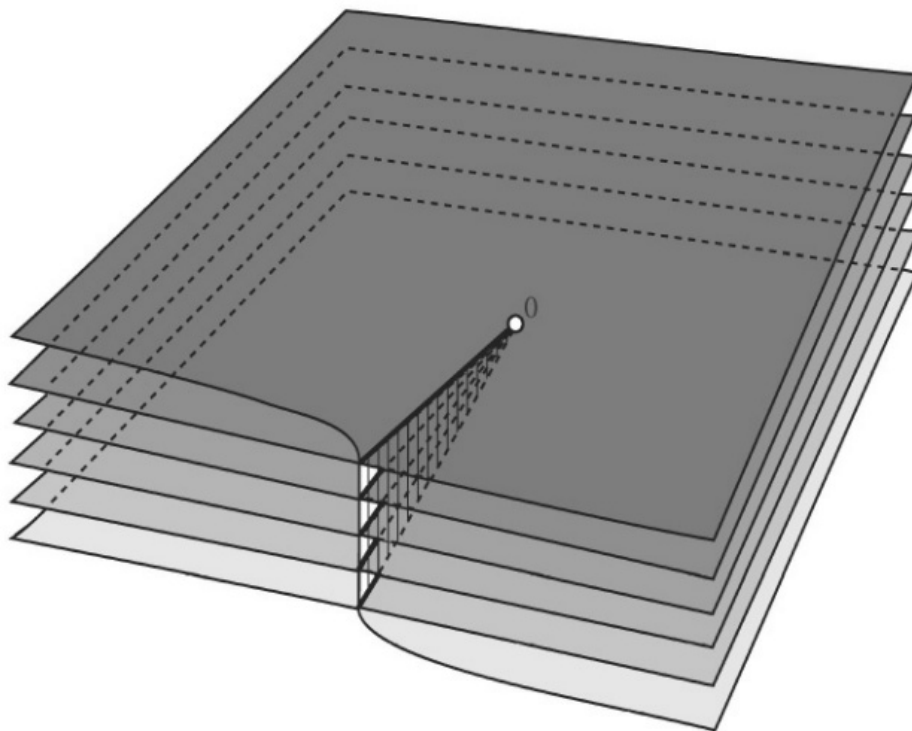


图 3.57: The Riemann surface of $\sqrt[6]{z}$.

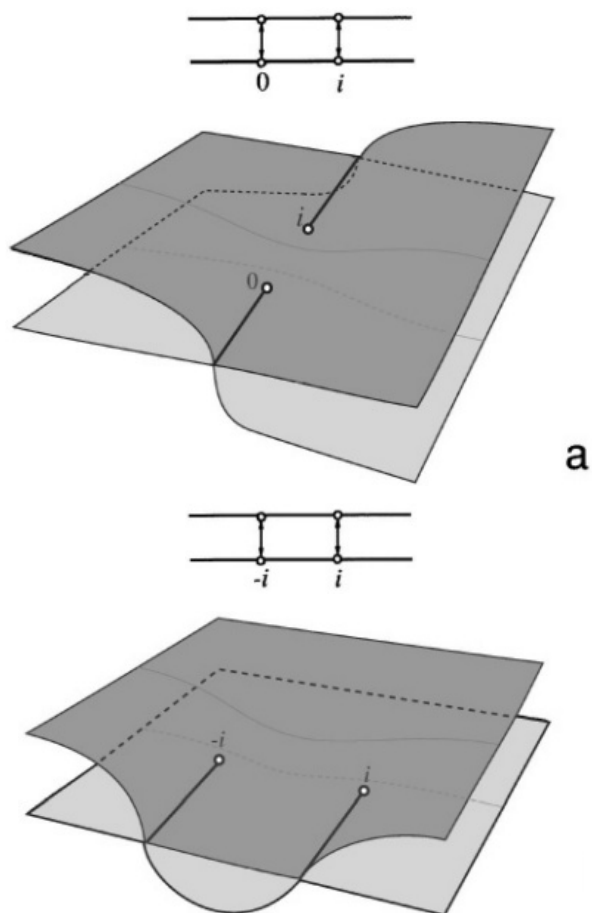


图 3.58: Riemann surfaces for $\sqrt{z(z-i)}$ and $\sqrt{z^2+1}$.

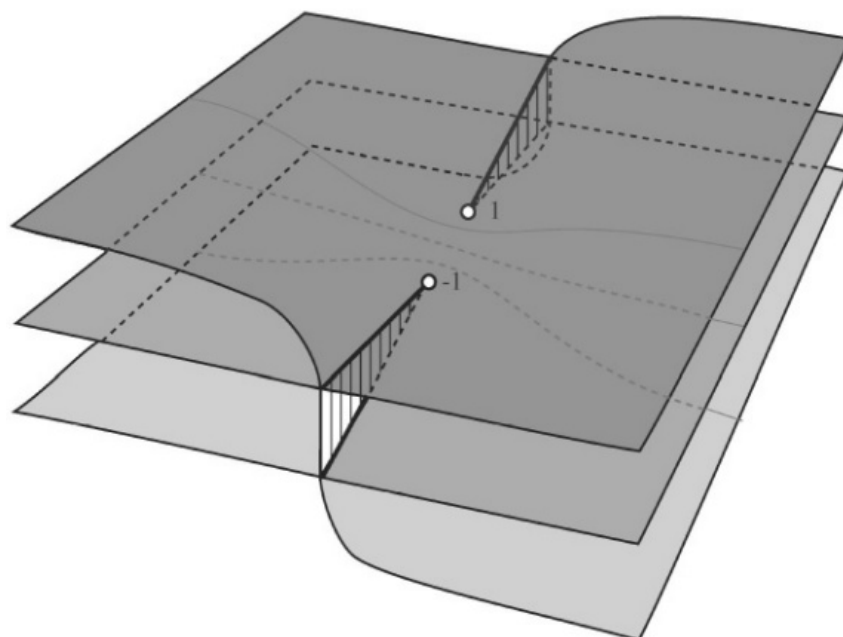


图 3.59: The Riemann surface of $\sqrt[3]{z^2 - 1}$.

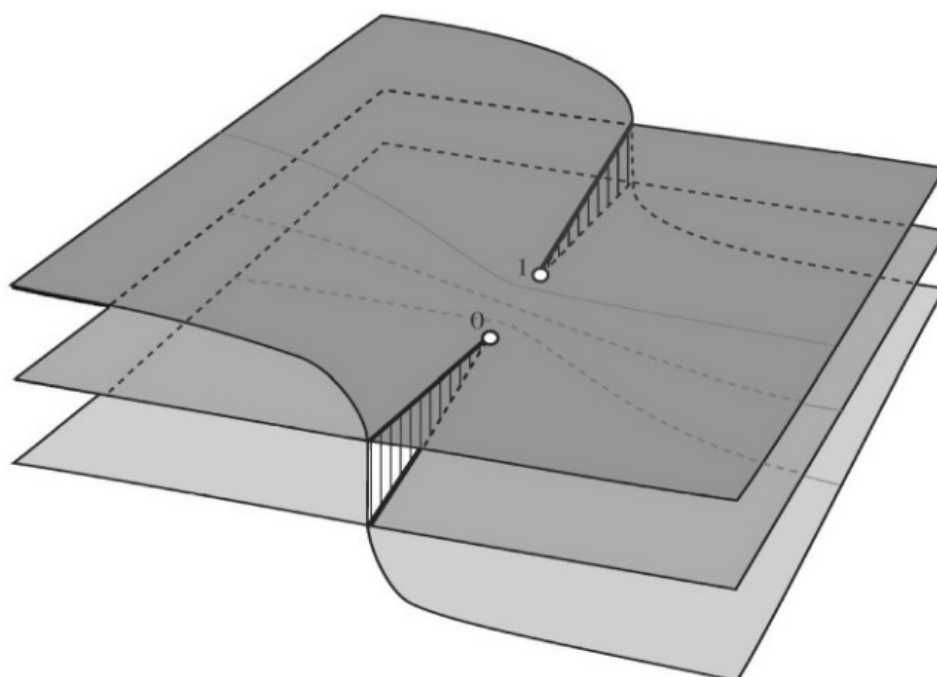


图 3.60: The Riemann surface of $\sqrt[3]{z(z-1)^2}$.

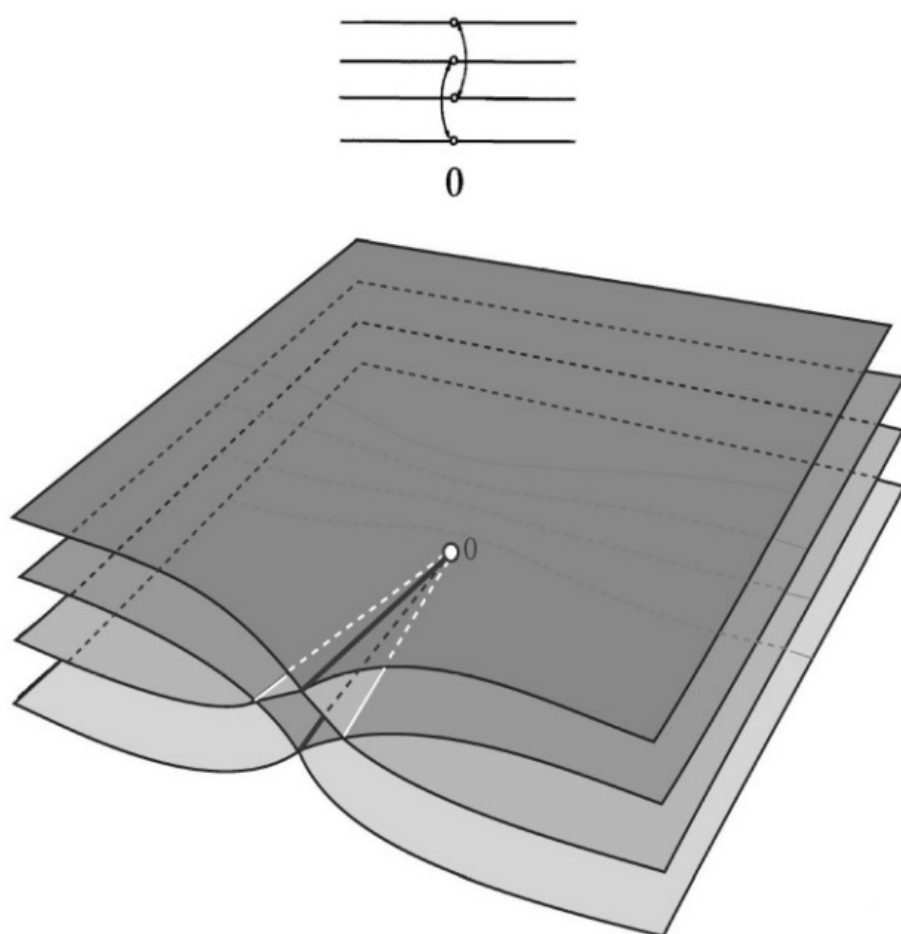


图 3.61

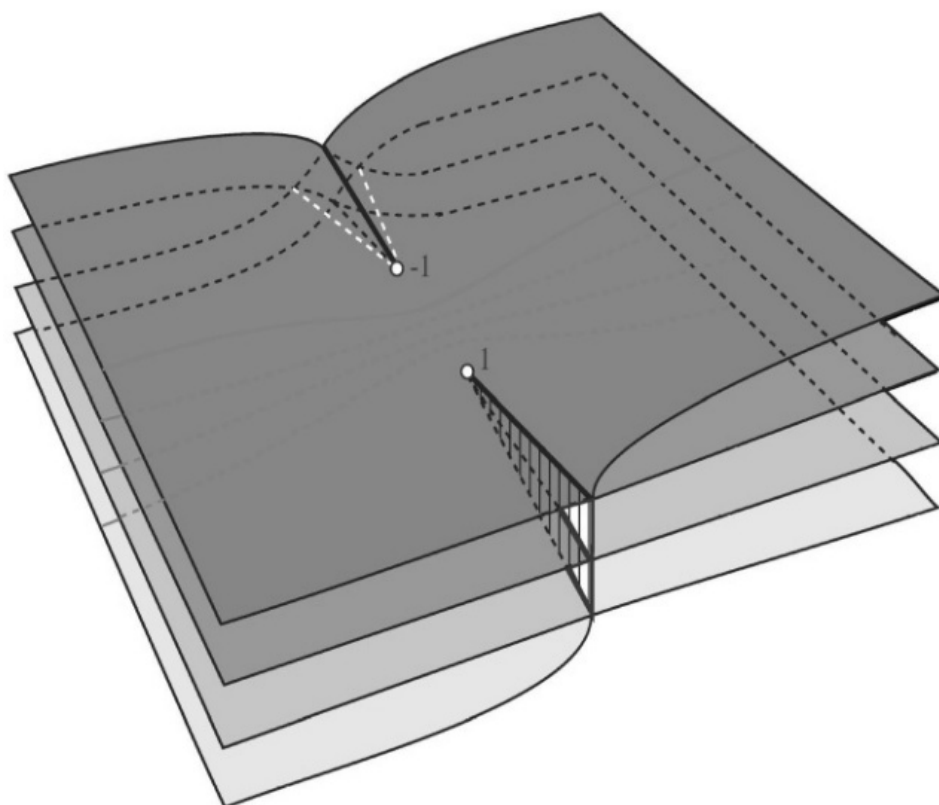


图 3.62: The Riemann surface associated with $\sqrt[4]{(z^2 - 1)^3(z + 1)^3}$.

$$\sqrt[4]{\frac{(z + i)^2}{z(z - 1)^3}}$$

$$\sqrt{z^2 - 1} + \sqrt[4]{z - 1}$$

$$\frac{\sqrt{z} + \sqrt{z}}{\sqrt[3]{z(z - 1)}}$$

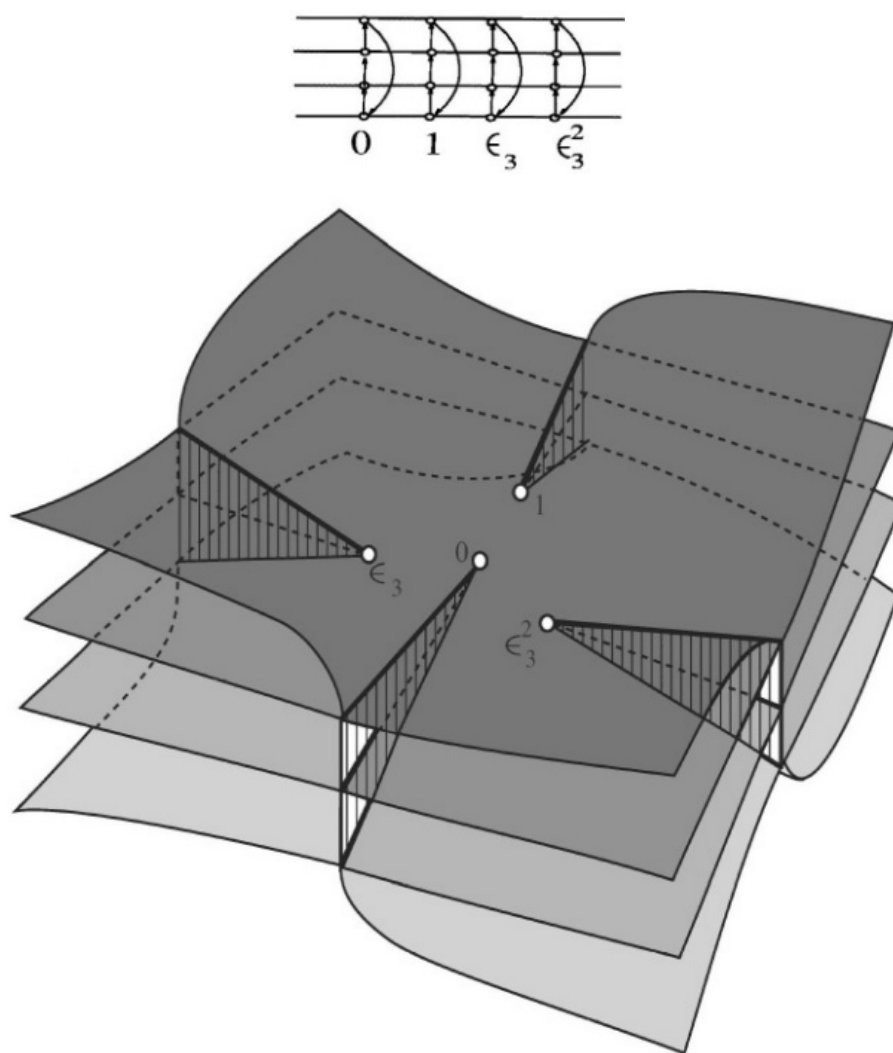


图 3.63: The Riemann surface of $\sqrt[4]{z(z^3 - 1)}$.

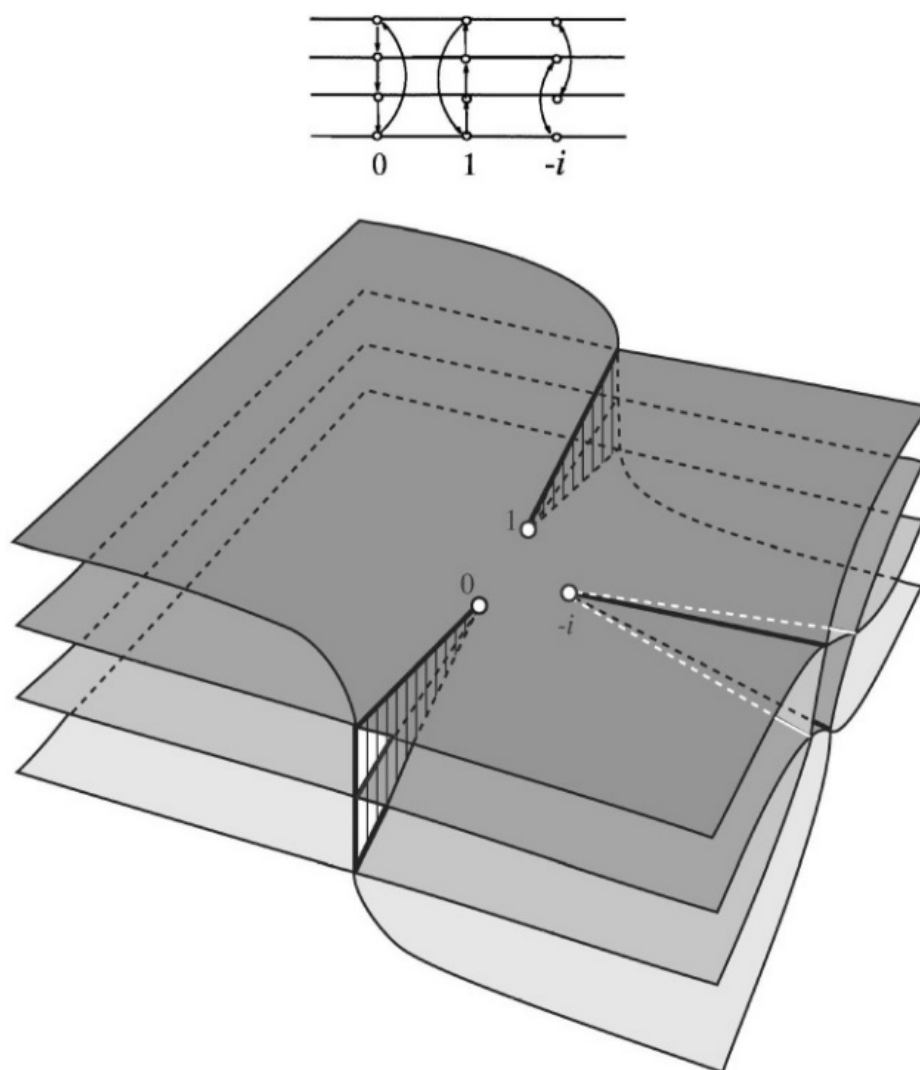


图 3.64: The Riemann surface associated with $\sqrt[4]{(z+i)^2/(z(z-1)^3)}$.

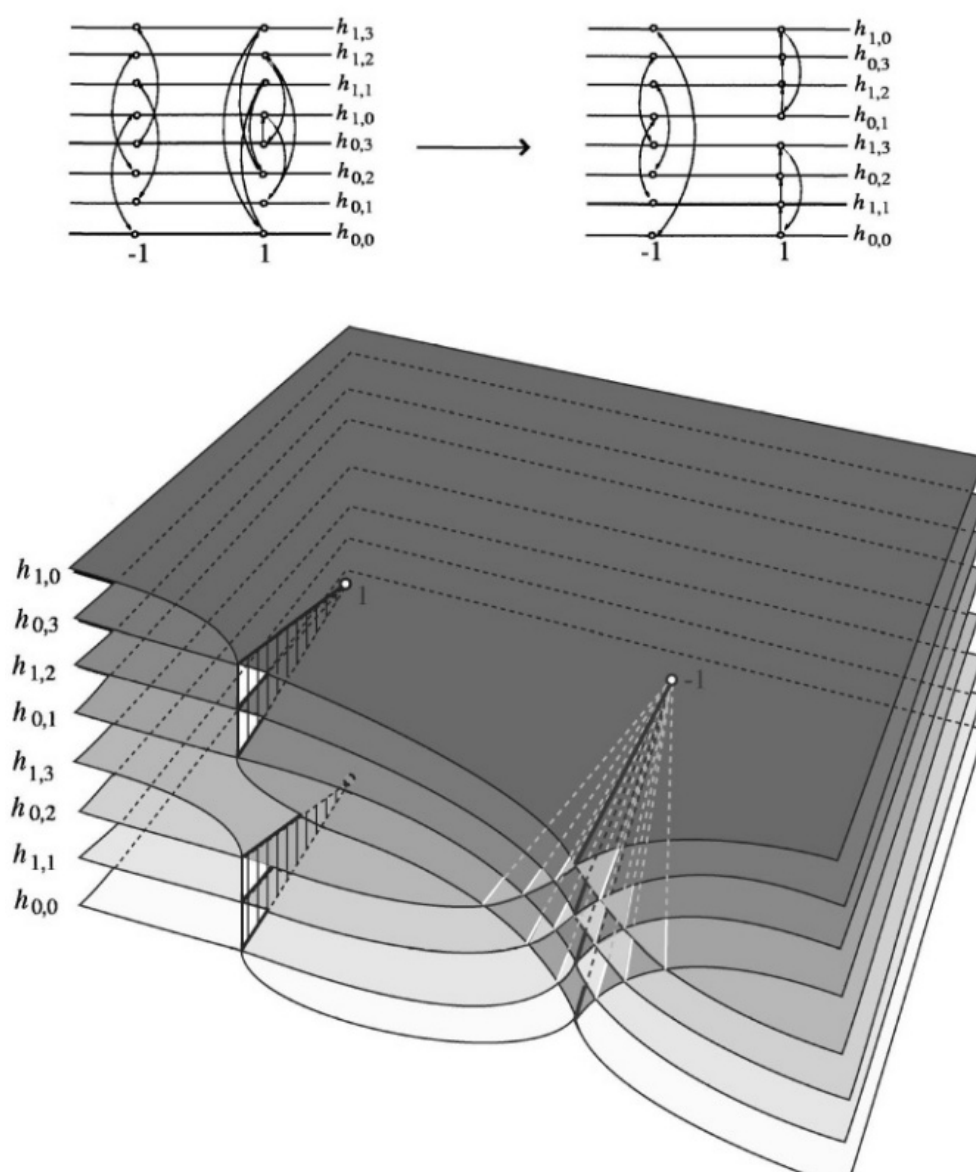


图 3.65: The Riemann surface of $\sqrt{z^2 - 1} + \sqrt[4]{z - 1}$.

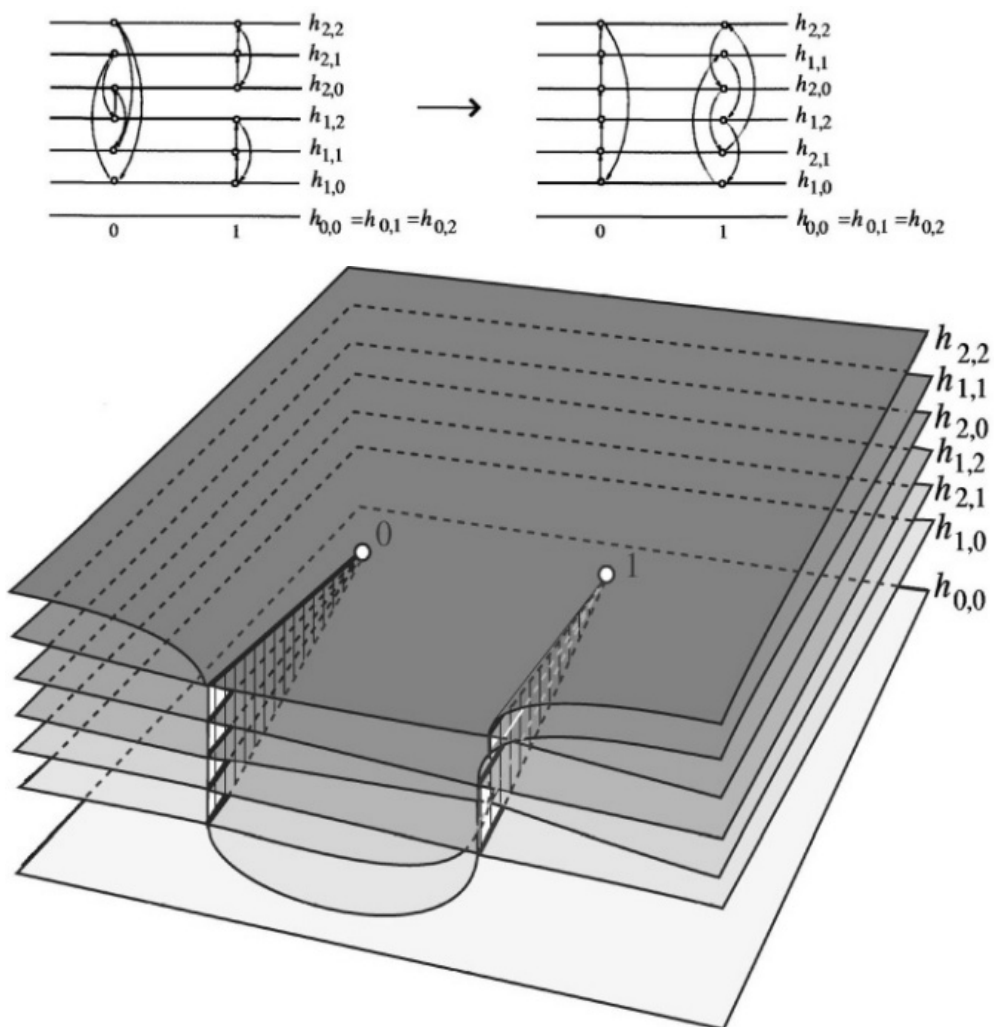


图 3.66: The Riemann surface associated with $(\sqrt{z} + \sqrt{z})/\sqrt[3]{z(z-1)}$.

第四章 Appendix by A. Khovanskii: Solvability of equations by explicit formulae (Liouville's theory, differential Galois theory, and topological obstructions)

This appendix is dedicated to the study of the solvability of differential equations by explicit formulae. This is a quite old problem: the first idea for solving it dates back to Abel. Today three approaches to this problem are known. The first belongs to Liouville. The second considers the problem from the point of view of Galois theory and is related to the names of Picard, Vessiot, Kolchin, and others. The third, topological approach was first introduced, for functions of one variable, in my thesis. I am infinitely grateful to my research director V. I. Arnold, who aroused my interest in this subject.

I had always believed that the topological approach could not be completely applied to the case of many variables. Only recently did I discover that this is not true and that, in the multidimensional case, one can obtain absolutely analogous results [25]–[27].

This appendix contains the subject of my lectures to the Moscow Mathematical Society and to the students of the École Normale Supérieure at the Independent University of Moscow (October 1994).

The section concerning functions of many variables was added for this appendix in autumn 2002.

I would like to thank T. V. Belokrinskaya for her help during the editing of this appendix and F. Aicardi for the translation into English.

A.1 Explicit solvability of equations

Some differential equations possess “explicit solutions.” If this is the case, the solution itself answers the question of solvability. In general, however, all attempts to find explicit solutions of equations turn out to be in vain. One therefore tries to prove that explicit solutions do not exist for some class of equations. We must define precisely what this means; otherwise it is unclear what we wish to prove. We choose the following approach: distinguish certain classes of functions, and say that an equation is explicitly solvable if its solution belongs to one of these classes. Different classes of functions give different notions of solvability.

To define a class of functions, give a list of basic functions and a list of allowed operations. The class is the set of all functions obtained from the basic functions by means of the allowed operations.

EXAMPLE 1. *The class of functions representable by radicals.*

The basic functions are constants and the identity function, whose value is equal to that of the independent variable.

The allowed operations are the arithmetic operations (addition, subtraction, multiplication, and division) and extraction of roots $\sqrt[n]{f}$, $n = 2, 3, \dots$, of a given function f .

For example,

$$g(x) = \sqrt[3]{5x} + 2\sqrt{x} + \sqrt[7]{x^3} + \frac{3}{5}$$

is representable by radicals.

The famous problem of solving algebraic equations by radicals concerns this class. Consider the algebraic equation

$$y^n + r_1(x)y^{n-1} + \cdots + r_n(x) = 0, \quad (\text{A.1})$$

in which the $r_i(x)$ are rational functions of one variable. The complete answer to the problem of solving equation (A.1) by radicals is Galois theory (see § A.8).

Even in the simplest class, that of Example 1, one encounters a difficulty: the functions under consideration are multivalued.

For example, let us determine exactly what is meant by the sum of two multivalued analytic functions $f(x)$ and $g(x)$. Consider an arbitrary point a , a germ f_a of $f(x)$ at a , and a germ g_a of $g(x)$ at a . The function $\varphi(x)$ defined by the germ $f_a + g_a$ is said to be representable as the sum of $f(x)$ and $g(x)$. This sum is not uniquely defined. For example, there are exactly two functions representable as $\sqrt{x} + \sqrt{x}$, namely $f_1 = 2\sqrt{x}$ and $f_2 \equiv 0$.

The closure of a class of multivalued functions with respect to addition is a class that contains, together with any two functions, all functions representable by their sum. The same convention applies to all operations on multivalued functions encountered in this chapter.

EXAMPLE 2. Elementary functions. The basic elementary functions are those learned at school and usually represented on calculator keyboards: constants, the identity function, the n th roots $\sqrt[n]{x}$, the exponential $\exp x$, the logarithm $\ln x$, and the trigonometric functions $\sin x$, $\cos x$, $\tan x$, $\arcsin x$, $\arccos x$, and $\arctan x$. The allowed operations are the arithmetic operations and composition.

Elementary functions are expressed by formulae, for instance

$$f(x) = \arctan(\exp(\sin x) + \cos x).$$

From the beginning of analysis, one learns that integrating elementary functions is far from easy. Liouville proved that indefinite integrals of elementary functions are not, in general, elementary.

EXAMPLE 3. Functions representable by quadratures. The basic functions in this class are the basic elementary functions. The allowed operations are the arithmetic operations, composition, and integration. A class is said to be closed with respect to integration if, together with every function f , it contains a function g such that $g' = f$.

For example,

$$\exp\left(\int^x \frac{dt}{\ln t}\right)$$

is representable by quadratures. But, as Liouville proved, this function is not elementary.

Examples 2 and 3 can be modified. A class of functions is said to be closed with respect to solutions of algebraic equations if, together with every set of functions $f_1, \dots, f_n$, it contains a function y satisfying

$$y^n + f_1 y^{n-1} + \cdots + f_n = 0.$$

EXAMPLE 4. Adding the operation of solving algebraic equations to the definition of elementary functions gives the class of generalized elementary functions.

EXAMPLE 5. The class of functions representable by generalized quadratures is obtained from the class of functions representable by quadratures by adding the operation of solving algebraic equations.

A.2 Liouville's theory

The first exact proofs of the non-solvability of certain equations, either by quadratures or by elementary functions, were obtained by Liouville in the middle of the nineteenth century. Here we briefly expound his results.

The reader can find a broader exposition of Liouville's method and of work on analogous subjects by Chebychev, Mordukai-Boltovski, Ostrovski, and Ritt in book [1].

First, Liouville showed that the classes of functions in Examples 2–5 can be constructed in a very simple way. The set of basic elementary functions seems very large, and the composition operation introduces algebraic difficulties into the definition. Liouville first proved that the lists of basic functions can be greatly reduced: in half the cases only constants remain, and in the other cases only constants and the identity function. Second, he proved that composition is superfluous in the list of allowed operations. All necessary operations can be defined using only arithmetic operations and differentiation. This fact plays an essential role in the algebraization of the problem of differential fields numerability.

We now formulate the corresponding definitions in differential algebra.

A field of functions F is called a *differential field* if it is closed under differentiation, i.e. if $g \in F$, then $g' \in F$. One may also consider abstract differential fields: fields equipped with an additional differentiation operation satisfying the Leibniz identity

$$(a \cdot b)' = a' \cdot b + a \cdot b'.$$

Suppose a differential field F contains a smaller differential field F_0 , so $F_0 \subseteq F$. An element $y \in F$ is said to be *algebraic over F_0* if it satisfies an algebraic equation

$$y^n + a_1 y^{n-1} + \cdots + a_n = 0,$$

where the coefficients a_i belong to F_0 . In particular:

- y is *radical over F_0* if $y^k \in F_0$;
- y is *integral over F_0* if $y' \in F_0$;
- y is *logarithmic over F_0* if $y' = a'/a$, where $a \in F_0$;
- y is *exponential integral over F_0* if $y' = ay$, where $a \in F_0$;
- y is *exponential over F_0* if $y' = a'y$, where $a \in F_0$.

The extension of F_0 by an element y , denoted $F_0\{y\}$, is the minimal differential field containing F_0 and y . The field $F_0\{y\}$ consists of rational functions in $y, y', \dots, y^{(k)}, \dots$ with coefficients in F_0 .

1. An element y is said to be *representable by radicals over F_0* if there is a sequence

$$F_0 \subseteq F_1 \subseteq \cdots \subseteq F_k$$

such that every extension $F_i \subseteq F_{i+1}$ is obtained by adding one radical to F_i , and $F_0\{y\}$ is contained in F_k . Such a sequence is called a tower.

2. An element y is said to be *elementary over F_0* when logarithmic and exponential elements may be added.
3. An element y is said to be *representable by quadratures over F_0* when the addition of integrals and exponential integrals is allowed.
4. An element y is called a *generalized elementary element over F_0* when algebraic, exponential, and loga-

rithmic elements may be added.

5. An element y is said to be *representable by generalized quadratures over F_0* when algebraic, integral, and exponential integral elements may be added.

定理 4.1 (Liouville)

A function is elementary (a generalized elementary function) if and only if it is an elementary (generalized elementary) element over the field of rational functions $\mathcal{R}$. A function is representable by quadratures (representable by generalized quadratures) if and only if it is representable by quadratures (representable by generalized quadratures) over the field of complex numbers $\mathbb{C}$.



For example, it follows from Theorem 1 that the basic elementary function $f(x) = \arctan x$ is representable by quadratures over the field $F_0 = \mathbb{C}$. Indeed, this becomes clear from the equations $f' \equiv \frac{1}{1+x^2}$, $x' \equiv 1$.

To prove, for example, the part of Theorem 1 which concerns functions representable by quadratures, it suffices to verify first that there exist analogous representations for all the basic elementary functions and, furthermore, that the class of functions representable by quadratures over the field $\mathbb{C}$ is closed with respect to composition.

Liouville constructed a nice theory about the solvability of equations. Let us show two examples of his results.

定理 4.2 (Liouville)

The indefinite integral $y(x)$ of the algebraic function $A(x)$ of one complex variable is representable by generalized elementary functions if and only if it is representable in the form

$$y(x) = \int^x A(t) dt = A_0(x) + \sum_{i=1}^k \lambda_i \ln A_i(x),$$

where the $A_i(x)$, for $i = 0, 1, \dots, k$, are algebraic functions.



A priori the integral of an algebraic function could be given by a very complicated formula. It could have the form

$$y = \exp(\exp(\exp(\exp(\exp(x)))))$$

Theorem 2 says that this does not happen. Either the integral of an algebraic function can be written in a simple way, or in general it is not a generalized elementary function.

定理 4.3 (Liouville)

The linear differential equation

$$y'' + p(x)y' + q(x)y = 0 \tag{A.11}$$

where $p(x)$ and $q(x)$ are rational functions, is solvable by generalized quadratures if and only if its solution can be written in the form

$$y = \exp\left(\int^x R(t) dt\right),$$

where $R(x)$ is an algebraic function.



A priori the solution of equation (A.11) could be expressed by very complicated formulae. Theorem 3 says that this is nowhere the case. Either the equation has sufficiently simple roots, or in general it cannot be solved by generalized quadratures.

Liouville found a series of results of this type. The common idea is the following: simple equations have either simple solutions, or in general have no solutions in a given class (by quadratures, by elementary functions, etc.).

The strategy of the proof in Liouville’s theory is the following: prove that if a simple equation has a solution which is represented by a complicated formula, then this formula can always be simplified.

Liouville, undoubtedly, was inspired by the results of Lagrange, Abel, and Galois on the non-solvability by radicals of algebraic equations. Differently from Galois theory, Liouville’s theory does not involve the notion of the group of automorphisms. Liouville, however, uses, in order to simplify his formulae, “infinitely small automorphisms.”

Let us return to Theorem 2 on the integrability of algebraic functions. The following corollary follows from this theorem.

推论 4.1

If the integral of an algebraic function A is a generalized elementary function, then the differential form $A(x) dx$ has some unavoidable singularities on the Riemann surface of the algebraic function A .



It is well known that on every algebraic curve with positive genus there exist non-singular differential forms (the so-called abelian differentials of the first type). It follows that algebraic functions whose Riemann surfaces have positive genus are not, in general, integrable by generalized elementary functions.

This was already known by Abel, who discovered it as he was proving the non-solvability by radicals of a generic equation of fifth degree. Observe also that Abel’s proof of non-solvability by radicals is based on topological arguments. I do not know whether the topological properties of the Riemann surfaces of functions representable by generalized quadratures are different from those of the Riemann surfaces of generalized elementary functions. Indeed, I am unable to prove through topological arguments that the integral of an algebraic function is not an elementary function: each one of such integrals is by definition a function representable by generalized quadratures. However, if an algebraic function depends on a parameter, its integral may depend on the parameter in an arbitrarily complicated manner. One can prove that the integral of an algebraic function, as a function of one parameter, may not be representable by generalized quadratures, and consequently may not be a generalized elementary function of the parameter (cf. the example in §A.9).

4.1 Picard–Vessiot’s theory

Consider the linear differential equation

$$y^{(n)} + r_1(x)y^{(n-1)} + \cdots + r_n(x)y = 0, \quad (\text{A.12})$$

in which the $r_i(x)$ are rational functions of a complex argument.

Near a non-singular point x_0 there exist n linearly independent solutions $y_1, \dots, y_n$ of equation (A.12). In this neighbourhood one can consider the function field $\mathcal{R}\{y_1, \dots, y_n\}$, obtained by adding to the field of

rational functions $\mathcal{R}$ all solutions y_i and all their derivatives $y_i^{(p)}$ up to order $n - 1$. (The derivatives of higher order are obtained from equation (A.12).)

The field of functions $\mathcal{R}\{y_1, \dots, y_n\}$ is a differential field, i.e., it is closed with respect to differentiation, as is the field $\mathcal{R}$ of rational functions. One calls an automorphism of the differential field F an automorphism σ of the field F which also preserves differentiation, i.e., $\sigma(g') = [\sigma(g)]'$. Consider an automorphism σ of the differential field $\mathcal{R}\{y_1, \dots, y_n\}$ which fixes all elements of the field $\mathcal{R}$. The set of all automorphisms of this type forms a group which is called the Galois group of equation (A.12). Every automorphism σ of the Galois group sends a solution of the equation to a solution of the equation. Hence to each such automorphism there corresponds a linear transformation M_σ of the space V^n of solutions. The automorphism σ is completely defined by the transformation M_σ , because the field $\mathcal{R}\{y_1, \dots, y_n\}$ is generated by the functions y_i . In general, not every linear transformation of the space V^n is an automorphism σ of the Galois group. The reason is that the automorphism σ preserves all differential relations holding among the solutions. The Galois group can be considered as a special group of linear transformations of the solutions. It turns out that this group is algebraic.

Thus the Galois group of an equation is the algebraic group of linear transformations of the space of solutions that preserves all differential relations between the solutions.

Picard began to translate systematically Galois theory to the case of linear differential equations. As in the original Galois theory, one also finds here a one-to-one correspondence (the Galois correspondence) between the intermediate differential fields and the algebraic subgroups of the Galois group.

Picard and Vessiot proved in 1910 that the solvability of an equation by quadratures and by generalized quadratures depends exclusively on its Galois group.

定理 4.4 (Picard–Vessiot)

A differential equation is solvable by quadratures if and only if its Galois group is soluble. A differential equation is solvable by generalized quadratures if and only if the connected component of unity in its Galois group is soluble.



The reader can find the basic results of differential Galois theory in the book [2]. In [3] the reader will find a brief exposition of the current state of this theory together with a rich bibliography.

Observe that from the Picard–Vessiot theorem it is not difficult to deduce that if equation (A.12) is solvable by generalized quadratures, then it has a solution of the form

$$y_1 = \exp \left(\int^x A_1(t) dt \right),$$

where $A_1(x)$ is an algebraic function. If the equation has an explicit solution y_1 , then one can decrease its order, taking as the new unknown function $z = (y/y_1)'$. The function z satisfies a differential equation having an explicit form and a lower order. If the initial equation was solvable, the new equation for the function z is also solvable. By the Picard–Vessiot theorem it must therefore have a solution of the type

$$z_1 = \exp \left(\int A_2(x) dx \right),$$

where A_2 is an algebraic function, and so on. We see in this way that if a linear equation is solvable by generalized quadratures, the formulae expressing the solutions are not exceedingly complicated. Here the Picard–Vessiot approach coincides with the Liouville approach. Moreover, the criterion of solvability by generalized quadra-

tures can be formulated without mentioning the Galois group. Indeed, equation (A.12) of order n is solvable by generalized quadratures if and only if it has a solution of the form $y_1 = \exp \left(\int^x A(x) dx \right)$, and the equation of order $n - 1$ for the function z is solvable by generalized quadratures.

This theorem was enunciated and proved by Murdakai–Boltovskii exactly in this form. Murdakai and Boltovskii obtained this result at the same time in 1910 using Liouville’s method, independently of the works of Picard and Vessiot. The Mordukai–Boltovskii theorem is a generalization of Liouville’s theorem (cf. Theorem 3 in the preceding section) to linear differential equations of any order.

4.2 Topological obstructions for the representation of functions by quadratures

There exists a third approach to the problem of the representability of a function by quadratures (cf. [4]–[10]). Consider the functions representable by quadratures as multi-valued analytic functions of a complex variable. It turns out that there are some topological restrictions on the kind of disposition on the complex plane of the Riemann surface of a function representable by quadratures. If the function does not satisfy these conditions, it cannot be represented by quadratures.

This approach, besides its geometrical evidence, possesses the following advantage. The topological obstructions are related to the character of the multi-valued function. They hold not only for functions representable by quadratures, but also for a wider class of functions. This class is obtained by adding to the functions representable by quadratures all the meromorphic functions and allowing the presence of such functions in all formulae. Hence the topological results on non-representability by quadratures are stronger than those of an algebraic nature. The reason for this is that the composition of two functions is not an algebraic operation. In differential algebra, instead of the composition of two functions one considers the differential equation that they satisfy. But, for instance, the Euler function Γ does not satisfy any algebraic differential equation; therefore it is useless to seek an equation satisfied, for example, by the function $\Gamma(\exp x)$. The only known results on the non-representability of functions by quadratures and, for instance, by the Euler function Γ are those obtained by our method.

On the other hand, by this method one cannot prove the non-representability by quadratures of an arbitrary meromorphic single-valued function.

Using differential Galois theory (and, to be precise, its linear-algebraic part, related to matrix algebraic groups and their differential invariants), one can prove that the sole reason for the non-solvability by quadratures of linear differential equations of Fuchs type (cf. §A.11) is of a topological nature. In other words, when there are no topological obstructions to solvability by quadratures for a differential equation of Fuchs type, this equation is solvable by quadratures.

The topological obstructions for the representation of a function by quadratures and by generalized quadratures are the following.

First, functions representable by generalized quadratures and, as a special case, by quadratures can have at most a countable set of singular points on the complex plane (cf. §A.5), even though for the simplest functions representable by quadratures the set of singular points may be everywhere dense.

Second, the monodromy group of a function representable by quadratures is necessarily soluble (cf. §A.7), whereas for the simplest functions representable by quadratures the monodromy group may already contain a continuum of elements.

There also exist analogous topological restrictions on the disposition of the Riemann surface for functions representable by generalized quadratures. However, these restrictions cannot be simply formulated: in this case the monodromy group is not considered as an abstract group, but as the group of permutations of the sheets of the Riemann surface. In other words, in the formulation of such restrictions not only the monodromy group intervenes, but also the monodromy pair of the function. The monodromy pair of a function consists of its monodromy group and of a stationary subgroup for some germ (cf. §A.9). We shall see this geometrical approach to the problem of solvability more precisely.

4.3 S-functions

We define a class of functions which will be the object of this section.

定义 4.1

An analytic multi-valued function of a complex variable is called an $\mathcal{S}$ -function if the set of its singular points is at most countable.



Let us make this definition more precise. Two regular germs f_a and g_b , defined at points a and b on the Riemann sphere S^2 , are said to be equivalent if the germ g_b is obtained from the germ f_a by a regular continuation along some curve. Every germ g_b equivalent to the germ f_a is called a regular germ of the analytic multi-valued function f generated by the germ f_a .

A point $b \in S^2$ is said to be singular for the germ f_a if there exists a curve $\gamma: [0, 1] \rightarrow S^2$, $\gamma(0) = a$, $\gamma(1) = b$, such that the germ f_a cannot be regularly continued along this curve, but for every t , $0 \leq t < 1$, this germ can be continued along the shortened curve $\gamma: [0, t] \rightarrow S^2$. It is easy to see that the sets of singular points for equivalent germs coincide.

A regular germ is called an $\mathcal{S}$ -germ if the set of its singular points is at most countable. An analytic multi-valued function is called an $\mathcal{S}$ -function if each one of its regular germs is an $\mathcal{S}$ -germ.

定理 4.5 (Closure of the class of $\mathcal{S}$ -functions)

See [6], [8], [10]. The class $\mathcal{S}$ of all $\mathcal{S}$ -functions is closed with respect to the following operations:

1. differentiation, i.e., if $f \in \mathcal{S}$, then $f' \in \mathcal{S}$;
2. integration, i.e., if $f \in \mathcal{S}$, then $\int f(x) dx \in \mathcal{S}$;
3. composition, i.e., if $g, f \in \mathcal{S}$, then $g \circ f \in \mathcal{S}$;
4. meromorphic operation, i.e., if $f_i \in \mathcal{S}$, $i = 1, \dots, n$, $F(x_1, \dots, x_n)$ is a meromorphic function of n variables and $f = F(f_1, \dots, f_n)$, then $f \in \mathcal{S}$.⁴
5. solution of algebraic equations, i.e., if $f_i \in \mathcal{S}$, $i = 1, \dots, n$, and $f^n + f_1 f^{n-1} + \dots + f_n = 0$, then $f \in \mathcal{S}$;
6. solution of linear differential equations, i.e., if $f_i \in \mathcal{S}$, $i = 1, \dots, n$, and $f^{(n)} + f_1 f^{(n-1)} + \dots + f_n = 0$, then $f \in \mathcal{S}$.



推论 4.2

If the multi-valued function f can be obtained from single-valued $\mathcal{S}$ -functions by the operations of integration, differentiation, meromorphic operations, compositions, and solutions of algebraic and linear differential equations, then the function f has at most a countable set of singular points. In particular, a function having an uncountable set of singular points is not representable by generalized quadratures. 

4.4 Monodromy group

The monodromy group of an $\mathcal{S}$ -function f with a set A of singular points is the group of all permutations of the sheets of the Riemann surface of f which are visited when one moves around the points of the set A .

More precisely, let F_a be the set of all germs of the $\mathcal{S}$ -function f at a point a not belonging to the set A of singular points. Consider a closed curve γ in $S^2 \setminus A$ beginning at the point a . The continuation of every germ of the set F_a along the curve γ leads to a germ of the set F_a .

Consequently, to every curve γ there corresponds a mapping of the set F_a into itself, and to homotopic curves in $S^2 \setminus A$ there corresponds the same mapping. To the composition of curves there corresponds the composition of mappings. One has thus defined a homomorphism τ of the fundamental group of the set $S^2 \setminus A$ into the group $S(F_a)$ of the bijective mappings of the set F_a into itself. One calls the monodromy group of the $\mathcal{S}$ -function f the image of the fundamental group $\pi_1(S^2 \setminus A, a)$ in the group $S(F_a)$ under the homomorphism τ .

We show some results which are useful in the study of functions representable by quadratures as functions of one complex variable.

Example. Consider the function $w(z) = \ln(1 - z^\alpha)$, where $\alpha > 0$ is an irrational number. The function w is an elementary function given by a very simple formula. However, its Riemann surface is very complicated. The set A of its singular points consists of the points $0, \infty$, and the points $a_k = e^{2\pi ki/\alpha}$, where k is any integer. Since α is irrational, the points a_k are densely distributed on the unit circle. It is not difficult to prove that the fundamental group $\pi_1(S^2 \setminus A)$ and the monodromy group of the function w are continuous. One can also prove that the image under the homomorphism τ of the fundamental group $\pi_1(S^2 \setminus (A \cup \{b\}))$ of the complement of $A \cup \{b\}$, where $b \neq a_k$ is an arbitrary point on the unit circle, is a proper subgroup of the monodromy group of the function w . (That the elimination of a single point can produce a radical change in the monodromy group makes all proofs essentially difficult.)

4.5 Obstructions for the representability of functions by quadratures

We have proved the following theorem.

定理 4.6

See [6], [8], [10]. The class of all $\mathcal{S}$ -functions having a soluble monodromy group is closed with respect to composition, meromorphic operations, integration, and differentiation. 

We thus obtain the following corollary.

推论 4.3 (Result on quadratures)

The monodromy group of a function f representable by quadratures is soluble. Moreover, the monodromy group of every function f which is obtained from single-valued $\mathcal{S}$ -functions by means of compositions, meromorphic operations, integration, and differentiation is also soluble.



We now see the application of this result to algebraic equations.

4.6 Solvability of algebraic equations

Consider the algebraic equation

$$y^n + r_1 y^{n-1} + \cdots + r_n = 0, \quad (\text{A.13})$$

where the r_i are rational functions of the complex variable x .

Near a non-singular point x_0 there are all the solutions $y_1, \dots, y_n$ of equation (A.13). In this neighbourhood one can consider the field of all functions $\mathcal{R}\{y_1, \dots, y_n\}$ that is obtained by adding to the field $\mathcal{R}$ all solutions y_i .

Consider the automorphisms σ of the field $\mathcal{R}\{y_1, \dots, y_n\}$ which fix every element of the field $\mathcal{R}$. The totality of these automorphisms forms a group which is called the Galois group of equation (A.13). Every automorphism σ of the Galois group transforms a solution of the equation into a solution of the equation; consequently, to every automorphism σ there corresponds a permutation S_σ of the solutions. The automorphism σ is completely defined by the permutation S_σ because the field $\mathcal{R}\{y_1, \dots, y_n\}$ is generated by the functions y_i . In general, not all permutations of the solutions can be continued to an automorphism σ of the Galois group: the reason is that the automorphisms σ preserve all relations existing among the solutions.

The Galois group of an equation is thus the permutation group of the solutions that preserves all relations among the solutions.

Every permutation S_γ of the set of solutions can be continued, as an automorphism of the monodromy group, to an automorphism of the entire field $\mathcal{R}\{y_1, \dots, y_n\}$. Indeed, with the functions $y_1, \dots, y_n$, every element of the field $\mathcal{R}\{y_1, \dots, y_n\}$ is continued meromorphically along the curve γ . This continuation gives the required automorphism, because during the continuation the arithmetic operations are preserved and every rational function returns to its preceding value because of uniqueness.

In this way the monodromy group of the equation is contained in the Galois group; in fact, the Galois group coincides with the monodromy group. Indeed, the functions of the field $\mathcal{R}\{y_1, \dots, y_n\}$ that are fixed under the action of the monodromy group are the single-valued functions. These functions are algebraic, but every algebraic single-valued function is a rational function. Therefore the monodromy group and the Galois group have the same field of invariants, and thus by Galois theory they coincide.

According to Galois theory, equation (A.13) is solvable by radicals over the field of rational functions if and only if its Galois group is soluble over this field. In other words, Galois theory proves the following theorems:

1. An algebraic function y whose monodromy group is soluble is representable by radicals.
2. An algebraic function y whose monodromy group is not soluble is not representable by radicals.

Our theorem makes result 2 stronger:

An algebraic function y whose monodromy group is not soluble cannot be represented through single-valued $\mathcal{S}$ -functions by means of meromorphic operations, compositions, integrations, and differentiations.

If an algebraic equation is not solvable by radicals, then it remains non-solvable using logarithms, exponentials, and the other meromorphic functions on the complex plane. A stronger version of this statement is given in §A.15.

4.7 The monodromy pair

The monodromy group of a function is not only an abstract group but is the group of transitive permutations of the sheets of its Riemann surface. Algebraically this object is given by a pair of groups: the permutation group and a subgroup of it, the stationary group of a certain element.

One calls the monodromy pair of an $\mathcal{S}$ -function a pair of groups consisting of the monodromy group of this function and the stationary subgroup of a sheet of the Riemann surface. The monodromy pair is defined correctly, i.e., this pair of groups up to isomorphism does not depend on the choice of the sheet.

定义 4.2

The pair of groups $[\Gamma, \Gamma_0]$ is called an almost soluble pair of groups if there exists a sequence of subgroups

$$\Gamma = \Gamma_1 \supseteq \cdots \supseteq \Gamma_m, \quad \Gamma_m \subset \Gamma_0,$$

such that for every i , $1 \leq i \leq m-1$, the group Γ_{i+1} is a normal divisor of the group Γ_i , and the quotient group Γ_i/Γ_{i+1} is either commutative or finite.



Any group Γ can be considered as the pair of groups $[\Gamma, e]$, where e is the unit subgroup (the group containing only the unit element). We say that the group Γ is almost soluble if the pair $[\Gamma, e]$ is almost soluble.

定理 4.7

See [6], [8], [10]. The class of all $\mathcal{S}$ -functions having an almost soluble monodromy pair is closed with respect to composition, meromorphic operations, integration, differentiation, and solutions of algebraic equations.



We thus obtain the following corollary.

推论 4.4 (Result on generalized quadratures)

The monodromy pair of a function f representable by generalized quadratures is almost soluble. Moreover, the monodromy pair of every function f which is obtained from single-valued $\mathcal{S}$ -functions by means of composition, meromorphic operations, integration, differentiation, and solutions of algebraic equations is also almost soluble.



Let us now consider some examples of functions not representable by generalized quadratures. Suppose the Riemann surface of a function f is a universal covering of $S^2 \setminus A$, where S^2 is the Riemann sphere and A is a finite set containing at least three points. Thus the function f cannot be expressed in terms of $\mathcal{S}$ -functions by means of generalized quadratures, compositions, and meromorphic operations. Indeed, the monodromy pair of this function consists of a free non-commutative group and its unit subgroup. One easily sees that such a pair

of groups is not almost soluble.

Example. Consider the function z which realizes the conformal transformation of the upper half-plane into the triangle with vanishing angles, bounded by three arcs of circles. The function z is the inverse of the modular Picard function. The Riemann surface of the function z is a universal covering of the sphere without three points; consequently, the function z cannot be expressed in terms of single-valued $\mathcal{S}$ -functions by means of generalized quadratures, compositions, and meromorphic operations.

Observe that the function z is strictly related to the elliptic integrals

$$K(k) = \int_0^1 \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}}; \quad K'(k) = \int_0^{1/k} \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}}.$$

Each of the functions $K(k)$, $K'(k)$, and $z(w)$ can be obtained from the others by quadratures. It follows that neither of the integrals $K(k)$ and $K'(k)$ can be expressed in terms of single-valued $\mathcal{S}$ -functions by means of generalized quadratures, compositions, and meromorphic operations.

In the following section we will generalize the example above, finding all polygons bounded by arcs of circles to which the upper half-plane can be sent by functions representable by generalized quadratures.

4.8 Mapping of the half-plane to a polygon bounded by arcs of circles

4.8.1 Application of the symmetry principle

In the complex plane consider a polygon G bounded by arcs of circles. By Riemann's theorem, there exists a function f_G sending the upper half-plane to the polygon G . This mapping was studied by Riemann, Schwarz, Christoffel, Klein, and others (cf., for example, [11]). Let us recall some classical results which will be useful.

Denote by $B = \{b_j\}$ the pre-image of the set of vertices of the polygon G under the mapping f_G , by $H(G)$ the group of conformal transformations of the sphere generated by the inversions with respect to the sides of the polygon, and by $L(G)$ the subgroup of homographic mappings (the quotients of two linear functions). $L(G)$ is a subgroup of index 2 of the group $H(G)$. From the Riemann–Schwarz symmetry principle one obtains the following results.

命题 4.1

1. The function f_G can be meromorphically continued along any curve avoiding the set B .
2. All germs of the multi-valued function f_G at a non-singular point $a \notin B$ are obtained by applying to a given germ f_a the group $L(G)$ of homographic mappings.
3. The monodromy group of the function f_G is isomorphic to the group $L(G)$.
4. The singularities of the function f_G are of the following types at the points b_j . If at the vertex a_j of the polygon G that corresponds to the point b_j the angle is equal to $\alpha_j \neq 0$, then the function f_G , through a homographic transformation, is put into the form

$$f_G(z) = (z - b_j)^{\beta_j} \varphi(z), \quad \beta_j = \frac{\alpha_j}{2\pi},$$

where the function φ is holomorphic in a neighbourhood of the point b_j . If the angle α_j is equal

to zero, then the function f_G is put by a homographic transformation into the form

$$f_G(z) = \ln(z) + \varphi(z),$$

where the function φ is holomorphic in a neighbourhood of b_j .



From our results it follows that if the function f_G is representable by generalized quadratures, then the group $L(G)$ and the group $H(G)$ are almost soluble.

4.8.2 Almost soluble groups of homographic and conformal mappings

Let π be the epimorphism of the group $SL(2)$ of matrices of order 2 with unit determinant onto the group L of homographic mappings,

$$\pi: \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \frac{az + b}{cz + d}.$$

Since $\ker \pi = \mathbb{Z}_2$, the group $\tilde{L} \subseteq L$ and the group $\pi^{-1}(\tilde{L}) = \Gamma \subseteq SL(2)$ are both almost soluble. The group Γ is a group of matrices; therefore Γ is almost soluble if and only if it has a normal subgroup Γ_0 of finite index which admits a triangular form. (This version of Lie's theorem is also true in higher dimensions and plays an important role in differential Galois theory.) Since the group Γ_0 consists of matrices of order 2, the group Γ_0 can be put into triangular form in one of the following three cases:

1. the group Γ_0 has only one one-dimensional eigenspace;
2. the group Γ_0 has two one-dimensional eigenspaces;
3. the group Γ_0 has a two-dimensional eigenspace.

Consider now the group of homographic mappings $\tilde{L} = \pi(\Gamma)$. The group $\tilde{L}$ of homographic mappings is almost soluble if and only if it has a normal subgroup $L_0 = \pi(\Gamma_0)$ of finite index, and the set of invariant points consists of either a unique point, two points, or the whole Riemann sphere.

The group of conformal mappings H contains the group L of index 2 (or of index 1) consisting of the homographic mappings. Hence an analogous proposition holds for an almost soluble group $\tilde{H}$ of conformal mappings.

引理 4.1

A group of conformal mappings of the sphere is almost soluble if and only if it satisfies at least one of these conditions:

1. the group has only one invariant point;
2. the group has an invariant set consisting of two points;
3. the group is finite.



This lemma follows from the preceding propositions because the set of invariant points for a normal divisor is invariant under the action of the group. It is well known that a finite group $\tilde{L}$ of homographic mappings of the sphere is sent by a homographic transformation of coordinates to a group of rotations.

It is not difficult to prove that if the product of two inversions with respect to two different circles corresponds under stereographic projection to a rotation of the sphere, then these circles correspond to great circles.

Hence every finite group $\tilde{H}$ of conformal mappings generated by inversions with respect to some circles is sent by a homographic transformation of coordinates to a group of motions of the sphere generated by reflections.

All the finite groups of motions of the sphere generated by reflections are well known. They are exactly the symmetry groups of the following objects:

1. the regular pyramid with a regular n -gon as base;
2. the n -dihedron, i.e., the solid made from two regular pyramids joining their bases;
3. the tetrahedron;
4. the cube or the octahedron;
5. the dodecahedron or the icosahedron.

All these groups of symmetries, except the group of the dodecahedron–icosahedron, are soluble. The sphere whose centre coincides with the centre of gravity of the solid is cut by the symmetry planes of the solid along a net of great circles. Lattices corresponding to the stated solids are called finite nets of great circles. The stereographic projections of these finite nets are shown in Figures 129–133.

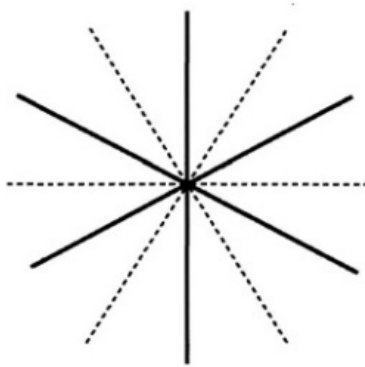


FIG. 129: PYRAMID

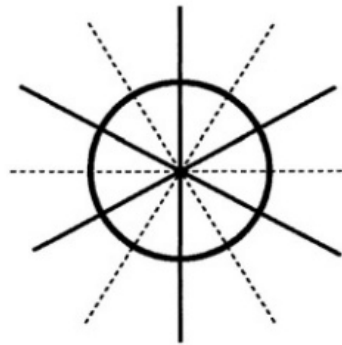


FIG. 130: 6-DIHEDRON

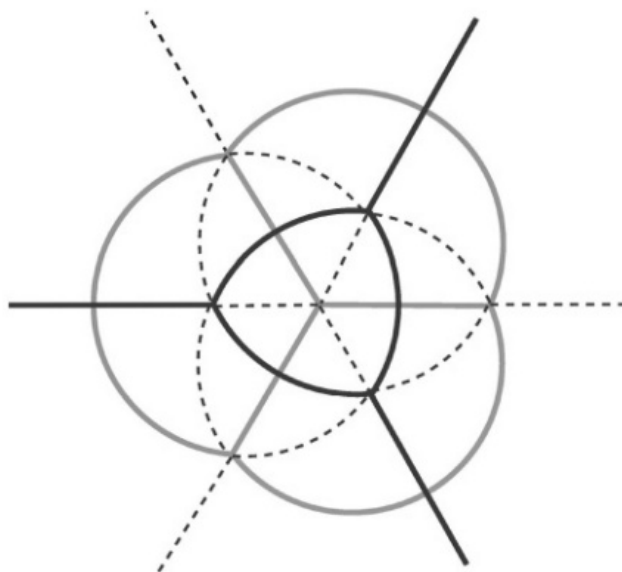


图 4.1: Figures 129–131: pyramid; 6-dihedron; tetrahedron–tetrahedron.

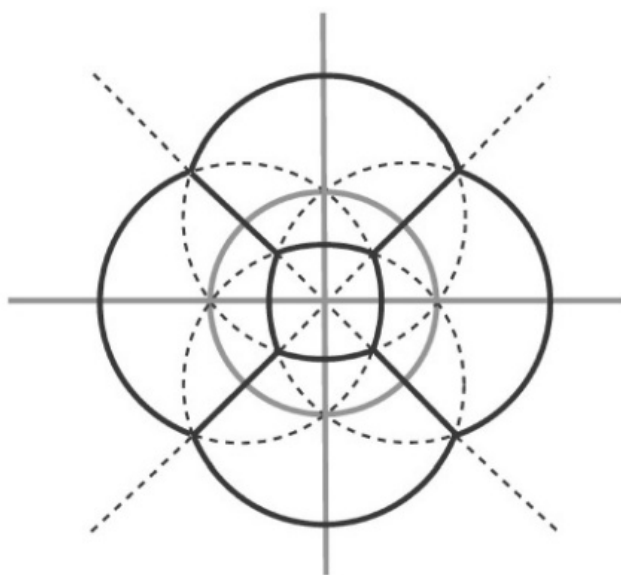


图 4.2: Figure 132: cube-octahedron.

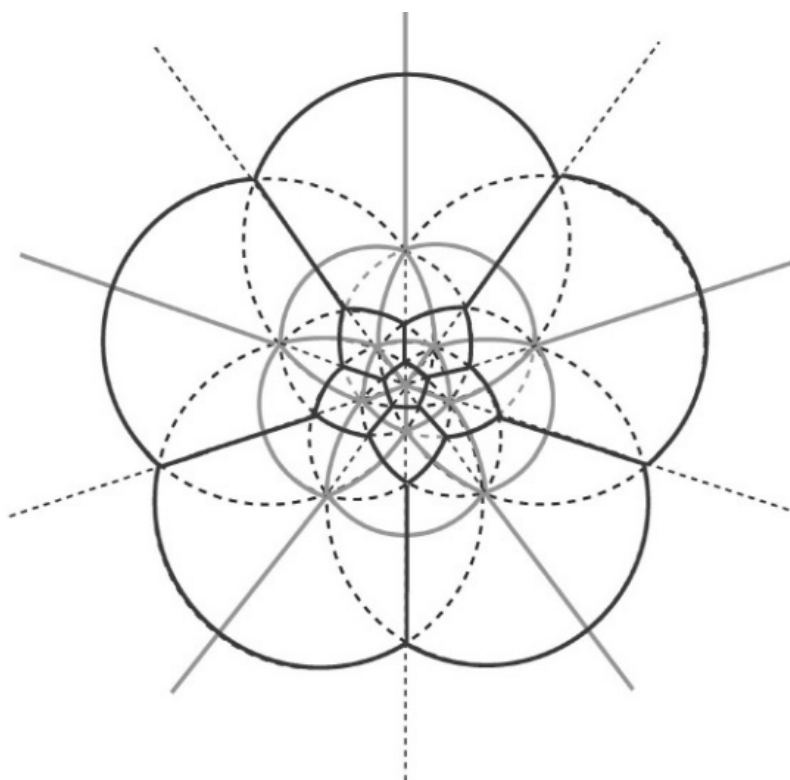


图 4.3: Figure 133: dodecahedron-icosahedron.

4.8.3 The integrable case

Let us come back to the problem of the representability of the function f_G by generalized quadratures.

We now consider the different possible cases and prove that the condition we have found is not only necessary but also sufficient for the representability of the function f_G by generalized quadratures.

First integrability case. The group $H(G)$ has an invariant point. This means that the continuations of the edges of the polygon G intersect at a point. Sending this point to infinity by a homographic transformation, we obtain the polygon $\bar{G}$ bounded by segments of straight lines (cf. Figure 134).

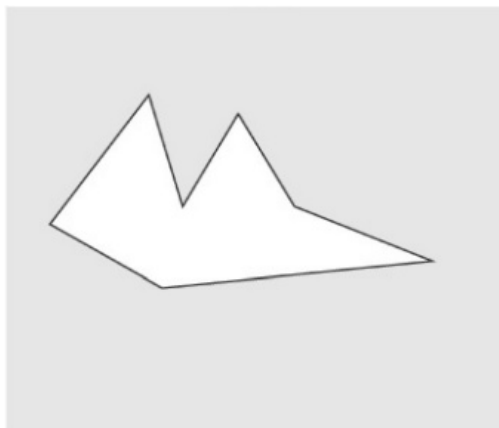


图 4.4: Figure 134.

All mappings in $L(\bar{G})$ have the form $z \mapsto az + b$. All germs of the function $\bar{f} = f_{\bar{G}}$ at a non-singular point c are obtained by applying to a given germ $\bar{f}_c$ the group $L(\bar{G})$, $\bar{f}_c \mapsto a\bar{f}_c + b$. The germ

$$R_c = \frac{\bar{f}_c''}{\bar{f}_c'}$$

is invariant under the action of the group $L(\bar{G})$. This means that the germ R_c is the germ of a single-valued function. A singular point b_j of the function R_c can only be a pole (cf. the proposition in §A.10.1). Thus the function R_c is rational. The equation

$$\frac{\bar{f}''}{\bar{f}'} = R$$

is integrable by quadratures. This case of integrability is well known. The function $\bar{f}$ in this case is called the Christoffel–Schwarz integral.

Second integrability case. The invariant set of the group $H(G)$ consists of two points. This means that there are two points with the following properties: for every side of the polygon G , these points either are obtained from one another by an inversion with respect to this side or belong to the continuation of this side. Sending one of these points to the origin and the other one to infinity by a homographic transformation, we obtain the polygon $\bar{G}$ bounded by arcs of circles with centre at the point 0 and by segments of rays coming from the point 0 (cf. Figure 135).

All transformations of the group $L(\bar{G})$ are of the form $z \mapsto az$, $z \mapsto b/z$. All germs of the function $\bar{f} = f_{\bar{G}}$

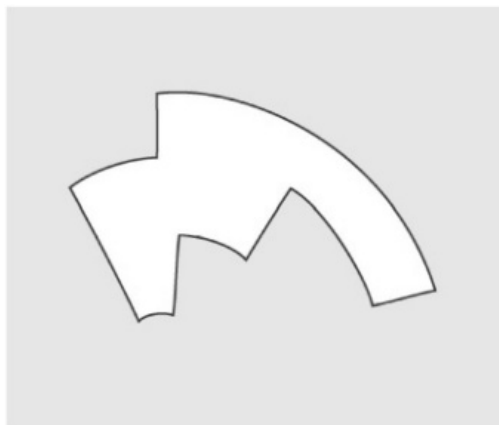


图 4.5: Figure 135.

at a non-singular point c are obtained by applying to a given germ $\bar{f}_c$ the transformations of the group $L(\bar{G})$:

$$\bar{f}_c \mapsto a\bar{f}_c, \quad \bar{f}_c \mapsto \frac{b}{\bar{f}_c}.$$

The germ $R_c = (\bar{f}'_c/\bar{f}_c)^2$ is invariant under the action of the group $L(\bar{G})$ and it is the germ of the single-valued function R . The only singularities of the function R are poles (cf. the proposition in §A.10.1). Thus the function R is rational. The equation

$$R = \left(\frac{\bar{f}'}{\bar{f}} \right)^2$$

is integrable by quadratures.

Third integrability case. The group $H(G)$ is finite. This means that the polygon G is sent by a homographic transformation to a polygon $\bar{G}$ whose sides lie on a finite net of great circles (see Figures 129–133). The group $L(G)$ is finite, and as a consequence the function f_G has a finite number of values. Since all singularities of the function f_G are of “jump” type (cf. the proposition in §A.10.1), the function f_G is an algebraic function.

Let us analyze the case in which the group $H(G)$ is finite and soluble. This happens if and only if the polygon G is sent by a homographic transformation to a polygon $\bar{G}$ whose sides lie on a net of great circles different from that of the dodecahedron–icosahedron. In this case the group $L(G)$ is soluble, and the function f_G is expressed in terms of rational functions by means of arithmetic operations and radicals (cf. §A.8).

From our results the following theorem follows.

定理 4.8 (Polygons bounded by arcs of circles)

See [6], [8], [10]. For an arbitrary polygon G not belonging to the three cases of integrability above, the function f_G not only is not representable by generalized quadratures, but cannot be expressed in terms of single-valued $\mathcal{S}$ -functions by means of generalized quadratures, compositions, and meromorphic operations.



4.9 Topological obstructions for the solvability of differential equations

4.9.1 The monodromy group of a linear differential equation and its relation with the Galois group

Consider the linear differential equation

$$y^{(n)} + r_1 y^{(n-1)} + \cdots + r_n y = 0, \quad (\text{A.14})$$

where the r_i are rational functions of the complex variable x . The poles of the functions r_i and ∞ are called the singular points of equation (A.14).

Near a non-singular point x_0 the solutions of the equation form a space V^n of dimension n . Consider now an arbitrary curve $\gamma(t)$ on the complex plane, beginning at x_0 and ending at the point x_1 and avoiding the singular points a_i . The solutions of the equation can be analytically continued along the curve, remaining solutions of the equation. Hence to every curve γ there corresponds a linear mapping M_γ of the space $V_{x_0}^n$ of solutions at the point x_0 into the space $V_{x_1}^n$ of solutions at the point x_1 .

If one changes the curve γ , avoiding the singular points and leaving its ends fixed, the mapping M_γ does not vary. Hence to a closed curve there corresponds a linear transformation of the space V^n into itself. The totality of these linear transformations of the space V^n forms a group which is called the monodromy group of equation (A.14). Thus the monodromy group of an equation is the group of linear transformations of the solutions which correspond to different turns around the singular points. The monodromy group of an equation characterizes the property of its solutions being multi-valued.

Near a non-singular point x_0 there are n linearly independent solutions $y_1, \dots, y_n$ of equation (A.14). In this neighbourhood one can consider the field of functions $\mathcal{R}\{y_1, \dots, y_n\}$ that is obtained by adding to the field of rational functions $\mathcal{R}$ all solutions y_i and all their derivatives.

Every transformation M_γ of the monodromy group of the space of solutions can be continued to an automorphism of the entire field $\mathcal{R}\{y_1, \dots, y_n\}$. Indeed, together with the functions $y_1, \dots, y_n$, every element of the field $\mathcal{R}\{y_1, \dots, y_n\}$ can be analytically continued along the curve γ . This continuation gives the required automorphism, because during continuation arithmetic operations and differentiation are preserved, and rational functions return to their initial values because of their uniqueness.

In this way the monodromy group of an equation is contained in its Galois group.

The field of invariants of the monodromy group is a subfield of $\mathcal{R}\{y_1, \dots, y_n\}$ consisting of the single-valued functions. Differently from the algebraic case, for differential equations the field of invariants under the action of the monodromy group can be bigger than the field of rational functions.

For example, for equation (A.14), in which all the coefficients $r_i(x)$ are polynomials, all solutions are single-valued. But of course the solutions of such equations are not always polynomials. The reason is that here the solutions of differential equations may grow exponentially while approaching the singular points. One knows an extension of the class of linear differential equations for which there are no similar complications, i.e., for which the solutions, while approaching the singular points, grow at most as some power. Differential equations which possess this property are called equations of Fuchs' type.

For differential equations of Fuchs' type the Frobenius theorem holds.

定理 4.9

For differential equations of Fuchs' type, the subfield of the differential field $\mathcal{R}\{y_1, \dots, y_n\}$ that consists of single-valued functions coincides with the field of rational functions.



According to differential Galois theory, it follows from the Frobenius theorem that the algebraic closure of the monodromy group M (i.e., the smallest algebraic group containing M) coincides with the Galois group.

Differential Galois theory thus gives the following criterion of solvability of differential equations of Fuchs' type.

定理 4.10

A differential equation of Fuchs' type is solvable by quadratures or by generalized quadratures if its monodromy group is, respectively, soluble or almost soluble.



Differential Galois theory provides at the same time two results:

1. if the monodromy group of a differential equation of Fuchs' type is soluble (almost soluble), then this equation is solvable by quadratures (by generalized quadratures);
2. if the monodromy group of a differential equation of Fuchs' type is not soluble (almost soluble), then this equation is not solvable by quadratures (by generalized quadratures).

Our theorem makes result 2 stronger. Indeed, it is easy to see that for almost every solution of differential equation (A.14) the monodromy pair is $[M, e]$, where M is the monodromy group of the equation and e its trivial subgroup. We thus have the following.

定理 4.11

See [6], [8]. If the monodromy group of differential equation (A.14) is not soluble (almost soluble), then almost every solution of this equation is not representable in terms of single-valued $\mathcal{S}$ -functions by means of compositions, meromorphic operations, integrations, differentiations, and solutions of algebraic equations.



Is the monodromy group of a given linear differential equation soluble (almost soluble)? This question turns out to be quite difficult. However, there exists an interesting example in which the answer to this question is very simple.

4.9.2 Systems of differential equations of Fuchs' type with small coefficients

Consider a system of linear differential equations of Fuchs' type, i.e., a system of the type

$$y' = A(x)y \quad (\text{A.15})$$

where $y = (y_1, \dots, y_n)$ is the unknown vector-valued function and A is an $n \times n$ matrix consisting of rational functions of the complex variable x , having the form

$$A(x) = \sum_{i=1}^k \frac{A_i}{x - a_i},$$

where the A_i are constant matrices.

If the matrices A_i can be simultaneously put into triangular form, then the system (A.15), like every triangular system, is solvable by quadratures. There are undoubtedly non-triangular systems which are solvable. However, if the matrices A_i are sufficiently small, then such systems do not exist. More precisely, we have obtained the following result.

Theorem 4 ([9]). A non-triangular system (A.15) with matrices A_i sufficiently small, $\|A_i\| < \varepsilon(a_1, \dots, a_k, n)$, is strictly non-solvable; that is, it is not solvable even using all single-valued $\mathcal{S}$ -functions, compositions, meromorphic operations, integrations, differentiations, and solutions of algebraic equations.

The proof of this theorem uses the Lappo-Danilevskij theory [12].

4.10 Algebraic functions of several variables

Up to now we have considered only single-valued functions. We are ready to make two observations concerning functions of several variables, the proofs of which do not require new notions and are obtained by the same method we used for single-valued functions.

Consider the algebraic equation

$$y^n + r_1 y^{n-1} + \dots + r_n = 0. \quad (\text{A.16})$$

Here the r_i are rational functions of k complex variables $x_1, \dots, x_k$.

1. According to Galois theory, equation (A.16), when it has a soluble monodromy group, is solvable by radicals. But if the monodromy group of equation (A.16) is not soluble, then not only is the equation not solvable by radicals, but it cannot be solved even by using radicals of entire functions of several variables, arithmetic operations, and compositions. This statement can be considered a variation of Abel's theorem about the non-solvability of algebraic equations of degree higher than four. (A stronger result is presented in §A.15.)
2. Equation (A.16) defines an algebraic function of k variables. What are the conditions for representing a function of k variables by algebraic functions of a smaller number of variables, using compositions and arithmetic operations? Hilbert's thirteenth problem consists of this question⁵. If one excludes the remarkable results [13, 14] on this subject⁶,

$$\lambda^n + z_1 \lambda^{n-1} + \dots + z_n = 0,$$

up to now there had been no proof that there exist algebraic functions of several variables which are not representable by algebraic functions of a single variable.

We know, however, the following result.

Theorem ([4], [5]). An entire function y of two variables (a, b) , defined by the equation

$$y^5 + ay + b = 0,$$

cannot be expressed in terms of entire functions of a single variable by means of compositions, additions, and subtractions.

The reason is the following. To every singular point p of an algebraic function one can associate a local monodromy group, that is, the group of permutations of the sheets of the Riemann surface obtained by going around the singularities of the function along curves lying in an arbitrarily small neighbourhood of p . For algebraic functions of one variable this local group is commutative; as a consequence, the local monodromy group of an algebraic function expressed by means of sums and differences of integer functions must be soluble. But the local monodromy group of the function

$$y^5 + ay + b = 0$$

near the point $(0, 0)$ is the group $S(5)$ of all permutations of five elements, which is not soluble. This explains the statement of the theorem.

Observe that if the operation of division is allowed, then the above argument no longer holds. Indeed, division is an operation that destroys continuity, and its application destroys locality. In fact, the function y satisfying

$$y^5 + ay + b = 0$$

can be expressed by means of division in terms of a function $g(x)$ of one variable, defined by the equation

$$g^5 + g + x = 0,$$

and the function of one variable $f(a) = \sqrt[4]{a}$. It is not difficult to see that

$$y(a, b) = g\left(\frac{b}{\sqrt[4]{a^5}}\right) \sqrt[4]{a}.$$

4.11 Functions of several complex variables representable by quadratures and generalized quadratures

The multidimensional case is more complicated than the one-dimensional case. We have to reformulate the basic definitions and, in particular, slightly change the definition of representability of functions by quadratures and by generalized quadratures. In this section we give a new formulation of the problem.

Suppose a class of basic functions and a set of allowed operations have been fixed. Is a given function (being, for instance, the solution of a given algebraic or differential equation, or the result of one of the other allowed operations) representable in terms of the basic functions by means of the allowed operations? First of all, we are interested in exactly this problem, but we give it a slightly different meaning. We consider the distinct single-valued branches of a multivalued function as single-valued functions on different domains; we also consider every multivalued function as the set of its single-valued branches. We apply the allowed operations (such as the arithmetic operations or composition) only to single-valued branches on different domains. Since our functions are analytic, it suffices to consider as domains only small neighbourhoods of points. The problem is now the following: is it possible to express a given germ of a function at a given point in terms of the germs of the basic functions by means of the allowed operations? Of course, here the answer depends on the choice of the single-valued germ of the multivalued function at that point. However, it happens that (for the class of basic functions in which we are interested) either the representation sought does not exist for any single-valued germ of the function at any point, or, on the contrary, all germs of the given multivalued function are expressed by the

same representation at almost all points. In the former case we say that no branches of the given multivalued function can be expressed in terms of the branches of the basic functions by means of the allowed operations; in the latter case we say that this representation exists.

First of all, observe the difference between this formulation of the problem and that of the problem expounded in §A.1. For analytic functions of a single variable there exists amongst the allowed operations, in fact, the operation of analytic continuation.

Consider the following example. Let f_1 be an analytic function, defined in a domain U of the plane $\mathbb{C}^1$, which cannot be continued beyond the boundary of U , and let f_2 be the analytic function in U defined by $f_2 = -f_1$. According to the definition in §A.1, the zero function is representable in the form $f_1 + f_2$ for all values of the argument. From this new point of view, the equation $f_1 + f_2 = 0$ is satisfied only inside U , not outside it. Previously we were not interested in the existence of a unique domain in which all required properties should hold for the single-valued branches of the multivalued function: one result of an operation could hold on one domain, and another result in another domain on the analytic continuations of the functions obtained. For the $\mathcal{S}$ -functions of a single variable one can obtain the needed topological limitations even with this extended notion of an operation on analytic multivalued functions. For functions of several variables we can no longer use this extended notion, and we must adopt a new formulation with some restrictions, which can seem less (but which is perhaps more) natural.

Let us start by giving precise definitions. Fix the standard space $\mathbb{C}^n$ with coordinate system $x_1, \dots, x_n$.

Definition.

1. The germ of a function φ at a point $a \in \mathbb{C}^n$ can be expressed in terms of the germs of the functions $f_1, \dots, f_n$ at a by means of integration if it satisfies the equation $d\varphi = \alpha$, where $\alpha = f_1 dx_1 + \dots + f_n dx_n$. For the germs of the given functions $f_1, \dots, f_n$, the germ φ exists if and only if the 1-form α is closed. The germ φ is thus defined up to additive constants.
2. The germ of a function φ at a point $a \in \mathbb{C}^n$ can be expressed in terms of the germs of the functions $f_1, \dots, f_n$ at a by means of the exponential and integrations if it satisfies the equation $d\varphi = \alpha\varphi$, where $\alpha = f_1 dx_1 + \dots + f_n dx_n$. For the germs of the given functions $f_1, \dots, f_n$, the germ φ exists if and only if the 1-form α is closed. The germ φ is thus defined up to multiplicative constants.
3. The germ of a function y at a point $a \in \mathbb{C}^n$ can be expressed in terms of the germs of the functions $f_0, \dots, f_k$ at a by means of a solution of an algebraic equation if the germ f_0 does not vanish and y satisfies the equation

$$f_0 y^k + f_1 y^{k-1} + \dots + f_k = 0.$$

Definition.

1. The class of germs of functions in $\mathbb{C}^n$ representable by quadratures (over the field of constants) is defined by the following choice: the germs of the basic functions are the germs of the constant functions (at every point of $\mathbb{C}^n$); the allowed operations are the arithmetic operations, integration, and raising to the power of the integral.

2. The class of germs of functions in $\mathbb{C}^n$ representable by generalized quadratures (over the field of constants) is defined by the following choice: the germs of the basic functions are the germs of the constant functions (at every point of $\mathbb{C}^n$); the allowed operations are the arithmetic operations, integration, raising to the power of the integral, and solving algebraic equations.

Notice that the above definitions can be translated almost literally to the case of abstract differential fields provided with n commutative differentiation operations $\partial/\partial x_1, \dots, \partial/\partial x_n$. In such a generalized form these definitions are due to Kolchin.

Now consider the class of germs of functions representable by quadratures and by generalized quadratures in spaces $\mathbb{C}^n$ of any dimension $n \geq 1$. Repeating the Liouville argument (cf. Theorem 1 in §A.2), it is not difficult to prove that the classes of germs of functions of several variables representable by quadratures and by generalized quadratures contain the germs of rational functions of several variables and the germs of all elementary basic functions; these classes of germs are closed with respect to composition. (Closure with respect to composition of a class of germs of functions representable by quadratures means the following: if $f_1, \dots, f_m$ are germs of functions representable by quadratures at a point $a \in \mathbb{C}^n$, and g is a germ of a function representable by quadratures at the point $b \in \mathbb{C}^m$, where $b = (f_1(a), \dots, f_m(a))$, then the germ $g(f_1, \dots, f_m)$ at $a \in \mathbb{C}^n$ is the germ of a function representable by quadratures.)

4.12 $\mathcal{SC}$ -germs

Does there exist a class of germs of functions of several variables sufficiently wide (containing the germs of functions representable by generalized quadratures and the germs of entire functions of several variables, and closed with respect to natural operations such as composition) for which the monodromy group is defined? In this section we define the class of $\mathcal{SC}$ -germs and state the theorem about the closure of this class with respect to natural operations; this gives an affirmative answer to the question posed. I discovered the class of $\mathcal{SC}$ -germs relatively recently: up to that time I believed the answer were negative.

In the case of functions of a single variable it was useful to introduce the class of $\mathcal{S}$ -functions. Let us start with a direct generalization of the class of $\mathcal{S}$ -functions to the multidimensional case.

A subspace $A \subset M$ in a connected k -dimensional analytic manifold M is said to be thin if there exists a countable set of open subsets $U_i \subset M$ and a countable set of analytic subspaces $A_i \subset U_i$ in these open subsets such that $A \subseteq \bigcup_i A_i$. An analytic multivalued function on M is called an $\mathcal{S}$ -function if the set of its singular points is thin. Let us make this definition more precise.

Two regular germs f_a and g_b , given at points a and b of M , are said to be equivalent if the germ g_b is obtained by regular continuation of the germ f_a along some curve. Every germ g_b equivalent to f_a is also called a regular germ of the analytic multivalued function f generated by f_a .

A point $b \in M$ is said to be singular for the germ f_a if there exists a curve $\gamma: [0, 1] \rightarrow M$, with $\gamma(0) = a$ and $\gamma(1) = b$, such that the germ f_a cannot be regularly continued along this curve, but for every t , $0 \leq t < 1$, this germ can be continued along the shortened curve $\gamma|_{[0, t]}: [0, t] \rightarrow M$. It is easy to see that the sets of singular points for equivalent germs coincide.

A regular germ is said to be an $\mathcal{S}$ -germ if the set of its singular points is thin. An analytic multivalued function is called an $\mathcal{S}$ -function if every one of its regular germs is an $\mathcal{S}$ -germ.

Remark. For functions of one complex variable we have already given two definitions of $\mathcal{S}$ -functions. The first is the definition above, and the second is given by the theorem in §A.5. These definitions obviously coincide.

For $\mathcal{S}$ -functions of several variables, the notions of monodromy group and monodromy pair carry over automatically.

Let us clarify why the multidimensional case is more complicated than the one-dimensional case.

Imagine the following situation. Let $f(x, y)$ be a multivalued analytic function of two variables with a set A of branch points, where $A \subset \mathbb{C}^2$ is an analytic curve in the complex plane. It can happen that at one of the points $a \in A$ there exists an analytic germ f_a of the multivalued analytic function f (by the definition of the set A of branch points, at a not every germ of f exists, yet some of them can exist). Now let $g_1(t)$ and $g_2(t)$ be two analytic functions of the complex variable t that define a mapping of the complex line $\mathbb{C}$ into the complex plane $\mathbb{C}^2$, such that the image of $\mathbb{C}$ is contained in A ; that is, $(g_1(t), g_2(t)) \in A$ for every $t \in \mathbb{C}$. Let b be the preimage of a under this mapping, so that $a = (g_1(b), g_2(b))$. What can we say about the multivalued analytic function on the complex line generated by the germ of $f(g_1, g_2)$ at b , obtained as the result of composing the germs of the rational functions g_1, g_2 at b with the germ of f at a ? It is clear that the analytic properties of this function depend essentially on the continuation of the germ f_a along the singular curve A .

Nothing like this may happen under the composition of functions of a single variable. Indeed, the set of singularities of an $\mathcal{S}$ -function of one variable consists of isolated points. If the image of complex space under an analytic mapping g is entirely contained in the set of singular points of a function f , then g is constant. It is evident that if g is constant, after f has been defined on its set of singular points, the function $f(g)$ turns out to be constant too.

In the one-dimensional case, for our purpose it suffices to study the values of an analytic multivalued function only in the complement of its singular points. In the multidimensional case we have to study the possibility of continuing those germs of functions which meet along their set of singularities (if, of course, the germ of the function is defined at an arbitrary point of the set of singularities). It happens that germs of multivalued functions sometimes are automatically continued along their set of singularities [24]; this allows us to overcome all difficulties.

An important role is played by the following definition.

Definition. A germ f_a of an analytic function at a point a of $\mathbb{C}^n$ is called an $\mathcal{SC}$ -germ if the following condition is satisfied. For every connected complex analytic manifold M , every analytic mapping $G: M \rightarrow \mathbb{C}^n$, and every preimage c of a , $G(c) = a$, there exists a thin subset $A \subset M$ such that, for every curve $\gamma: [0, 1] \rightarrow M$ beginning at c , $\gamma(0) = c$, and having no intersection with A except, at most, at its initial point; that is, $\gamma(t) \notin A$ for $t > 0$, the germ f_a can be analytically continued along the curve $G \circ \gamma: [0, 1] \rightarrow \mathbb{C}^n$.

Proposition. If the set of singular points of an $\mathcal{S}$ -function is an analytic set, then every germ of this function is an $\mathcal{SC}$ -germ.

This proposition follows directly from the results set out in [24].

It is evident that every $\mathcal{SC}$ -germ is the germ of an $\mathcal{S}$ -function. For the $\mathcal{SC}$ -germs, the notions of monodromy group and monodromy pair are thus well defined.

In the sequel we will need the notion of a holonomic system of linear differential equations. A system of N linear differential equations $L_j(y) = 0$, $j = 1, \dots, N$,

$$L_j(y) = \sum a_{i_1, \dots, i_n}^j \frac{\partial^{i_1 + \dots + i_n} y}{\partial x_1^{i_1} \dots \partial x_n^{i_n}} = 0,$$

for the unknown function y , whose coefficients $a_{i_1, \dots, i_n}^j$ are analytic functions of the n complex variables $x_1, \dots, x_n$, is said to be holonomic if its space of solutions has finite dimension.

Theorem on the closure of the class of $\mathcal{SC}$ -germs. The class of $\mathcal{SC}$ -germs in $\mathbb{C}^n$ is closed with respect to the following operations:

1. differentiation: if f is an $\mathcal{SC}$ -germ at a point $a \in \mathbb{C}^n$, then for every $i = 1, \dots, n$ the germ of the partial derivative $\partial f / \partial x_i$ is also an $\mathcal{SC}$ -germ at a ;
2. integration: if $df = f_1 dx_1 + \dots + f_n dx_n$, where $f_1, \dots, f_n$ are $\mathcal{SC}$ -germs at a point $a \in \mathbb{C}^n$, then f is also an $\mathcal{SC}$ -germ at a ;
3. composition with $\mathcal{SC}$ -germs of m variables: if $f_1, \dots, f_m$ are $\mathcal{SC}$ -germs at a point $a \in \mathbb{C}^n$ and g is an $\mathcal{SC}$ -germ at the point $(f_1(a), \dots, f_m(a)) \in \mathbb{C}^m$, then $g(f_1, \dots, f_m)$ is also an $\mathcal{SC}$ -germ at a ;
4. solutions of algebraic equations: if $f_0, \dots, f_k$ are $\mathcal{SC}$ -germs at a point $a \in \mathbb{C}^n$, the germ f_0 is not zero, and the germ y satisfies $f_0 y^k + f_1 y^{k-1} + \dots + f_k = 0$, then y is also an $\mathcal{SC}$ -germ at a ;
5. solutions of holonomic systems of linear differential equations: if the germ of a function y at a point $a \in \mathbb{C}^n$ satisfies the holonomic system of N linear differential equations

$$L_j(y) = \sum a_{i_1, \dots, i_n}^j \frac{\partial^{i_1 + \dots + i_n} y}{\partial x_1^{i_1} \dots \partial x_n^{i_n}} = 0,$$

all of whose coefficients $a_{i_1, \dots, i_n}^j$ are $\mathcal{SC}$ -germs at a , then y is also an $\mathcal{SC}$ -germ at a .

Corollary. If a germ of a function f can be obtained from the germs of single-valued $\mathcal{S}$ -functions having an analytic set of singular points by means of integrations, differentiations, meromorphic operations, compositions, solutions of algebraic equations, and solutions of holonomic systems of linear differential equations, then this germ of f is an $\mathcal{SC}$ -germ. In particular, a germ which is not an $\mathcal{SC}$ -germ cannot be represented by generalized quadratures.

4.13 Topological obstructions to representability by quadratures of functions of several variables

This section is dedicated to the topological obstructions to representability by quadratures and generalized quadratures of functions of several complex variables. These obstructions are analogous to those holding for functions of one variable considered in §§A.7–A.9.

Theorem 1. The class of all $\mathcal{SC}$ -germs in $\mathbb{C}^n$ having a soluble monodromy group is closed with respect to integration and differentiation. Moreover, this class is closed with respect to composition with $\mathcal{SC}$ -germs of m variables ($m \geq 1$) having soluble monodromy groups.

Result on quadratures. The monodromy group of any germ of a function f representable by quadratures is soluble. Moreover, every germ of a function representable by the germs of single-valued $\mathcal{S}$ -functions having an analytic set of singular points is also soluble by means of integrations, differentiations, and compositions.

Corollary. If the monodromy group of the algebraic equation

$$y^k + r_1 y^{k-1} + \cdots + r_k = 0,$$

in which the r_i are rational functions of n variables, is not soluble, then any germ of its solutions not only is not representable by radicals, but cannot be represented in terms of the germs of single-valued $\mathcal{S}$ -functions having an analytic set of singular points by means of integrations, differentiations, and compositions.

This corollary represents the strongest version of Abel's theorem.

Theorem 2. The class of all $\mathcal{SC}$ -germs in $\mathbb{C}^n$ having an almost soluble monodromy pair is closed with respect to integration, differentiation, and solutions of algebraic equations. Moreover, this class is closed with respect to composition with $\mathcal{SC}$ -germs of m variables ($m \geq 1$) having an almost soluble monodromy pair.

Result on generalized quadratures. The monodromy pair of a germ of a function f representable by generalized quadratures is almost soluble. Moreover, the monodromy pair of every germ of a function f representable in terms of the germs of single-valued $\mathcal{S}$ -functions having an analytic set of singular points by means of integrations, differentiations, compositions, and solutions of algebraic equations is also almost soluble.

4.14 Topological obstruction to the solvability of holonomic systems of linear differential equations

4.14.1 The monodromy group of a holonomic system of linear differential equations

Consider a holonomic system of N differential equations $L_j(y) = 0$, $j = 1, \dots, N$,

$$L_j(y) = \sum a_{i_1, \dots, i_n}^j \frac{\partial^{i_1 + \dots + i_n} y}{\partial x_1^{i_1} \cdots \partial x_n^{i_n}} = 0,$$

where y is the unknown function and the coefficients $a_{i_1, \dots, i_n}^j$ are rational functions of the n complex variables $x_1, \dots, x_n$.

It is known that for any holonomic system there exists a singular algebraic surface Σ in $\mathbb{C}^n$ with the following properties. Every solution of the system can be analytically continued along an arbitrary curve avoiding Σ . Let

V be the finite-dimensional space of solutions of a holonomic system near a point x_0 which lies outside Σ . Consider an arbitrary curve $\gamma(t)$ in $\mathbb{C}^n$ with initial point x_0 , not crossing Σ . The solutions of the system can be analytically continued along γ , remaining solutions of the system. Consequently, to every curve γ of this type there corresponds a linear transformation M_γ of the solution space V into itself. The totality of the linear transformations M_γ corresponding to all curves γ forms a group, called the monodromy group of the holonomic system.

Kolchin generalized Picard–Vessiot theory to holonomic systems of differential equations. From Kolchin’s theory we obtain two corollaries concerning solvability by quadratures of holonomic systems of differential equations. As in the one-dimensional case, a holonomic system is said to be regular if, when approaching the singular set Σ and infinity, its solutions grow at most as some power.

Theorem 1. A regular holonomic system of linear differential equations is soluble by quadratures and by generalized quadratures if its monodromy group is, respectively, soluble and almost soluble.

Kolchin’s theory proves at the same time two results.

1. If the monodromy group of a regular holonomic system of linear differential equations is soluble (almost soluble), then this system is solvable by quadratures (by generalized quadratures).
2. If the monodromy group of a regular holonomic system of linear differential equations is not soluble (is not almost soluble), then this system is not solvable by quadratures (by generalized quadratures).

Our theorem makes the second result stronger.

Theorem 2. If the monodromy group of a holonomic system of linear differential equations is not soluble (is not almost soluble), then every germ of almost all solutions of this system cannot be expressed in terms of the germs of single-valued $\mathcal{S}$ -functions having an analytic set of singular points by means of compositions, meromorphic operations, integrations, and differentiations (by means of compositions, meromorphic operations, integrations, differentiations, and solutions of algebraic equations).

4.14.2 Holonomic systems of linear differential equations with small coefficients

Consider a completely integrable system of linear differential equations of the form

$$dy = Ay, \tag{A.17}$$

where $y = (y_1, \dots, y_n)$ is the unknown vector function and A is an $n \times n$ matrix consisting of differential 1-forms with rational coefficients in $\mathbb{C}^n$, satisfying the complete integrability condition $dA + A \wedge A = 0$ and having the form

$$A = \sum_{i=1}^k A_i \frac{dl_i}{l_i},$$

where the A_i are constant matrices and the l_i are linear non-homogeneous functions in $\mathbb{C}^n$.

If the matrices A_i can be simultaneously put into triangular form, then system (A.17), like every completely integrable triangular system, is solvable by quadratures. There undoubtedly exist integrable non-triangular systems. However, when the matrices A_i are sufficiently small, such systems do not exist. More precisely, we have proved the following theorem.

Theorem 3. A completely integrable non-triangular system (A.17), with the moduli of the matrices A_i sufficiently small, is strictly non-solvable; that is, its solution cannot be represented even through the germs of all single-valued S -functions having an analytic set of singular points by means of compositions, meromorphic operations, integrations, differentiations, and solutions of algebraic equations.

The proof of this theorem uses a multidimensional variation of the Lappo-Danilevskij theorem [28].